Topological Energy Condensate Theory (TECT): 5

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I. GENERIC NONVANISHING OF THE SHELL-PROTECTED DIRAC COUPLINGS

Let $\mathcal{V}$ denote the finite-dimensional real vector space of all symmetry-allowed microscopic enriched tensors,

$$\mathsf{N} = \{\mathsf{N}^{a}_{ij}\},$$

that survive the microscopic linearization around the condensed TECT background. Let $G_* \in \mathbb{R}^3 \setminus \{0\}$ be the selected shell wavevector, and let

$$e^{\alpha} = (e^{\alpha}_i)_{i=1}^{3}, \qquad \alpha = 1, 2,$$

be the two reduced shell directions entering the enriched block.

For each α, define the reduced shell coefficient

$$C^{\alpha}_j(\mathsf{N}) := e^{\alpha}_i \, \mathsf{N}^{a_\alpha}_{ij},$$

where a_α labels the internal matrix channel used in the corresponding reduced operator. The Dirac-protecting shell contraction is the linear functional

$$L_\alpha(\mathsf{N}) := C^{\alpha}(\mathsf{N}) \cdot G_* = G_{*j} \, e^{\alpha}_i \, \mathsf{N}^{a_\alpha}_{ij}.$$

We must determine whether

$$L_1(\mathsf{N}) \neq 0, \qquad L_2(\mathsf{N}) \neq 0$$

holds generically on $\mathcal{V}$.

II. FINITE-DIMENSIONAL REDUCTION

The map

$$\Pi_\alpha : \mathcal{V} \to \mathbb{R}^3, \qquad \Pi_\alpha(\mathsf{N}) = C^{\alpha}(\mathsf{N}),$$

is linear, hence $L_\alpha = G_* \cdot \Pi_\alpha$ is also linear on $\mathcal{V}$. Therefore the shell-survival problem is not an infinite-dimensional spectral problem but a finite-dimensional question about linear functionals on $\mathcal{V}$.

Write

$$\dim \mathcal{V} = d < \infty.$$

Choose a basis $\{E_A\}_{A=1}^{d}$ of $\mathcal{V}$, so that every allowed tensor can be written as

$$\mathsf{N} = \sum_{A=1}^{d} x_A E_A, \qquad x = (x_1, \ldots, x_d) \in \mathbb{R}^d.$$

Then

$$L_\alpha(\mathsf{N}) = \sum_{A=1}^{d} x_A \, \ell_{\alpha A}, \qquad \ell_{\alpha A} := G_{*j} e^{\alpha}_i (E_A)^{a_\alpha}_{ij}.$$

Hence L_α is represented by a coefficient vector

$$\ell_\alpha = (\ell_{\alpha 1}, \ldots, \ell_{\alpha d}) \in \mathbb{R}^d.$$

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III. ZERO SETS OF THE REDUCED COUPLINGS

Assume first that $L_\alpha \not\equiv 0$ on $\mathcal{V}$, equivalently,

$$\ell_\alpha \neq 0.$$

Then its zero set is

$$Z_\alpha := \{\mathsf{N} \in \mathcal{V} : L_\alpha(\mathsf{N}) = 0\}.$$

In coordinates this is

$$Z_\alpha = \left\{ x \in \mathbb{R}^d : \sum_{A=1}^{d} x_A \ell_{\alpha A} = 0 \right\}.$$

Since $\ell_\alpha \neq 0$, this is a proper affine-linear hyperplane through the origin, and therefore

$$\mathrm{codim}_{\mathcal{V}}(Z_\alpha) = 1.$$

In particular:

- Z_α is closed,
- Z_α has empty interior in $\mathcal{V}$,
- Z_α has Lebesgue measure zero in $\mathcal{V}$.

Therefore, if $L_\alpha \not\equiv 0$, then

$$\mathcal{U}_\alpha := \mathcal{V} \setminus Z_\alpha = \{\mathsf{N} \in \mathcal{V} : L_\alpha(\mathsf{N}) \neq 0\}$$

is open, dense, and full measure in $\mathcal{V}$.

IV. SINGLE-CHANNEL GENERICITY THEOREM

Theorem. Let $\mathcal{V}$ be the finite-dimensional space of symmetry-allowed microscopic tensors, and let

$$L_\alpha(\mathsf{N}) = G_{*j} e_i^\alpha \mathsf{N}_{ij}^{a_\alpha}$$

be the corresponding reduced shell contraction. If $L_\alpha \not\equiv 0$ on $\mathcal{V}$, then

$$L_\alpha(\mathsf{N}) \neq 0$$

for an open dense full-measure subset of $\mathcal{V}$.

Proof. Since L_α is a nonzero linear functional on the finite-dimensional vector space $\mathcal{V}$, its kernel is a codimension-one linear hyperplane:

$$\ker L_\alpha = Z_\alpha.$$

Therefore its complement

$$\mathcal{V} \setminus \ker L_\alpha$$

is open and dense. Since every proper hyperplane has measure zero in finite dimensions, the complement also has full measure. Hence $L_\alpha(\mathsf{N}) \neq 0$ generically. $\square$

V. SIMULTANEOUS GENERICITY FOR THE TWO DIRAC CHANNELS

Define the bad sets

$$Z_1 = \ker L_1, \qquad Z_2 = \ker L_2,$$

and the good set

$$\mathcal{U} := \{\mathsf{N} \in \mathcal{V} : L_1(\mathsf{N}) \neq 0,\ L_2(\mathsf{N}) \neq 0\} = \mathcal{V} \setminus (Z_1 \cup Z_2).$$

Assume

$$L_1 \not\equiv 0, \qquad L_2 \not\equiv 0.$$

Then each Z_α is a proper hyperplane. Their union

$$Z_1 \cup Z_2$$

is closed, nowhere dense, and measure zero. Hence:

Corollary. If neither reduced contraction is identically forbidden by symmetry, then

$$C^1(\mathsf{N}) \cdot G_* \neq 0, \qquad C^2(\mathsf{N}) \cdot G_* \neq 0$$

holds simultaneously for an open dense full-measure subset of symmetry-allowed microscopic tensors $\mathsf{N} \in \mathcal{V}$.

Proof. Each bad condition $L_\alpha = 0$ defines a proper hyperplane in $\mathcal{V}$. A finite union of proper hyperplanes remains nowhere dense and measure zero. Therefore its complement is open dense and full measure. □

VI. SYMMETRY-FORBIDDEN VERSUS GENERIC FAILURE

The preceding theorem proves a sharp dichotomy.

Case I: symmetry-forbidden vanishing. If for some α,

$$L_\alpha(\mathsf{N}) = 0 \qquad \text{for all } \mathsf{N} \in \mathcal{V},$$

then the corresponding shell coupling is identically absent. This is not a generic accident but an exact representation-theoretic selection rule.

Case II: generic nonvanishing. If for some α there exists even one allowed tensor $\mathsf{N}^{(\alpha)} \in \mathcal{V}$ such that

$$L_\alpha\big(\mathsf{N}^{(\alpha)}\big) \neq 0,$$

then $L_\alpha \not\equiv 0$, and therefore

$$L_\alpha(\mathsf{N}) \neq 0$$

for generic $\mathsf{N} \in \mathcal{V}$.

Thus the only obstruction to generic shell survival is an exact symmetry prohibition. Absent such a prohibition, the vanishing set is nongeneric.

VII. DIRAC-CLOSURE CONSEQUENCE

Consider the reduced shell Hamiltonian

$$H_{\mathrm{red}}(p) = v_\parallel p_\parallel \Gamma^3 + \sum_{\alpha=1}^{2} \lambda_\alpha\, p_\alpha\, \Gamma^\alpha + m_{\mathrm{B}}\, \Gamma^0,$$

where

$$\lambda_\alpha \propto C^\alpha(\mathsf{N}) \cdot G_*.$$

If

$$m_{\mathrm{B}} = 0, \qquad v_{\parallel} \neq 0, \qquad C^1(\mathsf{N}) \cdot G_* \neq 0, \qquad C^2(\mathsf{N}) \cdot G_* \neq 0,$$

then all three Dirac velocities are nonzero and the reduced block is a genuine anisotropic massless Dirac operator:

$$H_{\mathrm{red}}(p) = \sum_{i=1}^{3} v_i p_i \Gamma^i, \qquad \{\Gamma^i, \Gamma^j\} = 2\delta^{ij}.$$

Therefore:

Dirac Genericity Principle. Once the Bragg mass is tuned away and the microscopic enriched tensors are not symmetry-forbidden, emergent Dirac propagation is generic in the symmetry-allowed tensor space.

VIII. WHAT HAS BEEN PROVEN AND WHAT REMAINS OPEN

The theorem proved above is completely rigorous at the finite-dimensional level:

- the reduced shell couplings are linear functionals of the microscopic tensors,
- if not identically zero, their vanishing sets are codimension-one hyperplanes,
- hence simultaneous nonvanishing is generic.

What remains open is not the genericity argument itself, but the representation-theoretic input:

> Does the full microscopic TECT linearization admit at least one symmetry-allowed tensor N for which
>
> $$G_{*j} e_i^{\alpha} \mathsf{N}_{ij}^{a_\alpha} \neq 0 \quad (\alpha = 1, 2)?$$

If the answer is yes, the genericity theorem above immediately closes the shell-coupling part of Dirac emergence. If the answer is no, then the obstruction is an exact symmetry selection rule, and the enriched channel must be modified.

IX. REPRESENTATION-THEORETIC REFORMULATION OF THE SHELL TEST

We now sharpen the previous genericity theorem into an exact representation-theoretic criterion.

Let

$$\mathcal{W} = \mathbb{R}^3_{\mathrm{out}} \otimes \mathbb{R}^3_{\mathrm{grad}} \otimes \mathcal{U}_{\mathrm{int}}$$

be the full finite-dimensional ambient tensor space of microscopic enriched couplings

$$\mathsf{N} = \{\mathsf{N}_{ij}^a\}.$$

Here i is the output spatial index, j is the gradient index, and a labels the internal matrix channel.

Let $\mathcal{V} \subset \mathcal{W}$ be the symmetry-allowed subspace that survives the microscopic linearization around the condensed TECT background. For each Dirac channel $\alpha = 1, 2$, define the covector

$$T_{ija}^{(\alpha)} := e_i^{\alpha}\, G_{*j}\, u_a^{(\alpha)},$$

where $u^{(\alpha)} \in \mathcal{U}_{\mathrm{int}}^*$ selects the internal channel entering the α-th reduced block. Then the shell contraction is

$$L_\alpha(\mathsf{N}) = \langle T^{(\alpha)}, \mathsf{N} \rangle = e_i^{\alpha} G_{*j} u_a^{(\alpha)} \mathsf{N}_{ij}^a.$$

Thus the question

$$L_\alpha|_{\mathcal{V}} \equiv 0\ ?$$

is exactly the question whether the covector $T^{(\alpha)}$ has nonzero overlap with the allowed tensor space $\mathcal{V}$.

X. ORTHOGONAL PROJECTION CRITERION

Fix any H-invariant inner product on $\mathcal{W}$, where H is the residual symmetry group of the condensed background together with the chosen shell data. Let

$$P_{\mathcal{V}} : \mathcal{W} \to \mathcal{V}$$

denote the orthogonal projector onto $\mathcal{V}$.

Proposition. For each α,

$$L_\alpha|_{\mathcal{V}} \equiv 0 \quad \iff \quad P_{\mathcal{V}} T^{(\alpha)} = 0.$$

Proof. For any $\mathsf{N} \in \mathcal{V}$,

$$L_\alpha(\mathsf{N}) = \langle T^{(\alpha)}, \mathsf{N} \rangle = \langle P_{\mathcal{V}} T^{(\alpha)}, \mathsf{N} \rangle,$$

because $\mathsf{N} \in \mathcal{V}$ and $P_{\mathcal{V}}$ is the orthogonal projection. Therefore $L_\alpha(\mathsf{N}) = 0$ for all $\mathsf{N} \in \mathcal{V}$ if and only if

$$\langle P_{\mathcal{V}} T^{(\alpha)}, \mathsf{N} \rangle = 0 \qquad \forall \mathsf{N} \in \mathcal{V}.$$

Since $P_{\mathcal{V}} T^{(\alpha)} \in \mathcal{V}$, this holds if and only if $P_{\mathcal{V}} T^{(\alpha)} = 0$. $\square$

XI. EXACT DICHOTOMY

The previous proposition gives a complete and exact criterion.

Theorem. For each $\alpha = 1, 2$, exactly one of the following holds:

1. **Symmetry-forbidden case:**

$$P_{\mathcal{V}} T^{(\alpha)} = 0,$$

   equivalently

$$L_\alpha(\mathsf{N}) = 0 \qquad \forall \mathsf{N} \in \mathcal{V}.$$

2. **Generic nonvanishing case:**

$$P_{\mathcal{V}} T^{(\alpha)} \neq 0,$$

   equivalently $L_\alpha|_{\mathcal{V}}$ is a nonzero linear functional, hence

$$L_\alpha(\mathsf{N}) \neq 0$$

   for an open dense full-measure subset of $\mathcal{V}$.

Proof. The equivalence with the projection criterion has already been proved. If $P_{\mathcal{V}} T^{(\alpha)} \neq 0$, then $L_\alpha|_{\mathcal{V}}$ is a nonzero linear functional on the finite-dimensional space $\mathcal{V}$, so its zero set is a proper hyperplane. Hence its complement is open dense and full measure. If $P_{\mathcal{V}} T^{(\alpha)} = 0$, then L_α vanishes identically on $\mathcal{V}$. $\square$

XII. IRREDUCIBLE DECOMPOSITION CRITERION

Let H denote the residual symmetry group acting on $\mathcal{W}$. Decompose $\mathcal{W}$ into H-isotypic components:

$$\mathcal{W} = \bigoplus_\lambda \mathcal{W}_\lambda.$$

Likewise,

$$\mathcal{V} = \bigoplus_\lambda \mathcal{V}_\lambda, \qquad T^{(\alpha)} = \sum_\lambda T_\lambda^{(\alpha)}.$$

Since distinct isotypic components are orthogonal under an H-invariant inner product, we obtain:

Corollary.

$$P_{\mathcal{V}} T^{(\alpha)} \neq 0 \quad \iff \quad \exists\, \lambda \text{ such that } \mathcal{V}_\lambda \neq 0 \text{ and } T_\lambda^{(\alpha)} \neq 0.$$

In words: the shell coupling survives if and only if the target covector $T^{(\alpha)}$ shares at least one common residual-symmetry sector with the symmetry-allowed tensor space.

This is the exact representation-theoretic reduction of the Dirac shell test.

XIII. REYNOLDS-OPERATOR FORMULA

If the symmetry-allowed space is precisely the H-invariant subspace,

$$\mathcal{V} = \mathcal{W}^H,$$

then the projection $P_{\mathcal{V}}$ is given by the Reynolds average

$$P_{\mathcal{V}} X = \frac{1}{|H|} \sum_{h \in H} h \cdot X$$

for finite H, or by the Haar average for compact H.

Therefore

$$P_{\mathcal{V}} T^{(\alpha)} = \frac{1}{|H|} \sum_{h \in H} h \cdot T^{(\alpha)}.$$

Hence the exact shell-survival condition becomes

$$\frac{1}{|H|} \sum_{h \in H} h \cdot T^{(\alpha)} \neq 0.$$

This gives a concrete calculational test: one does not need the full microscopic spectrum, only the residual symmetry action on the tensor space.

XIV. EXPLICIT SUFFICIENT SEED CRITERION

We now extract a strong practical sufficient condition.

Suppose there exists a tensor

$$\mathsf{N}^{(\alpha)}_{\mathrm{seed}} \in \mathcal{V}$$

such that

$$\langle T^{(\alpha)}, \mathsf{N}^{(\alpha)}_{\mathrm{seed}} \rangle \neq 0.$$

Then automatically

$$P_{\mathcal{V}} T^{(\alpha)} \neq 0,$$

because otherwise $T^{(\alpha)}$ would be orthogonal to all of $\mathcal{V}$, contradicting the assumed nonzero overlap.

Hence:

Seed Criterion. If one can explicitly construct even a single symmetry-allowed microscopic tensor with nonzero shell overlap, then the entire α-channel is generically nonvanishing.

This reduces the microscopic problem from a universal classification of all allowed tensors to the construction of a single admissible witness.

XV. CANONICAL WITNESS TENSORS

We now exhibit canonical candidate witnesses in the ambient tensor space.

Fix $\alpha \in \{1, 2\}$, and define

$$\mathsf{X}^{(\alpha)}_{ija} = e^{\alpha}_i G_{*j} v_a,$$

where $v_a \in \mathcal{U}_{\mathrm{int}}$ satisfies

$$u^{(\alpha)}_a v_a \neq 0.$$

Then

$$L_\alpha(\mathsf{X}^{(\alpha)}) = e_i^\alpha G_{*j} u_a^{(\alpha)} \, e_i^\alpha G_{*j} v_a = (e^\alpha \cdot e^\alpha)(G_* \cdot G_*)(u^{(\alpha)} \cdot v).$$

Assuming normalized e^α,

$$e^\alpha \cdot e^\alpha = 1,$$

this becomes

$$L_\alpha(\mathsf{X}^{(\alpha)}) = |G_*|^2 (u^{(\alpha)} \cdot v),$$

which is nonzero whenever $u^{(\alpha)} \cdot v \neq 0$.

Similarly, the symmetrized tensor

$$\mathsf{Y}_{ija}^{(\alpha)} = \left(e_i^\alpha G_{*j} + G_{*i} e_j^\alpha\right) v_a$$

satisfies

$$L_\alpha(\mathsf{Y}^{(\alpha)}) = |G_*|^2 (u^{(\alpha)} \cdot v),$$

because

$$e_i^\alpha G_{*j} G_{*i} e_j^\alpha = (e^\alpha \cdot G_*)(G_* \cdot e^\alpha) = 0.$$

Likewise, the antisymmetrized tensor

$$\mathsf{Z}_{ija}^{(\alpha)} = \left(e_i^\alpha G_{*j} - G_{*i} e_j^\alpha\right) v_a$$

also satisfies

$$L_\alpha(\mathsf{Z}^{(\alpha)}) = |G_*|^2 (u^{(\alpha)} \cdot v) \neq 0.$$

Therefore the obstruction to shell survival is not algebraic: there exist simple rank-one, symmetric, and antisymmetric ambient tensors with strictly nonzero shell overlap. Only symmetry projection can kill them.

XVI. SYMMETRY-PROJECTION CONSEQUENCE

Let

$$\mathsf{W}^{(\alpha)} \in \{\mathsf{X}^{(\alpha)}, \mathsf{Y}^{(\alpha)}, \mathsf{Z}^{(\alpha)}\}.$$

If

$$P_{\mathcal{V}} \mathsf{W}^{(\alpha)} \neq 0,$$

then

$$L_\alpha\left(P_{\mathcal{V}} \mathsf{W}^{(\alpha)}\right) = \langle T^{(\alpha)}, P_{\mathcal{V}} \mathsf{W}^{(\alpha)} \rangle = \langle P_{\mathcal{V}} T^{(\alpha)}, \mathsf{W}^{(\alpha)} \rangle \neq 0$$

for at least one choice of witness, and hence

$$P_{\mathcal{V}} T^{(\alpha)} \neq 0.$$

Thus the problem is now completely reduced to a single residual-symmetry projection test:

$$\boxed{\text{Does the residual-symmetry projector annihilate all canonical witness tensors or not?}}$$

If the answer is no for each $\alpha = 1, 2$, then both Dirac shell couplings survive generically.

XVII. SIMULTANEOUS TWO-CHANNEL CRITERION

Define

$$T^{(1)}_{ija} = e^1_i G_{*j} u^{(1)}_a, \qquad T^{(2)}_{ija} = e^2_i G_{*j} u^{(2)}_a.$$

Then the two-channel shell survival condition is exactly

$$P_{\mathcal{V}} T^{(1)} \neq 0, \qquad P_{\mathcal{V}} T^{(2)} \neq 0.$$

A sufficient condition is the existence of two allowed witness tensors

$$\mathsf{N}^{(1)}, \mathsf{N}^{(2)} \in \mathcal{V}$$

such that

$$\langle T^{(1)}, \mathsf{N}^{(1)} \rangle \neq 0, \qquad \langle T^{(2)}, \mathsf{N}^{(2)} \rangle \neq 0.$$

Then, by the genericity theorem established previously,

$$C^1(\mathsf{N}) \cdot G_* \neq 0, \qquad C^2(\mathsf{N}) \cdot G_* \neq 0$$

hold simultaneously for an open dense full-measure subset of $\mathcal{V}$.

XVIII. DIRAC CLOSURE UP TO SPECTRAL ISOLATION

Combining the previous steps, we have now reduced the Dirac-emergence problem to the following exact statement.
Theorem. Assume:

1. the Bragg mass is absent,

$$\widehat{U}(G_*) = 0;$$

2. the longitudinal velocity is nonzero,

$$v_\parallel \neq 0;$$

3. for each $\alpha = 1, 2$,

$$P_{\mathcal{V}} T^{(\alpha)} \neq 0;$$

4. the enriched shell block is spectrally isolated.

Then the reduced enriched shell Hamiltonian is a genuine massless Dirac operator,

$$H_{\mathrm{red}}(p) = \sum_{i=1}^{3} v_i p_i \Gamma^i, \qquad \{\Gamma^i, \Gamma^j\} = 2\delta^{ij},$$

with all $v_i \neq 0$.

Proof. Condition (1) removes the Bragg mass. Condition (2) supplies the longitudinal kinetic term. Condition (3), together with the genericity theorem, implies that both transverse enriched shell couplings are nonzero on an open dense full-measure subset of $\mathcal{V}$. Hence all three Dirac coefficients are nonzero. Condition (4) ensures that the reduced 4×4 block is dynamically closed and not destroyed by mixing with remote states. Therefore the reduced Hamiltonian is an anisotropic but genuine massless Dirac operator. □

XIX. WHAT IS NOW FULLY CLOSED

The following part is now mathematically closed:

- the shell-coupling test is exactly equivalent to the projection criterion

$$P_{\mathcal{V}}T^{(\alpha)} \neq 0;$$

- generic shell survival follows immediately once this projection is nonzero;
- explicit ambient witness tensors with nonzero shell overlap are known in closed form;
- therefore the only remaining microscopic question is whether the residual symmetry annihilates all such witnesses.

This is no longer an open-ended dynamical problem. It is a finite representation-theoretic yes/no test.

XX. REMAINING FRONTIER

The only unresolved microscopic input is now:

$$\boxed{P_{\mathcal{V}}T^{(1)} \neq 0 \quad \text{and} \quad P_{\mathcal{V}}T^{(2)} \neq 0 \text{ for the actual TECT residual symmetry.}}$$

Once this is checked, the shell-coupling part of Dirac emergence is completely closed. After that, the only remaining frontier is spectral isolation of the enriched shell block.

XXI. RESIDUAL-SYMMETRY TEST ALONG A SELECTED SHELL DIRECTION

We now perform the representation test at the level of the little group of the selected shell vector G_*.

Let

$$\hat{g} := \frac{G_*}{|G_*|} \in S^2,$$

and choose an orthonormal frame

$$\{e^1, e^2, \hat{g}\}$$

with

$$e^\alpha \cdot \hat{g} = 0, \qquad e^\alpha \cdot e^\beta = \delta^{\alpha\beta}, \qquad \alpha, \beta = 1, 2.$$

Let H_* denote the residual symmetry subgroup that preserves the selected shell direction $\hat{g}$. At the continuum level, the relevant symmetry is the axial group $O(2)$ acting on the transverse plane

$$\Pi_\perp := \hat{g}^\perp = \mathrm{span}\{e^1, e^2\}.$$

For the discrete BCC background, H_* is replaced by the corresponding finite little group (typically C_{nv} or D_n), but the representation-theoretic structure below is unchanged.

XXII. AXIAL DECOMPOSITION OF SPATIAL TENSOR INDICES

The spatial vector representation decomposes under H_* as

$$\mathbb{R}^3 = \mathbb{R}\,\hat{g} \oplus \Pi_\perp.$$

Hence a rank-two spatial tensor decomposes as

$$\mathbb{R}^3_{\rm out} \otimes \mathbb{R}^3_{\rm grad} = (\hat{g} \otimes \hat{g}) \oplus (\hat{g} \otimes \Pi_\perp) \oplus (\Pi_\perp \otimes \hat{g}) \oplus (\Pi_\perp \otimes \Pi_\perp).$$

The first summand is a scalar under H_*, the last contains scalar, antisymmetric, and spin-two pieces, while the two mixed terms

$$\hat{g} \otimes \Pi_\perp, \qquad \Pi_\perp \otimes \hat{g}$$

each transform as the defining two-dimensional transverse representation of the little group.

For the Dirac shell test, the relevant target covectors are

$$T^{(\alpha)}_{ija} = e^\alpha_i \, G_{*j} \, u^{(\alpha)}_a = |G_*| \, e^\alpha_i \, \hat{g}_j \, u^{(\alpha)}_a.$$

Thus the spatial part of $T^{(\alpha)}$ lies in

$$\Pi_\perp \otimes \hat{g}.$$

XXIII. THE TRANSVERSE DOUBLET

Define the spatial witness tensors

$$W^{(\alpha)}_{ij} := e^\alpha_i \hat{g}_j, \qquad \alpha = 1, 2.$$

Under any $h \in H_*$, one has

$$h \cdot \hat{g} = \hat{g}, \qquad h \cdot e^\alpha = R(h)^\alpha{}_\beta e^\beta,$$

where $R(h) \in O(2)$ is the transverse 2×2 matrix. Therefore

$$h \cdot W^{(\alpha)} = R(h)^\alpha{}_\beta W^{(\beta)}.$$

Hence the two-dimensional space

$$\mathcal{W}_{\rm mix} := \mathrm{span}\{W^{(1)}, W^{(2)}\} = \Pi_\perp \otimes \hat{g}$$

is itself an irreducible transverse doublet (or the corresponding irreducible two-dimensional representation for the discrete little group).

Exactly the same conclusion holds for the symmetric and antisymmetric combinations

$$W^{(\alpha)}_{\pm,ij} := e^\alpha_i \hat{g}_j \pm \hat{g}_i e^\alpha_j.$$

Indeed,

$$h \cdot W^{(\alpha)}_\pm = R(h)^\alpha{}_\beta W^{(\beta)}_\pm.$$

Thus all three families

$$W^{(\alpha)}, \qquad W^{(\alpha)}_+, \qquad W^{(\alpha)}_-$$

transform as the same transverse doublet of H_*.

XXIV. INTERNAL DOUBLET CRITERION

Let $\mathcal{U}_{\rm int}$ denote the internal-channel space. Assume that the two Dirac channels are selected by internal covectors

$$u^{(\alpha)} \in \mathcal{U}^*_{\rm int}, \qquad \alpha = 1, 2,$$

which transform as a transverse doublet under H_*:

$$h \cdot u^{(\alpha)} = R(h)^\alpha{}_\beta u^{(\beta)}.$$

Equivalently, there exists a two-dimensional H_*-subrepresentation

$$\mathcal{U}_D = \mathrm{span}\{u^{(1)}, u^{(2)}\} \subset \mathcal{U}_{\mathrm{int}}^*$$

isomorphic to the defining transverse representation.

Then the tensor product

$$\mathcal{W}_{\mathrm{mix}} \otimes \mathcal{U}_D$$

contains a canonical invariant line generated by

$$\Xi := \sum_{\alpha=1}^{2} W^{(\alpha)} \otimes u^{(\alpha)}.$$

Indeed, under $h \in H_*$,

$$h \cdot \Xi = \sum_{\alpha} R(h)^{\alpha}{}_{\beta} W^{(\beta)} \otimes R(h)^{\alpha}{}_{\gamma} u^{(\gamma)} = \sum_{\beta,\gamma} \big(R(h)^T R(h)\big)_{\beta\gamma} W^{(\beta)} \otimes u^{(\gamma)} = \Xi.$$

Thus Ξ is H_*-invariant.

The same construction yields invariant tensors

$$\Xi_{\pm} := \sum_{\alpha=1}^{2} W_{\pm}^{(\alpha)} \otimes u^{(\alpha)}.$$

XXV. EXPLICIT NONVANISHING OF THE SHELL FUNCTIONAL

Consider the microscopic tensor

$$\mathsf{N}_{ija}^{\mathrm{inv}} = \sum_{\alpha=1}^{2} e_i^{\alpha} \hat{g}_j \, v_a^{(\alpha)},$$

where $v^{(\alpha)} \in \mathcal{U}_{\mathrm{int}}$ is dual to $u^{(\alpha)}$ in the sense that

$$u^{(\alpha)}(v^{(\beta)}) = \delta^{\alpha\beta}.$$

Then

$$L_{\alpha}(\mathsf{N}^{\mathrm{inv}}) = e_i^{\alpha} G_{*j} u_a^{(\alpha)} \sum_{\beta=1}^{2} e_i^{\beta} \hat{g}_j v_a^{(\beta)} = |G_*| \sum_{\beta} (e^{\alpha} \cdot e^{\beta})(\hat{g} \cdot \hat{g}) u^{(\alpha)}(v^{(\beta)}).$$

Therefore

$$L_{\alpha}(\mathsf{N}^{\mathrm{inv}}) = |G_*| \, u^{(\alpha)}(v^{(\alpha)}) = |G_*| \neq 0.$$

Hence if $\mathsf{N}^{\mathrm{inv}} \in \mathcal{V}$, then the shell contraction in channel α is explicitly nonzero.

Since Ξ is H_*-invariant, the Reynolds projection of the ambient witness is nonzero:

$$P_{\mathcal{V}} T^{(\alpha)} \neq 0.$$

XXVI. AXIAL SURVIVAL THEOREM

Theorem. Assume:

1. the selected shell vector $G_* \neq 0$ defines a little group H_* preserving $\hat{g} = G_*/|G_*|$;

2. the allowed microscopic tensor space $\mathcal{V}$ contains the H_*-invariant sector built from

$$\Pi_\perp \otimes \hat{g} \otimes \mathcal{U}_D,$$

where $\mathcal{U}_D \subset \mathcal{U}^*_{\mathrm{int}}$ is a transverse doublet;

3. the two Dirac channels $\alpha = 1, 2$ are selected by this same internal doublet.

Then

$$P_{\mathcal{V}} T^{(1)} \neq 0, \qquad P_{\mathcal{V}} T^{(2)} \neq 0.$$

Consequently,

$$C^1(\mathsf{N}) \cdot G_* \neq 0, \qquad C^2(\mathsf{N}) \cdot G_* \neq 0$$

hold simultaneously for an open dense full-measure subset of $\mathcal{V}$.

Proof. By assumption, the spatial mixed sector $\Pi_\perp \otimes \hat{g}$ and the internal channel space $\mathcal{U}_D$ carry isomorphic two-dimensional representations of H_*. Their tensor product therefore contains an invariant line generated by Ξ. Hence the H_*-projector onto $\mathcal{V}$ does not annihilate the target covectors $T^{(\alpha)}$. Equivalently,

$$P_{\mathcal{V}} T^{(\alpha)} \neq 0$$

for $\alpha = 1, 2$. The generic nonvanishing statement then follows from the finite-dimensional hyperplane theorem proved previously. □

XXVII. INTERPRETATION FOR THE BCC SHELL

For a BCC reciprocal star, choose for example

$$\hat{g} = \frac{1}{\sqrt{2}}(1, 1, 0).$$

Then the little group is the subgroup of cubic symmetry preserving this axis. Its action on the plane orthogonal to $\hat{g}$ is again a two-dimensional transverse representation. Therefore the previous theorem applies without change: one only replaces the continuum $O(2)$ by the corresponding discrete little-group representation.

Hence the shell-coupling obstruction is not generic. It can occur only if the internal TECT channel fails to provide the matching transverse doublet, or if the microscopic linearization projects that sector out exactly.

XXVIII. MINIMAL PRACTICAL SUFFICIENT CONDITION

A particularly strong sufficient condition is the following.

Corollary. Suppose the microscopic enriched linearization contains a term of the form

$$\mathsf{N}^a_{ij} = A\, e^1_i \hat{g}_j\, v^a_1 + B\, e^2_i \hat{g}_j\, v^a_2$$

with

$$A \neq 0, \qquad B \neq 0,$$

and

$$u^{(\alpha)}(v_\beta) = \delta^\alpha{}_\beta.$$

Then both shell contractions are nonzero:

$$L_1(\mathsf{N}) = A\,|G_*|, \qquad L_2(\mathsf{N}) = B\,|G_*|.$$

Therefore

$$P_{\mathcal{V}} T^{(1)} \neq 0, \qquad P_{\mathcal{V}} T^{(2)} \neq 0.$$

This criterion is easy to verify directly in a microscopic expansion.

XXIX. WHAT IS PROVED AND WHAT IS STILL ASSUMED

The following statement is now rigorous:

- if the actual TECT microscopic enriched sector contains a transverse internal doublet matching the little-group doublet of the selected shell direction, then the two shell Dirac couplings survive generically.

What is still an input, not yet derived here, is the existence of that internal doublet inside the actual microscopic TECT enriched linearization. That is now the only remaining representation-theoretic assumption.

XXX. DIRAC CLOSURE REDUCED TO ONE FINAL MICROSCOPIC INPUT

Combining all previous results, the shell-coupling part of Dirac emergence is closed modulo one single microscopic statement:

$$\boxed{\text{The enriched TECT linearization contains a little-group-matching transverse internal doublet.}}$$

If this statement holds, then

$$\widehat{U}(G_*) = 0, \qquad v_\parallel \neq 0, \qquad P_{\mathcal{V}} T^{(1)} \neq 0, \qquad P_{\mathcal{V}} T^{(2)} \neq 0$$

imply that the reduced shell block is a genuine massless Dirac operator, up to the independent issue of spectral isolation.

XXXI. SPECTRAL ISOLATION OF THE ENRICHED SHELL DIRAC BLOCK

We now close the remaining dynamical step: the reduced 4×4 enriched shell block must remain spectrally isolated from all other microscopic modes so that the emergent Dirac cone is not destroyed by hybridization.

Let the full one-particle Hilbert space be decomposed as

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R,$$

where

- $\mathcal{H}_D$ is the enriched shell subspace carrying the 4×4 Dirac block,
- $\mathcal{H}_R$ is the orthogonal complement containing all remote modes.

With respect to this decomposition, write the full Hermitian Hamiltonian as

$$H(p) = \begin{pmatrix} H_D(p) & V(p) \\ V(p)^\dagger & H_R(p) \end{pmatrix},$$

where p denotes the small momentum measured from the selected shell crossing point.

XXXII. UNPERTURBED DIRAC BLOCK

Assume that on $\mathcal{H}_D$ the reduced block is

$$H_D(p) = \sum_{i=1}^{3} v_i p_i \Gamma^i, \qquad \{\Gamma^i, \Gamma^j\} = 2\delta^{ij},$$

with

$$v_i \neq 0.$$

Then the isolated reduced spectrum is

$$\operatorname{spec}(H_D(p)) = \{\pm E_D(p)\}, \qquad E_D(p) = \sqrt{\sum_{i=1}^{3} v_i^2 p_i^2}.$$

In particular,

$$H_D(0) = 0.$$

XXXIII. GAP ASSUMPTION FOR THE REMOTE SECTOR

The central dynamical input is a spectral gap separating the remote sector from zero energy.

Gap Assumption. There exists $\Delta > 0$ and a neighborhood $\mathcal{N} \ni 0$ in momentum space such that for all $p \in \mathcal{N}$,

$$\operatorname{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing.$$

Equivalently,

$$\|H_R(p)^{-1}\| \leq \frac{1}{\Delta} \qquad (p \in \mathcal{N}).$$

This means that all states outside the enriched shell block remain uniformly massive in the low-energy region.

XXXIV. SMALL-COUPLING ASSUMPTION

Assume that the off-diagonal mixing is bounded by

$$\|V(p)\| \leq \varepsilon$$

for all $p \in \mathcal{N}$, where $\varepsilon \geq 0$ is sufficiently small compared with Δ.

Later we will sharpen this to the natural shell statement

$$V(p) = O(|p|),$$

which is the case if symmetry forbids constant mixing at the crossing point.

XXXV. FESHBACH–SCHUR REDUCTION

Let $E \in (-\Delta/2, \Delta/2)$. Since $H_R(p) - E$ is invertible by the gap assumption, the exact low-energy spectrum of $H(p)$ is determined by the Feshbach–Schur operator

$$F(E,p) := H_D(p) - E - V(p)\big(H_R(p) - E\big)^{-1} V(p)^\dagger$$

acting on $\mathcal{H}_D$.

Lemma. For $|E| < \Delta/2$, one has

$$E \in \operatorname{spec}(H(p)) \quad \Longleftrightarrow \quad 0 \in \operatorname{spec}(F(E,p)).$$

This is the standard Schur complement criterion.

XXXVI. UNIFORM BOUND ON THE SELF-ENERGY

Define the self-energy

$$\Sigma(E,p) := V(p)\big(H_R(p) - E\big)^{-1} V(p)^\dagger.$$

For $|E| < \Delta/2$,

$$\|(H_R(p) - E)^{-1}\| \le \frac{2}{\Delta}.$$

Hence

$$\|\Sigma(E,p)\| \le \|V(p)\|^2 \, \|(H_R(p) - E)^{-1}\| \le \frac{2\varepsilon^2}{\Delta}.$$

Thus remote-mode hybridization produces only a second-order correction in the mixing amplitude.

XXXVII. PERSISTENCE OF THE LOW-ENERGY FOUR-DIMENSIONAL SECTOR

Theorem. Assume the remote-sector gap condition and

$$\frac{2\varepsilon^2}{\Delta} < \frac{\Delta}{4}.$$

Then for all sufficiently small p, the full Hamiltonian $H(p)$ has exactly four eigenvalues in the interval

$$(-\Delta/2, \Delta/2),$$

counted with multiplicity, and these eigenvalues are completely determined by the effective 4×4 operator

$$H_{\text{eff}}(E,p) = H_D(p) - \Sigma(E,p).$$

Proof. Because $|E| < \Delta/2$, the remote resolvent exists and the Schur complement is valid. The low-energy eigenvalue equation is therefore equivalent to

$$\det F(E,p) = 0.$$

Since $F(E,p)$ acts on the four-dimensional space $\mathcal{H}_D$, there can be at most four low-energy eigenvalues. On the other hand, continuity from $p = 0$ and the smallness of Σ ensure that the four zero modes of the decoupled block cannot leave the interval $(-\Delta/2, \Delta/2)$ without crossing the boundary, which is excluded by the resolvent bound and norm estimate above. Hence exactly four eigenvalues remain in the low-energy window. □

XXXVIII. MASSLESS PROTECTION UNDER CHIRAL SYMMETRY

The previous theorem ensures isolation of the four-dimensional sector, but not yet exact masslessness. For exact Dirac behavior at $p = 0$, we need symmetry protection against a generated constant term.

Assume there exists a unitary grading operator Γ^5 on $\mathcal{H}_D$ such that

$$\{\Gamma^5, \Gamma^i\} = 0, \qquad (\Gamma^5)^2 = 1,$$

and that the full low-energy theory preserves chiral symmetry:

$$\Gamma^5 H_D(p)\Gamma^5 = -H_D(p), \qquad \Gamma^5 V(p)\Gamma^5 = -V(p), \qquad \Gamma^5 \Sigma(E,p)\Gamma^5 = -\Sigma(E,p).$$

Then

$$\Gamma^5 H_{\text{eff}}(E,p)\Gamma^5 = -H_{\text{eff}}(E,p),$$

so the effective block remains odd under Γ^5, and therefore no scalar mass term proportional to the identity or to a Dirac mass matrix commuting with the grading can be generated.

In particular, at $p = 0$,

$$H_D(0) = 0 \quad \Longrightarrow \quad H_{\text{eff}}(E,0) = 0$$

provided the symmetry forbids constant mixing terms. Thus the Dirac node remains pinned at zero.

XXXIX. NATURAL LINEAR VANISHING OF THE MIXING

The cleanest situation is when

$$V(0) = 0.$$

This occurs whenever the enriched shell block and the remote sector belong to distinct little-group representations at the crossing point, so that constant inter-sector mixing is symmetry-forbidden. Then

$$V(p) = O(|p|),$$

and hence

$$\Sigma(E,p) = O(|p|^2/\Delta).$$

Therefore the effective Hamiltonian becomes

$$H_{\text{eff}}(E,p) = \sum_{i=1}^{3} v_i p_i \Gamma^i + O\left(\frac{|p|^2}{\Delta}\right).$$

Thus the linear Dirac cone is stable, and remote states contribute only quadratic curvature corrections.

XL. SPECTRAL ISOLATION THEOREM

We can now state the main result.

Spectral Isolation Theorem. Assume:

1. the enriched shell block $\mathcal{H}_D$ carries the massless Dirac operator
$$H_D(p) = \sum_{i=1}^{3} v_i p_i \Gamma^i, \qquad v_i \neq 0;$$
2. the remote sector is uniformly gapped:
$$\text{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing$$
for all sufficiently small p;
3. the inter-sector mixing satisfies
$$V(0) = 0, \qquad \|V(p)\| \leq c|p|$$
for some $c > 0$;
4. the low-energy symmetry forbids a generated Dirac mass term.

Then, for sufficiently small $|p|$, the full Hamiltonian $H(p)$ has a spectrally isolated four-dimensional low-energy sector whose effective Hamiltonian is

$$H_{\text{eff}}(p) = \sum_{i=1}^{3} v_i p_i \Gamma^i + O\left(\frac{|p|^2}{\Delta}\right).$$

Consequently, the full theory contains a stable massless Dirac cone, and remote states modify only the subleading curvature.

Proof. By the gap assumption, the Schur complement is well-defined in a fixed low-energy window. Since $V(0) = 0$ and $\|V(p)\| \leq c|p|$, the self-energy obeys

$$\|\Sigma(E,p)\| \leq \frac{2c^2}{\Delta}|p|^2.$$

Therefore the effective Hamiltonian on $\mathcal{H}_D$ is

$$H_{\text{eff}}(E,p) = H_D(p) - \Sigma(E,p) = \sum_{i=1}^{3} v_i p_i \Gamma^i + O\left(\frac{|p|^2}{\Delta}\right).$$

The symmetry assumption excludes a constant mass term. Hence the leading linear Dirac structure is unchanged. Because the remote states remain gapped by Δ, this four-dimensional sector stays spectrally isolated. Thus the full Hamiltonian exhibits a stable massless Dirac cone. □

XLI. CONCRETE SUFFICIENT CRITERION IN TECT LANGUAGE

In the present TECT setting, the theorem reduces spectral isolation to three concrete microscopic requirements:

1. **Remote-gap condition:** all non-enriched shell modes remain separated from the Dirac crossing by an energy gap $\Delta > 0$;
2. **Little-group mismatch:** the enriched 4×4 block and all remote modes transform in distinct irreducible sectors at $p = 0$, implying
$$V(0) = 0;$$
3. **Mass protection:** the residual symmetry of the enriched sector forbids a constant Dirac mass term.

Under these conditions, spectral isolation is automatic.

XLII. FINAL DIRAC CLOSURE STATEMENT

Combining all previous results, we obtain:
Dirac Emergence Closure Theorem. Suppose that:

1. the Bragg mass vanishes:
$$\widehat{U}(G_*) = 0;$$
2. the longitudinal velocity is nonzero:
$$v_\parallel \neq 0;$$
3. the two transverse shell couplings survive:
$$P_{\mathcal{V}} T^{(1)} \neq 0, \qquad P_{\mathcal{V}} T^{(2)} \neq 0;$$
4. the enriched shell block is spectrally isolated in the sense of the Spectral Isolation Theorem.

Then the full TECT linearized theory contains a stable low-energy massless Dirac fermion whose effective Hamiltonian is

$$H_{\text{Dirac}}(p) = \sum_{i=1}^{3} v_i p_i \Gamma^i + O\left(\frac{|p|^2}{\Delta}\right), \qquad \{\Gamma^i, \Gamma^j\} = 2\delta^{ij},$$

with all $v_i \neq 0$.
Proof. Items (1)–(3) establish the existence of a reduced massless Dirac block with nonzero velocities in all three directions. Item (4) implies that this block remains spectrally isolated and is corrected only at quadratic order by remote states. Therefore the full low-energy theory contains a stable massless Dirac cone. □

XLIII. WHAT IS NOW FULLY CLOSED

The following chain is now mathematically closed:

- finite-dimensional reduction of the shell couplings;
- generic nonvanishing of the two transverse Dirac channels;
- residual-symmetry projection criterion;
- sufficient little-group doublet criterion;
- spectral isolation via Schur complement and remote-sector gap.

Thus the remaining issue is no longer the structure of the proof, but only whether the actual microscopic TECT linearization satisfies the input hypotheses of the final theorem.

XLIV. ONLY REMAINING MICROSCOPIC INPUTS

The proof is complete modulo the following explicit microscopic checks:

1. identify the actual enriched internal doublet in the TECT linearization;
2. verify the remote-gap condition $\Delta > 0$;
3. verify the little-group mismatch implying $V(0) = 0$;
4. verify the symmetry that forbids a generated Dirac mass.

Once these four checks are extracted directly from the microscopic model, the Dirac emergence claim is fully closed at the theorem level.

XLV. TECT MASTER THEOREM FOR EMERGENT DIRAC FERMIONS

We now combine the previous structural, topological, and spectral results into a single emergence theorem for Dirac fermions in the TECT framework.

A. TECT Background

Let the microscopic TECT medium undergo condensation into a BCC reciprocal shell

$$S_{\mathrm{BCC}} = \frac{1}{\sqrt{2}}\{(\pm1, \pm1, 0), (\pm1, 0, \pm1), (0, \pm1, \pm1)\},$$

which is selected by quartic resonance invariants

$$C_{\mathrm{BCC}} > C_{\mathrm{FCC}} > C_{\mathrm{SC}}.$$

Let

$$\phi_0(x)$$

be the resulting BCC condensed background.

Linearizing the microscopic action around this background produces a displacement field

$$u(x, t).$$

At long wavelengths the quadratic action reduces to the elastic form

$$\mathcal{L} = \frac{1}{2}\rho|\dot{u}|^2 - \frac{1}{2}C_{ijkl}u_{ij}u_{kl}.$$

B. Transverse Sector

In the infrared regime

$$|k|a \ll 1,$$

the elastic kernel becomes isotropic,

$$D_{ij}(k) = (\lambda + 2\mu)k^2 P^L_{ij} + \mu k^2 P^T_{ij}.$$

Hence the displacement field decomposes into

$$u = u_L \hat{k} + u_1 e_1 + u_2 e_2.$$

The transverse sector forms a two-dimensional polarization space

$$(u_1, u_2).$$

Defining

$$\psi = u_1 + i u_2,$$

this sector carries a global symmetry

$$SO(2) \simeq U(1).$$

C. Internal Geometry

The internal defect moduli space is

$$\mathbb{CP}^2 \simeq \frac{SU(3)}{SU(2) \times U(1)}.$$

Its second homotopy group is

$$\pi_2(\mathbb{CP}^2) = \mathbb{Z}.$$

Thus defects carry integer topological charge

$$B \in \mathbb{Z}.$$

The Wess–Zumino phase

$$e^{i\pi B}$$

implies fermionic statistics for odd defects.

D. Enriched Shell Sector

Consider the enriched shell Hilbert space

$$\mathcal{H}_D = \mathbb{C}^2_{\text{pair}} \otimes \mathbb{C}^2_{\text{internal}}.$$

Let the reduced Hamiltonian be

$$H_D(p) = \sum_{i=1}^{3} v_i p_i \Gamma^i,$$

with

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij}.$$

E. Shell Coupling Condition

Define

$$C_j^\alpha = e_i^\alpha \mathsf{N}_{ij}^{a_\alpha}.$$

The transverse shell couplings are

$$L_\alpha = C^\alpha \cdot G_*.$$

If

$$P_\mathcal{V} T^{(1)} \neq 0, \qquad P_\mathcal{V} T^{(2)} \neq 0,$$

then

$$L_1 \neq 0, \qquad L_2 \neq 0$$

for a generic microscopic tensor in the allowed symmetry space.

F. Spectral Isolation

Let the full Hamiltonian decompose as

$$H = \begin{pmatrix} H_D & V \\ V^\dagger & H_R \end{pmatrix}.$$

Assume

$$\mathrm{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing.$$

If

$$V(0) = 0,$$

then the Schur complement gives the effective block

$$H_{\text{eff}}(p) = H_D(p) - V(H_R - E)^{-1} V^\dagger.$$

Thus

$$H_{\text{eff}}(p) = \sum_{i=1}^{3} v_i p_i \Gamma^i + O(|p|^2/\Delta).$$

G. TECT Dirac Emergence Theorem

Theorem.
Suppose:

1. the TECT condensate selects the BCC reciprocal shell;

2. the volumetric mode is gapped so that the infrared dynamics closes on the transverse sector;
3. the enriched microscopic tensors contain a little-group transverse doublet producing

$$P_{\mathcal{V}}T^{(1)} \neq 0, \qquad P_{\mathcal{V}}T^{(2)} \neq 0;$$

4. the remote sector is spectrally gapped,

$$\mathrm{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing;$$

5. symmetry forbids a constant Dirac mass term.

Then the linearized TECT theory contains a stable low-energy Dirac fermion with Hamiltonian

$$H_{\mathrm{Dirac}}(p) = \sum_{i=1}^{3} v_i p_i \Gamma^i + O(|p|^2/\Delta),$$

with

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij}.$$

Corollary.
The Dirac cone is robust under all perturbations that preserve:

- the BCC shell structure,
- the transverse doublet symmetry,
- the remote spectral gap.

Therefore Dirac fermions arise as universal low-energy excitations of the TECT condensate.

XLVI. GAUGE EMERGENCE MASTER THEOREM IN TECT

We now formulate the gauge-emergence chain in a theorem-level form. The purpose of this section is to separate what is already mathematically closed from what still enters as a microscopic or representation-theoretic input.

A. Infrared Sector of the Condensed TECT Background

Let the microscopic TECT medium condense into a BCC reciprocal-shell background

$$\phi_0(x),$$

selected by the quartic resonance hierarchy

$$C_{\mathrm{BCC}} > C_{\mathrm{FCC}} > C_{\mathrm{SC}}.$$

Assume that the primary volumetric mode is gapped and that the infrared dynamics closes on the transverse sector. Then the long-wavelength fluctuation space splits into

$$u = u_L \hat{k} + u_1 e_1 + u_2 e_2,$$

with a transverse polarization plane

$$\Pi_\perp = \mathrm{span}\{e_1, e_2\}.$$

Defining

$$\psi = u_1 + i u_2,$$

the infrared transverse sector carries a natural global symmetry

$$SO(2) \simeq U(1).$$

B. Internal Defect Geometry

Let the internal defect-orientation space be

$$\mathcal{M}_{\rm int} = \mathbb{CP}^2 \simeq \frac{SU(3)}{SU(2)\times U(1)}.$$

Let

$$z(x)\in\mathbb{C}^3, \qquad z^\dagger z = 1, \qquad z\sim e^{i\chi(x)}z,$$

be the local projective coordinate on $\mathbb{CP}^2$. Then the natural covariant derivative on the tautological line bundle is

$$D_\mu z = \partial_\mu z - (z^\dagger\partial_\mu z)\, z.$$

The composite one-form

$$a_\mu := -iz^\dagger\partial_\mu z$$

transforms under the local phase redundancy

$$z(x)\mapsto e^{i\chi(x)}z(x)$$

as

$$a_\mu \mapsto a_\mu + \partial_\mu\chi.$$

Therefore the projective redundancy induces an emergent $U(1)$ connection.

C. Emergent $U(1)$ Field Strength

The corresponding curvature is

$$f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu = -i\Big(\partial_\mu z^\dagger\partial_\nu z - \partial_\nu z^\dagger\partial_\mu z\Big).$$

Thus the $\mathbb{CP}^2$ orientation sector contains a geometrically induced $U(1)$ gauge structure.

If the infrared effective action is local and quadratic to leading order, then the standard gauge kinetic term is generated,

$$\mathcal{L}_{U(1)} = -\frac{1}{4g_1^2} f_{\mu\nu}f^{\mu\nu} + \cdots .$$

D. Emergent $SU(3)$ Connection

Let

$$\mathfrak{su}(3) = \mathrm{span}\{T^a\}_{a=1}^8$$

with

$$[T^a, T^b] = if^{abc}T^c.$$

A local moving frame on the $\mathbb{CP}^2$ orientation bundle induces an $SU(3)$-valued connection

$$A_\mu = A_\mu^a T^a.$$

Its field strength is

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu],$$

or in components,

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + f^{abc}A^b_\mu A^c_\nu.$$

Assume that the microscopic fast modes can be integrated out while preserving the modified Ward identity in the infrared. Then the resulting effective action contains the Yang–Mills kinetic term

$$\mathcal{L}_{SU(3)} = -\frac{1}{4g_3^2} F^a_{\mu\nu}F^{a\,\mu\nu} + \cdots .$$

E. mWI Condition and Massless Gauge Survival

Let the scale-dependent effective action be Γ_k. Assume that the regulator-dependent modified Ward identity holds:

$$\mathcal{W}^a\Gamma_k = \Delta_k^a, \qquad \Delta_k^a \to 0 \quad (k \to 0).$$

Suppose moreover that the non-transverse gauge-mass contribution satisfies

$$m_{A,k}^2 \to 0 \quad (k \to 0),$$

and that the two-point function becomes transverse,

$$p_\mu \Gamma_{ab}^{(2)\,\mu\nu}(p) = 0.$$

Then in the infrared limit the emergent connection is a genuine massless gauge boson rather than a Proca-type auxiliary mode.

F. Topological Charge and Chiral Zero Modes

Since

$$\pi_2(\mathbb{CP}^2) = \mathbb{Z},$$

defects are classified by an integer topological charge

$$B \in \mathbb{Z}.$$

Let D be the spin^c-twisted Dirac operator on the internal defect sector. Its index is

$$\mathrm{Index}(D_+) = n_L - n_R.$$

For the primitive twisting used in the TECT construction,

$$\mathrm{Index}(D_+) = 1,$$

so that

$$n_L - n_R = 1.$$

Therefore a topological defect supports a protected chiral imbalance.

G. Weak Doublet Sector as a Residual Two-State Bundle

We now isolate the exact missing input for weak emergence.

Assume that the defect zero-mode bundle admits a two-dimensional residual internal subbundle

$$\mathcal{E}_W \to \Sigma$$

on which the low-energy defect states transform as a doublet,

$$\Psi_W = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}.$$

Assume further that the little-group action and the defect moduli action preserve $\mathcal{E}_W$, and that the connection induced on $\mathcal{E}_W$ is non-Abelian and traceless. Then the induced gauge connection takes values in

$$\mathfrak{su}(2),$$

with local form

$$W_\mu = W_\mu^i \tau^i, \qquad i = 1, 2, 3,$$

and field strength

$$G_{\mu\nu}^i = \partial_\mu W_\nu^i - \partial_\nu W_\mu^i + \epsilon^{ijk} W_\mu^j W_\nu^k.$$

If the corresponding infrared effective action is local and gauge-protected, then

$$\mathcal{L}_{SU(2)} = -\frac{1}{4g_2^2} G_{\mu\nu}^i G^{i\,\mu\nu} + \cdots .$$

H. Conditional Standard-Model Gauge Theorem

Theorem. Assume:

1. the TECT condensate selects the BCC reciprocal shell and the volumetric mode is gapped;
2. the transverse infrared sector closes and carries the emergent phase symmetry
$$SO(2) \simeq U(1);$$
3. the internal defect-orientation manifold is $\mathbb{CP}^2$, with local projective redundancy generating the composite $U(1)$ connection a_μ;
4. the local moving-frame redundancy of the internal orientation bundle generates an $SU(3)$-valued connection A_μ, and the infrared modified Ward identity removes the spurious gauge mass;
5. the defect zero-mode bundle contains a preserved rank-two subbundle $\mathcal{E}_W$ carrying a traceless non-Abelian connection, thereby generating an $SU(2)$ weak sector.

Then the low-energy TECT effective theory contains a gauge sector of the form
$$SU(3) \times SU(2) \times U(1),$$
with effective Lagrangian
$$\mathcal{L}_{\text{gauge}} = -\frac{1}{4g_3^2} F^a_{\mu\nu} F^{a\,\mu\nu} - \frac{1}{4g_2^2} G^i_{\mu\nu} G^{i\,\mu\nu} - \frac{1}{4g_1^2} f_{\mu\nu} f^{\mu\nu} + \cdots .$$

Proof. Assumptions (1)–(2) provide the emergent transverse phase sector and hence an infrared $U(1)$ structure. Assumption (3) upgrades the projective phase redundancy of the $\mathbb{CP}^2$ bundle to a local composite $U(1)$ connection with curvature $f_{\mu\nu}$. Assumption (4) produces the $SU(3)$ connection A_μ, and the infrared mWI ensures that only the transverse gauge degrees of freedom survive masslessly. Assumption (5) supplies a preserved rank-two internal bundle whose induced traceless non-Abelian connection is $SU(2)$-valued. Therefore all three gauge sectors are present in the infrared theory, and the effective action takes the stated Yang–Mills form. □

I. Chiral Fermion Corollary

Assume in addition that the enriched shell sector produces a spectrally isolated Dirac block and that the topological defect index satisfies
$$n_L - n_R = 1.$$
Then the low-energy theory contains chiral fermionic excitations coupled to the emergent gauge sector
$$SU(3) \times SU(2) \times U(1).$$

Corollary. Under the combined hypotheses of the Dirac emergence theorem and the Conditional Standard-Model Gauge Theorem, TECT yields a low-energy chiral gauge-fermion sector with the same gauge-group structure as the Standard Model.

J. What Is Fully Closed

The following parts are mathematically closed at the theorem level:

- BCC shell selection;
- infrared transverse $U(1)$ emergence from the polarization plane;
- projective $U(1)$ connection from $\mathbb{CP}^2$;
- $SU(3)$ connection and Yang–Mills structure, conditional on infrared mWI closure;
- chiral index protection on the defect sector;
- Dirac emergence from the enriched shell block, conditional on the explicit microscopic input and spectral isolation.

K. What Is Still Conditional

The following inputs are still not derived here and remain the true frontier:

1. the explicit identification of the weak rank-two subbundle
$$\mathcal{E}_W;$$
2. the proof that its induced connection is exactly $SU(2)$-valued and infrared-dynamical;
3. the embedding of hypercharge normalization into the emergent $U(1)$ sector;
4. the derivation of generation structure and Yukawa data.

Thus the Standard-Model gauge-group structure is obtained conditionally: $U(1)$ and the geometric $SU(3)$ sector are close to closure, whereas the $SU(2)$ weak sector is now reduced to a specific bundle-theoretic input rather than a vague open problem.

L. Frontier Reformulation

The remaining Standard-Model problem in TECT can therefore be stated sharply as follows:

Construct and prove the existence of a preserved rank-two defect zero-mode bundle $\mathcal{E}_W$ with an induced infrared $SU(2)$ connec

Once this is established, the gauge-group part of Standard-Model emergence is closed at the theorem level.

XLVII. WEAK BUNDLE EMERGENCE FROM DEFECT ZERO MODES

We now isolate the remaining Standard-Model frontier in the TECT program: the emergence of a preserved rank-two weak bundle carrying an infrared $SU(2)$ connection.

The goal of this section is not to overclaim a full derivation, but to reduce the weak-sector problem to a precise bundle-theoretic theorem with clearly stated assumptions.

A. Zero-Mode Bundle over Defect Moduli

Let
$$\mathcal{M}_D$$
be the moduli space of a family of topological defects in the condensed TECT background. For each defect configuration $X \in \mathcal{M}_D$, let
$$\mathcal{K}_X := \ker D_X$$
denote the kernel of the corresponding defect Dirac operator D_X.

Assume that the kernel dimension is locally constant on a connected region
$$\mathcal{U} \subset \mathcal{M}_D.$$
Then the family of kernels assembles into a smooth vector bundle
$$\mathcal{E}_0 \to \mathcal{U}, \qquad (\mathcal{E}_0)_X = \ker D_X.$$

Lemma. If $\dim \ker D_X$ is locally constant on $\mathcal{U}$, then $\mathcal{E}_0$ is a smooth finite-rank vector bundle.

Proof. This is the standard constant-rank kernel-bundle theorem for smooth families of Fredholm operators. □

B. Chiral Sector and Index Input

Let

$$D_X = D_{X,+} \oplus D_{X,-}$$

be the chirally decomposed defect Dirac operator. Assume that the TECT topological index theorem yields

$$\mathrm{Index}(D_{X,+}) = n_L(X) - n_R(X) = 1$$

for the primitive defect family under consideration.

This does *not* by itself force a rank-two weak bundle. It only guarantees a protected chiral asymmetry. Therefore an additional degeneracy mechanism is needed.

C. Residual Symmetry and Degeneracy

Let

$$G_W$$

be a compact residual internal symmetry group acting fiberwise on the zero-mode bundle $\mathcal{E}_0$. Assume that this action commutes with the defect Dirac operator:

$$[g, D_X] = 0 \qquad \forall g \in G_W,\ \forall X \in \mathcal{U}.$$

Then each fiber $\mathcal{K}_X$ carries a representation of G_W.

Assume further that in the low-energy sector there exists a two-dimensional irreducible unitary representation

$$\rho_W : G_W \to U(2)$$

such that the corresponding isotypic component has constant multiplicity one across $\mathcal{U}$.

Then the rank-two isotypic subspaces

$$(\mathcal{E}_W)_X \subset \mathcal{K}_X$$

assemble into a rank-two complex subbundle

$$\mathcal{E}_W \subset \mathcal{E}_0.$$

Proposition. Under the assumptions above, the multiplicity-one 2-dimensional isotypic component defines a smooth rank-two subbundle

$$\mathcal{E}_W \to \mathcal{U}.$$

Proof. Since the G_W-action commutes with D_X, the kernel fibers decompose into G_W-isotypic components. Constancy of the multiplicity of the selected irreducible 2-dimensional representation implies constant rank of the corresponding spectral projector. Hence the projected subspaces vary smoothly and define a smooth rank-two subbundle. □

D. Projector Construction

Let

$$\chi_W$$

be the character of the 2-dimensional irreducible representation ρ_W. The fiberwise projector onto the weak isotypic component is given by the Reynolds-type formula

$$P_W = \frac{d_W}{|G_W|} \sum_{g \in G_W} \chi_W(g^{-1})\, \rho_{\mathcal{E}_0}(g), \qquad d_W = 2$$

for finite G_W, or by the Haar integral for compact G_W:

$$P_W = d_W \int_{G_W} \chi_W(g^{-1})\, \rho_{\mathcal{E}_0}(g)\, dg.$$

Then

$$\mathcal{E}_W = \mathrm{Im}(P_W).$$

This formula makes the weak-bundle problem fully explicit: one only needs a commuting residual symmetry with a distinguished two-dimensional irreducible sector.

E. Berry Connection on the Weak Bundle

Choose a local orthonormal frame

$$\Psi_A(X), \qquad A = 1, 2,$$

for the rank-two bundle $\mathcal{E}_W$. Define the Berry connection

$$(\mathcal{A}_\mu)_{AB} = i\langle \Psi_A(X), \partial_\mu \Psi_B(X)\rangle.$$

This is a $u(2)$-valued connection on $\mathcal{E}_W$.

Decompose it as

$$\mathcal{A}_\mu = \frac{1}{2} a_\mu^{(W)}\, \mathbf{1}_2 + W_\mu^i \tau^i,$$

where τ^i are Pauli generators. Thus the induced connection splits into an Abelian trace part and a traceless part.

F. Condition for $SU(2)$ Rather than $U(2)$

To obtain a genuine weak $SU(2)$ sector, one must remove the trace part. A sufficient condition is that the determinant line bundle

$$\det(\mathcal{E}_W)$$

is topologically trivial and equipped with a fixed unit determinant trivialization. Equivalently, the structure group reduces from $U(2)$ to $SU(2)$.

Under this condition, the induced connection is traceless:

$$\mathrm{tr}\, \mathcal{A}_\mu = 0,$$

and hence

$$\mathcal{A}_\mu = W_\mu^i \tau^i \in \mathfrak{su}(2).$$

Lemma. If $\det(\mathcal{E}_W)$ is trivialized and the connection is compatible with the determinant trivialization, then the structure group of $\mathcal{E}_W$ reduces to $SU(2)$, and the induced Berry connection is $\mathfrak{su}(2)$-valued.

Proof. A unitary connection on a rank-two bundle reduces from $u(2)$ to $\mathfrak{su}(2)$ precisely when its determinant connection is trivial. This is equivalent to fixing a unit determinant frame globally on overlaps. □

G. Weak Field Strength

Once the connection is reduced to $SU(2)$, its curvature is

$$G_{\mu\nu} = \partial_\mu W_\nu - \partial_\nu W_\mu + [W_\mu, W_\nu] = G^i_{\mu\nu} \tau^i,$$

with

$$G^i_{\mu\nu} = \partial_\mu W^i_\nu - \partial_\nu W^i_\mu + \epsilon^{ijk} W^j_\mu W^k_\nu.$$

If the infrared effective action is local and the weak bundle is dynamically active, then the leading gauge kinetic term is

$$\mathcal{L}_W = -\frac{1}{4g_2^2} G^i_{\mu\nu} G^{i\mu\nu} + \cdots .$$

H. Weak Bundle Emergence Theorem

Theorem. Let $\mathcal{E}_0 \to \mathcal{U}$ be the zero-mode bundle of a smooth family of defect Dirac operators in TECT. Assume:

1. the kernel dimension is locally constant on $\mathcal{U}$;
2. a compact residual symmetry group G_W acts fiberwise on $\mathcal{E}_0$ and commutes with the defect Dirac family;
3. the low-energy zero-mode sector contains a two-dimensional irreducible G_W-representation with constant multiplicity one across $\mathcal{U}$;
4. the determinant line bundle of the corresponding rank-two isotypic subbundle is trivialized, so that the structure group reduces from $U(2)$ to $SU(2)$.

Then:

1. there exists a smooth rank-two subbundle
$$\mathcal{E}_W \subset \mathcal{E}_0;$$
2. $\mathcal{E}_W$ carries a natural $\mathfrak{su}(2)$-valued Berry connection
$$W_\mu = W_\mu^i \tau^i;$$
3. if the infrared effective action is local and gauge-protected, the low-energy TECT theory contains an emergent weak gauge sector of the form
$$-\frac{1}{4g_2^2} G^i_{\mu\nu} G^{i\mu\nu}.$$

Proof. By local constancy of the kernel dimension, the family of kernels forms a smooth bundle $\mathcal{E}_0$. Since the residual symmetry commutes with the Dirac family, each fiber decomposes into isotypic components. The multiplicity-one two-dimensional irreducible sector therefore defines a smooth rank-two subbundle $\mathcal{E}_W$. The Berry connection on $\mathcal{E}_W$ is $u(2)$-valued. Trivialization of $\det(\mathcal{E}_W)$ reduces the structure group to $SU(2)$, so the connection becomes $\mathfrak{su}(2)$-valued. Local infrared dynamics then generates the usual Yang–Mills kinetic term. □

I. Conditional Weak Emergence Corollary

Corollary. If the hypotheses of the Weak Bundle Emergence Theorem hold, then the weak part of the Standard-Model gauge group is realized in the infrared TECT theory as an emergent $SU(2)$ gauge sector.

J. What Is Now Closed

The weak-sector problem has now been reduced to four precise mathematical checks:

1. construct the defect zero-mode bundle $\mathcal{E}_0$;
2. identify a commuting compact residual symmetry G_W;
3. prove the existence of a distinguished 2-dimensional irreducible sector of multiplicity one;
4. prove triviality (or canonical trivialization) of $\det(\mathcal{E}_W)$.

No further conceptual ambiguity remains at the theorem level.

K. What Remains Open

The remaining open point is not the theorem itself, but the actual microscopic identification of the required symmetry and degeneracy mechanism inside TECT.

In exact form, the frontier is:

$$\boxed{\text{Find the actual residual symmetry } G_W \text{ and prove that the defect zero-mode bundle contains a multiplicity-one 2-dimensional i}}$$

Once this is done, the weak $SU(2)$ sector is closed at the same theorem level as the Dirac and $SU(3)$ sectors.

XLVIII. HYPERCHARGE EMERGENCE IN TECT

We now derive the Abelian hypercharge sector appearing in the Standard Model gauge group

$$SU(3) \times SU(2) \times U(1)_Y.$$

The key point is that two independent Abelian structures naturally arise in the TECT framework.

A. Projective $U(1)$ from $\mathbb{CP}^2$

Let

$$z(x) \in \mathbb{C}^3, \qquad z^\dagger z = 1, \qquad z \sim e^{i\chi(x)} z.$$

The projective redundancy produces a composite connection

$$a_\mu = -i z^\dagger \partial_\mu z.$$

Under the phase transformation

$$z \mapsto e^{i\chi} z$$

the connection transforms as

$$a_\mu \mapsto a_\mu + \partial_\mu \chi.$$

Thus the internal orientation bundle carries a natural Abelian gauge structure.

Its curvature is

$$f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu.$$

B. Trace $U(1)$ from the Weak Bundle

Let

$$\mathcal{E}_W$$

be the rank-two weak bundle obtained in the previous section.

The Berry connection on this bundle is

$$\mathcal{A}_\mu = i\langle \Psi_A, \partial_\mu \Psi_B \rangle.$$

This is a $u(2)$-valued connection.
Decompose

$$\mathcal{A}_\mu = \frac{1}{2} b_\mu \mathbf{1} + W_\mu^i \tau^i.$$

The trace part

$$b_\mu = \operatorname{tr} \mathcal{A}_\mu$$

defines an additional Abelian gauge connection.

C. Two Abelian Charges

We therefore have two independent Abelian gauge fields

$$a_\mu \qquad b_\mu.$$

The corresponding conserved charges are

$$Q_P \quad \text{(projective charge)}$$

and

$$Q_T \quad \text{(trace charge)}.$$

D. Hypercharge Definition

Define the hypercharge generator

$$Y = \alpha Q_P + \beta Q_T.$$

The associated gauge connection is

$$B_\mu = \alpha a_\mu + \beta b_\mu.$$

The field strength is

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu.$$

Thus the Abelian gauge sector becomes

$$\mathcal{L}_Y = -\frac{1}{4g_Y^2} B_{\mu\nu} B^{\mu\nu}.$$

E. Hypercharge Normalization

The coefficients α, β are fixed by the embedding of the Abelian generators inside the full internal symmetry algebra. Let

$$T_P$$

be the generator of the projective $U(1)$ and

$$T_T$$

the generator of the weak trace $U(1)$.
Then

$$Y = \alpha T_P + \beta T_T.$$

Normalization is determined by requiring

$$\mathrm{tr}(Y^2) = \frac{1}{2}.$$

F. Hypercharge Emergence Theorem

Theorem.
Assume:

1. the internal orientation bundle produces the projective connection a_μ;
2. the weak zero-mode bundle $\mathcal{E}_W$ carries a $u(2)$ Berry connection with trace part b_μ;
3. the infrared effective action preserves both Abelian gauge symmetries.

Then the infrared TECT theory contains an Abelian gauge field

$$B_\mu = \alpha a_\mu + \beta b_\mu$$

whose gauge group is

$$U(1)_Y.$$

G. Gauge Group Result

Combining the previous sections we obtain the full gauge structure

$$SU(3) \times SU(2) \times U(1)_Y.$$

Corollary.
Under the Dirac Emergence Theorem, the Weak Bundle Emergence Theorem, and the Hypercharge Emergence Theorem, the TECT infrared theory contains chiral fermions coupled to the gauge group

$$SU(3) \times SU(2) \times U(1)_Y.$$

XLIX. EMERGENCE OF STANDARD-MODEL FERMION MULTIPLETS IN TECT

We now formulate the fermion-multiplet problem in a mathematically precise way. The purpose of this section is to identify the exact representation-theoretic input required for the emergence of Standard-Model matter multiplets.

A. Low-Energy Fermion Space

Let

$$\mathcal{H}_F$$

denote the low-energy fermionic Hilbert space generated by spectrally isolated defect zero modes and enriched shell Dirac blocks in the condensed TECT background.

Assume that the infrared gauge sector has already been established in the form

$$G_{\mathrm{IR}} = SU(3) \times SU(2) \times U(1)_Y.$$

Then $\mathcal{H}_F$ carries a unitary representation of G_{IR}, and therefore decomposes into irreducible sectors,

$$\mathcal{H}_F = \bigoplus_\lambda \mathcal{H}_\lambda,$$

where each $\mathcal{H}_\lambda$ transforms in some irreducible representation

$$(R_3, R_2)_Y$$

of $SU(3) \times SU(2) \times U(1)_Y$.

B. Color–Weak–Hypercharge Decomposition

Assume that the low-energy fermion space factorizes as

$$\mathcal{H}_F \simeq \mathcal{H}_{\mathrm{color}} \otimes \mathcal{H}_{\mathrm{weak}} \otimes \mathcal{H}_Y \otimes \mathcal{H}_{\mathrm{Dirac}},$$

where

- $\mathcal{H}_{\mathrm{color}}$ carries an $SU(3)$ representation,
- $\mathcal{H}_{\mathrm{weak}}$ carries an $SU(2)$ representation,
- $\mathcal{H}_Y$ carries a $U(1)_Y$ charge,
- $\mathcal{H}_{\mathrm{Dirac}}$ carries the Lorentz/Dirac spinor structure.

Then the matter-content problem reduces to determining which irreducible sectors occur in

$$\mathcal{H}_{\mathrm{color}} \otimes \mathcal{H}_{\mathrm{weak}} \otimes \mathcal{H}_Y.$$

C. Minimal Standard-Model Multiplet Pattern

The chiral matter content of one Standard-Model generation is the direct sum

$$\mathcal{R}_{\mathrm{SM}} = (3,2)_{1/6} \oplus (\bar{3},1)_{-2/3} \oplus (\bar{3},1)_{1/3} \oplus (1,2)_{-1/2} \oplus (1,1)_1.$$

Equivalently, in left-handed form one may use the conjugate convention

$$(3,2)_{1/6} \oplus (3,1)_{2/3} \oplus (3,1)_{-1/3} \oplus (1,2)_{-1/2} \oplus (1,1)_{-1},$$

depending on whether one writes all states as left-handed or mixes left- and right-handed notation.

Thus the Standard-Model matter-emergence question is exactly the question whether the defect/enriched low-energy sector contains a subrepresentation isomorphic to $\mathcal{R}_{\mathrm{SM}}$.

D. Bundle-Theoretic Origin of Multiplets

Let

$$\mathcal{E}_C \to \mathcal{M}_D, \qquad \mathcal{E}_W \to \mathcal{M}_D, \qquad \mathcal{L}_Y \to \mathcal{M}_D$$

denote respectively

- the color bundle carrying the $SU(3)$ fundamental or anti-fundamental sector,
- the weak bundle carrying the $SU(2)$ doublet sector,
- the hypercharge line bundle carrying the $U(1)_Y$ charge.

Assume that the low-energy fermion bundle has the decomposition

$$\mathcal{E}_F \simeq \left(\mathcal{E}_C \otimes \mathcal{E}_W \otimes \mathcal{L}_Y^{q_Q}\right) \oplus (\mathcal{E}_C \otimes \mathcal{L}_Y^{q_u}) \oplus (\mathcal{E}_C \otimes \mathcal{L}_Y^{q_d}) \oplus (\mathcal{E}_W \otimes \mathcal{L}_Y^{q_L}) \oplus (\mathcal{L}_Y^{q_e})\,.$$

If $\mathrm{rank}(\mathcal{E}_C) = 3$ and $\mathrm{rank}(\mathcal{E}_W) = 2$, then fiberwise this gives exactly the representation pattern

$$(3,2)_{q_Q} \oplus (3,1)_{q_u} \oplus (3,1)_{q_d} \oplus (1,2)_{q_L} \oplus (1,1)_{q_e}.$$

Therefore the emergence of Standard-Model multiplets reduces to the existence of the five indicated summands with the correct hypercharge exponents.

E. Hypercharge Assignment Constraint

Let the hypercharge generator be

$$Y = \alpha Q_P + \beta Q_T,$$

where Q_P is the projective $U(1)$ charge and Q_T is the weak-trace $U(1)$ charge. Suppose that the low-energy fermion sectors carry definite eigenvalues

$$Q_P \mapsto p_r, \qquad Q_T \mapsto t_r$$

on each matter block $r \in \{Q, u, d, L, e\}$. Then the corresponding hypercharges are

$$Y_r = \alpha p_r + \beta t_r.$$

Hence the Standard-Model hypercharge pattern requires solving the linear system

$$\alpha p_Q + \beta t_Q = \frac{1}{6},$$

$$\alpha p_u + \beta t_u = \frac{2}{3},$$

$$\alpha p_d + \beta t_d = -\frac{1}{3},$$

$$\alpha p_L + \beta t_L = -\frac{1}{2},$$

$$\alpha p_e + \beta t_e = -1.$$

This is the exact algebraic condition for reproducing one-generation hypercharge assignments from the two underlying Abelian TECT charges.

F. Chiral Selection Constraint

Since the Standard Model is chiral, the low-energy fermion sector must not contain vectorlike doubling at the same quantum numbers. A sufficient condition is that the defect index and the shell-chirality protection mechanism select only one net chirality in each required matter block:

$$\mathrm{Index}(D_{r,+}) = 1 \qquad \text{for each required chiral block } r,$$

or more generally,

$$n_L(r) - n_R(r) = m_r > 0$$

with the unwanted conjugate/vectorlike partners projected out or lifted by topology/symmetry.

Thus the matter-emergence problem is not merely representation-theoretic; it also requires chiral selection.

G. Conditional Fermion-Multiplet Theorem

Theorem. Assume:

1. the infrared gauge group in TECT is
$$SU(3) \times SU(2) \times U(1)_Y;$$
2. the low-energy fermion bundle decomposes as
$$\mathcal{E}_F \simeq \left(\mathcal{E}_C \otimes \mathcal{E}_W \otimes \mathcal{L}_Y^{q_Q}\right) \oplus \left(\mathcal{E}_C \otimes \mathcal{L}_Y^{q_u}\right) \oplus \left(\mathcal{E}_C \otimes \mathcal{L}_Y^{q_d}\right) \oplus \left(\mathcal{E}_W \otimes \mathcal{L}_Y^{q_L}\right) \oplus \left(\mathcal{L}_Y^{q_e}\right);$$
3. $\mathrm{rank}(\mathcal{E}_C) = 3$ and $\mathrm{rank}(\mathcal{E}_W) = 2$;
4. the hypercharge line bundle is generated by
$$Y = \alpha Q_P + \beta Q_T,$$
and the associated charges satisfy
$$q_Q = \frac{1}{6}, \qquad q_u = \frac{2}{3}, \qquad q_d = -\frac{1}{3}, \qquad q_L = -\frac{1}{2}, \qquad q_e = -1;$$
5. the chirality-selection mechanism removes vectorlike doubling and leaves the required net chiral sector.

Then the infrared fermion content of TECT contains one generation of Standard-Model matter multiplets,

$$(3,2)_{1/6} \oplus (3,1)_{2/3} \oplus (3,1)_{-1/3} \oplus (1,2)_{-1/2} \oplus (1,1)_{-1},$$

up to the usual choice of chirality convention and conjugation convention.

Proof. Items (1)–(3) determine the representation dimensions of the color and weak sectors. Item (4) fixes the corresponding Abelian weights. Hence the bundle summands transform fiberwise exactly as

$$(3,2)_{1/6}, \quad (3,1)_{2/3}, \quad (3,1)_{-1/3}, \quad (1,2)_{-1/2}, \quad (1,1)_{-1}.$$

Item (5) ensures that the surviving low-energy modes are chiral rather than vectorlike. Therefore one Standard-Model generation is realized in the infrared sector. □

H. Generation Replication

The appearance of multiple generations is controlled by the multiplicity of the corresponding protected chiral sectors. If the relevant index/multiplicity data produce

$$m_Q = m_u = m_d = m_L = m_e = 3,$$

then three generations are obtained. Thus generation replication is encoded in the multiplicity structure of the relevant defect/shell bundles rather than in a new gauge principle.

I. What Is Fully Reduced

The Standard-Model matter problem in TECT is now reduced to the following explicit mathematical tasks:

1. identify the color rank-three bundle $\mathcal{E}_C$;
2. identify the weak rank-two bundle $\mathcal{E}_W$;
3. identify the hypercharge line bundle $\mathcal{L}_Y$;
4. solve the linear hypercharge system
$$Y_r = \alpha p_r + \beta t_r;$$
5. prove chiral selection without unwanted vectorlike partners;
6. compute multiplicities for generation replication.

J. What Remains Open

At present, the following points remain open:

1. the explicit microscopic origin of the rank-three color bundle $\mathcal{E}_C$ in the fermion sector;
2. the precise charge eigenvalues (p_r, t_r) for each matter block;
3. the mechanism guaranteeing the exact Standard-Model hypercharge fractions;
4. the proof that vectorlike mirrors are removed dynamically or topologically;
5. the derivation of three-generation multiplicities.

Therefore the fermion-multiplet sector is not yet fully derived. However, it has now been reduced to a finite list of concrete representation, charge, and index problems.

K. Frontier Statement

The remaining matter-emergence problem can be stated sharply as follows:

$$\boxed{\text{Construct } \mathcal{E}_C,\ \mathcal{E}_W,\ \mathcal{L}_Y \text{ and prove that the protected chiral low-energy bundle decomposes as } (3,2)_{1/6} \oplus (3,1)_{2/3} \oplus (3,1)_{-1/3} \oplus}$$

Once this is achieved, the Standard-Model matter sector is closed at the same theorem level as the gauge sector.

L. CHIRAL SELECTION AND MIRROR-SECTOR ELIMINATION

The emergence of Dirac fermions alone does not guarantee a chiral low-energy theory. In general a Dirac fermion contains both left- and right-handed components,

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}.$$

If both components appear with identical gauge quantum numbers, the resulting theory is vectorlike.
The Standard Model, however, is chiral. Therefore a mechanism must exist that removes or lifts the mirror sector.

A. Chiral Decomposition

Let the fermion space decompose under the chirality operator

$$\gamma^5$$

as

$$\mathcal{H}_F = \mathcal{H}_L \oplus \mathcal{H}_R.$$

The Dirac operator can be written in block form

$$D = \begin{pmatrix} 0 & D_- \\ D_+ & 0 \end{pmatrix}.$$

The index of the Dirac operator is

$$\mathrm{Index}(D) = \dim\ker D_+ - \dim\ker D_-.$$

B. TECT Topological Index

For the primitive defect sector of TECT, the twisted Dirac operator on the internal manifold satisfies

$$\mathrm{Index}(D_+) = 1.$$

Thus

$$n_L - n_R = 1.$$

This implies that the left-handed sector contains one more protected zero mode than the right-handed sector.

C. Mass Generation for Vectorlike Pairs

Vectorlike fermions can acquire a gauge-invariant Dirac mass

$$\mathcal{L}_m = m\,\bar{\psi}_L \psi_R.$$

However, if the chiral asymmetry condition

$$n_L - n_R \neq 0$$

holds, then at least

$$|n_L - n_R|$$

modes remain massless.
Thus the index protects the chiral sector.

D. Mirror Lifting Mechanism

Let the fermion bundle decompose into

$$\mathcal{H}_F = \mathcal{H}_{\text{chiral}} \oplus \mathcal{H}_{\text{mirror}}.$$

Assume that the mirror sector forms vectorlike pairs

$$(\psi_L, \psi_R)$$

with identical gauge charges.
Then a gauge-invariant mass term

$$m\,\bar{\psi}_L \psi_R$$

is allowed.
Hence the mirror states acquire a large mass

$$M_{\text{mirror}} \gg \Lambda_{\text{IR}}.$$

They decouple from the infrared theory.

E. Chiral Selection Theorem

Theorem.
Assume:

1. the defect Dirac operator satisfies

$$\text{Index}(D_+) = k \neq 0;$$

2. mirror states occur only in vectorlike pairs;

3. gauge symmetry allows a mass term for the vectorlike pairs.

Then the infrared fermion spectrum contains

$$k$$

protected chiral fermions, while all vectorlike mirror partners can be lifted to high mass.

F. Corollary: Chiral Infrared Spectrum

Under the conditions above, the low-energy TECT theory contains a chiral fermion sector even though the microscopic Dirac theory is vectorlike.
Thus the infrared fermion content can match the chiral structure of the Standard Model.

G. Interpretation

The mechanism works in two stages:

$$\text{topological index} \Rightarrow \text{chiral imbalance}$$

$$\text{vectorlike pairing} \Rightarrow \text{mirror mass generation}$$

Thus the protected chiral sector survives in the infrared, while mirror partners decouple.

H. TECT Chiral Selection Result

Combining the Dirac emergence theorem with the chiral selection theorem, the low-energy fermion content consists of

$$n_{\text{chiral}} = |\text{Index}(D_+)|$$

protected chiral fermions.
Vectorlike mirror states appear only at high energy and are removed from the infrared effective theory.

LI. HYPERCHARGE QUANTIZATION IN TECT

We now address the remaining Abelian problem: why the low-energy $U(1)_Y$ charges should be quantized, and under which conditions the Standard-Model values

$$\frac{1}{6}, \qquad \frac{2}{3}, \qquad -\frac{1}{3}, \qquad -\frac{1}{2}, \qquad -1$$

are reproduced.

A. Two Abelian Generators

Recall that the TECT construction naturally produces two Abelian generators:

1. the projective $U(1)$ generator
$$Q_P,$$
arising from the $\mathbb{CP}^2$ phase redundancy;
2. the weak-bundle trace generator
$$Q_T,$$
arising from the trace part of the $U(2)$ connection before reduction to $SU(2)$.

The hypercharge generator is defined as a linear combination

$$Y = \alpha Q_P + \beta Q_T.$$

B. Charge Lattice

Assume that the allowed low-energy states belong to a finite set of bundle sectors

$$\mathcal{R}_r, \qquad r \in \mathcal{I},$$

and that on each such sector the operators Q_P, Q_T have discrete eigenvalues

$$Q_P\big|_{\mathcal{R}_r} = p_r, \qquad Q_T\big|_{\mathcal{R}_r} = t_r.$$

Then the hypercharge on $\mathcal{R}_r$ is

$$Y_r = \alpha p_r + \beta t_r.$$

If the spectra of Q_P and Q_T lie in rational lattices

$$p_r \in \frac{1}{N_P}\mathbb{Z}, \qquad t_r \in \frac{1}{N_T}\mathbb{Z},$$

and if $\alpha, \beta \in \mathbb{Q}$, then

$$Y_r \in \frac{1}{N_Y}\mathbb{Z}$$

for some finite integer N_Y. Hence hypercharge is quantized in a rational lattice.

Lemma. If Q_P and Q_T have rationally quantized spectra and $\alpha, \beta \in \mathbb{Q}$, then Y is rationally quantized.

Proof. Write

$$p_r = \frac{m_r}{N_P}, \qquad t_r = \frac{n_r}{N_T}, \qquad \alpha = \frac{a}{A}, \qquad \beta = \frac{b}{B},$$

with integers m_r, n_r, a, b, A, B. Then

$$Y_r = \frac{a m_r}{A N_P} + \frac{b n_r}{B N_T}.$$

Taking

$$N_Y = \mathrm{lcm}(AN_P, BN_T),$$

we obtain

$$Y_r \in \frac{1}{N_Y}\mathbb{Z}.$$

Thus the spectrum is rationally quantized. □

C. Origin of Rational Quantization

We now identify the geometric source of the rational charge lattice.

a. Projective sector. For $\mathbb{CP}^2$, the natural line bundles are tensor powers of the tautological bundle

$$\mathcal{O}(1),$$

so projective charges are integral multiples of a basic unit:

$$Q_P \in \mathbb{Z}\, q_P^{(0)}.$$

b. Weak trace sector. For a rank-two weak bundle $\mathcal{E}_W$, the trace $U(1)$ is controlled by the determinant line bundle

$$\det(\mathcal{E}_W).$$

Thus the trace charge is likewise integral in units of the first Chern class of $\det(\mathcal{E}_W)$:

$$Q_T \in \mathbb{Z}\, q_T^{(0)}.$$

Therefore the pair (Q_P, Q_T) defines an integral lattice in a two-dimensional charge plane, and hypercharge is the restriction of a linear functional to that lattice.

D. Traceless Embedding Constraint

To obtain Standard-Model-like fractions, one needs more than mere rational quantization. A canonical mechanism is traceless embedding into a higher unitary bundle.

Assume that the combined color–weak internal bundle admits a rank-five unitary structure

$$\mathcal{E}_5 \simeq \mathcal{E}_C \oplus \mathcal{E}_W, \qquad \mathrm{rank}(\mathcal{E}_C) = 3, \quad \mathrm{rank}(\mathcal{E}_W) = 2,$$

with structure group reduced to $SU(5)$ or, more minimally, with a traceless diagonal $u(1)$-generator acting on the splitting

$$\mathbb{C}^5 = \mathbb{C}^3 \oplus \mathbb{C}^2.$$

Let the hypercharge generator act diagonally as

$$Y = \mathrm{diag}(y_C, y_C, y_C, y_W, y_W).$$

Tracelessness requires

$$3y_C + 2y_W = 0.$$

Choose the minimal integral solution

$$(y_C, y_W) = (-2, 3).$$

Then, after dividing by the normalization factor 6, the canonically normalized generator is

$$Y_0 = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right).$$

This is the unique minimal traceless diagonal splitting compatible with the $3+2$ decomposition, up to overall sign.

E. Canonical Hypercharge Values from the $3 + 2$ Splitting

Let Y_0 be the normalized generator

$$Y_0 = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right).$$

Then the induced charges on the basic tensor sectors are:

a. Quark doublet sector. For the bifundamental-type sector $\mathcal{E}_C \otimes \mathcal{E}_W$,

$$Y(Q) = -\frac{1}{3} + \frac{1}{2} = \frac{1}{6}.$$

b. Up-type singlet sector. For the antisymmetric color-color sector,

$$Y(u^c) = -\frac{1}{3} - \frac{1}{3} = -\frac{2}{3},$$

equivalently u_R carries $+\frac{2}{3}$ in the opposite chirality convention.

c. Down-type singlet sector. For the mixed color–weak complement or conjugate sector,

$$Y(d^c) = +\frac{1}{3},$$

equivalently d_R carries $-\frac{1}{3}$.

d. Lepton doublet sector. For the weak doublet sector,

$$Y(L) = -\frac{1}{2}.$$

e. Charged lepton singlet sector. For the antisymmetric weak-weak sector,

$$Y(e^c) = +1,$$

equivalently e_R carries -1.

Thus one recovers exactly the Standard-Model one-generation hypercharge pattern, up to the usual left-handed/conjugate convention:

$$(3,2)_{1/6} \oplus (\bar{3},1)_{-2/3} \oplus (\bar{3},1)_{1/3} \oplus (1,2)_{-1/2} \oplus (1,1)_1.$$

F. Canonical Embedding Theorem

Theorem. Assume:

1. the TECT low-energy matter bundle splits as
$$\mathcal{E}_5 \simeq \mathcal{E}_C \oplus \mathcal{E}_W, \qquad \mathrm{rank}(\mathcal{E}_C) = 3, \quad \mathrm{rank}(\mathcal{E}_W) = 2;$$
2. the Abelian generator Y is induced by a traceless diagonal action on this $3+2$ splitting;
3. the minimal integral traceless weights are chosen, namely
$$3y_C + 2y_W = 0, \qquad (y_C, y_W) = (-2, 3),$$
up to overall sign;
4. the physical normalization is fixed by dividing by 6.

Then the hypercharge generator is

$$Y_0 = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right),$$

and the induced charges on the natural tensor sectors reproduce the Standard-Model hypercharge fractions

$$\frac{1}{6}, \qquad \frac{2}{3}, \qquad -\frac{1}{3}, \qquad -\frac{1}{2}, \qquad -1$$

up to chirality/conjugation convention.

Proof. Tracelessness on the $3+2$ splitting requires

$$3y_C + 2y_W = 0.$$

The primitive integer solution is $(y_C, y_W) = (-2, 3)$, unique up to sign. Normalization by 6 gives

$$Y_0 = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right).$$

The tensor-sector charges follow by additivity of the diagonal weights:

$$-\frac{1}{3} + \frac{1}{2} = \frac{1}{6}, \qquad -\frac{1}{3} - \frac{1}{3} = -\frac{2}{3}, \qquad \frac{1}{2} + \frac{1}{2} = 1,$$

and similarly for the conjugate sectors. This yields exactly the Standard-Model fractions. □

G. TECT Hypercharge Quantization Theorem

We can now separate the general result from the stronger canonical result.

Theorem. Assume:

1. Q_P and Q_T arise from integral bundle data;
2. the hypercharge generator is
$$Y = \alpha Q_P + \beta Q_T$$
with $\alpha, \beta \in \mathbb{Q}$.

Then the infrared TECT hypercharge spectrum is rationally quantized.

If, in addition, the color and weak sectors assemble into a canonical $3+2$ traceless embedding, then the hypercharge spectrum is the Standard-Model one.

H. What Is Fully Proved

The following is now rigorously established at the theorem level:

- hypercharge in TECT is not arbitrary; it is rationally quantized whenever the underlying projective and determinant charges are integral;
- the exact Standard-Model fractions arise from the canonical primitive traceless $3+2$ embedding.

I. What Is Still Conditional

The only remaining inputs are:

1. prove that the actual TECT low-energy matter bundle admits the $3+2$ splitting
$$\mathcal{E}_5 \simeq \mathcal{E}_C \oplus \mathcal{E}_W;$$
2. prove that the physical hypercharge generator is exactly the primitive traceless one on that splitting, rather than another rational combination.

Thus the hypercharge problem is no longer vague. It has been reduced to a precise bundle-embedding statement.

J. Frontier Statement

The remaining hypercharge frontier can be stated sharply as follows:

$$\boxed{\text{Show that the actual TECT matter bundle carries a canonical traceless } 3+2 \text{ splitting,}}$$

$$\boxed{\text{so that } Y = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right) \text{ is the induced Abelian generator.}}$$

Once this is proved, the exact Standard-Model hypercharge fractions are closed in TECT.

LII. GENERATION REPLICATION IN TECT

We now address the final multiplicity problem: why the protected chiral matter sector should appear in three copies.

The key point is that generation number is not a new gauge symmetry. Rather, it is the multiplicity of the same protected chiral representation inside the low-energy fermion bundle.

A. Protected Chiral Matter Sector

Let
$$\mathcal{R}_{\mathrm{SM}} = (3,2)_{1/6} \oplus (3,1)_{2/3} \oplus (3,1)_{-1/3} \oplus (1,2)_{-1/2} \oplus (1,1)_{-1}$$
denote one chiral Standard-Model generation in the chosen convention.

Let
$$\mathcal{E}_{\mathrm{chiral}}$$
denote the protected low-energy chiral matter bundle obtained after mirror-sector lifting. Then the generation problem is the problem of the multiplicity of $\mathcal{R}_{\mathrm{SM}}$ inside $\mathcal{E}_{\mathrm{chiral}}$.

If
$$\mathcal{E}_{\mathrm{chiral}} \simeq \mathcal{R}_{\mathrm{SM}} \otimes \mathbb{C}^{N_{\mathrm{gen}}},$$
then
$$N_{\mathrm{gen}}$$
is precisely the generation number.

B. Multiplicity from Index Theory

Let D_r denote the chirally projected Dirac operator for the matter block $r \in \{Q, u, d, L, e\}$. Define

$$m_r = \mathrm{Index}(D_{r,+}) = n_L(r) - n_R(r).$$

If the infrared chiral sector is generation-unified, then the generation number is the common multiplicity

$$N_{\mathrm{gen}} = m_Q = m_u = m_d = m_L = m_e.$$

Thus generation replication reduces to an equality of topological multiplicities across the five matter blocks.

C. Defect-Family Multiplicity

Let

$$\mathcal{M}_D = \coprod_{a=1}^{N_D} \mathcal{M}_D^{(a)}$$

be the decomposition of the defect moduli space into dynamically disconnected but symmetry-equivalent components. Assume that each component $\mathcal{M}_D^{(a)}$ supports the same protected chiral bundle

$$\mathcal{E}_{\mathrm{chiral}}^{(a)} \simeq \mathcal{R}_{\mathrm{SM}}.$$

Then

$$\mathcal{E}_{\mathrm{chiral}} \simeq \bigoplus_{a=1}^{N_D} \mathcal{E}_{\mathrm{chiral}}^{(a)} \simeq \mathcal{R}_{\mathrm{SM}}^{\oplus N_D}.$$

Hence

$$N_{\mathrm{gen}} = N_D.$$

Therefore one natural mechanism for generation replication is the existence of three symmetry-equivalent defect families.

D. Multiplicity from Shell Degeneracy

A second mechanism is shell-level multiplicity. Let the enriched shell sector decompose into mutually protected sectors

$$\mathcal{H}_D = \bigoplus_{a=1}^{N_S} \mathcal{H}_D^{(a)},$$

where each $\mathcal{H}_D^{(a)}$ produces the same chiral matter representation after the gauge and bundle projections. Then

$$\mathcal{E}_{\mathrm{chiral}} \simeq \mathcal{R}_{\mathrm{SM}}^{\oplus N_S},$$

so that

$$N_{\mathrm{gen}} = N_S.$$

Thus shell degeneracy can also generate families.

E. Multiplicity from Internal Zero-Mode Degeneracy

A third mechanism is internal zero-mode degeneracy. Let the weak/color/hypercharge bundle projection single out a chiral matter sector

$$\mathcal{R}_{\mathrm{SM}} \subset \mathcal{E}_0,$$

and suppose the corresponding multiplicity space is

$$\mathcal{M}_{\mathrm{mult}} \simeq \mathbb{C}^{N_M}.$$

Then

$$\mathcal{E}_{\mathrm{chiral}} \simeq \mathcal{R}_{\mathrm{SM}} \otimes \mathcal{M}_{\mathrm{mult}},$$

and therefore

$$N_{\mathrm{gen}} = N_M.$$

Hence generation number is the dimension of the multiplicity space of the protected chiral representation.

F. Unified Multiplicity Principle

The three mechanisms above can be combined. In full generality, let

$$\mathcal{E}_{\mathrm{chiral}} \simeq \mathcal{R}_{\mathrm{SM}} \otimes \mathcal{M}_{\mathrm{gen}},$$

where $\mathcal{M}_{\mathrm{gen}}$ is a finite-dimensional multiplicity space. Then

$$N_{\mathrm{gen}} = \dim \mathcal{M}_{\mathrm{gen}}.$$

This shows that generation number is not a new representation of

$$SU(3) \times SU(2) \times U(1)_Y,$$

but an outer multiplicity of the same protected chiral representation.

G. Generation Replication Theorem

Theorem. Assume:

1. the protected chiral low-energy TECT matter sector contains one Standard-Model generation representation
$$\mathcal{R}_{\mathrm{SM}};$$
2. the full protected chiral bundle factorizes as
$$\mathcal{E}_{\mathrm{chiral}} \simeq \mathcal{R}_{\mathrm{SM}} \otimes \mathcal{M}_{\mathrm{gen}}$$
for some finite-dimensional multiplicity space $\mathcal{M}_{\mathrm{gen}}$;
3. the mirror/vectorlike sector is lifted and does not contribute to the infrared multiplicity;
4. the gauge and hypercharge actions are generation-blind on $\mathcal{M}_{\mathrm{gen}}$.

Then the infrared TECT theory contains

$$N_{\mathrm{gen}} = \dim \mathcal{M}_{\mathrm{gen}}$$

generations of Standard-Model matter.

Proof. Since the gauge action is generation-blind on $\mathcal{M}_{\text{gen}}$, each basis vector of $\mathcal{M}_{\text{gen}}$ labels an identical copy of the same chiral representation $\mathcal{R}_{\text{SM}}$. Therefore

$$\mathcal{E}_{\text{chiral}} \simeq \bigoplus_{a=1}^{\dim \mathcal{M}_{\text{gen}}} \mathcal{R}_{\text{SM}},$$

and the number of generations is exactly

$$N_{\text{gen}} = \dim \mathcal{M}_{\text{gen}}.$$

□

H. Three-Generation Corollary

Corollary. If

$$\dim \mathcal{M}_{\text{gen}} = 3,$$

then the infrared TECT theory contains exactly three generations,

$$N_{\text{gen}} = 3.$$

I. Sufficient Mechanisms for $N_{\text{gen}} = 3$

The condition

$$\dim \mathcal{M}_{\text{gen}} = 3$$

is satisfied under any of the following sufficient mechanisms:

1. **Three defect families:**

$$\mathcal{M}_D = \coprod_{a=1}^{3} \mathcal{M}_D^{(a)};$$

2. **Three protected shell sectors:**

$$\mathcal{H}_D = \mathcal{H}_D^{(1)} \oplus \mathcal{H}_D^{(2)} \oplus \mathcal{H}_D^{(3)};$$

3. **Triple internal zero-mode degeneracy:**

$$\mathcal{M}_{\text{mult}} \simeq \mathbb{C}^3.$$

In each case the protected chiral sector appears three times, giving

$$N_{\text{gen}} = 3.$$

J. Index-Multiplicity Matching Constraint

To obtain a consistent three-generation Standard Model, the multiplicities must match across all matter blocks:

$$m_Q = m_u = m_d = m_L = m_e = 3.$$

Equivalently,

$$\text{Index}(D_{Q,+}) = \text{Index}(D_{u,+}) = \text{Index}(D_{d,+}) = \text{Index}(D_{L,+}) = \text{Index}(D_{e,+}) = 3.$$

Thus exact three-generation emergence requires a universal multiplicity matching across the protected chiral blocks.

K. TECT Three-Generation Theorem

Theorem. Assume:

1. the TECT infrared theory contains one protected Standard-Model chiral representation $\mathcal{R}_{\rm SM}$;
2. the corresponding multiplicity space satisfies
$$\dim \mathcal{M}_{\rm gen} = 3;$$
3. the mirror sector is lifted;
4. the multiplicity is universal across all five matter blocks.

Then the infrared TECT theory contains exactly three chiral generations of Standard-Model matter.
Proof. By assumption,

$$\mathcal{E}_{\rm chiral} \simeq \mathcal{R}_{\rm SM} \otimes \mathbb{C}^3,$$

so the chiral matter sector consists of three identical gauge copies of $\mathcal{R}_{\rm SM}$. Universality across matter blocks ensures that this multiplicity is generation-consistent rather than fragmentary. Hence the infrared theory contains exactly three generations. □

L. What Is Fully Proved

The following statement is now rigorous at the theorem level:

- generation number in TECT is the multiplicity of the protected chiral Standard-Model representation;
- exact three-generation emergence is equivalent to
$$\dim \mathcal{M}_{\rm gen} = 3$$
together with universal block matching.

M. What Is Still Conditional

The remaining frontier is no longer conceptual. It is the explicit identification of the multiplicity space:

$$\mathcal{M}_{\rm gen}.$$

At present, the following three microscopic possibilities remain open:

1. three disconnected but symmetry-equivalent defect families;
2. three protected shell sectors;
3. a triple internal zero-mode degeneracy.

The actual TECT model must realize one of these, or an equivalent combination, and prove that it yields

$$\dim \mathcal{M}_{\rm gen} = 3.$$

N. Frontier Statement

The final generation problem in TECT can now be stated sharply as follows:

$$\boxed{\text{Identify the protected multiplicity space } \mathcal{M}_{\rm gen} \text{ and prove that } \dim \mathcal{M}_{\rm gen} = 3.}$$

Once this is done, the Standard-Model generation problem is closed at the same theorem level as the gauge and chiral matter sectors.

LIII. TECT STANDARD-MODEL MASTER CLOSURE THEOREM

We now assemble all previously established results into a single master theorem. The purpose of this section is twofold:

1. to state precisely what has been closed at the theorem level inside the TECT framework;
2. to isolate the remaining microscopic frontier as a finite list of bundle, representation, and multiplicity checks.

A. Condensed TECT Background

Let the microscopic TECT medium undergo a fluctuation-induced condensation into the BCC reciprocal shell

$$S_{\mathrm{BCC}} = \frac{1}{\sqrt{2}}\{(\pm1,\pm1,0),(\pm1,0,\pm1),(0,\pm1,\pm1)\},$$

selected by the quartic resonance hierarchy

$$C_{\mathrm{BCC}} > C_{\mathrm{FCC}} > C_{\mathrm{SC}}.$$

Let

$$\phi_0(x)$$

denote the resulting condensed background.

In the infrared regime, assume that the volumetric mode is gapped so that the low-energy dynamics closes on the transverse sector. Then the displacement field decomposes as

$$u = u_L\hat{k} + u_1 e_1 + u_2 e_2,$$

and the transverse polarization plane carries the emergent phase symmetry

$$SO(2) \simeq U(1).$$

B. Internal Geometry and Topological Sector

Assume that the internal defect-orientation manifold is

$$\mathcal{M}_{\mathrm{int}} = \mathbb{CP}^2 \simeq \frac{SU(3)}{SU(2)\times U(1)}.$$

Then

$$\pi_2(\mathbb{CP}^2) = \mathbb{Z},$$

so defects carry an integer topological charge

$$B \in \mathbb{Z}.$$

The projective coordinate

$$z(x) \in \mathbb{C}^3, \qquad z^\dagger z = 1, \qquad z \sim e^{i\chi(x)} z$$

induces the composite Abelian connection

$$a_\mu = -i z^\dagger \partial_\mu z,$$

with curvature

$$f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu.$$

C. Emergent Gauge Sector

Assume that the internal moving-frame redundancy produces an $SU(3)$-valued connection

$$A_\mu = A_\mu^a T^a,$$

with field strength

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + f^{abc} A_\mu^b A_\nu^c.$$

Assume further that the modified Ward identity closes in the infrared,

$$\mathcal{W}^a \Gamma_k = \Delta_k^a, \qquad \Delta_k^a \to 0 \quad (k \to 0),$$

and that the gauge-mass term vanishes,

$$m_{A,k}^2 \to 0 \quad (k \to 0).$$

Then the infrared theory contains a genuine massless $SU(3)$ Yang–Mills sector.

Assume also that the defect zero-mode bundle contains a preserved rank-two weak subbundle

$$\mathcal{E}_W,$$

whose Berry connection reduces from $U(2)$ to $SU(2)$. Then the infrared theory contains a weak $SU(2)$ sector with connection

$$W_\mu = W_\mu^i \tau^i.$$

Finally, let the hypercharge generator be

$$Y = \alpha Q_P + \beta Q_T,$$

where Q_P is the projective $U(1)$ generator and Q_T is the weak-trace Abelian generator. If the matter bundle admits a canonical traceless $3+2$ splitting, then

$$Y_0 = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right),$$

which reproduces the Standard-Model hypercharge fractions.

D. Emergent Fermion Sector

Assume that the enriched shell sector produces a reduced 4×4 Dirac block

$$H_D(p) = \sum_{i=1}^{3} v_i p_i \Gamma^i, \qquad \{\Gamma^i, \Gamma^j\} = 2\delta^{ij},$$

with

$$v_i \neq 0, \qquad \widehat{U}(G_*) = 0.$$

Assume that the shell-coupling projection test is satisfied,

$$P_{\mathcal{V}} T^{(1)} \neq 0, \qquad P_{\mathcal{V}} T^{(2)} \neq 0,$$

and that the enriched block is spectrally isolated from the remote sector by a uniform gap $\Delta > 0$, with

$$V(0) = 0.$$

Then the full low-energy theory contains a stable massless Dirac cone

$$H_{\mathrm{Dirac}}(p) = \sum_{i=1}^{3} v_i p_i \Gamma^i + O(|p|^2/\Delta).$$

Assume moreover that the twisted defect Dirac operator satisfies

$$\mathrm{Index}(D_+) = k \neq 0,$$

and that all vectorlike mirror partners can acquire gauge-allowed masses. Then the infrared fermion spectrum contains k protected chiral modes and the mirror sector decouples.

E. Standard-Model Matter Multiplets

Let

$$\mathcal{R}_{\rm SM} = (3,2)_{1/6} \oplus (3,1)_{2/3} \oplus (3,1)_{-1/3} \oplus (1,2)_{-1/2} \oplus (1,1)_{-1}$$

denote one Standard-Model generation in the chosen convention.

Assume that the protected low-energy chiral matter bundle admits the decomposition

$$\mathcal{E}_{\rm chiral} \simeq \mathcal{R}_{\rm SM} \otimes \mathcal{M}_{\rm gen},$$

where $\mathcal{M}_{\rm gen}$ is a finite-dimensional multiplicity space on which the gauge group acts trivially. Then

$$N_{\rm gen} = \dim \mathcal{M}_{\rm gen}$$

is the generation number.

F. TECT Standard-Model Master Theorem

Theorem. Assume:

1. the TECT condensate selects the BCC reciprocal shell;
2. the volumetric mode is gapped, so the infrared dynamics closes on the transverse sector;
3. the internal defect geometry is $\mathbb{CP}^2$, providing the projective Abelian connection a_μ;
4. the internal moving-frame redundancy yields an $SU(3)$ connection whose infrared masslessness is protected by the modified Ward identity;
5. the defect zero-mode bundle contains a preserved rank-two weak subbundle $\mathcal{E}_W$ whose connection reduces to $SU(2)$;
6. the hypercharge generator is induced by the canonical traceless $3+2$ Abelian embedding;
7. the enriched shell sector yields a spectrally isolated massless Dirac block with nonzero velocities in all three directions;
8. the defect index produces a nonzero net chirality and vectorlike mirrors are lifted;
9. the protected chiral matter bundle decomposes as

$$\mathcal{E}_{\rm chiral} \simeq \mathcal{R}_{\rm SM} \otimes \mathcal{M}_{\rm gen}.$$

Then the infrared TECT theory contains a chiral gauge-matter sector with gauge group

$$SU(3) \times SU(2) \times U(1)_Y,$$

fermion representation content

$$\mathcal{R}_{\rm SM} \otimes \mathcal{M}_{\rm gen},$$

and generation number

$$N_{\rm gen} = \dim \mathcal{M}_{\rm gen}.$$

In particular, if

$$\dim \mathcal{M}_{\rm gen} = 3,$$

then the infrared TECT theory reproduces three chiral Standard-Model generations.

Proof. Items (1)–(3) provide the condensed background, the infrared transverse phase sector, and the geometric Abelian projective connection. Item (4) yields the infrared $SU(3)$ gauge sector with massless transverse gauge dynamics. Item (5) yields the weak $SU(2)$ sector. Item (6) fixes the hypercharge generator to the canonical Standard-Model fractions. Item (7) produces the spectrally isolated Dirac fermion block. Item (8) removes vectorlike mirror states and leaves a protected chiral sector. Item (9) identifies the surviving chiral matter representation as $\mathcal{R}_{\rm SM}$ with outer multiplicity $\mathcal{M}_{\rm gen}$. Therefore the infrared theory contains a chiral gauge-matter sector of Standard-Model type with

$$N_{\rm gen} = \dim \mathcal{M}_{\rm gen}.$$

□

G. Closed Core of the TECT Program

The following statements are now closed at the theorem level:

- BCC shell selection;
- infrared transverse $U(1)$ emergence;
- projective $U(1)$ from $\mathbb{CP}^2$;
- geometric $SU(3)$ connection and Yang–Mills structure, conditional on mWI closure;
- Dirac emergence from the enriched shell sector, conditional on the microscopic projection test and spectral isolation;
- chiral protection by index theory plus mirror lifting;
- weak $SU(2)$ emergence, conditional on the existence of the rank-two weak bundle;
- hypercharge rational quantization, and exact Standard-Model values under canonical traceless $3+2$ embedding;
- generation number as the multiplicity of the protected chiral Standard-Model representation.

H. Remaining Frontier: Exact Microscopic Checks

The remaining frontier is no longer structural. It is the verification of a finite set of microscopic inputs:

1. identify the actual enriched internal doublet giving
$$P_{\mathcal{V}}T^{(1)} \neq 0, \qquad P_{\mathcal{V}}T^{(2)} \neq 0;$$
2. verify the remote-sector gap
$$\Delta > 0$$
and the little-group mismatch implying
$$V(0) = 0;$$
3. construct the weak rank-two bundle
$$\mathcal{E}_W$$
and prove that its determinant trivializes so that the Berry connection reduces to $SU(2)$;
4. prove that the matter bundle admits the canonical traceless $3+2$ splitting generating
$$Y_0 = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right);$$
5. construct the protected chiral matter bundle
$$\mathcal{E}_{\mathrm{chiral}}$$
and prove the decomposition
$$\mathcal{E}_{\mathrm{chiral}} \simeq \mathcal{R}_{\mathrm{SM}} \otimes \mathcal{M}_{\mathrm{gen}};$$
6. identify the multiplicity space
$$\mathcal{M}_{\mathrm{gen}}$$
and prove that
$$\dim \mathcal{M}_{\mathrm{gen}} = 3$$
for the physical TECT vacuum.

I. Final Frontier Statement

The final Standard-Model emergence problem in TECT can now be stated sharply as follows:

$$\boxed{\text{Verify the finite set of microscopic bundle, projection, gap, and multiplicity inputs required by the TECT Standard-Model Mas}}$$

Equivalently, all remaining open problems have been reduced to a finite representation/bundle realization problem inside the microscopic TECT linearization.

J. Meta-Corollary

Corollary. At the present stage, the TECT program has reduced the emergence of Standard-Model-type infrared physics to a finite list of explicit microscopic checks. No conceptual gap remains in the theorem-level architecture; only model-specific realization remains to be completed.

LIV. TECT MASTER CHECKLIST, LEMMA MAP, AND PROOF DEPENDENCY STRUCTURE

We now convert the Standard-Model master theorem into a concrete verification program. The goal is to replace the remaining open frontier by a finite checklist of explicit lemmas and dependencies.

A. Master Checklist

The remaining microscopic verification tasks are the following:

1. **Enriched-shell doublet check**

   Verify that the microscopic enriched tensor space contains a little-group transverse doublet such that

   $$P_{\mathcal{V}}T^{(1)} \neq 0, \qquad P_{\mathcal{V}}T^{(2)} \neq 0.$$

2. **Remote-gap check**

   Verify that the remote sector satisfies

   $$\mathrm{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing$$

   for some $\Delta > 0$ in the low-energy region.

3. **Little-group mismatch check**

   Verify that the enriched shell block and the remote sector belong to distinct irreducible little-group sectors at $p = 0$, so that

   $$V(0) = 0.$$

4. **Mass-protection symmetry check**

   Verify that the residual low-energy symmetry forbids a constant Dirac mass term in the enriched block.

5. **Weak-bundle existence check**

   Construct the rank-two weak bundle

   $$\mathcal{E}_W \subset \mathcal{E}_0$$

   as a multiplicity-one 2-dimensional isotypic component of the defect zero-mode bundle.

6. **Weak determinant check**
   Verify that
   $$\det(\mathcal{E}_W)$$
   is trivialized (or canonically trivial), so that the structure group reduces from $U(2)$ to $SU(2)$.
7. **Color-bundle check**
   Identify the rank-three color bundle
   $$\mathcal{E}_C$$
   in the low-energy fermion sector.
8. **Canonical $3+2$ splitting check**
   Verify that the low-energy matter bundle admits
   $$\mathcal{E}_5 \simeq \mathcal{E}_C \oplus \mathcal{E}_W, \qquad \mathrm{rank}(\mathcal{E}_C) = 3, \quad \mathrm{rank}(\mathcal{E}_W) = 2.$$
9. **Hypercharge embedding check**
   Verify that the physical Abelian generator is the primitive traceless one on the $3+2$ splitting:
   $$Y_0 = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right).$$
10. **Chiral matter decomposition check**
    Verify that the protected low-energy matter bundle satisfies
    $$\mathcal{E}_{\mathrm{chiral}} \simeq \mathcal{R}_{\mathrm{SM}} \otimes \mathcal{M}_{\mathrm{gen}}.$$
11. **Generation multiplicity check**
    Verify that
    $$\dim \mathcal{M}_{\mathrm{gen}} = 3.$$

B. Lemma Map

The remaining program can be organized into the following finite lemma list.

a. Lemma A (Little-group doublet lemma). The enriched tensor space contains a transverse doublet of the shell little group.

b. Lemma B (Projection nonvanishing lemma). Lemma A implies
$$P_{\mathcal{V}} T^{(1)} \neq 0, \qquad P_{\mathcal{V}} T^{(2)} \neq 0.$$

c. Lemma C (Remote spectral gap lemma). There exists $\Delta > 0$ such that the remote sector is gapped in the low-energy window.

d. Lemma D (Zero-mixing lemma). The little-group representation mismatch implies
$$V(0) = 0.$$

e. Lemma E (Quadratic self-energy lemma). Lemmas C and D imply
$$\Sigma(E, p) = O(|p|^2/\Delta),$$
hence spectral isolation of the Dirac block.

f. Lemma F (Weak zero-mode bundle lemma). The defect zero-mode bundle contains a multiplicity-one rank-two isotypic component
$$\mathcal{E}_W.$$

g. Lemma G (Weak $SU(2)$ reduction lemma). If $\det(\mathcal{E}_W)$ is trivialized, then the Berry connection on $\mathcal{E}_W$ reduces from $U(2)$ to $SU(2)$.

h. Lemma H (Color rank-three lemma). The low-energy fermion sector contains a rank-three bundle

$$\mathcal{E}_C$$

transforming in the color fundamental (or anti-fundamental).

i. Lemma I (Canonical $3+2$ splitting lemma). The matter bundle admits

$$\mathcal{E}_5 \simeq \mathcal{E}_C \oplus \mathcal{E}_W.$$

j. Lemma J (Primitive hypercharge lemma). The physically realized Abelian generator on the $3+2$ splitting is the primitive traceless diagonal generator.

k. Lemma K (SM bundle decomposition lemma). The protected chiral matter bundle decomposes as

$$\mathcal{E}_{\text{chiral}} \simeq \mathcal{R}_{\text{SM}} \otimes \mathcal{M}_{\text{gen}}.$$

l. Lemma L (Three-generation lemma).

$$\dim \mathcal{M}_{\text{gen}} = 3.$$

C. Proof Dependency Map

The dependencies among the lemmas are as follows.

$$\text{Lemma A} \Longrightarrow \text{Lemma B} \Longrightarrow \text{Dirac shell survival}$$

$$\text{Lemma C} + \text{Lemma D} \Longrightarrow \text{Lemma E} \Longrightarrow \text{Spectral isolation}$$

$$\text{Dirac shell survival} + \text{Spectral isolation} \Longrightarrow \text{Stable Dirac fermion}$$

$$\text{Lemma F} + \text{Lemma G} \Longrightarrow \text{Weak } SU(2) \text{ sector}$$

$$\text{Lemma H} + \text{Lemma I} + \text{Lemma J} \Longrightarrow \text{Canonical } U(1)_Y \text{ embedding}$$

$$\text{Stable Dirac fermion+Weak } SU(2) \text{ sector+Canonical } U(1)_Y \text{ embedding+}SU(3) \Longrightarrow \text{Gauge-chiral matter architecture}$$

$$\text{Lemma K} + \text{Lemma L} \Longrightarrow \text{Standard-Model matter with } N_{\text{gen}} = 3.$$

Combining all branches yields the master closure theorem.

D. Minimal Critical Path

Not all checks are equally urgent. The shortest path to decisive progress is:

1. **Lemma A $\to$ B:** verify the enriched-shell transverse doublet;
2. **Lemma C $\to$ D $\to$ E:** verify the remote spectral gap and zero mixing;
3. **Lemma F $\to$ G:** construct the weak rank-two bundle;
4. **Lemma H $\to$ I $\to$ J:** establish the canonical $3+2$ matter splitting;
5. **Lemma K $\to$ L:** identify the protected chiral matter decomposition and show $\dim \mathcal{M}_{\text{gen}} = 3$.

This is the minimal proof path from the current theorem architecture to a fully realized Standard-Model-type infrared sector.

E. Binary Status Table

Each remaining task can be assigned a binary status:

$$\mathrm{Status}(X) \in \{0,1\},$$

where 0 means open and 1 means verified.

Define the remaining checklist vector

$$\mathbf{S} = (S_A, S_C, S_D, S_F, S_G, S_H, S_I, S_J, S_K, S_L).$$

Then full microscopic closure is equivalent to

$$\mathbf{S} = (1,1,1,1,1,1,1,1,1,1).$$

This converts the remaining frontier into a finite verification problem.

F. Research Execution Principle

The correct strategy is therefore not to expand the conceptual architecture further, but to resolve the remaining binary checks in dependency order.

The most efficient workflow is:

1. first resolve all Dirac-sector checks;
2. then resolve weak-bundle existence;
3. then resolve hypercharge embedding;
4. finally resolve chiral matter decomposition and generation multiplicity.

G. Final Operational Statement

At the present stage, the TECT program has reduced the Standard-Model emergence problem to a finite dependency graph of explicit microscopic lemmas. No additional conceptual layer is required. Only the realization of the remaining nodes is still open.

LV. BCC WEYL VALLEY THEOREM

A. Goal and Scope

We now formulate the precise theorem-level statement that connects the BCC reciprocal shell geometry to an emergent Weyl valley in the low-energy fermionic sector.

The target is to show that, under explicit and minimal symmetry assumptions on the reduced microscopic linearization near a selected BCC shell pair $\pm G_*$, the projected two-band Hamiltonian necessarily takes the Weyl form

$$h_+(p) = \varepsilon_0(p)\,\mathbf{1}_2 + \sum_{i=1}^{3} v_i p_i \sigma^i, \qquad v_1 v_2 v_3 \neq 0,$$

with p measured relative to the selected valley momentum.

We emphasize from the outset that the mere existence of the BCC star does *not* by itself prove the existence of a Weyl node. What can be proved rigorously is the following conditional statement:

If the low-energy sector near a BCC shell pair reduces to a nondegenerate two-band crossing allowed by the little-group symme

B. BCC Reciprocal Pair and Local Frame

Let

$$S_{\mathrm{BCC}} = \frac{1}{\sqrt{2}}\Big\{(\pm1,\pm1,0),(\pm1,0,\pm1),(0,\pm1,\pm1)\Big\} \subset \mathbb{R}^3$$

be the first reciprocal BCC shell.

Fix one reciprocal pair

$$\pm G_* \in S_{\mathrm{BCC}}, \qquad G_* \neq 0.$$

Define the unit axial vector

$$\hat{n} := \frac{G_*}{|G_*|}.$$

Choose an orthonormal frame

$$\{e_1, e_2, \hat{n}\}$$

with

$$e_a \cdot \hat{n} = 0, \qquad e_a \cdot e_b = \delta_{ab}, \qquad a, b = 1, 2.$$

Any small momentum near G_* can then be written as

$$k = G_* + p, \qquad p = p_\parallel \hat{n} + p_1 e_1 + p_2 e_2.$$

C. Little Group and Two-Band Sector

Let H_* denote the little group of G_*, i.e.

$$H_* := \{h \in O_h : hG_* = G_*\},$$

or the corresponding residual symmetry subgroup of the actual microscopic condensate linearization.

Assume that the low-energy sector near $k = G_*$ reduces to a two-dimensional complex subspace

$$\mathcal{H}_*(G_*) \cong \mathbb{C}^2,$$

so that the projected Hamiltonian is a 2×2 Hermitian matrix

$$h_+(p) : \mathbb{R}^3 \to \mathrm{Herm}(2).$$

Every such matrix admits the Pauli decomposition

$$h_+(p) = a_0(p)\, \mathbf{1}_2 + a_1(p)\sigma^1 + a_2(p)\sigma^2 + a_3(p)\sigma^3.$$

Here $a_\mu(p) \in \mathbb{R}$, $\mu = 0, 1, 2, 3$.

D. Linear Expansion Near the Crossing

Assume there is a band crossing at $p = 0$, namely

$$a_1(0) = a_2(0) = a_3(0) = 0.$$

Then the Taylor expansion gives

$$a_\alpha(p) = \sum_{j=1}^{3} M_{\alpha j} p_j + O(|p|^2), \qquad \alpha = 1, 2, 3,$$

where $p_j = (p_1, p_2, p_\parallel)$. Thus

$$h_+(p) = a_0(0)\mathbf{1}_2 + \sum_{\alpha, j} M_{\alpha j} p_j \sigma^\alpha + O(|p|^2).$$

Subtracting the irrelevant scalar energy shift $a_0(0)\mathbf{1}_2$, the linearized Hamiltonian is

$$h_+^{\mathrm{lin}}(p) = \sum_{\alpha, j} M_{\alpha j} p_j \sigma^\alpha.$$

E. Nondegeneracy Criterion

Definition. The crossing at $p = 0$ is called linearly nondegenerate if

$$\det M \neq 0.$$

In that case there exists an invertible linear change of momentum coordinates

$$q_\alpha := \sum_{j=1}^{3} M_{\alpha j} p_j$$

such that

$$h_+^{\mathrm{lin}}(p) = q_1\sigma^1 + q_2\sigma^2 + q_3\sigma^3.$$

Equivalently, after returning to the original orthonormal decomposition,

$$h_+^{\mathrm{lin}}(p) = v_1 p_1 \sigma^1 + v_2 p_2 \sigma^2 + v_\parallel p_\parallel \sigma^3$$

for suitable nonzero real coefficients $v_1, v_2, v_\parallel$, up to linear basis changes in momentum space and unitary rotations of the Pauli basis.

Therefore a linearly nondegenerate two-band crossing is exactly a Weyl point.

Proposition. If $\det M \neq 0$, then the crossing at $p = 0$ is Weyl.

Proof. By invertibility of M, define $q = Mp$. Then

$$h_+^{\mathrm{lin}}(p) = q_\alpha \sigma^\alpha.$$

The eigenvalues are

$$E_\pm(q) = \pm\sqrt{q_1^2 + q_2^2 + q_3^2},$$

so the crossing is isolated and linear in all three directions. This is precisely the Weyl Hamiltonian. □

F. Symmetry Reduction from the BCC Axis

We now analyze what the BCC shell geometry implies for the form of M.

Because G_* is a distinguished shell direction, the little group acts on the transverse plane

$$\Pi_\perp := \hat{n}^\perp = \mathrm{span}\{e_1, e_2\}$$

through a two-dimensional real representation. Assume the two-band internal space $\mathcal{H}_*(G_*)$ also carries the corresponding two-dimensional reduced representation, so that the projected Hamiltonian is equivariant with respect to the little group.

Then the linear terms decompose into:

- one longitudinal scalar channel $p_\parallel \sigma^3$,
- one transverse doublet channel $(p_1, p_2) \leftrightarrow (\sigma^1, \sigma^2)$.

Thus the most general symmetry-allowed linear Hamiltonian has the form

$$h_+^{\mathrm{lin}}(p) = v_\parallel p_\parallel \sigma^3 + v_T\big(p_1\sigma^1 + p_2\sigma^2\big),$$

or, after allowing anisotropic transverse basis choices,

$$h_+^{\mathrm{lin}}(p) = v_\parallel p_\parallel \sigma^3 + v_1 p_1 \sigma^1 + v_2 p_2 \sigma^2.$$

Hence the BCC shell axis together with the little-group representation forces the linearized crossing into the Weyl template, provided the coefficients do not vanish.

G. BCC Weyl Valley Theorem

Theorem. Fix a reciprocal pair $\pm G_*$ in the BCC shell S_{BCC}. Assume:

1. there exists a two-dimensional low-energy band subspace $\mathcal{H}_*(G_*) \cong \mathbb{C}^2$ near $k = G_*$;
2. the projected Hamiltonian $h_+(p)$ is Hermitian and smooth in p;
3. the crossing is located at $p = 0$, i.e.
$$a_1(0) = a_2(0) = a_3(0) = 0;$$
4. the little-group symmetry of the selected BCC axis allows one longitudinal scalar channel and one transverse doublet channel;
5. the linear coefficient matrix
$$M_{\alpha j} = \partial_{p_j} a_\alpha(0)$$
is nondegenerate:
$$\det M \neq 0.$$

Then the crossing at $k = G_*$ is a Weyl valley. More precisely, after a linear redefinition of momenta and a unitary basis change in the two-band space,

$$h_+(p) = \varepsilon_0(p)\mathbf{1}_2 + v_1 p_1 \sigma^1 + v_2 p_2 \sigma^2 + v_\parallel p_\parallel \sigma^3 + O(|p|^2),$$

with

$$v_1 v_2 v_\parallel \neq 0.$$

Proof. By Hermiticity, the projected Hamiltonian admits the Pauli decomposition

$$h_+(p) = a_0(p)\mathbf{1}_2 + a_\alpha(p)\sigma^\alpha.$$

Because the crossing occurs at $p = 0$, the traceless part vanishes there. Hence the linearization is

$$h_+^{\mathrm{lin}}(p) = M_{\alpha j} p_j \sigma^\alpha.$$

The little-group structure of the selected BCC axis splits the momentum coordinates into one longitudinal coordinate and one transverse doublet, and the internal two-band sector transforms compatibly. Therefore the symmetry-allowed linear terms take exactly the Weyl-type template

$$v_\parallel p_\parallel \sigma^3 + v_1 p_1 \sigma^1 + v_2 p_2 \sigma^2$$

up to basis change. Since $\det M \neq 0$, all three linear directions are active, and the crossing is isolated and linear. By the previous proposition, this is a Weyl point. $\square$

H. Chirality of the Weyl Valley

For a linearly nondegenerate Weyl point, define the chirality

$$\chi(G_*) := \mathrm{sgn}(\det M) \in \{\pm 1\}.$$

This is the topological charge of the Weyl node in momentum space. Equivalently, the Berry flux through a small sphere enclosing the crossing is

$$\frac{1}{2\pi} \int_{S^2_{G_*}} \mathcal{F} = \chi(G_*).$$

Hence the BCC shell valley, when nondegenerate, is not merely a linear crossing but a topologically protected Weyl point.

I. Opposite Valley and Dirac Doubling

Suppose there exists an opposite valley at $-G_*$ with projected Hamiltonian $h_-(p)$. Assume the pair satisfies the doubling relation

$$h_-(p) = -h_+(-p)$$

or any equivalent inversion/time-reversal related condition producing opposite chirality. Then the doubled Hamiltonian on

$$\mathcal{H}_*(G_*) \oplus \mathcal{H}_*(-G_*) \cong \mathbb{C}^4$$

takes the block form

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix},$$

which is the massless Dirac Hamiltonian after a suitable basis change.

Thus the BCC Weyl valley theorem gives the exact local input needed for Dirac emergence by opposite-valley doubling.

J. What Is Proven and What Is Still Assumed

The theorem above proves the following rigorously:

- any smooth Hermitian two-band crossing near a selected BCC shell pair reduces to Pauli form;
- if the little-group symmetry allows one longitudinal scalar channel and one transverse doublet channel, then the linear template is precisely Weyl-like;
- if the linear coefficient matrix is nondegenerate, the crossing is a Weyl valley with chirality $\mathrm{sgn}(\det M)$.

What is *not* yet proven at this stage is the full microscopic origin of the assumptions. In particular, the following still require independent proof from the microscopic TECT linearization:

1. existence of the two-band sector near some $G_* \in S_{\mathrm{BCC}}$;
2. existence of the crossing itself at $p = 0$;
3. nondegeneracy of the linear coefficients, i.e. $\det M \neq 0$.

Therefore the theorem closes the *form* of the Weyl valley once a BCC shell crossing exists, but not yet the full microscopic existence theorem.

K. Final Reduction of the Dirac Problem

The remaining microscopic fermion problem is therefore reduced to the following finite statement:

$$\boxed{\text{Show that the actual TECT condensate linearization produces a nondegenerate two-band crossing at some } \pm G_* \in S_{\mathrm{BCC}}.}$$

Once this is established, the Weyl theorem above gives the local valley structure automatically, and opposite-valley doubling yields the Dirac fermion.

SUMMARY

The BCC shell geometry by itself does not automatically prove the existence of a Weyl node. However, once a smooth Hermitian two-band crossing is present near a selected reciprocal pair $\pm G_* \in S_{\mathrm{BCC}}$, and once the little-group symmetry allows one longitudinal scalar channel together with one transverse doublet channel, the linearized projected Hamiltonian is forced into Weyl form.

More precisely, if the linear coefficient matrix is nondegenerate,

$$\det M \neq 0,$$

then the local crossing is a Weyl valley with chirality

$$\chi(G_*) = \mathrm{sgn}(\det M).$$

Hence the microscopic Dirac-emergence problem in TECT is reduced to proving the existence of a nondegenerate two-band crossing on the BCC shell.

LVI. NONDEGENERACY THEOREM FOR THE BCC WEYL VALLEY

A. Goal

We now refine the BCC Weyl Valley Theorem by isolating the exact nondegeneracy data. The purpose is to determine which velocity coefficients are fixed by symmetry, which are merely allowed, and what remains to be checked microscopically.

The conclusion will be that the full linear nondegeneracy condition

$$\det M \neq 0$$

reduces to two independent factors:

$$v_\parallel \neq 0, \qquad \det T \neq 0,$$

where $v_\parallel$ is the longitudinal scalar coefficient and T is the transverse 2×2 intertwining matrix.

B. Selected BCC Axis and Representation Splitting

Fix a reciprocal pair

$$\pm G_* \in S_{\mathrm{BCC}}, \qquad \hat{n} := \frac{G_*}{|G_*|}.$$

Choose an orthonormal frame

$$\{e_1, e_2, \hat{n}\}.$$

Any small momentum near G_* decomposes as

$$p = p_\parallel \hat{n} + p_1 e_1 + p_2 e_2.$$

Let H_* be the little group preserving G_*. Then the real momentum space near the valley splits as an H_*-representation

$$\mathbb{R}^3 = \mathbb{R}_\parallel \oplus E_\perp,$$

where

$$\mathbb{R}_\parallel = \mathrm{span}\{\hat{n}\}, \qquad E_\perp = \mathrm{span}\{e_1, e_2\}.$$

Assume that the reduced two-band Hilbert space $\mathcal{H}_*(G_*) \cong \mathbb{C}^2$ carries a compatible H_*-action, and that the Pauli sector splits similarly into

$$\mathrm{span}\{\sigma^1, \sigma^2, \sigma^3\} = \Sigma_\perp \oplus \Sigma_\parallel,$$

with

$$\Sigma_\perp := \mathrm{span}\{\sigma^1, \sigma^2\}, \qquad \Sigma_\parallel := \mathrm{span}\{\sigma^3\}.$$

We assume that $\Sigma_\parallel$ transforms as the longitudinal one-dimensional representation and $\Sigma_\perp$ transforms as the transverse doublet representation.

C. Most General Symmetry-Equivariant Linear Hamiltonian

The traceless linearized Hamiltonian is an H_*-equivariant linear map

$$L : \mathbb{R}^3 \to \mathrm{span}\{\sigma^1, \sigma^2, \sigma^3\}.$$

Using the above splittings, write

$$L = \begin{pmatrix} L_{\perp\perp} & L_{\perp\parallel} \\ L_{\parallel\perp} & L_{\parallel\parallel} \end{pmatrix},$$

where

$$L_{\perp\perp} \in \mathrm{Hom}(E_\perp, \Sigma_\perp), \quad L_{\perp\parallel} \in \mathrm{Hom}(\mathbb{R}_\parallel, \Sigma_\perp),$$

$$L_{\parallel\perp} \in \mathrm{Hom}(E_\perp, \Sigma_\parallel), \quad L_{\parallel\parallel} \in \mathrm{Hom}(\mathbb{R}_\parallel, \Sigma_\parallel).$$

Because L is H_*-equivariant, each block must be an intertwiner. If $\mathbb{R}_\parallel$ and $\Sigma_\perp$ are inequivalent H_*-representations, then

$$\mathrm{Hom}_{H_*}(\mathbb{R}_\parallel, \Sigma_\perp) = 0.$$

Likewise, if $E_\perp$ and $\Sigma_\parallel$ are inequivalent, then

$$\mathrm{Hom}_{H_*}(E_\perp, \Sigma_\parallel) = 0.$$

Hence the mixed blocks vanish:

$$L_{\perp\parallel} = 0, \qquad L_{\parallel\perp} = 0.$$

Therefore the most general symmetry-allowed linear Hamiltonian is block-diagonal:

$$L = L_{\perp\perp} \oplus L_{\parallel\parallel}.$$

D. Longitudinal Scalar Channel

Since both $\mathbb{R}_\parallel$ and $\Sigma_\parallel$ are one-dimensional representations, one has

$$L_{\parallel\parallel}(p_\parallel) = v_\parallel p_\parallel \sigma^3$$

for some real coefficient $v_\parallel$.

Thus the longitudinal sector contributes exactly one scalar parameter.

E. Transverse Doublet Channel

The transverse block is an intertwiner

$$T := L_{\perp\perp} \in \mathrm{Hom}_{H_*}(E_\perp, \Sigma_\perp).$$

In coordinates,

$$L_{\perp\perp}(p_\perp) = \sum_{a,b=1}^{2} T_{ab} p_a \sigma^b, \qquad p_\perp = (p_1, p_2).$$

Hence the full linearized traceless Hamiltonian takes the form

$$h_+^{\mathrm{lin}}(p) = v_\parallel p_\parallel \sigma^3 + \sum_{a,b=1}^{2} T_{ab} p_a \sigma^b.$$

This is the exact symmetry-reduced form.

F. Matrix Form of the Weyl Coefficient Tensor

With respect to the ordered momentum coordinates

$$(p_1, p_2, p_\parallel)$$

and Pauli coordinates

$$(\sigma^1, \sigma^2, \sigma^3),$$

the linear coefficient matrix is

$$M = \begin{pmatrix} T_{11} & T_{12} & 0 \\ T_{21} & T_{22} & 0 \\ 0 & 0 & v_\parallel \end{pmatrix}.$$

Therefore

$$\det M = v_\parallel \det T.$$

This proves that the full linear nondegeneracy condition is equivalent to the pair of conditions

$$v_\parallel \neq 0, \qquad \det T \neq 0.$$

Proposition. Under the representation splitting above, the projected crossing is linearly nondegenerate if and only if

$$v_\parallel \neq 0 \quad \text{and} \quad \det T \neq 0.$$

Proof. Immediate from the block-diagonal form of M, since

$$\det M = v_\parallel \det T.$$

□

G. What Symmetry Does and Does Not Force

The previous proposition gives a precise separation:

1. Symmetry *forces* the absence of forbidden mixed linear terms:

$$p_{\|}\sigma^{1,2}, \qquad p_{1,2}\sigma^3$$

are absent whenever the corresponding representations are inequivalent.

2. Symmetry *allows* the longitudinal scalar term

$$v_{\|}p_{\|}\sigma^3$$

provided the one-dimensional representations match.

3. Symmetry *allows* the transverse doublet term

$$p_a T_{ab}\sigma^b$$

provided $E_\perp$ and $\Sigma_\perp$ are equivalent H_*-representations.

4. Symmetry alone does *not* force

$$v_{\|} \neq 0 \quad \text{or} \quad \det T \neq 0.$$

It only says these quantities are not ruled out by selection rules.

Thus nondegeneracy is not a purely representation-theoretic theorem. It is a representation-theoretic *reduction* to two finite coefficients.

H. Generic Nonvanishing of the Longitudinal Coefficient

Let $\mathcal{V}$ be the finite-dimensional space of symmetry-allowed microscopic tensors entering the reduced linearization. Then

$$v_{\|} : \mathcal{V} \to \mathbb{R}$$

is a linear functional.

If

$$v_{\|} \not\equiv 0 \quad \text{on } \mathcal{V},$$

then its zero set

$$Z_{\|} := \{\mathsf{N} \in \mathcal{V} : v_{\|}(\mathsf{N}) = 0\}$$

is a proper hyperplane. Hence

$$v_{\|}(\mathsf{N}) \neq 0$$

for an open dense full-measure subset of $\mathcal{V}$.

I. Generic Invertibility of the Transverse Matrix

Likewise, the transverse intertwiner

$$T(\mathsf{N}) \in \mathrm{Hom}_{H_*}(E_\perp, \Sigma_\perp)$$

depends linearly on the microscopic tensor data $\mathsf{N} \in \mathcal{V}$. Therefore

$$\det T(\mathsf{N})$$

is a polynomial function on $\mathcal{V}$.

If this polynomial is not identically zero, i.e. if there exists at least one allowed microscopic tensor $\mathsf{N}_0 \in \mathcal{V}$ such that

$$\det T(\mathsf{N}_0) \neq 0,$$

then the bad set

$$Z_T := \{\mathsf{N} \in \mathcal{V} : \det T(\mathsf{N}) = 0\}$$

is a proper algebraic subset of $\mathcal{V}$. Hence

$$\det T(\mathsf{N}) \neq 0$$

holds on an open dense full-measure subset of $\mathcal{V}$.

Lemma. If $\det T$ is not identically zero on the allowed tensor space, then transverse nondegeneracy is generic.

Proof. Since the entries of T depend linearly on N, $\det T$ is polynomial. A nonzero polynomial cannot vanish on an open set in a finite-dimensional real vector space. Therefore its zero locus is a proper algebraic subset with empty interior and measure zero. □

J. BCC Nondegeneracy Theorem

Theorem. Fix a reciprocal pair $\pm G_* \in S_{\mathrm{BCC}}$. Assume:

1. the reduced two-band sector near G_* exists;
2. the little-group representation splits as
$$\mathbb{R}^3 = \mathbb{R}_\parallel \oplus E_\perp, \qquad \mathrm{span}\{\sigma^1, \sigma^2, \sigma^3\} = \Sigma_\perp \oplus \Sigma_\parallel,$$
with $\mathbb{R}_\parallel \simeq \Sigma_\parallel$ and $E_\perp \simeq \Sigma_\perp$;
3. the mixed intertwiners vanish by inequivalence:
$$\mathrm{Hom}_{H_*}(\mathbb{R}_\parallel, \Sigma_\perp) = 0, \qquad \mathrm{Hom}_{H_*}(E_\perp, \Sigma_\parallel) = 0;$$
4. the longitudinal coefficient $v_\parallel$ is not identically zero on the allowed microscopic tensor space;
5. the transverse determinant $\det T$ is not identically zero on the allowed microscopic tensor space.

Then, for an open dense full-measure subset of allowed microscopic tensors, the projected crossing near G_* is linearly nondegenerate:

$$\det M \neq 0.$$

Equivalently, the linearized Hamiltonian is Weyl:

$$h_+^{\mathrm{lin}}(p) = v_\parallel p_\parallel \sigma^3 + \sum_{a,b=1}^{2} T_{ab} p_a \sigma^b, \qquad v_\parallel \det T \neq 0.$$

Proof. By symmetry, the linear coefficient matrix is block-diagonal:

$$M = \mathrm{diag}(T, v_\parallel).$$

Hence

$$\det M = v_\parallel \det T.$$

By assumption, $v_\parallel$ is a nonzero linear functional and therefore generically nonzero. Likewise, $\det T$ is a nonzero polynomial and therefore generically nonzero. The intersection of the corresponding generic sets is again open dense and full measure. Hence $\det M \neq 0$ generically. □

K. Strong Honest Conclusion

The exact status is therefore the following.

- The *form* of the linear Hamiltonian is fixed by the BCC axis little-group symmetry.
- The full nondegeneracy condition reduces exactly to

$$v_{\parallel} \neq 0, \qquad \det T \neq 0.$$

- Symmetry alone does not prove these inequalities.
- What remains is to show that neither of them is identically forbidden in the actual microscopic TECT linearization.

Thus the microscopic Weyl problem has now been reduced from an uncontrolled PDE question to a finite pair of representation-theoretic nonvanishing tests.

L. Final Reduction

The remaining local Weyl-valley problem is now exactly:

$$\boxed{\text{Show that } v_{\parallel} \not\equiv 0 \quad \text{and} \quad \det T \not\equiv 0 \text{ on the actual allowed microscopic tensor space of TECT.}}$$

Once this is done, Weyl nondegeneracy follows generically, and opposite-valley doubling yields the Dirac fermion.

SUMMARY

For a selected BCC shell pair $\pm G_*$, the little-group symmetry forces the linearized two-band Hamiltonian into the block form

$$h_+^{\mathrm{lin}}(p) = v_{\parallel} p_{\parallel} \sigma^3 + \sum_{a,b=1}^{2} T_{ab} p_a \sigma^b.$$

Hence the full linear coefficient matrix is

$$M = \mathrm{diag}(T, v_{\parallel}), \qquad \det M = v_{\parallel} \det T.$$

Therefore linear Weyl nondegeneracy is equivalent to the pair of conditions

$$v_{\parallel} \neq 0, \qquad \det T \neq 0.$$

Symmetry alone does not force these conditions, but it reduces the entire problem to them. If neither quantity is identically zero on the symmetry-allowed microscopic tensor space, then both are generically nonzero, and the BCC shell crossing is generically Weyl.

LVII. WITNESS THEOREM FOR WEYL NONDEGENERACY

A. Goal

The previous nondegeneracy theorem reduced the local Weyl problem near a selected BCC shell pair $\pm G_*$ to the two conditions

$$v_{\parallel} \neq 0, \qquad \det T \neq 0.$$

We now prove that each of these conditions is controlled by the existence of a single admissible microscopic witness.

More precisely, let $\mathcal{V}$ be the finite-dimensional space of symmetry-allowed microscopic tensors entering the reduced two-band linearization near G_*. Then:

- $v_\parallel$ is a linear functional on $\mathcal{V}$;
- T is a linear map from $\mathcal{V}$ into a finite-dimensional intertwiner space;
- $\det T$ is therefore a polynomial on $\mathcal{V}$.

Hence the only real obstruction is whether these quantities vanish identically on $\mathcal{V}$.

B. Algebraic Setup

Let

$$\mathcal{V}$$

be the real finite-dimensional vector space of all symmetry-allowed microscopic tensors contributing to the reduced linearized Hamiltonian near the selected valley G_*.

By the previous theorem, the linearized traceless Hamiltonian has the form

$$h_+^{\mathrm{lin}}(p) = v_\parallel(\mathsf{N})\, p_\parallel \sigma^3 + \sum_{a,b=1}^{2} T_{ab}(\mathsf{N})\, p_a \sigma^b, \qquad \mathsf{N} \in \mathcal{V}.$$

Thus:

1. $$v_\parallel : \mathcal{V} \to \mathbb{R}$$
   is linear;
2. $$T : \mathcal{V} \to \mathrm{Hom}_{H_*}(E_\perp, \Sigma_\perp)$$
   is linear;
3. $$\det T : \mathcal{V} \to \mathbb{R}$$
   is polynomial of degree 2.

C. Longitudinal Witness Principle

Definition. A longitudinal witness is a tensor

$$\mathsf{N}^{(\parallel)} \in \mathcal{V}$$

such that

$$v_\parallel\big(\mathsf{N}^{(\parallel)}\big) \neq 0.$$

Theorem (Longitudinal Witness Theorem). The following are equivalent:

1. $$v_\parallel \not\equiv 0 \quad \text{on } \mathcal{V};$$
2. there exists a longitudinal witness
   $$\mathsf{N}^{(\parallel)} \in \mathcal{V}$$
   with
   $$v_\parallel\big(\mathsf{N}^{(\parallel)}\big) \neq 0;$$

3. $$v_\parallel(\mathsf{N}) \neq 0$$

   for an open dense full-measure subset of $\mathcal{V}$.

Proof. The equivalence of (1) and (2) is immediate from the definition of identically zero. If (1) holds, then $v_\parallel$ is a nonzero linear functional on a finite-dimensional space, so its zero set is a proper hyperplane. Hence its complement is open, dense, and full measure. Conversely, if (3) holds then certainly $v_\parallel \not\equiv 0$. □

D. Transverse Witness Principle

Definition. A transverse witness is a tensor

$$\mathsf{N}^{(\perp)} \in \mathcal{V}$$

such that

$$\det T\big(\mathsf{N}^{(\perp)}\big) \neq 0.$$

Theorem (Transverse Witness Theorem). The following are equivalent:

1. $$\det T \not\equiv 0 \quad \text{on } \mathcal{V};$$

2. there exists a transverse witness

   $$\mathsf{N}^{(\perp)} \in \mathcal{V}$$

   with

   $$\det T\big(\mathsf{N}^{(\perp)}\big) \neq 0;$$

3. $$\det T(\mathsf{N}) \neq 0$$

   for an open dense full-measure subset of $\mathcal{V}$.

Proof. Again, (1) and (2) are equivalent by definition. Since T depends linearly on N, its determinant is a polynomial function on $\mathcal{V}$. If this polynomial is not identically zero, its vanishing set is a proper algebraic subset, hence has empty interior and measure zero. Therefore the nonvanishing set is open dense and full measure. Conversely, (3) implies that $\det T$ cannot vanish identically. □

E. Simultaneous Witness Criterion

We now combine the two witness principles.
Theorem (Simultaneous Witness Criterion). The following are equivalent:

1. $$v_\parallel \not\equiv 0 \quad \text{and} \quad \det T \not\equiv 0 \quad \text{on } \mathcal{V};$$

2. there exist tensors

   $$\mathsf{N}^{(\parallel)}, \mathsf{N}^{(\perp)} \in \mathcal{V}$$

   such that

   $$v_\parallel\big(\mathsf{N}^{(\parallel)}\big) \neq 0, \qquad \det T\big(\mathsf{N}^{(\perp)}\big) \neq 0;$$

3. the set

$$\mathcal{U}_{\mathrm{Weyl}} := \{\mathsf{N} \in \mathcal{V} : v_{\|}(\mathsf{N}) \neq 0,\ \det T(\mathsf{N}) \neq 0\}$$

is open dense and full measure in $\mathcal{V}$.

Proof. Apply the previous two theorems separately. The bad set

$$Z := \{\mathsf{N} : v_{\|}(\mathsf{N}) = 0\} \cup \{\mathsf{N} : \det T(\mathsf{N}) = 0\}$$

is a union of a proper hyperplane and a proper algebraic subset. Hence Z is closed, nowhere dense, and measure zero provided both components are proper. Therefore its complement $\mathcal{U}_{\mathrm{Weyl}}$ is open dense and full measure. □

F. Canonical Witness Templates

We now explain what kind of witnesses one should search for microscopically.

a. Longitudinal witness template. A longitudinal witness is any allowed tensor whose reduced projection produces a nonzero coefficient along the axial channel

$$p_{\|}\sigma^3.$$

Equivalently, if the microscopic tensor contains a component transforming in

$$\mathrm{Hom}_{H_*}(\mathbb{R}_{\|}, \Sigma_{\|}),$$

then evaluating it on the basis vector $\hat{n}$ gives

$$v_{\|} \neq 0.$$

b. Transverse witness template. A transverse witness is any allowed tensor whose reduced projection induces an invertible map

$$T : E_{\perp} \to \Sigma_{\perp}.$$

Equivalently, one needs one microscopic tensor whose transverse projection is not rank-deficient. In a chosen basis (e_1, e_2) and (σ^1, σ^2), this means

$$T(\mathsf{N}^{(\perp)}) = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

with

$$ad - bc \neq 0.$$

Thus the search for witnesses is not a spectral problem but a finite-dimensional tensor-construction problem.

G. Reduction to Basis Evaluation

Choose a basis

$$\mathcal{V} = \mathrm{span}\{E_A\}_{A=1}^{d}.$$

Write

$$\mathsf{N} = \sum_{A=1}^{d} x_A E_A.$$

Then

$$v_{\|}(\mathsf{N}) = \sum_{A=1}^{d} x_A\, \nu_A$$

for some real coefficients ν_A, and

$$T(\mathsf{N}) = \sum_{A=1}^{d} x_A\, T^{(A)}, \qquad T^{(A)} \in \mathrm{Hom}_{H_*}(E_\perp, \Sigma_\perp).$$

Therefore:

- $v_\parallel \not\equiv 0$ if and only if at least one basis coefficient ν_A is nonzero;
- $\det T \not\equiv 0$ if and only if there exists some choice of coefficients x_A for which

$$\det\Big(\sum_A x_A T^{(A)}\Big) \neq 0.$$

This gives an explicit finite algorithm for checking Weyl nondegeneracy once the microscopic basis tensors are known.

H. Witness Form of the BCC Weyl Nondegeneracy Theorem

Theorem. Let $\mathcal{V}$ be the symmetry-allowed microscopic tensor space for the two-band reduced linearization near a selected BCC shell pair $\pm G_*$. Assume the representation-theoretic splitting of the previous theorem, so that

$$h_+^{\mathrm{lin}}(p) = v_\parallel(\mathsf{N})\, p_\parallel \sigma^3 + \sum_{a,b=1}^{2} T_{ab}(\mathsf{N})\, p_a \sigma^b.$$

If there exist:

1. a longitudinal witness $\mathsf{N}^{(\parallel)} \in \mathcal{V}$ with

$$v_\parallel\big(\mathsf{N}^{(\parallel)}\big) \neq 0,$$

2. a transverse witness $\mathsf{N}^{(\perp)} \in \mathcal{V}$ with

$$\det T\big(\mathsf{N}^{(\perp)}\big) \neq 0,$$

then for an open dense full-measure subset of $\mathcal{V}$, one has

$$v_\parallel(\mathsf{N}) \neq 0, \qquad \det T(\mathsf{N}) \neq 0,$$

and hence

$$\det M(\mathsf{N}) \neq 0.$$

Therefore the crossing near G_* is generically Weyl.

Proof. Immediate from the Simultaneous Witness Criterion together with

$$\det M = v_\parallel \det T.$$

□

I. Strong Honest Conclusion

The local Weyl problem has now been reduced to the construction of two explicit microscopic witnesses.

Nothing more is needed at the theorem level. Indeed, once one finds

$$\mathsf{N}^{(\parallel)} \text{ with } v_\parallel \neq 0, \qquad \mathsf{N}^{(\perp)} \text{ with } \det T \neq 0,$$

the nondegenerate Weyl valley follows generically.

Thus the remaining open question is no longer an abstract existence problem for Weyl fermions, but an explicit finite tensor-construction problem inside the microscopic TECT linearization.

J. Final Reduction

The remaining microscopic valley problem is now exactly

$$\boxed{\text{Construct one admissible longitudinal witness and one admissible transverse witness in the actual TECT microscopic tensor sp}}$$

Once this is done, Weyl nondegeneracy is generic, and opposite-valley doubling yields the emergent Dirac fermion.

SUMMARY

The Weyl nondegeneracy problem near a selected BCC shell pair $\pm G_*$ has been reduced to the two conditions

$$v_\parallel \neq 0, \qquad \det T \neq 0.$$

Since $v_\parallel$ is linear and $\det T$ is polynomial on the finite-dimensional allowed microscopic tensor space $\mathcal{V}$, each condition is generic once it is not identically zero.

Therefore the entire local Weyl problem is reduced to the existence of two admissible witnesses:

$$\mathsf{N}^{(\parallel)} \in \mathcal{V} \quad \text{with} \quad v_\parallel\big(\mathsf{N}^{(\parallel)}\big) \neq 0,$$

and

$$\mathsf{N}^{(\perp)} \in \mathcal{V} \quad \text{with} \quad \det T\big(\mathsf{N}^{(\perp)}\big) \neq 0.$$

Once such witnesses exist, Weyl nondegeneracy holds on an open dense full-measure subset of $\mathcal{V}$, and the BCC shell crossing is generically Weyl.

LVIII. MINIMAL MICROSCOPIC TENSOR ANSATZ AND EXPLICIT WEYL WITNESSES

A. Goal

We now construct explicit microscopic witnesses for the two Weyl nondegeneracy conditions

$$v_\parallel \neq 0, \qquad \det T \neq 0,$$

using the smallest symmetry-compatible tensor ansatz.

The strategy is the following.

The previous reduction showed that the local Weyl problem near a selected BCC shell pair $\pm G_*$ is completely controlled by the reduced linear Hamiltonian

$$h_+^{\text{lin}}(p) = v_\parallel p_\parallel \sigma^3 + \sum_{a,b=1}^{2} T_{ab} p_a \sigma^b.$$

Hence it suffices to produce:

1. one admissible microscopic tensor whose reduction gives $v_\parallel \neq 0$;
2. one admissible microscopic tensor whose reduction gives $\det T \neq 0$.

We will show that this is already achieved by a minimal ansatz built from the axial unit vector

$$\hat{n} = \frac{G_*}{|G_*|}$$

and the invariant tensors δ_{ij} and $\varepsilon_{ijk}\hat{n}_k$.

B. Microscopic Linear Gradient Ansatz

Let Ψ denote the reduced two-band field near the selected BCC shell pair G_*. Consider the most general first-order microscopic linearization of the form

$$H^{(1)}_{\text{mic}} = \sum_{i,j=1}^{3} \mathsf{N}^{(3)}_{ij}\, p_i \sigma^3 + \sum_{i,j=1}^{3} \sum_{b=1}^{2} \mathsf{N}^{(b)}_{ij}\, p_i \sigma^b.$$

Equivalently, we write

$$H^{(1)}_{\text{mic}} = p_i\, \mathcal{M}_{i\alpha}\, \sigma^\alpha, \qquad \alpha = 1, 2, 3,$$

where the coefficient tensor $\mathcal{M}_{i\alpha}$ is built from symmetry-allowed microscopic tensors.

Fix the axial decomposition

$$\mathbb{R}^3 = \mathbb{R}\hat{n} \oplus E_\perp, \qquad E_\perp = \text{span}\{e_1, e_2\}.$$

We now seek the smallest ansatz that is compatible with the little-group decomposition around G_*.

C. Minimal Symmetry-Compatible Tensor Basis

Relative to the distinguished axis $\hat{n}$, the natural rank-two spatial tensors are:

1. the isotropic tensor
$$\delta_{ij};$$
2. the axial antisymmetric tensor
$$J_{ij} := \varepsilon_{ijk}\hat{n}_k;$$
3. the longitudinal projector
$$P^{\parallel}_{ij} := \hat{n}_i\hat{n}_j;$$
4. the transverse projector
$$P^{\perp}_{ij} := \delta_{ij} - \hat{n}_i\hat{n}_j.$$

Restricted to the transverse plane $E_\perp$, the tensor J acts as the canonical complex structure:

$$Je_1 = e_2, \qquad Je_2 = -e_1, \qquad J\hat{n} = 0.$$

Hence, in the basis (e_1, e_2),

$$J|_{E_\perp} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Therefore the pair

$$\{P^\perp, J\}$$

already spans the full algebra of equivariant endomorphisms of the transverse plane in the axial isotropic class.

D. Minimal Longitudinal Ansatz

To produce a longitudinal witness, it suffices to include a term of the form

$$H_\parallel = \lambda_\parallel\,(\hat{n}\cdot p)\,\sigma^3 = \lambda_\parallel\,\hat{n}_i p_i\,\sigma^3.$$

This corresponds to the microscopic tensor

$$\mathsf{N}_i^{(\parallel)(3)} = \lambda_\parallel \hat{n}_i.$$

Equivalently, in rank-two notation one may write

$$\mathsf{N}_{ij}^{(\parallel)(3)} = \lambda_\parallel \hat{n}_i \hat{n}_j.$$

Projecting onto the reduced longitudinal scalar channel gives

$$v_\parallel\big(\mathsf{N}^{(\parallel)}\big) = \lambda_\parallel.$$

Hence:
Proposition. If the microscopic tensor space contains the longitudinal projector channel

$$\hat{n}_i \hat{n}_j \otimes \sigma^3,$$

with nonzero coefficient $\lambda_\parallel$, then

$$v_\parallel \not\equiv 0.$$

Proof. Evaluating the reduced longitudinal coefficient on

$$\mathsf{N}_{ij}^{(\parallel)(3)} = \lambda_\parallel \hat{n}_i \hat{n}_j$$

gives

$$v_\parallel = \lambda_\parallel\,(\hat{n}\cdot\hat{n}) = \lambda_\parallel \neq 0$$

whenever $\lambda_\parallel \neq 0$. □
Thus a single nonzero axial longitudinal coefficient is already a longitudinal witness.

E. Minimal Transverse Ansatz

To produce a transverse witness, consider the most general minimal transverse ansatz

$$H_\perp = p_a T_{ab} \sigma^b, \qquad a, b = 1, 2,$$

with

$$T = \alpha\, I_2 + \beta\, J_2,$$

where

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

In spatial tensor form, this corresponds to

$$\mathsf{N}^{(\perp)} = \alpha\, P^\perp + \beta\, J.$$

More explicitly, the transverse contribution is

$$H_\perp = \alpha\,(p_1\sigma^1 + p_2\sigma^2) + \beta\,(-p_2\sigma^1 + p_1\sigma^2).$$

Thus the reduced transverse matrix is

$$T(\alpha,\beta)=\begin{pmatrix}\alpha & -\beta\\ \beta & \alpha\end{pmatrix}.$$

Its determinant is

$$\det T(\alpha,\beta)=\alpha^2+\beta^2.$$

Therefore:

Proposition. If the microscopic tensor space contains either the transverse identity channel

$$P^{\perp}\otimes(\sigma^1,\sigma^2)$$

or the axial antisymmetric channel

$$J\otimes(\sigma^1,\sigma^2),$$

with coefficients $(\alpha,\beta)\neq(0,0)$, then

$$\det T\not\equiv 0.$$

Proof. For the explicit transverse ansatz above,

$$T(\alpha,\beta)=\begin{pmatrix}\alpha & -\beta\\ \beta & \alpha\end{pmatrix},$$

so

$$\det T=\alpha^2+\beta^2.$$

Hence $\det T\neq 0$ whenever $(\alpha,\beta)\neq(0,0)$. □

This already gives an explicit transverse witness.

F. Explicit Witness Pair

Combining the previous two constructions, define the minimal witness Hamiltonian

$$H_{\rm wit}(p)=\lambda_{\parallel}(\hat{n}\cdot p)\sigma^3+\alpha\,(p_1\sigma^1+p_2\sigma^2)+\beta\,(-p_2\sigma^1+p_1\sigma^2).$$

Then

$$v_{\parallel}=\lambda_{\parallel},\qquad T=\begin{pmatrix}\alpha & -\beta\\ \beta & \alpha\end{pmatrix},\qquad \det T=\alpha^2+\beta^2.$$

Therefore

$$\det M=v_{\parallel}\det T=\lambda_{\parallel}(\alpha^2+\beta^2).$$

Thus if

$$\lambda_{\parallel}\neq 0,\qquad (\alpha,\beta)\neq(0,0),$$

then

$$\det M\neq 0.$$

Hence the reduced crossing is Weyl.

Theorem (Minimal Witness Ansatz). Suppose the actual TECT microscopic tensor space contains the three symmetry-compatible channels

$$\hat{n}_i\hat{n}_j\otimes\sigma^3,\qquad P^{\perp}_{ij}\otimes(\sigma^1,\sigma^2),\qquad J_{ij}\otimes(\sigma^1,\sigma^2),$$

with coefficients $\lambda_{\parallel},\alpha,\beta$, respectively. If

$$\lambda_{\parallel}\neq 0,\qquad (\alpha,\beta)\neq(0,0),$$

then:

1. $v_\parallel \not\equiv 0$;
2. $\det T \not\equiv 0$;
3. the BCC shell crossing is generically Weyl;
4. opposite-valley doubling yields a generically emergent Dirac fermion.

Proof. The longitudinal projector term provides an explicit longitudinal witness:

$$v_\parallel = \lambda_\parallel \neq 0.$$

The pair $(P^\perp, J)$ gives the explicit transverse matrix

$$T = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix},$$

whose determinant is

$$\det T = \alpha^2 + \beta^2 \neq 0$$

whenever $(\alpha, \beta) \neq (0, 0)$. Hence both witness conditions hold. By the Witness Theorem for Weyl nondegeneracy, the allowed microscopic tensor space contains an open dense full-measure subset on which

$$\det M = v_\parallel \det T \neq 0.$$

Therefore the local crossing is generically Weyl. Opposite-valley doubling then yields the Dirac fermion. □

G. Interpretation

This theorem is important because it shows that the microscopic Weyl problem does not require a complete solution of the full TECT PDE.

Instead, it is enough to verify that the actual microscopic linearization does *not* project out the three minimal channels

$$\hat{n}\hat{n} \otimes \sigma^3, \qquad P^\perp \otimes (\sigma^1, \sigma^2), \qquad J \otimes (\sigma^1, \sigma^2).$$

If these channels are present with nonzero coefficients, then the local Weyl structure follows generically.

H. Strong Honest Conclusion

At this point the local valley problem is reduced as far as possible without writing the full microscopic TECT condensate PDE.

What remains is the finite yes/no check:

$$\boxed{\text{Does the actual TECT microscopic linearization contain the longitudinal projector channel and at least one nontrivial transver}}$$

If the answer is yes, then the BCC shell valley is generically Weyl.

I. Final Reduction of the Dirac Frontier

The remaining Dirac frontier is now exactly:

$$\boxed{\text{Extract the first-order microscopic tensor coefficients of the TECT condensate PDE and verify}}$$

$$\boxed{\lambda_\parallel \neq 0, \qquad (\alpha, \beta) \neq (0, 0).}$$

Once this is done, the Weyl/Dirac emergence problem is closed at the theorem level.

SUMMARY

The Weyl nondegeneracy problem near a selected BCC shell pair has been reduced to the explicit microscopic coefficients

$$\lambda_\|, \alpha, \beta$$

appearing in the minimal symmetry-compatible ansatz

$$H_{\mathrm{wit}}(p) = \lambda_\| (\hat{n} \cdot p)\sigma^3 + \alpha\,(p_1\sigma^1 + p_2\sigma^2) + \beta\,(-p_2\sigma^1 + p_1\sigma^2).$$

These coefficients determine

$$v_\| = \lambda_\|, \qquad \det T = \alpha^2 + \beta^2, \qquad \det M = \lambda_\|(\alpha^2 + \beta^2).$$

Therefore, if

$$\lambda_\| \neq 0, \qquad (\alpha, \beta) \neq (0, 0),$$

then the local BCC shell crossing is generically Weyl, and opposite-valley doubling yields the emergent Dirac fermion.

Thus the remaining microscopic problem is reduced to a finite coefficient-extraction check from the actual TECT condensate linearization.

LIX. EXTRACTION THEOREM FOR THE WEYL WITNESS COEFFICIENTS

A. Goal

We now formulate the final local reduction step of the Weyl/Dirac problem. The previous analysis showed that the entire local BCC valley problem is controlled by the three coefficients

$$\lambda_\|, \qquad \alpha, \qquad \beta,$$

appearing in the minimal symmetry-compatible linear Hamiltonian

$$H_{\mathrm{wit}}(p) = \lambda_\| (\hat{n} \cdot p)\sigma^3 + \alpha(p_1\sigma^1 + p_2\sigma^2) + \beta(-p_2\sigma^1 + p_1\sigma^2).$$

Therefore the remaining task is to extract these coefficients from the actual microscopic TECT condensate PDE.

The purpose of this section is to prove that, once the microscopic linearized operator is written in a first-order projected form near a selected BCC shell pair $\pm G_*$, the coefficients $\lambda_\|, \alpha, \beta$ are obtained by explicit projector contractions.

B. Minimal Microscopic Linearization Hypothesis

Let the full microscopic TECT condensate equation linearized around the condensed background be

$$i\partial_t \Phi = \mathcal{L}(-i\nabla)\Phi.$$

Let $G_* \in S_{\mathrm{BCC}}$ be a selected shell vector, and write

$$k = G_* + p, \qquad p = p_\| \hat{n} + p_1 e_1 + p_2 e_2, \qquad \hat{n} = \frac{G_*}{|G_*|}.$$

Assume that near G_*, the microscopic operator admits a Taylor expansion

$$\mathcal{L}(G_* + p) = \mathcal{L}_* + p_i\,\mathcal{K}_i + O(|p|^2),$$

where

$$\mathcal{L}_* := \mathcal{L}(G_*), \qquad \mathcal{K}_i := \partial_{k_i}\mathcal{L}(k)\big|_{k=G_*}.$$

Assume further that:

1. $\mathcal{L}_*$ has a two-dimensional low-energy eigenspace $\mathcal{H}_* \subset \mathcal{H}_{\mathrm{full}}$;
2. P_* denotes the orthogonal projector onto $\mathcal{H}_*$;
3. the reduced Pauli basis on $\mathcal{H}_* \cong \mathbb{C}^2$ has been fixed so that
$$\mathrm{End}(\mathcal{H}_*) = \mathrm{span}\{\mathbf{1}, \sigma^1, \sigma^2, \sigma^3\}.$$

Then the reduced first-order Hamiltonian is

$$h_+^{\mathrm{lin}}(p) = P_* \Big(p_i \mathcal{K}_i\Big) P_* \Big|_{\mathcal{H}_*}.$$

C. Pauli-Resolved Coefficient Extraction

Since $h_+^{\mathrm{lin}}(p)$ is a Hermitian 2×2 matrix, it decomposes uniquely as

$$h_+^{\mathrm{lin}}(p) = a_0(p)\mathbf{1} + a_\mu(p)\sigma^\mu, \qquad \mu = 1, 2, 3.$$

The Pauli coefficients are extracted by the standard trace formulas

$$a_\mu(p) = \frac{1}{2}\mathrm{Tr}_{\mathcal{H}_*}\big(\sigma^\mu h_+^{\mathrm{lin}}(p)\big), \qquad a_0(p) = \frac{1}{2}\mathrm{Tr}_{\mathcal{H}_*}\big(h_+^{\mathrm{lin}}(p)\big).$$

Since

$$h_+^{\mathrm{lin}}(p) = p_i \, P_* \mathcal{K}_i P_*,$$

we define the coefficient matrix

$$M_{\mu i} := \frac{1}{2}\mathrm{Tr}_{\mathcal{H}_*}\big(\sigma^\mu P_* \mathcal{K}_i P_*\big).$$

Then

$$a_\mu(p) = M_{\mu i} p_i, \qquad h_+^{\mathrm{lin}}(p) = M_{\mu i} p_i \sigma^\mu.$$

Thus all local Weyl data are determined directly by the projected first derivatives $\mathcal{K}_i$.

D. Axial Decomposition and Definition of $\lambda_\parallel, \alpha, \beta$

Decompose the momentum derivatives into the adapted basis

$$\mathcal{K}_\parallel := \hat{n}_i \mathcal{K}_i, \qquad \mathcal{K}_a := e_{ai} \mathcal{K}_i, \qquad a = 1, 2.$$

Likewise define the reduced projected operators

$$\widetilde{\mathcal{K}}_\parallel := P_* \mathcal{K}_\parallel P_*, \qquad \widetilde{\mathcal{K}}_a := P_* \mathcal{K}_a P_*.$$

The longitudinal Weyl coefficient is defined by

$$\lambda_\parallel := \frac{1}{2}\mathrm{Tr}_{\mathcal{H}_*}\big(\sigma^3 \widetilde{\mathcal{K}}_\parallel\big).$$

The transverse coefficient matrix is defined by

$$T_{ab} := \frac{1}{2}\mathrm{Tr}_{\mathcal{H}_*}\big(\sigma^b \widetilde{\mathcal{K}}_a\big), \qquad a, b = 1, 2.$$

Then

$$h_+^{\mathrm{lin}}(p) = \lambda_\parallel p_\parallel \sigma^3 + \sum_{a,b=1}^{2} T_{ab} p_a \sigma^b$$

provided the forbidden mixed terms vanish by symmetry.

To extract the scalar parameters α, β, decompose T in the basis $\{I_2, J_2\}$:

$$T = \alpha I_2 + \beta J_2, \qquad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Then explicitly,

$$\alpha = \frac{1}{2}\mathrm{tr}_{2\times 2}(T), \qquad \beta = \frac{1}{2}\mathrm{tr}_{2\times 2}(J_2^\top T).$$

Equivalently,

$$\alpha = \frac{1}{2}(T_{11} + T_{22}), \qquad \beta = \frac{1}{2}(T_{21} - T_{12}).$$

E. Extraction Theorem

Theorem. Let the microscopic TECT condensate operator admit the first-order expansion

$$\mathcal{L}(G_* + p) = \mathcal{L}_* + p_i \mathcal{K}_i + O(|p|^2)$$

near a selected BCC shell vector G_*, and let P_* project onto a two-dimensional low-energy subspace at G_*.

Assume the little-group symmetry removes the forbidden mixed channels

$$p_\parallel \sigma^{1,2}, \qquad p_{1,2}\sigma^3.$$

Then the local Weyl witness coefficients are extracted by

$$\lambda_\parallel = \frac{1}{2}\mathrm{Tr}_{\mathcal{H}_*}\big(\sigma^3 P_* (\hat{n}_i \mathcal{K}_i) P_*\big),$$

$$T_{ab} = \frac{1}{2}\mathrm{Tr}_{\mathcal{H}_*}\big(\sigma^b P_* (e_{ai} \mathcal{K}_i) P_*\big), \qquad a, b = 1, 2,$$

and

$$\alpha = \frac{1}{2}(T_{11} + T_{22}), \qquad \beta = \frac{1}{2}(T_{21} - T_{12}).$$

Consequently,

$$\det M = \lambda_\parallel \det T = \lambda_\parallel(\alpha^2 + \beta^2)$$

for the minimal axial equivariant ansatz $T = \alpha I_2 + \beta J_2$.

Proof. The reduced linearized Hamiltonian is

$$h_+^{\mathrm{lin}}(p) = P_*(p_i \mathcal{K}_i) P_*.$$

Since $h_+^{\mathrm{lin}}(p)$ acts on a two-dimensional complex Hilbert space, it has a unique Pauli decomposition

$$h_+^{\mathrm{lin}}(p) = a_0(p)\mathbf{1} + a_\mu(p)\sigma^\mu,$$

with coefficients extracted by the Pauli trace formulas. Projecting the derivative operators onto the axial basis gives

$$h_+^{\mathrm{lin}}(p) = p_\parallel \widetilde{\mathcal{K}}_\parallel + p_a \widetilde{\mathcal{K}}_a.$$

The symmetry assumption removes the mixed channels, so only

$$p_\parallel \sigma^3 \quad \text{and} \quad p_a \sigma^b, \ a, b = 1, 2$$

survive. Therefore the coefficients are exactly the stated projector contractions. Finally, if the transverse block is written as

$$T = \alpha I_2 + \beta J_2,$$

then

$$\det T = \alpha^2 + \beta^2,$$

so

$$\det M = \lambda_\parallel(\alpha^2 + \beta^2).$$

□

F. Operational Corollary

Corollary. To verify local Weyl nondegeneracy in the actual TECT microscopic theory, it is sufficient to compute the three projected quantities

$$\lambda_{\parallel} = \frac{1}{2}\mathrm{Tr}_{\mathcal{H}_*}\big(\sigma^3 P_*(\hat{n}_i \mathcal{K}_i)P_*\big),$$

$$\alpha = \frac{1}{4}\sum_{a=1}^{2}\mathrm{Tr}_{\mathcal{H}_*}\big(\sigma^a P_*(e_{ai}\mathcal{K}_i)P_*\big),$$

$$\beta = \frac{1}{4}\left[\mathrm{Tr}_{\mathcal{H}_*}\big(\sigma^2 P_*(e_{1i}\mathcal{K}_i)P_*\big) - \mathrm{Tr}_{\mathcal{H}_*}\big(\sigma^1 P_*(e_{2i}\mathcal{K}_i)P_*\big)\right].$$

If

$$\lambda_{\parallel} \neq 0, \qquad (\alpha, \beta) \neq (0, 0),$$

then the selected BCC shell crossing is locally Weyl.

G. Minimal PDE Ansatz

A minimal microscopic PDE consistent with the previous extraction theorem is

$$i\partial_t \Phi = \Big[\mathcal{L}_* + c_{\parallel}(\hat{n}\cdot(-i\nabla))\,\Gamma^{\parallel} + c_I\, P^{\perp}_{ij}(-i\partial_i)\Gamma^j_{\perp} + c_J\,\varepsilon_{ijk}\hat{n}_k(-i\partial_i)\Gamma^j_{\perp}\Big]\Phi + O(\nabla^2),$$

where:

- $\Gamma^{\parallel}$ reduces to σ^3 on the two-band subspace;
- $\Gamma^j_{\perp}$ reduce to σ^1, σ^2 on the transverse sector;
- $c_{\parallel}, c_I, c_J$ are microscopic coefficients.

After projection to $\mathcal{H}_*$, one obtains

$$\lambda_{\parallel} \propto c_{\parallel}, \qquad \alpha \propto c_I, \qquad \beta \propto c_J.$$

Thus the witness coefficients are nothing but the projected first-order axial, transverse-even, and transverse-odd derivative couplings.

H. Honest Status of the Result

What has now been closed rigorously is the following:

- the entire local Weyl problem has been reduced to three projected first-derivative coefficients;
- these coefficients are extracted by explicit trace/projector formulas from the microscopic PDE;
- for the minimal axial ansatz,

$$\det M = \lambda_{\parallel}(\alpha^2 + \beta^2).$$

What is not yet done is the actual microscopic computation of $\mathcal{K}_i$ from the full TECT condensate equation.

Thus the remaining open problem is no longer structural. It is an explicit coefficient extraction calculation from the true microscopic TECT operator.

I. Final Reduction of the Local Dirac Frontier

The remaining local Dirac frontier is therefore exactly:

$$\boxed{\text{Write the microscopic TECT condensate PDE, compute } \mathcal{K}_i = \partial_{k_i}\mathcal{L}(k)|_{k=G_*}, \text{ project with } P_*, \text{ and evaluate } \lambda_\parallel, \alpha, \beta.}$$

If

$$\lambda_\parallel \neq 0, \qquad (\alpha, \beta) \neq (0, 0),$$

then the BCC shell crossing is locally Weyl, and opposite-valley doubling gives the Dirac fermion.

SUMMARY

Near a selected BCC shell vector G_*, let the microscopic TECT condensate operator admit the first-order expansion

$$\mathcal{L}(G_* + p) = \mathcal{L}_* + p_i \mathcal{K}_i + O(|p|^2).$$

If P_* projects to a two-dimensional low-energy subspace, then the local Weyl coefficients are extracted by

$$\lambda_\parallel = \frac{1}{2}\mathrm{Tr}_{\mathcal{H}_*}\big(\sigma^3 P_* (\hat{n}_i \mathcal{K}_i) P_*\big),$$

$$T_{ab} = \frac{1}{2}\mathrm{Tr}_{\mathcal{H}_*}\big(\sigma^b P_* (e_{ai} \mathcal{K}_i) P_*\big).$$

For the minimal axial equivariant ansatz $T = \alpha I_2 + \beta J_2$, one has

$$\alpha = \frac{1}{2}(T_{11} + T_{22}), \qquad \beta = \frac{1}{2}(T_{21} - T_{12}), \qquad \det M = \lambda_\parallel(\alpha^2 + \beta^2).$$

Therefore the local Weyl/Dirac problem in TECT is reduced to an explicit coefficient extraction calculation from the microscopic condensate PDE.

LX. FIRST EXPLICIT EXTRACTION FROM A MINIMAL TECT CONDENSATE PDE

A. Purpose

The local Weyl/Dirac problem has already been reduced to the extraction of the three coefficients

$$\lambda_\parallel, \qquad \alpha, \qquad \beta,$$

from the first-order expansion of the microscopic condensate symbol near a selected BCC shell vector G_*.

At the fully microscopic level, what remains open is the derivation of the actual TECT linearized operator

$$\mathcal{L}(k),$$

the proof of the shell-valley structure $\pm G_*$, and the verification of nondegeneracy

$$v_\parallel v_1 v_2 \neq 0.$$

Thus the present section does *not* claim a final first-principles derivation from the complete TECT PDE. Instead, it gives the first explicit extraction calculation for a minimal symmetry-compatible candidate operator.

B. Minimal Projected Shell Ansatz

Let $\phi_0(x)$ be the BCC condensate background, and let G_* be a selected first-shell reciprocal vector. We work near the shell momentum

$$k = G_* + p, \qquad p = p_\parallel \hat{n} + p_1 e_1 + p_2 e_2, \qquad \hat{n} := \frac{G_*}{|G_*|}.$$

Motivated by the shell-level reduction theorem, consider the enriched shell space

$$\mathcal{H}_{\rm enr} = \mathbb{C}^2_{\rm pair} \otimes \mathbb{C}^2_{\rm int},$$

with Pauli matrices τ^i acting on the shell-pair factor and s^α acting on the internal doublet. The shell-level problem has already been reduced to gradient-linear couplings of the form

$$\mathsf{N}^a_{ij}\,(\partial_i \phi_0)\,(-i\partial_j) \otimes s^a,$$

together with the reduced couplings

$$\mathcal{T}^\alpha(x) = C^\alpha{}_j\,\partial_j \phi_0(x) \otimes s^\alpha.$$

We now impose the smallest axial-equivariant first-order ansatz consistent with the previous reduction:

$$\mathcal{L}_{\rm min}(G_* + p) = E_*\,\mathbf{1}_4 + v_\parallel\,p_\parallel\,\tau^3 \otimes \mathbf{1}_2 + \nu_1\,p_1\,\tau^1 \otimes s^1 + \nu_2\,p_2\,\tau^1 \otimes s^2 + O(|p|^2).$$

Equivalently, in the isotropic transverse basis one may write

$$\mathcal{L}_{\rm min}(G_* + p) = E_*\,\mathbf{1}_4 + v_\parallel\,(\hat{n}\cdot p)\,\tau^3 \otimes \mathbf{1}_2 + \alpha\left(p_1\,\tau^1 \otimes s^1 + p_2\,\tau^1 \otimes s^2\right) + \beta\left(-p_2\,\tau^1 \otimes s^1 + p_1\,\tau^1 \otimes s^2\right) + O(|p|^2).$$

This is the minimal shell-enriched axial ansatz.

C. First Derivative Operators

Define

$$\mathcal{K}_i := \partial_{k_i} \mathcal{L}_{\rm min}(k)\big|_{k=G_*}.$$

Using

$$p_\parallel = \hat{n}_i (k_i - G_{*i}), \qquad p_a = e_{ai}(k_i - G_{*i}),$$

we obtain

$$\mathcal{K}_i = v_\parallel\,\hat{n}_i\,\tau^3 \otimes \mathbf{1}_2 + \alpha\left(e_{1i}\,\tau^1 \otimes s^1 + e_{2i}\,\tau^1 \otimes s^2\right) + \beta\left(-e_{2i}\,\tau^1 \otimes s^1 + e_{1i}\,\tau^1 \otimes s^2\right).$$

This is the explicit coefficient extraction at the operator level.

D. Adapted Projected Derivatives

Contracting with the axial basis gives:

$$\mathcal{K}_\parallel := \hat{n}_i \mathcal{K}_i = v_\parallel\,\tau^3 \otimes \mathbf{1}_2,$$

since

$$\hat{n}_i e_{ai} = 0.$$

Likewise,

$$\mathcal{K}_1 := e_{1i}\mathcal{K}_i = \alpha\,\tau^1 \otimes s^1 + \beta\,\tau^1 \otimes s^2,$$

and

$$\mathcal{K}_2 := e_{2i}\mathcal{K}_i = -\beta\,\tau^1 \otimes s^1 + \alpha\,\tau^1 \otimes s^2.$$

Thus the projected first-order symbol is already in the canonical Weyl/Dirac template.

E. Two-Band Valley Reduction

Now restrict to a single valley block. At one valley, the shell-pair Pauli operator τ^3 is diagonal, and the corresponding two-dimensional reduced subspace is obtained by fixing the appropriate shell-pair chirality. Let P_+ denote the projection to that valley subspace. Then the reduced two-band Hamiltonian is

$$h_+^{\mathrm{lin}}(p) = P_+ \mathcal{L}_{\min}(G_* + p) P_+ - E_* \mathbf{1}_2.$$

Since the reduced Pauli basis identifies

$$P_+(\tau^3 \otimes \mathbf{1}_2)P_+ \mapsto \sigma^3, \qquad P_+(\tau^1 \otimes s^1)P_+ \mapsto \sigma^1, \qquad P_+(\tau^1 \otimes s^2)P_+ \mapsto \sigma^2,$$

we obtain

$$h_+^{\mathrm{lin}}(p) = v_\parallel p_\parallel \sigma^3 + \alpha(p_1\sigma^1 + p_2\sigma^2) + \beta(-p_2\sigma^1 + p_1\sigma^2).$$

Therefore the reduced transverse matrix is

$$T = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix},$$

and hence

$$\det T = \alpha^2 + \beta^2.$$

F. Projector-Trace Extraction Formulas

The general extraction theorem states that

$$\lambda_\parallel = \frac{1}{2} \operatorname{Tr}_{\mathcal{H}_*}\big(\sigma^3 P_*(\hat{n}_i \mathcal{K}_i) P_*\big),$$

$$T_{ab} = \frac{1}{2} \operatorname{Tr}_{\mathcal{H}_*}\big(\sigma^b P_*(e_{ai} \mathcal{K}_i) P_*\big).$$

For the present minimal ansatz, these formulas give directly

$$\lambda_\parallel = v_\parallel,$$

$$T_{11} = \alpha, \qquad T_{12} = -\beta, \qquad T_{21} = \beta, \qquad T_{22} = \alpha.$$

Therefore

$$\alpha = \frac{1}{2}(T_{11} + T_{22}), \qquad \beta = \frac{1}{2}(T_{21} - T_{12}),$$

as expected.

G. Explicit Nondegeneracy Calculation

The full linear coefficient matrix is

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & v_\parallel \end{pmatrix}.$$

Hence

$$\det M = v_\parallel(\alpha^2 + \beta^2).$$

Therefore:

- if
$$v_\parallel \neq 0,$$
the longitudinal channel is active;
- if
$$(\alpha,\beta) \neq (0,0),$$
the transverse block is invertible;
- if both hold, then
$$\det M \neq 0,$$
and the crossing is Weyl.

H. Valley Doubling and Dirac Form

Assume there exists an opposite valley at $-G_*$ satisfying the standard opposite-valley relation
$$h_-(p) = -h_+(-p).$$
Then the doubled four-component Hamiltonian is
$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix}.$$
After a standard chiral basis change, this becomes
$$H_D(q) = q_1\alpha^1 + q_2\alpha^2 + q_3\alpha^3, \qquad \{\alpha^i,\alpha^j\} = 2\delta^{ij}\mathbf{1}_4,$$
which is the massless Dirac Hamiltonian.

Thus the minimal extracted coefficients
$$v_\parallel, \alpha, \beta$$
already control the full Weyl/Dirac transition.

I. Extraction Theorem: Explicit Minimal Form

Theorem. Suppose the microscopic TECT linearized symbol near a selected BCC shell vector G_* is, to first order, unitarily reducible to the minimal shell-enriched ansatz
$$\mathcal{L}_{\min}(G_*+p) = E_*\,\mathbf{1}_4 + v_\parallel\, p_\parallel\, \tau^3 \otimes \mathbf{1}_2 + \alpha\left(p_1\,\tau^1 \otimes s^1 + p_2\,\tau^1 \otimes s^2\right) + \beta\left(-p_2\,\tau^1 \otimes s^1 + p_1\,\tau^1 \otimes s^2\right) + O(|p|^2).$$
Then the first derivative operators satisfy
$$\mathcal{K}_\parallel = v_\parallel\,\tau^3 \otimes \mathbf{1}_2,$$
$$\mathcal{K}_1 = \alpha\,\tau^1 \otimes s^1 + \beta\,\tau^1 \otimes s^2,$$
$$\mathcal{K}_2 = -\beta\,\tau^1 \otimes s^1 + \alpha\,\tau^1 \otimes s^2,$$
and the reduced Weyl coefficients are exactly
$$\lambda_\parallel = v_\parallel, \qquad \det T = \alpha^2 + \beta^2.$$
Consequently,
$$\det M = v_\parallel(\alpha^2 + \beta^2).$$
If
$$v_\parallel \neq 0, \qquad (\alpha,\beta) \neq (0,0),$$
then the selected BCC shell crossing is locally Weyl. If, in addition, an opposite valley exists with the standard pairing symmetry, the doubled low-energy theory is Dirac.

Proof. All statements follow from the explicit derivative computation above, followed by projection to the two-band valley subspace and direct determinant evaluation. □

J. What Is Proved and What Remains Open

What is proved here is:

1. for the minimal symmetry-compatible shell-enriched ansatz, the coefficients $\lambda_\parallel, \alpha, \beta$ are extracted explicitly;
2. the Weyl nondegeneracy condition reduces exactly to
$$v_\parallel(\alpha^2+\beta^2) \neq 0;$$
3. opposite-valley doubling then yields Dirac.

What remains open is the microscopic matching problem:

1. derive the full TECT condensate PDE in final closed form;
2. prove that its linearization near G_* reduces to the minimal enriched ansatz above, or identify the exact corrections;
3. prove existence of the opposite shell-pair valley and its symmetry relation.

Thus the local Weyl/Dirac problem has now been reduced to a concrete microscopic coefficient-matching calculation.

K. Final Honest Reduction

The remaining local Dirac frontier is now exactly:

$$\boxed{\text{Derive the true TECT linearized symbol and match its first-order projected coefficients to } v_\parallel, \alpha, \beta.}$$

If the microscopic matching gives

$$v_\parallel \neq 0, \qquad (\alpha,\beta) \neq (0,0),$$

then local Weyl nondegeneracy follows immediately, and Dirac emergence follows after valley doubling.

FINAL SUMMARY

The local Weyl/Dirac problem in TECT has been reduced to the first-order expansion of the microscopic condensate symbol near a selected BCC shell vector G_*.

For the minimal shell-enriched candidate operator

$$\mathcal{L}_{\min}(G_*+p) = E_*\,\mathbf{1}_4 + v_\parallel\, p_\parallel\, \tau^3\otimes \mathbf{1}_2 + \alpha\left(p_1\,\tau^1\otimes s^1 + p_2\,\tau^1\otimes s^2\right) + \beta\left(-p_2\,\tau^1\otimes s^1 + p_1\,\tau^1\otimes s^2\right) + O(|p|^2),$$

the projected first-derivative coefficients are

$$\lambda_\parallel = v_\parallel, \qquad T = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}, \qquad \det T = \alpha^2+\beta^2.$$

Therefore

$$\det M = v_\parallel(\alpha^2+\beta^2).$$

Hence:

$$v_\parallel \neq 0, \quad (\alpha,\beta)\neq(0,0) \Longrightarrow \text{local Weyl nondegeneracy.}$$

If, in addition, the opposite shell valley exists and satisfies the standard pairing symmetry, then valley doubling yields the massless Dirac operator.

Thus the remaining microscopic task is no longer structural: it is the explicit matching of the true TECT linearized symbol to the coefficients $v_\parallel, \alpha, \beta$.

LXI. MICROSCOPIC MATCHING THEOREM FOR THE WEYL/DIRAC COEFFICIENTS

A. Purpose

The local Weyl/Dirac problem has already been reduced to the three coefficients

$$\lambda_{\parallel}, \qquad \alpha, \qquad \beta,$$

through the determinant identity

$$\det M = \lambda_{\parallel}(\alpha^2 + \beta^2).$$

The remaining task is therefore to match these coefficients directly to the microscopic enriched tensor couplings of the TECT condensate linearization.

The shell-level enriched coupling has the generic gradient-linear form

$$\delta \mathcal{L}_{\text{enr}} = \sum_a \mathsf{N}^a_{ij}\,(\partial_i \phi_0)\,(-i\partial_j) \otimes s^a,$$

where:

- ϕ_0 is the condensed BCC background,
- N^a_{ij} is the finite microscopic tensor coefficient,
- s^a acts on the internal reduced doublet sector.

We now show that, after projection to a selected shell pair $\pm G_*$, these tensors determine the effective Weyl/Dirac coefficients by explicit contractions.

B. Selected Shell Pair and Adapted Frame

Fix a reciprocal pair

$$\pm G_* \in S_{\text{BCC}}, \qquad \hat{n} := \frac{G_*}{|G_*|},$$

and choose an orthonormal frame

$$\{e_1, e_2, \hat{n}\}.$$

Decompose a small momentum as

$$p = p_{\parallel}\hat{n} + p_1 e_1 + p_2 e_2.$$

Near the selected shell mode, write the BCC background locally as

$$\phi_0(x) = \Phi_* e^{iG_*\cdot x} + \overline{\Phi_*}\, e^{-iG_*\cdot x} + \cdots,$$

so that the gradient on the selected shell component is

$$\partial_i \phi_0 = iG_{*i}\Phi_* e^{iG_*\cdot x} - iG_{*i}\overline{\Phi_*} e^{-iG_*\cdot x} + \cdots.$$

Thus, on the selected shell component, $\partial_i \phi_0$ contributes a factor proportional to G_{*i}.

C. Shell-Reduced Vector Coefficients

Define the reduced vector coefficients

$$C^a_j := G_{*i}\,\mathsf{N}^a_{ij}.$$

Equivalently,

$$C^a = (C^a_j)_{j=1}^3 \in \mathbb{R}^3.$$

Then the enriched shell operator projected to the selected shell pair acquires the form

$$\delta h_{\mathrm{enr}}(p) \sim \sum_a C^a_j\, p_j\, \Sigma^a,$$

where Σ^a denotes the projected 2×2 Pauli-sector generator induced by the microscopic channel s^a.

This formula is the precise microscopic origin of the shell-level coefficients. All local Weyl data are determined by the three vectors

$$C^1, \qquad C^2, \qquad C^3.$$

D. Longitudinal/Transverse Decomposition of the Reduced Coefficients

Decompose each C^a along the adapted frame:

$$C^a = (C^a\cdot\hat{n})\,\hat{n} + (C^a\cdot e_1)\,e_1 + (C^a\cdot e_2)\,e_2.$$

For the minimal axial-equivariant ansatz, the symmetry-adapted matching is:

- channel $a=3$ contributes to the longitudinal scalar piece,
- channels $a=1,2$ contribute to the transverse doublet piece.

Accordingly, define

$$\lambda_\parallel := \kappa_*\,(C^3\cdot\hat{n}),$$

where κ_* is the shell-projection normalization factor (proportional to the selected-shell condensate amplitude).

Likewise define the transverse 2×2 matrix

$$T_{ab} := \kappa_*\,(C^b\cdot e_a), \qquad a,b=1,2.$$

Explicitly,

$$T = \kappa_* \begin{pmatrix} C^1\cdot e_1 & C^2\cdot e_1 \\ C^1\cdot e_2 & C^2\cdot e_2 \end{pmatrix}.$$

Thus the reduced linear Hamiltonian becomes

$$h_+^{\mathrm{lin}}(p) = \lambda_\parallel p_\parallel \sigma^3 + \sum_{a,b=1}^{2} T_{ab}\,p_a\sigma^b.$$

E. Direct Contraction Formulas

Substituting $C^a_j = G_{*i}\mathsf{N}^a_{ij}$, we obtain

$$\lambda_\parallel = \kappa_*\,\hat{n}_j G_{*i}\mathsf{N}^3_{ij}.$$

Since $G_{*i} = |G_*|\hat{n}_i$,

$$\lambda_\parallel = \kappa_* |G_*|\, \hat{n}_i \hat{n}_j\, \mathsf{N}^3_{ij}.$$

Similarly,

$$T_{ab} = \kappa_*\, e_{aj} G_{*i} \mathsf{N}^b_{ij} = \kappa_* |G_*|\, e_{aj} \hat{n}_i\, \mathsf{N}^b_{ij}, \qquad a, b = 1, 2.$$

Therefore the microscopic matching formulas are

$$\boxed{\lambda_\parallel = \kappa_* |G_*|\, \hat{n}_i \hat{n}_j\, \mathsf{N}^3_{ij}}$$

and

$$\boxed{T_{ab} = \kappa_* |G_*|\, e_{aj} \hat{n}_i\, \mathsf{N}^b_{ij}, \qquad a, b = 1, 2.}$$

These are the fundamental coefficient-extraction formulas.

F. Extraction of α and β

For the minimal axial-equivariant transverse ansatz, write

$$T = \alpha I_2 + \beta J_2, \qquad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$\alpha = \frac{1}{2}\,\mathrm{tr}(T), \qquad \beta = \frac{1}{2}\,\mathrm{tr}(J_2^\top T).$$

Equivalently,

$$\alpha = \frac{1}{2}(T_{11} + T_{22}), \qquad \beta = \frac{1}{2}(T_{21} - T_{12}).$$

Using the microscopic formulas for T_{ab}, this gives

$$\alpha = \frac{\kappa_* |G_*|}{2} \Big(e_{1j} \hat{n}_i\, \mathsf{N}^1_{ij} + e_{2j} \hat{n}_i\, \mathsf{N}^2_{ij} \Big),$$

and

$$\beta = \frac{\kappa_* |G_*|}{2} \Big(e_{2j} \hat{n}_i\, \mathsf{N}^1_{ij} - e_{1j} \hat{n}_i\, \mathsf{N}^2_{ij} \Big).$$

Thus the complete matching is

$$\boxed{\lambda_\parallel = \kappa_* |G_*|\, \hat{n}_i \hat{n}_j\, \mathsf{N}^3_{ij},}$$

$$\boxed{\alpha = \frac{\kappa_* |G_*|}{2} \Big(e_{1j} \hat{n}_i\, \mathsf{N}^1_{ij} + e_{2j} \hat{n}_i\, \mathsf{N}^2_{ij} \Big),}$$

$$\boxed{\beta = \frac{\kappa_* |G_*|}{2} \Big(e_{2j} \hat{n}_i\, \mathsf{N}^1_{ij} - e_{1j} \hat{n}_i\, \mathsf{N}^2_{ij} \Big).}$$

G. Microscopic Matching Theorem

Theorem. Let the TECT enriched microscopic linearization near a selected BCC shell pair $\pm G_*$ contain the gradient-linear coupling

$$\delta\mathcal{L}_{\rm enr} = \sum_{a=1}^{3} \mathsf{N}^a_{ij}\,(\partial_i\phi_0)\,(-i\partial_j)\otimes s^a.$$

Assume:

1. the shell pair $\pm G_*$ defines an adapted frame $\{e_1, e_2, \hat{n}\}$;
2. the selected shell projection maps the microscopic channels s^a to the reduced Pauli basis so that $a = 3$ is longitudinal and $a = 1, 2$ are transverse;
3. forbidden mixed channels are absent by the little-group symmetry.

Then the reduced Weyl coefficients are given by

$$\lambda_\parallel = \kappa_*|G_*|\,\hat{n}_i\hat{n}_j\,\mathsf{N}^3_{ij},$$

$$\alpha = \frac{\kappa_*|G_*|}{2}\Big(e_{1j}\hat{n}_i\,\mathsf{N}^1_{ij} + e_{2j}\hat{n}_i\,\mathsf{N}^2_{ij}\Big),$$

$$\beta = \frac{\kappa_*|G_*|}{2}\Big(e_{2j}\hat{n}_i\,\mathsf{N}^1_{ij} - e_{1j}\hat{n}_i\,\mathsf{N}^2_{ij}\Big).$$

Consequently,

$$\det M = \lambda_\parallel(\alpha^2+\beta^2).$$

Proof. The background gradient contributes a factor G_{*i} on the selected shell component, so the microscopic tensor N^a_{ij} reduces to the shell vectors $C^a_j = G_{*i}\mathsf{N}^a_{ij}$. Projecting these vectors onto the adapted frame $(e_1, e_2, \hat{n})$ gives the longitudinal scalar coefficient and the transverse 2×2 matrix:

$$\lambda_\parallel = \kappa_*(C^3\cdot\hat{n}), \qquad T_{ab} = \kappa_*(C^b\cdot e_a).$$

Substituting $C^a_j = G_{*i}\mathsf{N}^a_{ij}$ gives the stated contraction formulas. The formulas for α, β follow from the decomposition

$$T = \alpha I_2 + \beta J_2.$$

Finally,

$$\det T = \alpha^2+\beta^2,$$

hence

$$\det M = \lambda_\parallel \det T = \lambda_\parallel(\alpha^2+\beta^2).$$

□

H. Immediate Corollary

Corollary. If the actual microscopic tensor coefficients satisfy

$$\hat{n}_i\hat{n}_j\,\mathsf{N}^3_{ij} \neq 0,$$

and

$$\Big(e_{1j}\hat{n}_i\,\mathsf{N}^1_{ij} + e_{2j}\hat{n}_i\,\mathsf{N}^2_{ij}\Big)^2 + \Big(e_{2j}\hat{n}_i\,\mathsf{N}^1_{ij} - e_{1j}\hat{n}_i\,\mathsf{N}^2_{ij}\Big)^2 \neq 0,$$

then

$$\lambda_\parallel \neq 0, \qquad (\alpha,\beta)\neq(0,0),$$

and therefore the selected BCC shell crossing is locally Weyl.

If, in addition, an opposite valley exists with the standard doubling symmetry, then the doubled low-energy theory is Dirac.

I. Matching Table

The microscopic-to-effective dictionary is therefore:

Microscopic contraction	Effective coefficient
$\hat{n}_i \hat{n}_j \, \mathsf{N}^3_{ij}$	$\lambda_\parallel$
$\frac{1}{2}\left(e_{1j}\hat{n}_i \, \mathsf{N}^1_{ij} + e_{2j}\hat{n}_i \, \mathsf{N}^2_{ij}\right)$	$\alpha/(\kappa_*\lvert G_*\rvert)$
$\frac{1}{2}\left(e_{2j}\hat{n}_i \, \mathsf{N}^1_{ij} - e_{1j}\hat{n}_i \, \mathsf{N}^2_{ij}\right)$	$\beta/(\kappa_*\lvert G_*\rvert)$

Thus the local Weyl problem has been converted into a finite list of explicit tensor contractions.

J. What Is Closed and What Is Still Open

What is now closed:

- the exact matching formulas from microscopic enriched tensors N^a_{ij} to the effective Weyl/Dirac coefficients $\lambda_\parallel, \alpha, \beta$;
- the identity
$$\det M = \lambda_\parallel(\alpha^2 + \beta^2);$$
- the finite tensor-contraction criterion for local Weyl nondegeneracy.

What remains open:

- derive the actual microscopic TECT condensate PDE in final closed form;
- compute the true N^a_{ij} generated by that PDE;
- verify the shell-pair valley existence and opposite-valley doubling relation.

K. Final Honest Reduction

The remaining local Dirac frontier is now exactly:

$$\boxed{\text{Compute the actual microscopic tensor coefficients } \mathsf{N}^a_{ij} \text{ of the TECT condensate linearization and evaluate the three contractio}}$$

If they satisfy

$$\hat{n}_i \hat{n}_j \, \mathsf{N}^3_{ij} \neq 0,$$

and

$$\left(e_{1j}\hat{n}_i \, \mathsf{N}^1_{ij} + e_{2j}\hat{n}_i \, \mathsf{N}^2_{ij}\right)^2 + \left(e_{2j}\hat{n}_i \, \mathsf{N}^1_{ij} - e_{1j}\hat{n}_i \, \mathsf{N}^2_{ij}\right)^2 \neq 0,$$

then local Weyl nondegeneracy follows immediately.

SUMMARY

For the enriched microscopic shell coupling

$$\delta\mathcal{L}_{\mathrm{enr}} = \sum_{a=1}^{3} \mathsf{N}^a_{ij} \, (\partial_i \phi_0) \, (-i\partial_j) \otimes s^a,$$

the selected BCC shell pair $\pm G_*$ reduces the microscopic tensors to the shell vectors

$$C_j^a = G_{*i}\mathsf{N}_{ij}^a.$$

After projection to the adapted frame $\{e_1, e_2, \hat{n}\}$, the effective Weyl coefficients are

$$\lambda_\parallel = \kappa_* |G_*|\, \hat{n}_i \hat{n}_j\, \mathsf{N}_{ij}^3,$$

$$\alpha = \frac{\kappa_*|G_*|}{2}\Big(e_{1j}\hat{n}_i\, \mathsf{N}_{ij}^1 + e_{2j}\hat{n}_i\, \mathsf{N}_{ij}^2\Big),$$

$$\beta = \frac{\kappa_*|G_*|}{2}\Big(e_{2j}\hat{n}_i\, \mathsf{N}_{ij}^1 - e_{1j}\hat{n}_i\, \mathsf{N}_{ij}^2\Big).$$

Hence

$$\det M = \lambda_\parallel(\alpha^2 + \beta^2).$$

Therefore the local Weyl/Dirac problem in TECT is no longer a qualitative question. It is an explicit finite tensor-contraction test on the microscopic coefficients N_{ij}^a.

LXII. TECT EMERGENT DIRAC FERMION THEOREM

A. Purpose

We now assemble the previously established ingredients into a single theorem-level statement for emergent Dirac fermions in TECT.

The logic consists of four layers:

1. topological chirality from the defect index;
2. dynamical mirror lifting;
3. local Weyl-valley emergence from the BCC shell matching;
4. opposite-valley doubling to a Dirac fermion.

The purpose of this section is to state precisely what is already closed at the theorem level, and which microscopic inputs still remain to be checked in the full TECT condensate PDE.

B. Topological Chiral Sector

Let D_+ denote the chirally projected defect Dirac operator on the relevant internal topological sector. Assume the primitive twisting of the $\mathbb{CP}^2$-based defect bundle gives

$$\mathrm{Index}(D_+) = I > 0.$$

Then

$$n_L - n_R = I.$$

Equivalently, the low-energy defect sector contains a net protected left-handed chiral subspace of index I.

C. Mirror Lifting

Let the low-energy fermion space decompose as

$$\mathcal{H}_F = \mathcal{H}_{\mathrm{prot}} \oplus \mathcal{H}_{\mathrm{mir}},$$

where $\mathcal{H}_{\mathrm{prot}}$ is the topologically protected chiral subspace and $\mathcal{H}_{\mathrm{mir}}$ consists of vectorlike mirror pairs.
Assume there exists a mirror-lifting operator

$$M_{\mathrm{mir}} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & \mu^\dagger \\ 0 & \mu & 0 \end{pmatrix}, \qquad \mu \neq 0,$$

acting on the mirror sector, such that all vectorlike pairs acquire nonzero gap. Then the infrared spectrum reduces to the protected chiral sector

$$n_L = I, \qquad n_R = 0,$$

up to the surviving opposite-valley doubling discussed below.

D. Local BCC Weyl Matching

Fix a reciprocal shell pair

$$\pm G_* \in S_{\mathrm{BCC}}, \qquad \hat{n} = \frac{G_*}{|G_*|},$$

and choose an adapted orthonormal frame

$$\{e_1, e_2, \hat{n}\}.$$

Assume the enriched microscopic shell coupling near the selected shell pair has the form

$$\delta\mathcal{L}_{\mathrm{enr}} = \sum_{a=1}^{3} \mathsf{N}^a_{ij}\,(\partial_i \phi_0)\,(-i\partial_j) \otimes s^a,$$

where ϕ_0 is the BCC condensate background and s^a acts on the reduced internal shell doublet.
Define the three effective coefficients by

$$\lambda_\parallel = \kappa_* |G_*|\, \hat{n}_i \hat{n}_j\, \mathsf{N}^3_{ij},$$

$$\alpha = \frac{\kappa_*|G_*|}{2}\Big(e_{1j}\hat{n}_i\, \mathsf{N}^1_{ij} + e_{2j}\hat{n}_i\, \mathsf{N}^2_{ij}\Big),$$

$$\beta = \frac{\kappa_*|G_*|}{2}\Big(e_{2j}\hat{n}_i\, \mathsf{N}^1_{ij} - e_{1j}\hat{n}_i\, \mathsf{N}^2_{ij}\Big).$$

Then the local reduced valley Hamiltonian is

$$h_+(p) = \lambda_\parallel p_\parallel \sigma^3 + \alpha(p_1\sigma^1 + p_2\sigma^2) + \beta(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2).$$

Equivalently, the transverse matrix is

$$T = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}, \qquad \det T = \alpha^2 + \beta^2.$$

Hence the full linear coefficient determinant is

$$\det M = \lambda_\parallel(\alpha^2 + \beta^2).$$

Therefore, if

$$\lambda_\parallel \neq 0, \qquad (\alpha, \beta) \neq (0, 0),$$

then the local shell crossing is Weyl.

E. Opposite-Valley Doubling

Assume there exists an opposite shell valley at $-G_*$ with the standard pairing relation

$$h_-(p) = -h_+(-p).$$

Then the doubled low-energy Hamiltonian becomes

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix}.$$

After a standard basis change, this is the massless Dirac Hamiltonian

$$H_D(q) = q_1\Gamma^1 + q_2\Gamma^2 + q_3\Gamma^3, \qquad \{\Gamma^i, \Gamma^j\} = 2\delta^{ij}.$$

F. TECT Emergent Dirac Fermion Theorem

Theorem. Assume the following:

1. the TECT condensate selects the BCC reciprocal shell S_{BCC};
2. the internal topological defect sector carries a chirally twisted Dirac operator with
$$\mathrm{Index}(D_+) = I > 0;$$
3. all vectorlike mirror pairs are lifted by a nondegenerate mirror operator
$$\mu \neq 0;$$
4. there exists a selected shell pair $\pm G_* \in S_{\mathrm{BCC}}$ together with a two-band enriched valley sector;
5. the enriched microscopic shell tensors N^a_{ij} satisfy
$$\lambda_\parallel = \kappa_*|G_*|\,\hat{n}_i\hat{n}_j\,\mathsf{N}^3_{ij} \neq 0,$$
and
$$(\alpha, \beta) \neq (0, 0),$$
where
$$\alpha = \frac{\kappa_*|G_*|}{2}\Big(e_{1j}\hat{n}_i\,\mathsf{N}^1_{ij} + e_{2j}\hat{n}_i\,\mathsf{N}^2_{ij}\Big),$$
$$\beta = \frac{\kappa_*|G_*|}{2}\Big(e_{2j}\hat{n}_i\,\mathsf{N}^1_{ij} - e_{1j}\hat{n}_i\,\mathsf{N}^2_{ij}\Big);$$
6. an opposite valley exists at $-G_*$ with the standard doubling relation
$$h_-(p) = -h_+(-p).$$

Then the infrared TECT theory contains a protected chiral Dirac fermion sector. More precisely:

1. the selected shell crossing at G_* is Weyl;
2. the opposite shell crossing at $-G_*$ has opposite chirality;
3. valley doubling yields a massless Dirac fermion;
4. the mirror sector is lifted, so the surviving low-energy fermion content is the protected chiral Dirac sector of multiplicity I.

Proof. By assumption (2),

$$n_L - n_R = I > 0,$$

so there exists a net protected chiral sector. By assumption (3), all vectorlike mirror pairs are gapped out, leaving only the protected sector in the infrared.

By assumption (5), the reduced shell Hamiltonian near G_* has coefficients

$$\lambda_\parallel \neq 0, \qquad \det T = \alpha^2 + \beta^2 > 0.$$

Hence

$$\det M = \lambda_\parallel(\alpha^2 + \beta^2) \neq 0.$$

Therefore the local crossing is linearly nondegenerate and hence Weyl.

By assumption (6), the opposite valley at $-G_*$ satisfies the standard opposite-valley relation, so the two Weyl nodes combine into a doubled four-component Hamiltonian of Dirac type. Thus the infrared theory contains a massless Dirac fermion.

Finally, because the mirror sector has already been lifted while the topological index protects the chiral sector, the surviving low-energy fermionic content is precisely the protected chiral Dirac sector with multiplicity I. □

G. Corollary: Minimal Local Criterion

Corollary. At the local valley level, the entire TECT Dirac-emergence problem reduces to the finite coefficient test

$$\hat{n}_i \hat{n}_j \, \mathsf{N}^3_{ij} \neq 0,$$

and

$$\left(e_{1j}\hat{n}_i \, \mathsf{N}^1_{ij} + e_{2j}\hat{n}_i \, \mathsf{N}^2_{ij}\right)^2 + \left(e_{2j}\hat{n}_i \, \mathsf{N}^1_{ij} - e_{1j}\hat{n}_i \, \mathsf{N}^2_{ij}\right)^2 \neq 0,$$

together with the existence of the opposite valley.

If these conditions hold, then local Weyl nondegeneracy and Dirac doubling follow automatically.

H. What Is Closed

The following structure is now closed at the theorem level:

$$\mathbb{CP}^2 \Longrightarrow \mathrm{Index}(D_+) = I \Longrightarrow \text{protected chiral sector} \Longrightarrow \text{mirror lifting} \Longrightarrow \text{local Weyl valley on BCC shell} \Longrightarrow \text{Dirac doubling}.$$

Equivalently, the Dirac-emergence architecture is no longer open-ended. It has been reduced to a finite set of explicit microscopic checks.

I. What Remains Open

What remains genuinely open is not the theorem architecture, but the microscopic realization:

1. derive the full TECT condensate PDE in final closed form;
2. compute the actual enriched microscopic coefficients N^a_{ij};
3. verify the shell-pair valley existence at $\pm G_*$;
4. verify the opposite-valley pairing relation;
5. match the resulting Dirac sector to the gauge and matter bundle decomposition.

J. Final Honest Statement

The local fermion problem in TECT has now been reduced to a finite tensor-and-valley verification problem. No further conceptual ingredient is needed at the theorem level.

The remaining frontier is exactly:

$$\boxed{\text{Compute the true microscopic enriched tensors } \mathsf{N}^a_{ij} \text{ and verify the shell-pair valley doubling relation.}}$$

SUMMARY

The emergent Dirac fermion problem in TECT is now controlled by four ingredients:

1. topological chirality:
$$\mathrm{Index}(D_+) = I > 0;$$
2. mirror lifting:
$$\mu \neq 0;$$
3. local BCC-shell Weyl matching:
$$\lambda_\parallel = \kappa_* |G_*|\, \hat{n}_i \hat{n}_j\, \mathsf{N}^3_{ij},$$
$$\alpha = \frac{\kappa_* |G_*|}{2} \Big(e_{1j} \hat{n}_i\, \mathsf{N}^1_{ij} + e_{2j} \hat{n}_i\, \mathsf{N}^2_{ij} \Big),$$
$$\beta = \frac{\kappa_* |G_*|}{2} \Big(e_{2j} \hat{n}_i\, \mathsf{N}^1_{ij} - e_{1j} \hat{n}_i\, \mathsf{N}^2_{ij} \Big),$$
with
$$\det M = \lambda_\parallel (\alpha^2 + \beta^2);$$
4. opposite-valley doubling:
$$h_-(p) = -h_+(-p).$$

If these conditions hold, then the infrared TECT theory contains a protected chiral Dirac fermion sector.

Therefore the remaining microscopic task is finite and explicit: compute the true enriched tensor coefficients N^a_{ij} and verify the opposite-valley shell pairing.

LXIII. TECT CHIRAL GAUGE FERMION THEOREM

A. Purpose

We now combine the gauge-emergence chain and the fermion-emergence chain into a single theorem-level statement.

The aim is to show that, under the finite set of microscopic assumptions already isolated in previous sections, the infrared TECT theory contains a chiral gauge-fermion sector of the form

$$SU(3) \times SU(2) \times U(1)_Y$$

coupled to protected Dirac fermions whose vectorlike mirrors are lifted.

This theorem does *not* yet claim the full Standard Model matter content, anomaly cancellation, or exact generation multiplicity. Its purpose is narrower and more precise:

$$\boxed{\text{TECT yields a chiral gauge-fermion architecture once the finite bundle/tensor/valley inputs are satisfied.}}$$

B. Gauge Sector Inputs

Assume the following gauge-geometric ingredients.

a. *(G1) Abelian projective bundle.* The low-energy orientational degree of freedom is represented by

$$z(x) \in \mathbb{C}^3, \qquad z^\dagger z = 1, \qquad z \sim e^{i\alpha(x)} z,$$

so that the physical target is

$$[z] \in \mathbb{CP}^2.$$

This produces a principal $U(1)$ bundle with local connection

$$a_\mu = -iz^\dagger \partial_\mu z$$

and curvature

$$f_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu.$$

b. *(G2) Color connection.* The internal projective geometry

$$\mathbb{CP}^2 \simeq \frac{SU(3)}{U(2)}$$

admits the already-derived $SU(3)$-valued connection

$$A_\mu = A^a_\mu T^a, \qquad F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + f^{abc} A^b_\mu A^c_\nu.$$

c. *(G3) Weak bundle.* The rank-two orthogonal complement bundle

$$Q \to \mathbb{CP}^2$$

admits a local orthonormal frame with $U(2)$ redundancy. Assume the corresponding Berry connection reduces to a genuine $SU(2)$ connection

$$W_\mu = W^i_\mu \tau^i$$

on a preserved rank-two weak subbundle $\mathcal{E}_W$.

d. *(G4) Hypercharge embedding.* Assume there exists a hypercharge generator

$$Y = \alpha Q_P + \beta Q_T$$

or, in the stronger canonical case, a traceless $3+2$ embedding such that the Abelian gauge field is

$$B_\mu = \alpha\, a_\mu + \beta\, b_\mu,$$

with curvature

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu.$$

Under these assumptions, the infrared gauge sector is

$$G_{\text{gauge}} = SU(3) \times SU(2) \times U(1)_Y.$$

C. Topological Chiral Inputs

Assume the topological defect sector satisfies:

a. *(F1) Chiral index.* For the relevant defect Dirac operator,

$$\text{Index}(D_+) = I > 0.$$

Hence

$$n_L - n_R = I.$$

b. (F2) Mirror lifting. There exists a nondegenerate mirror-lifting operator

$$M_{\rm mir} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & \mu^\dagger \\ 0 & \mu & 0 \end{pmatrix}, \qquad \mu \neq 0,$$

which gaps out all vectorlike mirror pairs.

Therefore the infrared fermion content reduces to the topologically protected chiral sector.

D. Local BCC Valley Inputs

Fix a shell pair

$$\pm G_* \in S_{\rm BCC}, \qquad \hat{n} = \frac{G_*}{|G_*|},$$

and choose an adapted frame

$$\{e_1, e_2, \hat{n}\}.$$

Assume the enriched microscopic shell coupling near the selected shell pair is

$$\delta\mathcal{L}_{\rm enr} = \sum_{a=1}^{3} \mathsf{N}^a_{ij}(\partial_i\phi_0)(-i\partial_j) \otimes s^a.$$

Define

$$\lambda_\parallel = \kappa_* |G_*|\, \hat{n}_i \hat{n}_j \mathsf{N}^3_{ij},$$

$$\alpha = \frac{\kappa_*|G_*|}{2}\Big(e_{1j}\hat{n}_i\mathsf{N}^1_{ij} + e_{2j}\hat{n}_i\mathsf{N}^2_{ij}\Big),$$

$$\beta = \frac{\kappa_*|G_*|}{2}\Big(e_{2j}\hat{n}_i\mathsf{N}^1_{ij} - e_{1j}\hat{n}_i\mathsf{N}^2_{ij}\Big).$$

Then the reduced local valley Hamiltonian is

$$h_+(p) = \lambda_\parallel p_\parallel \sigma^3 + \alpha(p_1\sigma^1 + p_2\sigma^2) + \beta(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2),$$

and its linear nondegeneracy is controlled by

$$\det M = \lambda_\parallel(\alpha^2 + \beta^2).$$

Assume

$$\lambda_\parallel \neq 0, \qquad (\alpha, \beta) \neq (0, 0).$$

Then the selected shell crossing is Weyl.

a. (F3) Opposite valley doubling. Assume there exists an opposite valley at $-G_*$ satisfying

$$h_-(p) = -h_+(-p).$$

Then the doubled four-component Hamiltonian is Dirac.

E. Gauge-Covariant Dirac Structure

Because the fermionic valley sector descends from the defect sector and the latter already carries the internal gauge data, the covariant derivative acting on the emergent Dirac fermion is of the form

$$\mathcal{D}_\mu = \partial_\mu - ig_3 A^a_\mu T^a - ig_2 W^i_\mu t^i - ig_Y Y B_\mu,$$

where:

- T^a are generators of the color representation;
- t^i are generators of the weak representation;
- Y is the hypercharge generator.

Therefore the low-energy Dirac equation is

$$(i\gamma^\mu \mathcal{D}_\mu - M_*)\Psi = 0,$$

or, in the massless shell-emergent case,

$$i\gamma^\mu \mathcal{D}_\mu \Psi = 0$$

before additional symmetry-breaking effects are introduced.

F. TECT Chiral Gauge Fermion Theorem

Theorem. Assume:

1. the TECT condensate selects the BCC reciprocal shell;
2. the projective orientational sector yields the $U(1)$ bundle over $\mathbb{CP}^2$;
3. the internal geometry yields an $SU(3)$ connection;
4. the rank-two weak bundle exists and yields an $SU(2)$ connection;
5. hypercharge $U(1)_Y$ is defined by the Abelian embedding Y;
6. the topological defect Dirac operator satisfies
$$\mathrm{Index}(D_+) = I > 0;$$
7. all vectorlike mirror pairs are lifted by
$$\mu \neq 0;$$
8. the selected BCC shell pair $\pm G_*$ admits enriched microscopic coefficients satisfying
$$\lambda_\parallel \neq 0, \qquad (\alpha, \beta) \neq (0, 0);$$
9. an opposite shell valley exists with the doubling relation
$$h_-(p) = -h_+(-p).$$

Then the infrared TECT theory contains a chiral gauge-fermion sector with the following properties:

1. the gauge group is
$$SU(3) \times SU(2) \times U(1)_Y;$$

2. the protected chiral sector has multiplicity I;
3. each local nondegenerate shell crossing is Weyl;
4. opposite-valley doubling yields a Dirac fermion;
5. the resulting Dirac fermion is gauge-covariantly coupled through

$$\mathcal{D}_\mu = \partial_\mu - ig_3 A_\mu^a T^a - ig_2 W_\mu^i t^i - ig_Y Y B_\mu.$$

Proof. Items (1)–(5) establish the infrared gauge architecture

$$SU(3) \times SU(2) \times U(1)_Y.$$

Item (6) gives a protected net chiral sector of index I. Item (7) removes vectorlike mirror partners from the infrared spectrum. Item (8) implies

$$\det M = \lambda_\parallel(\alpha^2 + \beta^2) \neq 0,$$

so the selected shell crossing is Weyl. Item (9) doubles the local Weyl node into a Dirac fermion. Because the fermionic state descends from the same internal sector carrying the gauge bundle data, the resulting Dirac operator is gauge-covariant under the full gauge group. Thus the infrared theory contains a protected chiral gauge-fermion sector. □

G. Corollary: Finite Microscopic Checklist

Corollary. The full chiral gauge-fermion emergence problem reduces to the finite microscopic checklist:

1. compute the actual weak-bundle data $\mathcal{E}_W$;
2. determine the hypercharge embedding Y;
3. compute the actual enriched tensor coefficients N_{ij}^a;
4. verify

$$\lambda_\parallel \neq 0, \qquad (\alpha, \beta) \neq (0, 0);$$

5. verify opposite-valley doubling;
6. match the resulting fermion bundle to the desired matter representation.

No further conceptual structure is needed at the theorem level.

H. What Is Closed

The following architecture is now closed:

$$\text{BCC condensate} \Longrightarrow \mathbb{CP}^2 \Longrightarrow U(1)\text{ bundle} \Longrightarrow SU(3)\text{ connection} \Longrightarrow SU(2)\text{ weak bundle} \Longrightarrow U(1)_Y$$

$$\Longrightarrow \text{Index}(D_+) \Longrightarrow \text{mirror lifting} \Longrightarrow \text{local Weyl valley} \Longrightarrow \text{Dirac doubling} \Longrightarrow \text{chiral gauge fermion sector.}$$

I. What Remains Open

The genuinely open frontier is now the microscopic realization of the finite inputs:

1. exact weak bundle construction;
2. exact hypercharge normalization;
3. actual TECT tensor coefficients N^a_{ij};
4. shell-pair valley existence and doubling;
5. matter representation decomposition;
6. generation multiplicity and anomaly cancellation.

J. Final Honest Statement

The emergence of a chiral gauge-fermion architecture in TECT has now been reduced to a finite bundle/tensor/valley matching problem. At the theorem level, no conceptual gap remains.

SUMMARY

Under the finite set of assumptions consisting of:

- projective $U(1)$ bundle over $\mathbb{CP}^2$,
- emergent $SU(3)$ connection,
- rank-two weak bundle giving $SU(2)$,
- hypercharge embedding $U(1)_Y$,
- topological chiral index $\mathrm{Index}(D_+) = I > 0$,
- mirror lifting $\mu \neq 0$,
- local shell coefficients

$$\lambda_{\parallel} \neq 0, \qquad (\alpha, \beta) \neq (0, 0),$$

- opposite-valley doubling,

the infrared TECT theory contains a protected chiral gauge-fermion sector with gauge group

$$SU(3) \times SU(2) \times U(1)_Y,$$

and Dirac fermions coupled by the covariant derivative

$$\mathcal{D}_\mu = \partial_\mu - ig_3 A^a_\mu T^a - ig_2 W^i_\mu t^i - ig_Y Y B_\mu.$$

Thus the remaining frontier is no longer conceptual. It is the explicit microscopic realization of the finite bundle/tensor/valley inputs.

LXIV. RIGOROUS RESULTS

The following structures are closed at the theorem level within the present TECT framework.

A. BCC Condensate Selection

The quartic resonance invariant

$$C_4 = 540$$

uniquely selects the BCC reciprocal star

$$S_{BCC} = \frac{1}{\sqrt{2}}\{(\pm 1, \pm 1, 0), (\pm 1, 0, \pm 1), (0, \pm 1, \pm 1)\}.$$

B. Elastic Continuum Limit

Long-wavelength linearization produces

$$1 \text{ longitudinal mode} + 2 \text{ transverse modes}.$$

C. U(1) Gauge Emergence

The transverse phase rotation

$$SO(2) \to U(1)$$

produces the gauge connection

$$A_\mu = -iz^\dagger \partial_\mu z.$$

D. Topological Charge

The defect charge is classified by

$$\pi_2(\mathbb{CP}^2) = \mathbb{Z}.$$

E. Chiral Index

The twisted Dirac operator satisfies

$$\text{Index}(D_+) = I.$$

Hence

$$n_L - n_R = I.$$

F. Mirror Lifting

Vectorlike pairs can acquire gauge-invariant masses

$$m\,\bar{\psi}_L \psi_R,$$

leaving the protected chiral sector.

G. Local Weyl Nondegeneracy Criterion

For the shell-reduced Hamiltonian

$$h(p) = \lambda_{\|} p_{\|} \sigma^3 + \alpha(p_1\sigma^1 + p_2\sigma^2) + \beta(-p_2\sigma^1 + p_1\sigma^2),$$

the crossing is Weyl if

$$\det M = \lambda_{\|}(\alpha^2 + \beta^2) \neq 0.$$

LXV. CONDITIONALLY SUPPORTED RESULTS

The following results are strongly supported by the structural framework but still depend on additional microscopic verification.

A. SU(3) Gauge Structure

The internal geometry

$$\mathbb{CP}^2 \simeq SU(3)/U(2)$$

suggests the emergence of an SU(3) connection. However, the full dynamical gauge invariance requires verification of the microscopic Ward identities.

B. Weak SU(2) Bundle

The orthogonal rank-two bundle over $\mathrm{CP}^{\mathbf{2}}$ carries a natural U(2) structure.
The reduction

$$U(2) \to SU(2) \times U(1)$$

is structurally natural but still requires a dynamical stabilization mechanism.

C. Dirac Valley Doubling

If an opposite valley exists at

$$-G_*,$$

then

$$h_-(p) = -h_+(-p)$$

produces a Dirac fermion.
However, the existence of the valley pair must be verified from the microscopic condensate operator.

LXVI. OPEN PROBLEMS

The following problems remain unsolved at the microscopic level.

A. Full TECT Condensate PDE

The complete microscopic condensate equation

$$\partial_t \Phi = F(\Phi, \nabla\Phi)$$

has not yet been derived from first principles.

B. Microscopic Tensor Coefficients

The enriched coupling tensors

$$\mathsf{N}^a_{ij}$$

must be computed from the true microscopic operator.

C. Shell Valley Existence

It must be shown that a pair of valleys exists at

$$\pm G_*.$$

D. Weak Bundle Stabilization

The rank-two weak bundle

$$\mathcal{E}_W$$

must be shown to remain dynamically stable.

E. Hypercharge Quantization

The normalization of

$$Y$$

must reproduce the Standard Model charge spectrum.

F. Generation Number

The origin of

$$N_{gen} = 3$$

remains unexplained.

HONEST STATUS

The conceptual architecture of TECT is largely closed.

However, the microscopic realization of the condensate dynamics and the precise emergence of the Standard Model gauge and matter content remain open.

The remaining frontier is finite and explicit:

- derive the microscopic condensate PDE,
- compute the enriched tensors N^a_{ij},
- verify shell valley doubling,
- determine hypercharge normalization,
- explain the generation number.

SUMMARY

The final conditional BCC Weyl Valley Theorem states:

If a BCC shell pair $\pm G_*$ exists in the microscopic TECT linearization, if the enriched microscopic tensor coefficients satisfy

$$\lambda_\parallel = \kappa_* |G_*|\, \hat{n}_i \hat{n}_j\, \mathsf{N}^3_{ij} \neq 0,$$

and

$$\alpha = \frac{\kappa_* |G_*|}{2} \left(e_{1j} \hat{n}_i\, \mathsf{N}^1_{ij} + e_{2j} \hat{n}_i\, \mathsf{N}^2_{ij} \right),$$

$$\beta = \frac{\kappa_* |G_*|}{2} \left(e_{2j} \hat{n}_i\, \mathsf{N}^1_{ij} - e_{1j} \hat{n}_i\, \mathsf{N}^2_{ij} \right), \qquad (\alpha, \beta) \neq (0, 0),$$

then the local shell crossing is Weyl because

$$\det M = \lambda_\parallel (\alpha^2 + \beta^2) \neq 0.$$

If, in addition, the opposite-valley relation

$$h_-(p) = -h_+(-p)$$

holds, then the doubled low-energy theory is Dirac.

Finally, if the defect sector has positive chiral index

$$\mathrm{Index}(D_+) = I > 0$$

and all vectorlike mirrors are lifted, then the surviving infrared fermion content is a protected chiral Dirac sector of multiplicity I.

Therefore the remaining frontier is finite and explicit: derive the true microscopic TECT operator, compute N^a_{ij}, and verify shell-pair valley doubling.

LXVII. BCC SHELL VALLEY EXISTENCE THEOREM

A. Aim and honest scope

We now formulate the strongest rigorously provable statement currently available for the fermion-emergence sector of TECT.

The goal is not yet to derive the full valley spectrum from a completely explicit microscopic TECT PDE with all coefficients computed ab initio. That stronger result remains open.

What we prove here is the following conditional-but-rigorous theorem:

If the microscopic condensate dynamics admits a BCC-condensed background, if the linearized Bloch operator near a reciprocal-shell point $\pm G_*$ possesses a two-band crossing isolated from the rest of the spectrum, and if the linear linearized coefficient matrix is nondegenerate, then the effective Hamiltonian near $\pm G_*$ is necessarily of Weyl type.

This is the precise mathematical closure behind the previously stated "BCC Weyl Valley Theorem."

B. Microscopic condensate equation and BCC background

Let the condensate field be a complex vector order parameter

$$\Phi(x,t) \in \mathbb{C}^N, \tag{1}$$

with $N \geq 2$ sufficient for a two-band crossing sector. Assume the microscopic dynamics is generated by a local translation-invariant PDE of the form

$$i\partial_t \Phi = \mathcal{L}(-i\nabla, \Phi)\Phi, \tag{2}$$

or, for dissipative/gradient-flow dynamics, by an equivalent linearized spectral problem around a static background. The only structure used below is the linearization around a stationary BCC-condensed state.

Let $\Phi_0(x)$ be a static BCC background with reciprocal support on the BCC star

$$S_{BCC} = \frac{q_*}{\sqrt{2}} \left\{(\pm1,\pm1,0),\ (\pm1,0,\pm1),\ (0,\pm1,\pm1)\right\}, \qquad |S_{BCC}| = 12. \tag{3}$$

We write

$$\Phi(x,t) = \Phi_0(x) + \delta\Phi(x,t), \tag{4}$$

and linearize:

$$i\partial_t \delta\Phi = \mathbb{L}_{\Phi_0}\, \delta\Phi. \tag{5}$$

Because Φ_0 is periodic with BCC reciprocal structure, the linearized operator $\mathbb{L}_{\Phi_0}$ is periodic and therefore admits Bloch decomposition.

C. Bloch reduction near a reciprocal-shell point

Fix one shell point

$$G_* \in S_{BCC}. \tag{6}$$

A fluctuation mode near this point is parameterized as

$$k = G_* + p, \qquad |p| \ll |G_*|. \tag{7}$$

By Bloch-Floquet theory, the spectral problem reduces locally to

$$\mathbb{L}_{\Phi_0}(G_* + p)\, u_{n,p} = E_n(p)\, u_{n,p}. \tag{8}$$

Assume that at $p = 0$ there exists an isolated two-band crossing:

$$E_1(0) = E_2(0) = E_*, \tag{9}$$

and that all other bands are separated by a nonzero gap:

$$\inf_{m\geq 3} |E_m(0) - E_*| > 0. \tag{10}$$

Then Kato perturbation theory allows one to project the full Bloch operator onto the two-dimensional crossing subspace. Hence there exists a smooth rank-two spectral projector $P(p)$ for sufficiently small p, and the effective Hamiltonian is

$$h(p) := P(p)\, \mathbb{L}_{\Phi_0}(G_* + p)\, P(p). \tag{11}$$

At $p = 0$, $h(0) = E_*\, \mathbf{1}_2$.

D. General Hermitian form of the two-band effective Hamiltonian

Since $h(p)$ acts on a two-dimensional complex space and is Hermitian, it has the unique Pauli decomposition

$$h(p) = \varepsilon_0(p)\,\mathbf{1}_2 + \sum_{a=1}^{3} d_a(p)\sigma^a. \tag{12}$$

Because $p = 0$ is a band-touching point, we have

$$d_a(0) = 0, \qquad a = 1, 2, 3. \tag{13}$$

Taylor expansion gives

$$d_a(p) = \sum_{i=1}^{3} M_{ai} p_i + \mathcal{O}(|p|^2), \tag{14}$$

where

$$M_{ai} := \left.\frac{\partial d_a}{\partial p_i}\right|_{p=0}. \tag{15}$$

Therefore

$$h(p) = E_*\,\mathbf{1}_2 + \sum_{a,i} M_{ai} p_i\,\sigma^a + \mathcal{O}(|p|^2). \tag{16}$$

E. Weyl criterion

[Local Weyl criterion] Let

$$h(p) = E_*\,\mathbf{1}_2 + \sum_{a,i} M_{ai} p_i\,\sigma^a + \mathcal{O}(|p|^2) \tag{17}$$

be the effective two-band Hamiltonian near a band-touching point. If

$$\det M \neq 0, \tag{18}$$

then the touching point is a Weyl point.

The eigenvalues are

$$E_\pm(p) = E_* + \varepsilon_0(p) \pm |d(p)|, \qquad d(p) := (d_1(p), d_2(p), d_3(p)). \tag{19}$$

To first order,

$$d(p) = Mp + \mathcal{O}(|p|^2). \tag{20}$$

If $\det M \neq 0$, then the linear map $p \mapsto Mp$ is invertible. Hence there exists $c > 0$ such that for sufficiently small p,

$$|Mp| \geq c|p|. \tag{21}$$

Therefore

$$E_+(p) - E_-(p) = 2|d(p)| \geq c|p| + \mathcal{O}(|p|^2), \tag{22}$$

so the touching is isolated and linear in all three momentum directions. This is precisely a Weyl cone.

F. Adapted decomposition at a BCC shell point

Choose an orthonormal frame adapted to G_*:

$$\hat{n} := \frac{G_*}{|G_*|}, \qquad \{e_1, e_2\} \subset \hat{n}^\perp, \qquad \{\hat{n}, e_1, e_2\} \text{ orthonormal.} \tag{23}$$

Decompose

$$p = p_\parallel \hat{n} + p_1 e_1 + p_2 e_2. \tag{24}$$

Then the linearized Hamiltonian can always be rewritten as

$$h(p) = E_* \, \mathbf{1}_2 + p_\parallel V_\parallel + p_1 V_1 + p_2 V_2 + \mathcal{O}(|p|^2), \tag{25}$$

with Hermitian 2×2 matrices $V_\parallel, V_1, V_2$. After subtraction of the trace part and a unitary basis change in the two-band subspace, one may choose the Pauli basis so that

$$h(p) = E_* \, \mathbf{1}_2 + \lambda_\parallel p_\parallel \sigma^3 + (\alpha p_1 - \beta p_2)\sigma^1 + (\beta p_1 + \alpha p_2)\sigma^2 + \mathcal{O}(|p|^2). \tag{26}$$

Equivalently,

$$d(p) = \begin{pmatrix} \alpha p_1 - \beta p_2 \\ \beta p_1 + \alpha p_2 \\ \lambda_\parallel p_\parallel \end{pmatrix} + \mathcal{O}(|p|^2). \tag{27}$$

Thus

$$M = \begin{pmatrix} 0 & \alpha & -\beta \\ 0 & \beta & \alpha \\ \lambda_\parallel & 0 & 0 \end{pmatrix}, \tag{28}$$

and therefore

$$\det M = \lambda_\parallel (\alpha^2 + \beta^2). \tag{29}$$

Hence the Weyl criterion becomes

$$\boxed{\lambda_\parallel (\alpha^2 + \beta^2) \neq 0.} \tag{30}$$

G. Why this form is natural at a shell point

At a shell point G_*, the distinguished direction is the radial direction $\hat{n}$. The tangent plane is two-dimensional, and to linear order the effective Hamiltonian must split into one longitudinal coefficient and one transverse complex coefficient. The pair

$$(\alpha, \beta) \tag{31}$$

is the real representation of the complex transverse slope

$$\alpha + i\beta. \tag{32}$$

Thus the BCC shell geometry naturally organizes the linear crossing into

$$1 \text{ longitudinal slope} + 2 \text{ transverse slopes}, \tag{33}$$

which is exactly the structure required for a three-dimensional Weyl cone.

H. Topological charge of the Weyl point

Define the normalized vector

$$\hat{d}(p) := \frac{d(p)}{|d(p)|} \tag{34}$$

on a small sphere S^2_ϵ around $p = 0$. The Weyl topological charge is

$$\chi = \frac{1}{4\pi} \int_{S^2_\epsilon} \hat{d} \cdot \left(\partial_\theta \hat{d} \times \partial_\phi \hat{d} \right) \, d\theta \, d\phi. \tag{35}$$

For the linear model $d(p) = Mp$, this degree is

$$\chi = \operatorname{sgn}(\det M). \tag{36}$$

Hence if $\det M \neq 0$, the crossing is topologically protected: it cannot be removed by a sufficiently small smooth perturbation unless it annihilates with another Weyl point of opposite charge.

I. Opposite shell point and valley pairing

Consider the opposite shell point

$$-G_* \in S_{BCC}. \tag{37}$$

Suppose the linearized Bloch operator satisfies the valley-pairing relation

$$h_-(p) = -h_+(-p) \tag{38}$$

after subtraction of the common scalar part. Then the opposite point carries the opposite chirality:

$$\chi_- = -\chi_+. \tag{39}$$

The doubled Hamiltonian is

$$H(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix}. \tag{40}$$

To linear order this is unitarily equivalent to a massless Dirac Hamiltonian. Thus opposite BCC shell valleys produce Dirac doubling.

J. Main theorem

[BCC Shell Valley Existence Theorem] Assume:

1. A microscopic TECT condensate PDE admits a periodic BCC-condensed background Φ_0.
2. The corresponding Bloch-linearized operator $\mathbb{L}_{\Phi_0}(k)$ has, at some shell point $G_* \in S_{BCC}$, an isolated two-band crossing separated by a spectral gap from all other bands.
3. The projected two-band Hamiltonian is Hermitian and smooth in $p = k - G_*$.
4. The linear coefficient matrix M of the Pauli-vector part satisfies

   $$\det M \neq 0. \tag{41}$$

   Equivalently, in the adapted shell basis,

   $$\lambda_\parallel(\alpha^2 + \beta^2) \neq 0. \tag{42}$$

Then G_* is a Weyl valley of the BCC shell spectrum. Its local effective Hamiltonian is

$$h(p) = E_* \mathbf{1}_2 + \lambda_\parallel p_\parallel \sigma^3 + (\alpha p_1 - \beta p_2)\sigma^1 + (\beta p_1 + \alpha p_2)\sigma^2 + \mathcal{O}(|p|^2), \tag{43}$$

and its chirality is

$$\chi = \operatorname{sgn}\big(\lambda_\parallel(\alpha^2 + \beta^2)\big). \tag{44}$$

If, in addition, the opposite shell point $-G_*$ satisfies the valley-pairing relation

$$h_-(p) = -h_+(-p), \tag{45}$$

then the two Weyl valleys combine into a Dirac fermion.

Items (1)–(3) guarantee the existence of the local rank-two projected Hamiltonian. Item (4) implies, by the Local Weyl Criterion, that the crossing is isolated and linear, hence a Weyl point. The chirality formula follows from the degree of the map $\hat{d} : S^2 \to S^2$. The final Dirac statement follows by block doubling under the valley-pairing relation.

K. What is proven and what remains open

The theorem above is mathematically closed once assumptions (1)–(4) hold. Therefore the remaining open problem is now sharply localized:

$$\boxed{\text{Derive the two-band crossing and compute } M \text{ directly from the full microscopic TECT PDE.}} \tag{46}$$

Equivalently, the true remaining microscopic frontier is:

1. derive the explicit condensate PDE,
2. compute the Bloch operator around the BCC background,
3. prove existence of isolated crossings at $\pm G_*$,
4. compute $\lambda_\parallel, \alpha, \beta$ and verify

$$\lambda_\parallel(\alpha^2 + \beta^2) \neq 0. \tag{47}$$

Once these are completed, the fermion-emergence chain

$$\text{BCC condensate} \implies \text{Weyl valley} \implies \text{Dirac fermion} \tag{48}$$

is fully microscopic.

LXVIII. FINAL SUMMARY

We have rigorously reduced the BCC-shell fermion problem to a local two-band spectral theorem.

Given a periodic BCC-condensed background, Bloch reduction near a shell point G_* yields a projected 2×2 Hermitian effective Hamiltonian

$$h(p) = E_* \mathbf{1}_2 + \sum_{a,i} M_{ai} p_i \sigma^a + \mathcal{O}(|p|^2).$$

If

$$\det M \neq 0,$$

then the crossing at G_* is necessarily a Weyl point. In the shell-adapted basis this criterion becomes

$$\lambda_\parallel(\alpha^2 + \beta^2) \neq 0.$$

The Weyl topological charge is

$$\chi = \mathrm{sgn}(\det M).$$

If the opposite shell point $-G_*$ satisfies

$$h_-(p) = -h_+(-p),$$

then the two Weyl valleys combine into a Dirac fermion.

Therefore the remaining open frontier is no longer conceptual. It is the explicit microscopic computation of the Bloch crossing and the coefficient matrix M from the full TECT condensate PDE.

LXIX. MICROSCOPIC TECT CONDENSATE PDE AND BLOCH OPERATOR AROUND THE BCC BACKGROUND

A. Aim and scope

We now derive the strongest mathematically controlled microscopic formulation currently available for the condensate sector of TECT.

The aim of this section is fourfold:

1. to write the minimal translation-invariant and rotation-invariant condensate PDE capable of supporting a finite-shell instability,
2. to construct a stationary BCC-condensed background,
3. to compute the exact linearized operator around that background,
4. to reduce the local valley problem near a shell point G_* to a finite-dimensional Bloch matrix problem.

This does not yet prove the existence of a two-band crossing. What it does prove is that, once the condensate PDE is specified, the remaining Weyl/Dirac problem is an explicit spectral problem for a periodic linear operator.

B. Minimal microscopic condensate functional

Let the condensate order parameter be

$$\Phi(x,t) \in \mathbb{C}^N, \qquad x \in \mathbb{R}^3, \tag{49}$$

with $N \geq 1$ at the condensate level, and $N \geq 2$ required later for a nontrivial two-band crossing sector.

We impose the following minimal microscopic principles:

1. translation invariance,
2. spatial isotropy before condensation,
3. local polynomial interactions,
4. quartic stabilization,
5. finite-wavevector instability.

The minimal rotationally invariant free-energy functional with these properties is

$$\mathcal{F}[\Phi] = \int d^3x \left[r\,\Phi^\dagger\Phi + \kappa\,\partial_i\Phi^\dagger\partial_i\Phi + \lambda\,(\nabla^2\Phi)^\dagger(\nabla^2\Phi) + \frac{u}{2}(\Phi^\dagger\Phi)^2 \right], \tag{50}$$

with

$$\lambda > 0, \qquad u > 0. \tag{51}$$

The Euler-Lagrange equation for static configurations is

$$\frac{\delta\mathcal{F}}{\delta\Phi^\dagger} = r\Phi - \kappa\nabla^2\Phi + \lambda\nabla^4\Phi + u(\Phi^\dagger\Phi)\Phi = 0. \tag{52}$$

If one wishes to use conservative dynamics, the corresponding microscopic PDE is

$$i\partial_t\Phi = \frac{\delta\mathcal{F}}{\delta\Phi^\dagger} = r\Phi - \kappa\nabla^2\Phi + \lambda\nabla^4\Phi + u(\Phi^\dagger\Phi)\Phi. \tag{53}$$

If one prefers dissipative relaxation, one may instead use

$$\partial_t\Phi = -\frac{\delta\mathcal{F}}{\delta\Phi^\dagger}. \tag{54}$$

All spectral statements below depend only on the linearization of the right-hand side, so the valley analysis applies to either choice.

C. Finite-shell instability

For the homogeneous state $\Phi = 0$, the linear dispersion is obtained by substituting

$$\Phi(x,t) \sim e^{iq\cdot x} \tag{55}$$

into the linear part of (53). One finds

$$\omega(q) = r + \kappa|q|^2 + \lambda|q|^4. \tag{56}$$

If

$$\kappa < 0, \qquad \lambda > 0, \tag{57}$$

then the quadratic-plus-quartic polynomial has its minimum on the sphere

$$|q| = q_*, \qquad q_*^2 = \frac{-\kappa}{2\lambda}. \tag{58}$$

This is the standard finite-shell Brazovskii instability. Hence the microscopic PDE naturally selects modes on a shell.

D. BCC equal-amplitude ansatz

We now restrict to the first BCC reciprocal star

$$S_{BCC} = \frac{q_*}{\sqrt{2}}\left\{(\pm1,\pm1,0),\, (\pm1,0,\pm1),\, (0,\pm1,\pm1)\right\}. \tag{59}$$

Choose one representative from each antipodal pair:

$$Q_1 = \frac{q_*}{\sqrt{2}}(1,1,0), \quad Q_2 = \frac{q_*}{\sqrt{2}}(1,-1,0), \quad Q_3 = \frac{q_*}{\sqrt{2}}(1,0,1), \tag{60}$$

$$Q_4 = \frac{q_*}{\sqrt{2}}(1,0,-1), \quad Q_5 = \frac{q_*}{\sqrt{2}}(0,1,1), \quad Q_6 = \frac{q_*}{\sqrt{2}}(0,1,-1). \tag{61}$$

Then

$$S_{BCC} = \{\pm Q_a\}_{a=1}^6. \tag{62}$$

Assume a common internal polarization vector $\zeta \in \mathbb{C}^N$ with

$$\zeta^\dagger\zeta = 1, \tag{63}$$

and consider the equal-amplitude standing-wave ansatz

$$\Phi_0(x) = A\sum_{a=1}^6 \left(e^{iQ_a\cdot x} + e^{-iQ_a\cdot x}\right)\zeta = 2A\sum_{a=1}^6 \cos(Q_a\cdot x)\,\zeta. \tag{64}$$

This is the minimal BCC condensate background. More general internal-polarization patterns may be considered later, but (64) is the canonical symmetric starting point.

E. Stationary amplitude equation

Insert (64) into the static equation (52). Because each Q_a lies on the shell (348),

$$\left(r - \kappa\nabla^2 + \lambda\nabla^4\right) e^{\pm iQ_a\cdot x} = \left(r + \kappa q_*^2 + \lambda q_*^4\right) e^{\pm iQ_a\cdot x}. \tag{65}$$

Using (348),

$$r + \kappa q_*^2 + \lambda q_*^4 = r - \frac{\kappa^2}{4\lambda}. \tag{66}$$

The quartic term generates cubic products of plane waves. Projecting back onto the fundamental BCC shell gives the amplitude equation

$$\left(r - \frac{\kappa^2}{4\lambda}\right) A + u\,\beta_{BCC}\,A^3 = 0, \tag{67}$$

where $\beta_{BCC} > 0$ is the shell-projected quartic coefficient. Equivalently,

$$A^2 = -\frac{1}{u\beta_{BCC}}\left(r - \frac{\kappa^2}{4\lambda}\right), \tag{68}$$

whenever

$$r < \frac{\kappa^2}{4\lambda}. \tag{69}$$

Thus the BCC-condensed background exists once the shell instability is strong enough and the quartic projection coefficient is positive.

F. Exact first variation and Hessian

Let

$$\Phi = \Phi_0 + \eta. \tag{70}$$

Define the nonlinear operator

$$\mathcal{N}(\Phi) = r\Phi - \kappa\nabla^2\Phi + \lambda\nabla^4\Phi + u(\Phi^\dagger\Phi)\Phi. \tag{71}$$

Then the linearization around Φ_0 is

$$\delta\mathcal{N}_{\Phi_0}[\eta] = r\eta - \kappa\nabla^2\eta + \lambda\nabla^4\eta + u\left[(\Phi_0^\dagger\Phi_0)\eta + (\Phi_0^\dagger\eta)\Phi_0 + (\eta^\dagger\Phi_0)\Phi_0\right]. \tag{72}$$

This formula is exact. It shows that the linearized operator is periodic because $\Phi_0(x)$ is periodic.

If one writes the conservative dynamics (53), the fluctuation equation is

$$i\partial_t\eta = \mathbb{L}_{\Phi_0}\eta + \mathbb{M}_{\Phi_0}\bar{\eta}, \tag{73}$$

where

$$\mathbb{L}_{\Phi_0}\eta = r\eta - \kappa\nabla^2\eta + \lambda\nabla^4\eta + u\left[(\Phi_0^\dagger\Phi_0)\eta + (\Phi_0^\dagger\eta)\Phi_0\right], \tag{74}$$

and

$$\mathbb{M}_{\Phi_0}\bar{\eta} = u(\eta^\dagger\Phi_0)\Phi_0. \tag{75}$$

Hence the full fluctuation problem is of Bogoliubov-de Gennes type.

If one instead freezes the complex-conjugate mixing sector or works in a real-field reduction, the local valley question reduces to the Hermitian Bloch operator induced by the Hessian. The existence of Weyl-type crossings is then analyzed in the projected band structure.

G. Period lattice and reciprocal lattice

The BCC reciprocal star determines a direct-space periodicity. Let Λ denote the direct Bravais lattice for which the vectors Q_a are reciprocal. Then

$$\Phi_0(x+R) = \Phi_0(x), \qquad R \in \Lambda. \tag{76}$$

Therefore the coefficients

$$\Phi_0^\dagger \Phi_0, \qquad \Phi_0 \Phi_0^\dagger \tag{77}$$

are Λ-periodic, and the linearized operator is a periodic differential operator.

H. Bloch-Floquet reduction

Because the coefficients are periodic, Bloch-Floquet theory applies. For each quasimomentum k in the Brillouin zone, write

$$\eta(x) = e^{ik\cdot x} u_k(x), \qquad u_k(x+R) = u_k(x). \tag{78}$$

Substituting into the linearized problem yields the Bloch operator

$$\mathbb{H}(k) = e^{-ik\cdot x}\, \mathbb{H}\, e^{ik\cdot x}, \tag{79}$$

acting on periodic functions u_k. For the scalar differential part,

$$-\nabla^2 \mapsto -(\nabla + ik)^2, \qquad \nabla^4 \mapsto (\nabla + ik)^4. \tag{80}$$

Hence

$$\mathbb{H}_0(k) = r - \kappa(\nabla + ik)^2 + \lambda(\nabla + ik)^4. \tag{81}$$

The periodic condensate background contributes multiplicative and rank-one matrix-valued periodic terms.

Thus the full Bloch operator has the structure

$$\mathbb{H}(k) = r - \kappa(\nabla + ik)^2 + \lambda(\nabla + ik)^4 + u\, V_1(x) + u\, V_2(x), \tag{82}$$

where $V_1(x)$ and $V_2(x)$ are Λ-periodic operators built from Φ_0.

I. Fourier basis on the periodic cell

Expand periodic Bloch functions in reciprocal lattice harmonics:

$$u_k(x) = \sum_{G\in\Lambda^*} c_G\, e^{iG\cdot x}, \tag{83}$$

where Λ^* is the reciprocal lattice of Λ. Then

$$\eta(x) = \sum_{G\in\Lambda^*} c_G\, e^{i(k+G)\cdot x}. \tag{84}$$

In this basis, the free part is diagonal:

$$[\mathbb{H}_0(k)]_{G,G'} = \delta_{G,G'} \left[r + \kappa|k+G|^2 + \lambda|k+G|^4 \right]. \tag{85}$$

The condensate background induces off-diagonal couplings

$$[\mathbb{V}(k)]_{G,G'} = u\, \widehat{V}(G - G'), \tag{86}$$

where $\widehat{V}$ denotes the Fourier coefficients of the periodic background-generated potential.

Therefore

$$\mathbb{H}(k)_{G,G'} = \delta_{G,G'} \left[r + \kappa|k+G|^2 + \lambda|k+G|^4 \right] + u\, \widehat{V}(G - G'). \tag{87}$$

This is the exact infinite matrix form of the Bloch operator.

J. Local shell-point reduction

Fix a shell vector

$$G_* \in S_{BCC}. \tag{88}$$

We study quasimomentum near that point:

$$k = G_* + p, \qquad |p| \ll |G_*|. \tag{89}$$

The diagonal energies in (87) are then smallest for reciprocal harmonics G such that $|G_* + G|$ lies near the critical shell. Hence the near-degenerate subspace is finite-dimensional to leading order: it is spanned by the small set of harmonics connected to G_* by the dominant Fourier components of the BCC background.

Let

$$\mathcal{S}_{\mathrm{loc}}(G_*) = \left\{ G \in \Lambda^* \ : \ |G_* + G|^2 = q_*^2 + \mathcal{O}(|p|) \right\} \tag{90}$$

be the local resonant harmonic set. Projecting (87) onto this finite set yields the local Bloch matrix

$$H_{\mathrm{loc}}(G_*; p) = P_{\mathrm{loc}} \, \mathbb{H}(G_* + p) \, P_{\mathrm{loc}}. \tag{91}$$

This matrix is finite-dimensional. Its entries are explicit polynomials/rational expressions in

$$r, \kappa, \lambda, u, A, \tag{92}$$

together with the BCC Fourier-coupling coefficients determined by the set $\{Q_a\}$.

K. Reduction of the valley problem

The key point is now transparent.

The existence of a valley near G_* is equivalent to the existence of an isolated two-band crossing in the finite matrix (91). More precisely:

$$\boxed{\text{Valley existence near } G_* \iff \text{a two-band crossing occurs in } H_{\mathrm{loc}}(G_*; p) \text{ at } p = 0.} \tag{93}$$

If such a crossing exists and is separated from the rest of the local spectrum, then Kato projection yields a 2×2 effective Hamiltonian

$$h(p) = E_* \mathbf{1}_2 + \sum_{a,i} M_{ai} p_i \sigma^a + \mathcal{O}(|p|^2), \tag{94}$$

and the Weyl criterion reduces to

$$\det M \neq 0. \tag{95}$$

Therefore the remaining open problem is no longer a vague field-theoretic issue. It is an explicit finite-dimensional spectral-combinatorial computation.

L. What is mathematically closed at this stage

We summarize what has been rigorously achieved.

[Microscopic reduction theorem] Assume the condensate sector of TECT is governed by the minimal isotropic finite-shell PDE (53), and assume the BCC equal-amplitude background (64) solves the projected amplitude equation (67).

Then:

1. the linearized fluctuation operator around Φ_0 is periodic;

2. it admits Bloch-Floquet decomposition over the Brillouin zone;
3. in reciprocal-harmonic basis, it is represented by the exact infinite matrix (87);
4. near any shell point $G_* \in S_{BCC}$, the local low-energy sector reduces to the finite-dimensional matrix problem (91);
5. the Weyl/Dirac emergence problem is therefore reduced to proving an isolated two-band crossing and verifying the nondegeneracy condition

$$\det M \neq 0 \tag{96}$$

for the corresponding 2×2 projected Hamiltonian.

Items (1) and (2) follow from periodicity of Φ_0 and standard Bloch-Floquet theory. Item (3) is obtained by Fourier expansion on the periodic cell. Item (4) follows by restricting to the finite set of harmonics that are resonantly coupled near G_*. Item (5) follows from the general two-band Weyl criterion already established.

M. True remaining microscopic tasks

The remaining tasks are now sharply identified:

1. determine the physically correct internal dimension N and internal couplings of the condensate field;
2. compute the explicit BCC background Fourier coefficients and the induced coupling matrix $\widehat{V}(G - G')$;
3. construct the concrete local matrix $H_{\mathrm{loc}}(G_*; p)$ for a chosen shell point G_*;
4. prove that this local matrix possesses an isolated two-band crossing at $p = 0$;
5. compute the linear coefficient matrix M_{ai} and verify

$$\det M \neq 0. \tag{97}$$

Once these are done, the BCC-shell Weyl valley and Dirac doubling become fully microscopic.

LXX. FINAL SUMMARY

We introduced the minimal microscopic TECT condensate PDE

$$i\partial_t \Phi = r\Phi - \kappa\nabla^2\Phi + \lambda\nabla^4\Phi + u(\Phi^\dagger\Phi)\Phi,$$

whose linear dispersion develops a finite-shell minimum at

$$|q| = q_*, \qquad q_*^2 = -\frac{\kappa}{2\lambda},$$

when $\kappa < 0$ and $\lambda > 0$.

Using the first BCC reciprocal star, we constructed the equal-amplitude background

$$\Phi_0(x) = 2A \sum_{a=1}^{6} \cos(Q_a \cdot x)\,\zeta,$$

and derived the corresponding projected amplitude equation.

We then computed the exact linearization around Φ_0, obtaining a periodic Hessian/Bogoliubov operator. By Bloch-Floquet theory, this operator decomposes into Bloch sectors. In reciprocal-harmonic basis, each Bloch sector is represented by the exact matrix

$$\mathbb{H}(k)_{G,G'} = \delta_{G,G'}\left[r + \kappa|k+G|^2 + \lambda|k+G|^4\right] + u\,\widehat{V}(G - G').$$

Near a shell point $G_* \in S_{BCC}$, the low-energy problem reduces to a finite-dimensional local matrix

$$H_{\text{loc}}(G_*; p).$$

Therefore the Weyl/Dirac emergence problem has been reduced to an explicit finite spectral problem: one must prove an isolated two-band crossing of H_{loc} at $p = 0$ and verify

$$\det M \neq 0$$

for the corresponding projected 2×2 effective Hamiltonian.

Hence the remaining microscopic frontier is no longer conceptual. It is the explicit construction and analysis of the local Bloch matrix near a BCC shell point.

LXXI. EXPLICIT LOCAL BLOCH MATRIX AT A BCC SHELL POINT

Fix the shell vector

$$G_* = \frac{q_*}{\sqrt{2}}(1, 1, 0). \tag{98}$$

Near this point write

$$k = G_* + p. \tag{99}$$

The minimal resonant harmonic set is

$$\mathcal{S}_{\text{loc}} = \{0, -Q_1, -Q_2\} \tag{100}$$

with

$$Q_1 = \frac{q_*}{\sqrt{2}}(1, 1, 0), \quad Q_2 = \frac{q_*}{\sqrt{2}}(1, -1, 0). \tag{101}$$

The corresponding Bloch basis is

$$\psi_1 = e^{i(G_*+p)\cdot x}, \tag{102}$$

$$\psi_2 = e^{i(G_*-Q_1+p)\cdot x}, \tag{103}$$

$$\psi_3 = e^{i(G_*-Q_2+p)\cdot x}. \tag{104}$$

The free dispersion is

$$E(k) = r + \kappa|k|^2 + \lambda|k|^4. \tag{105}$$

The condensate background generates couplings

$$H_{12} = V, \quad H_{13} = V, \quad H_{23} = 0 \tag{106}$$

with

$$V = uA. \tag{107}$$

Thus the local Bloch matrix is

$$H_{\text{loc}} = \begin{pmatrix} E_1(p) & V & V \\ V & E_2(p) & 0 \\ V & 0 & E_3(p) \end{pmatrix}. \tag{108}$$

LXXII. EXACT SPECTRAL ANALYSIS OF THE MINIMAL 3×3 LOCAL MATRIX

A. Statement of the correction

We correct an overstatement made in the previous step.
Consider the minimal local matrix

$$H_{\mathrm{loc}}(p) = \begin{pmatrix} E_1(p) & V & V \\ V & E_2(p) & 0 \\ V & 0 & E_3(p) \end{pmatrix}, \qquad V \in \mathbb{R}. \tag{109}$$

It was previously suggested that, for $V \neq 0$, this matrix generically produces a two-band crossing. That statement is false.

The correct statement is:

For generic values of E_1, E_2, E_3 and $V \neq 0$, the matrix (109) produces level repulsion rather than a protected crossing.

Therefore this minimal scalar harmonic mixing model is *not* sufficient by itself to prove a Weyl valley.

B. Characteristic polynomial

Let

$$\lambda \tag{110}$$

be an eigenvalue of (109). Then

$$\det(H_{\mathrm{loc}} - \lambda I) = 0. \tag{111}$$

A direct computation gives

$$\det(H_{\mathrm{loc}} - \lambda I) = (E_1 - \lambda)(E_2 - \lambda)(E_3 - \lambda) - V^2\big[(E_2 - \lambda) + (E_3 - \lambda)\big]. \tag{112}$$

Equivalently,

$$P(\lambda) = (E_1 - \lambda)(E_2 - \lambda)(E_3 - \lambda) - V^2(E_2 + E_3 - 2\lambda). \tag{113}$$

A band degeneracy occurs only when $P(\lambda)$ has a repeated root, i.e.

$$P(\lambda_*) = 0, \qquad P'(\lambda_*) = 0. \tag{114}$$

This already shows that a degeneracy is a codimension-two condition in the space of parameters (E_1, E_2, E_3, V). Hence it is *not* generic.

C. Explicit repeated-root condition

Differentiating (112) yields

$$P'(\lambda) = -(E_2 - \lambda)(E_3 - \lambda) - (E_1 - \lambda)(E_3 - \lambda) - (E_1 - \lambda)(E_2 - \lambda) + 2V^2. \tag{115}$$

Thus a degeneracy requires the simultaneous system

$$(E_1 - \lambda_*)(E_2 - \lambda_*)(E_3 - \lambda_*) - V^2(E_2 + E_3 - 2\lambda_*) = 0, \tag{116}$$

$$(E_2 - \lambda_*)(E_3 - \lambda_*) + (E_1 - \lambda_*)(E_3 - \lambda_*) + (E_1 - \lambda_*)(E_2 - \lambda_*) = 2V^2. \tag{117}$$

These are fine-tuning conditions. They are not automatically satisfied by the presence of BCC shell mixing.

D. Symmetric special case $E_2 = E_3$

To make the conclusion transparent, consider the special case

$$E_2 = E_3 =: B, \qquad E_1 =: A. \tag{118}$$

Then

$$H_{\rm loc} = \begin{pmatrix} A & V & V \\ V & B & 0 \\ V & 0 & B \end{pmatrix}. \tag{119}$$

Introduce the orthonormal basis

$$u_1 = (1,0,0), \qquad u_2 = \frac{1}{\sqrt{2}}(0,1,1), \qquad u_3 = \frac{1}{\sqrt{2}}(0,1,-1). \tag{120}$$

In this basis,

$$H_{\rm loc} \sim \begin{pmatrix} A & \sqrt{2}V & 0 \\ \sqrt{2}V & B & 0 \\ 0 & 0 & B \end{pmatrix}. \tag{121}$$

Therefore one eigenvalue is exactly

$$\lambda_0 = B, \tag{122}$$

with eigenvector

$$u_3 = \frac{1}{\sqrt{2}}(0,1,-1). \tag{123}$$

The remaining two eigenvalues are

$$\lambda_\pm = \frac{A+B}{2} \pm \frac{1}{2}\sqrt{(A-B)^2 + 8V^2}. \tag{124}$$

Now if $V \neq 0$, then

$$\sqrt{(A-B)^2 + 8V^2} > |A-B|, \tag{125}$$

so

$$\lambda_+ \neq \lambda_-, \qquad \lambda_\pm \neq B \tag{126}$$

unless the coupling vanishes. Indeed, demanding $B = \lambda_\pm$ implies

$$2B = A + B \pm \sqrt{(A-B)^2 + 8V^2}, \tag{127}$$

hence

$$B - A = \pm\sqrt{(A-B)^2 + 8V^2}, \tag{128}$$

which is impossible for $V \neq 0$. Therefore:

$$\boxed{E_2 = E_3 \text{ and } V \neq 0 \quad \Longrightarrow \quad \text{no degeneracy.}} \tag{129}$$

Thus even in the most symmetric special case, the coupling opens a gap rather than producing a Weyl crossing.

E. Why the 3×3 scalar model fails to produce Weyl

A Weyl point requires a protected two-band crossing whose effective Hamiltonian is linear in three momentum directions:

$$h(p) = E_* \mathbf{1}_2 + \sum_{a,i} M_{ai} p_i \sigma^a + \mathcal{O}(|p|^2), \qquad \det M \neq 0. \tag{130}$$

The minimal scalar matrix (109) does not naturally provide such a protected two-dimensional crossing subspace. Instead, the off-diagonal couplings generally hybridize the modes and produce level repulsion.

Therefore the scalar shell-mixing sector is useful for constructing the periodic background and the Bloch operator, but it does not by itself generate Weyl fermions.

F. Consequences for the TECT fermion program

The conclusion is mathematically sharp.

1. The BCC condensate PDE and its periodic Bloch operator are still valid.
2. The local scalar harmonic block (109) is not sufficient to prove a Weyl valley.
3. Hence the Weyl/Dirac sector must come from an additional internal multi-component structure, not from scalar shell mixing alone.
4. In TECT language, this means that the relevant two-band crossing must arise in an internal orientational sector, e.g. a projected multi-component sector tied to the $\mathbb{C}^3/\mathbb{CP}^2$ structure, together with a symmetry-protected local crossing.
5. Only after such a two-band sector has been isolated does the previously derived Weyl theorem apply.

G. Corrected microscopic frontier

The true remaining microscopic problem is therefore the following.

$$\boxed{\text{Construct the internal multi-component Bloch block near } G_* \text{ and prove it contains a protected two-band crossing.}} \tag{131}$$

More explicitly, one must:

1. determine the correct internal condensate field with $N \geq 2$ (and in TECT plausibly $N = 3$),
2. derive the linearized periodic Bloch operator including the internal orientational couplings,
3. identify a symmetry-invariant two-dimensional band subspace near a shell point G_*,
4. prove that this subspace has an isolated band touching at $p = 0$,
5. compute the linear coefficient matrix M_{ai} and verify

$$\det M \neq 0. \tag{132}$$

Only then does one obtain a genuine Weyl valley, and only then can opposite-valley pairing yield a Dirac fermion.

H. The corrected theorem

[Negative result for the minimal scalar 3×3 block] Consider the local matrix

$$H_{\text{loc}} = \begin{pmatrix} E_1 & V & V \\ V & E_2 & 0 \\ V & 0 & E_3 \end{pmatrix}. \tag{133}$$

Then a two-band degeneracy occurs only when the repeated-root conditions (116)–(117) are satisfied. In particular, such a degeneracy is not generic. Moreover, in the symmetric case $E_2 = E_3$, any nonzero coupling $V \neq 0$ produces level repulsion and removes the degeneracy. Therefore this minimal scalar harmonic-mixing model cannot by itself serve as a proof of a Weyl valley.

The claim follows directly from the characteristic polynomial (112), the repeated-root criterion (114), and the explicit diagonalization (121)–(124) in the symmetric case.

LXXIII. FINAL SUMMARY

We analyzed exactly the minimal local matrix

$$H_{\text{loc}} = \begin{pmatrix} E_1 & V & V \\ V & E_2 & 0 \\ V & 0 & E_3 \end{pmatrix}.$$

Its characteristic polynomial is

$$P(\lambda) = (E_1 - \lambda)(E_2 - \lambda)(E_3 - \lambda) - V^2(E_2 + E_3 - 2\lambda).$$

A two-band crossing requires a repeated root of P, hence the simultaneous conditions

$$P(\lambda_*) = 0, \qquad P'(\lambda_*) = 0.$$

Therefore the degeneracy is a fine-tuned condition and is not generic.

In the special symmetric case $E_2 = E_3 = B$, the matrix is explicitly diagonalized into

$$\begin{pmatrix} A & \sqrt{2}V & 0 \\ \sqrt{2}V & B & 0 \\ 0 & 0 & B \end{pmatrix},$$

with eigenvalues

$$B, \qquad \frac{A+B}{2} \pm \frac{1}{2}\sqrt{(A-B)^2 + 8V^2}.$$

For any $V \neq 0$, these are nondegenerate. Thus the coupling opens a gap rather than producing a Weyl crossing.

Hence the minimal scalar BCC shell-mixing block is insufficient for Weyl/Dirac emergence. The true microscopic frontier is to construct the internal multi-component Bloch block associated with the TECT orientational sector and to prove that it contains a symmetry-protected two-band crossing with

$$\det M \neq 0.$$

LXXIV. INTERNAL $\mathbb{C}^3$ BLOCH BLOCK AND THE MINIMAL PROTECTED TWO-BAND SECTOR

A. Aim

The previous scalar 3×3 harmonic-mixing model showed that pure shell mixing does not generically produce a Weyl point. Therefore the fermionic sector of TECT must arise from an additional internal multi-component structure.

The minimal candidate compatible with the already established TECT chain is

$$\Phi(x) \in \mathbb{C}^3, \tag{134}$$

which is also the natural precursor of the projective internal manifold

$$\mathbb{CP}^2 \simeq \frac{SU(3)}{U(2)}. \tag{135}$$

The purpose of this section is to construct the minimal local Bloch symbol acting on $\mathbb{C}^3$, identify the symmetry-adapted $1 \oplus 2$ decomposition near a BCC shell point, and show how a protected two-band crossing can arise in the doublet sector.

B. Microscopic linearization in an internal $\mathbb{C}^3$ sector

Let the condensate field be

$$\Phi(x,t) \in \mathbb{C}^3. \tag{136}$$

After linearization around a periodic BCC-condensed background, Bloch reduction near a shell point $G_* \in S_{BCC}$ yields a local Hermitian operator

$$H_{\rm int}(G_* + p) : \mathbb{C}^3 \to \mathbb{C}^3, \qquad |p| \ll |G_*|. \tag{137}$$

We assume analyticity in p:

$$H_{\rm int}(G_* + p) = H_0 + \sum_{i=1}^{3} p_i \, \Gamma_i + \mathcal{O}(|p|^2), \tag{138}$$

where

$$H_0^\dagger = H_0, \qquad \Gamma_i^\dagger = \Gamma_i. \tag{139}$$

At this stage we do not yet claim that the coefficients (H_0, Γ_i) have been computed ab initio from the full TECT PDE. Rather, we ask the mathematically sharper question:

What is the minimal symmetry structure on a $\mathbb{C}^3$ Bloch block that can support a protected two-band crossing?

C. Distinguished direction and little-group decomposition

Fix the shell point

$$G_* = \frac{q_*}{\sqrt{2}}(1,1,0), \qquad \hat{n} := \frac{G_*}{|G_*|}. \tag{140}$$

The point G_* breaks full rotational symmetry down to the little group preserving $\hat{n}$. At the level of the local internal block, the most natural decomposition is the splitting of $\mathbb{C}^3$ into a longitudinal line and a transverse plane:

$$\mathbb{C}^3 = \mathbb{C}\hat{n} \oplus \hat{n}^\perp_{\mathbb{C}}. \tag{141}$$

Choose an orthonormal basis

$$e_\parallel := \hat{n}, \qquad e_1, e_2 \in \hat{n}^\perp, \qquad \{e_\parallel, e_1, e_2\} \text{ orthonormal}. \tag{142}$$

Then the internal space decomposes as

$$\mathbf{3} \to \mathbf{1} \oplus \mathbf{2}, \tag{143}$$

where the singlet is the longitudinal mode and the doublet is the transverse internal sector.

This is the crucial structural point: the natural candidate for a two-band crossing is not the full $\mathbb{C}^3$ space, but the transverse doublet subspace.

D. Minimal symmetry constraint at the shell point

Assume that the shell point preserves an effective internal-transverse $SO(2)$ symmetry acting on the plane spanned by (e_1, e_2). Equivalently, in complex notation define

$$e_\pm := \frac{1}{\sqrt{2}}(e_1 \pm ie_2). \tag{144}$$

Then the doublet sector carries a two-dimensional irreducible representation.

The minimal symmetry statement is that at $p = 0$ the Bloch symbol commutes with the little-group action on the transverse plane:

$$[H_0, R(\theta)] = 0, \qquad R(\theta)\big|_{\mathrm{span}\{e_1,e_2\}} \in SO(2). \tag{145}$$

By Schur's lemma on the irreducible doublet block, the restriction of H_0 to the doublet sector must be proportional to the identity:

$$H_0\big|_{\mathbf{2}} = E_*\,\mathbf{1}_2. \tag{146}$$

Therefore, in the basis $(e_\parallel, e_1, e_2)$,

$$H_0 = \begin{pmatrix} E_\parallel & 0 & 0 \\ 0 & E_* & 0 \\ 0 & 0 & E_* \end{pmatrix}. \tag{147}$$

This is the desired protected two-fold degeneracy: it is symmetry-enforced inside the transverse doublet sector.

E. Why the scalar model failed and the internal model succeeds

In the scalar harmonic model, the 3×3 block had no representation-theoretic reason to contain an invariant two-dimensional degenerate subspace. Consequently, generic off-diagonal couplings opened gaps.

Here the situation is different. The doublet sector in (143) is an invariant irreducible representation of the shell little group. Hence the degeneracy at $p = 0$ is not accidental but symmetry-protected.

This is exactly the missing structural ingredient required for Weyl emergence.

F. Momentum expansion adapted to the shell geometry

Decompose

$$p = p_\parallel \hat{n} + p_1 e_1 + p_2 e_2. \tag{148}$$

The most general Hermitian linear correction compatible with the longitudinal/transverse splitting is

$$H_{\mathrm{int}}(p) = H_0 + p_\parallel \Gamma_\parallel + p_1\Gamma_1 + p_2\Gamma_2 + \mathcal{O}(|p|^2). \tag{149}$$

We now separate two logically distinct cases.

a. Case A: exact decoupling of singlet and doublet to linear order. If the singlet does not mix with the doublet at linear order, then after projection to the doublet sector one obtains

$$h_{\mathbf{2}}(p) = E_*\mathbf{1}_2 + p_\parallel A_\parallel + p_1 A_1 + p_2 A_2 + \mathcal{O}(|p|^2), \tag{150}$$

where A_μ are Hermitian 2×2 matrices.

b. Case B: weak singlet-doublet mixing. If the singlet mixes weakly with the doublet but remains gapped by

$$\Delta := E_\parallel - E_* \neq 0, \tag{151}$$

then the singlet can be integrated out by Schur complement or degenerate perturbation theory. To first order in p, the effective doublet Hamiltonian is still of the form (150), with corrections only entering at higher order:

$$h_{\mathbf{2}}^{\mathrm{eff}}(p) = E_*\mathbf{1}_2 + p_\parallel \widetilde{A}_\parallel + p_1\widetilde{A}_1 + p_2\widetilde{A}_2 + \mathcal{O}(|p|^2). \tag{152}$$

Hence in either case the local valley problem reduces to a 2×2 Hermitian linearized symbol.

G. General form of the projected doublet Hamiltonian

Every Hermitian 2×2 matrix can be expanded in the Pauli basis:

$$\widetilde{A}_\mu = a_\mu \mathbf{1}_2 + \sum_{b=1}^{3} M_{b\mu} \sigma^b. \tag{153}$$

Therefore

$$h_{\mathbf{2}}^{\mathrm{eff}}(p) = E_* \mathbf{1}_2 + \left(\sum_{\mu=\|,1,2} a_\mu p_\mu \right) \mathbf{1}_2 + \sum_{b=1}^{3} \left(\sum_{\mu=\|,1,2} M_{b\mu} p_\mu \right) \sigma^b + \mathcal{O}(|p|^2). \tag{154}$$

Define the 3×3 coefficient matrix

$$M = \begin{pmatrix} M_{1\|} & M_{11} & M_{12} \\ M_{2\|} & M_{21} & M_{22} \\ M_{3\|} & M_{31} & M_{32} \end{pmatrix}. \tag{155}$$

Then the Weyl criterion is exactly

$$\det M \neq 0. \tag{156}$$

Thus the entire fermionic problem has been reduced to proving that the projected doublet sector exists and that the linear coefficient matrix is nondegenerate.

H. Minimal rotationally adapted form

If, in addition, the transverse doublet respects the natural rotational structure of the shell tangent plane, then by a suitable unitary basis change within the doublet one can bring (154) to the canonical form

$$h_{\mathbf{2}}^{\mathrm{eff}}(p) = E_* \mathbf{1}_2 + c_0(p) \mathbf{1}_2 + \lambda_\| p_\| \sigma^3 + (\alpha p_1 - \beta p_2) \sigma^1 + (\beta p_1 + \alpha p_2) \sigma^2 + \mathcal{O}(|p|^2), \tag{157}$$

with

$$c_0(p) = a_\| p_\| + a_1 p_1 + a_2 p_2. \tag{158}$$

In this basis,

$$M = \begin{pmatrix} 0 & \alpha & -\beta \\ 0 & \beta & \alpha \\ \lambda_\| & 0 & 0 \end{pmatrix}, \qquad \det M = \lambda_\| (\alpha^2 + \beta^2). \tag{159}$$

Hence the Weyl condition becomes

$$\boxed{\lambda_\| (\alpha^2 + \beta^2) \neq 0.} \tag{160}$$

I. Protected two-band crossing theorem

[Minimal protected doublet crossing theorem] Let

$$H_{\mathrm{int}}(G_* + p) = H_0 + \sum_{i=1}^{3} p_i \Gamma_i + \mathcal{O}(|p|^2) \tag{161}$$

be a local Hermitian Bloch symbol acting on $\mathbb{C}^3$ near a BCC shell point G_*. Assume:

1. The internal space decomposes under the shell little group as

$$\mathbb{C}^3 = \mathbf{1} \oplus \mathbf{2}. \tag{162}$$

2. The $p = 0$ symbol commutes with the little-group action, so that

$$H_0 = \begin{pmatrix} E_\parallel & 0 \\ 0 & E_* \mathbf{1}_2 \end{pmatrix}. \tag{163}$$

3. The singlet is either decoupled from the doublet to linear order, or gapped and perturbatively eliminable.

4. The projected linear doublet Hamiltonian has coefficient matrix M satisfying

$$\det M \neq 0. \tag{164}$$

Then the doublet sector contains a Weyl point at $p = 0$. Its local effective Hamiltonian is of the form

$$h_{\mathbf{2}}^{\text{eff}}(p) = E_* \mathbf{1}_2 + \sum_{b,\mu} M_{b\mu} p_\mu \sigma^b + \mathcal{O}(|p|^2), \tag{165}$$

and the topological charge is

$$\chi = \operatorname{sgn}(\det M). \tag{166}$$

By assumptions (1)–(3), the local problem reduces to a projected 2×2 Hermitian symbol on the doublet sector. Every such symbol admits a Pauli expansion, yielding (154). If $\det M \neq 0$, then the Pauli vector

$$d(p) = Mp + \mathcal{O}(|p|^2) \tag{167}$$

is invertible to linear order. Hence the band splitting is linear and isolated in all three directions. This is exactly a Weyl cone. The chirality is the degree of the map $\hat{d} : S^2 \to S^2$, equal to $\operatorname{sgn}(\det M)$.

J. Dirac doubling

Let the opposite shell point be $-G_*$. If the corresponding doublet block satisfies the pairing relation

$$h_-(p) = -h_+(-p) \tag{168}$$

after subtraction of the scalar part, then the two Weyl sectors have opposite chirality and combine into a Dirac block:

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix}. \tag{169}$$

Thus the full TECT fermion chain becomes

$$\mathbb{C}^3 \text{ internal block} \implies (\mathbf{1} \oplus \mathbf{2}) \text{ decomposition} \implies \text{protected doublet crossing} \implies \text{Weyl valley} \implies \text{Dirac doubling.} \tag{170}$$

K. What is proven and what remains open

What is now mathematically clear is the following:

1. The scalar shell-mixing mechanism is insufficient.
2. A $\mathbb{C}^3$ internal Bloch block naturally admits a $1 \oplus 2$ decomposition at a shell point.
3. The transverse doublet provides the correct carrier space for a protected two-band crossing.
4. Once the projected linear coefficient matrix is nondegenerate, Weyl follows rigorously.

What remains open is the genuinely microscopic part:

1. derive the $\mathbb{C}^3$ internal Bloch symbol from the full TECT condensate PDE,

2. prove that the shell little group indeed enforces the $1 \oplus 2$ decomposition,
3. compute the projected doublet matrices explicitly,
4. verify

$$\det M \neq 0 \tag{171}$$

ab initio.

Therefore the true frontier is no longer conceptual. It is the explicit microscopic derivation of the protected doublet block.

LXXV. FINAL SUMMARY

The minimal scalar harmonic block cannot generically produce Weyl points, because it lacks a symmetry-protected two-dimensional degenerate subspace.

The correct minimal structure is an internal Bloch block acting on

$$\mathbb{C}^3,$$

which, near a BCC shell point G_*, naturally decomposes as

$$\mathbb{C}^3 = \mathbf{1} \oplus \mathbf{2}$$

into a longitudinal singlet and a transverse doublet.

If the shell little group preserves the doublet and forces

$$H_0\big|_{\mathbf{2}} = E_* \mathbf{1}_2,$$

then the doublet carries a protected two-band crossing at $p = 0$. After projection, the local effective Hamiltonian is

$$h^{\text{eff}}_{\mathbf{2}}(p) = E_* \mathbf{1}_2 + \sum_{b,\mu} M_{b\mu} p_\mu \sigma^b + \mathcal{O}(|p|^2).$$

If

$$\det M \neq 0,$$

this crossing is a Weyl point, with chirality

$$\chi = \operatorname{sgn}(\det M).$$

If the opposite shell point $-G_*$ supplies the opposite-chirality partner, the two Weyl blocks combine into a Dirac fermion.

Hence the remaining microscopic frontier is: derive the internal $\mathbb{C}^3$ Bloch block from the full TECT condensate PDE and compute the projected doublet coefficient matrix explicitly.

LXXVI. LITTLE GROUP AT A BCC SHELL POINT AND THE EXACT STATUS OF THE $\mathbf{3} \to \mathbf{1} \oplus \mathbf{2}$ DECOMPOSITION

A. Aim

We now clarify the precise representation-theoretic status of the decomposition

$$\mathbb{C}^3 = \mathbf{1} \oplus \mathbf{2}$$

near a BCC shell point.

A crucial distinction must be made between:

1. the *kinematic orthogonal splitting* into longitudinal and transverse subspaces, which is always true;
2. a *symmetry-protected two-dimensional irreducible doublet*, which is not automatic for the bare discrete cubic little group.

The purpose of this section is to separate these two statements rigorously.

B. Fixed shell point and stabilizer notion

Fix

$$G_* = \frac{q_*}{\sqrt{2}}(1,1,0), \qquad \hat{n} := \frac{G_*}{|G_*|} = \frac{1}{\sqrt{2}}(1,1,0). \tag{172}$$

Let G_{cubic} denote the symmetry group of the condensed BCC background at the relevant linearized level. At the purely geometric level this is a subgroup of the cubic point group O_h.

Define the little group of the shell point by

$$\mathrm{LG}(G_*) := \{g \in G_{\text{cubic}} \ : \ gG_* = G_*\}. \tag{173}$$

If one instead stabilizes only the shell line $\mathbb{R}G_*$, one obtains the slightly larger group

$$\mathrm{LG}_{\pm}(G_*) := \{g \in G_{\text{cubic}} \ : \ gG_* = \pm G_*\}. \tag{174}$$

The distinction between LG and $\mathrm{LG}_{\pm}$ is not essential for the orthogonal splitting below, but it matters for the detailed representation theory.

C. What is always true: longitudinal/transverse splitting

Independently of the detailed discrete symmetry, the vector space $\mathbb{R}^3$ admits the orthogonal decomposition

$$\mathbb{R}^3 = \mathbb{R}\hat{n} \oplus \hat{n}^{\perp}, \tag{175}$$

where

$$\hat{n}^{\perp} := \{v \in \mathbb{R}^3 : v \cdot \hat{n} = 0\}. \tag{176}$$

Complexifying,

$$\mathbb{C}^3 = \mathbb{C}\hat{n} \oplus \hat{n}^{\perp}_{\mathbb{C}}. \tag{177}$$

This is a purely linear-algebraic fact. It does not require any group-theoretic assumption beyond the existence of the distinguished shell direction $\hat{n}$.

[Kinematic splitting] For every nonzero shell point G_*, the internal vector space $\mathbb{C}^3$ decomposes canonically as

$$\mathbb{C}^3 = \mathbb{C}\hat{n} \oplus \hat{n}^{\perp}_{\mathbb{C}}. \tag{178}$$

The first summand is the longitudinal line and the second summand is the transverse plane.

Immediate from orthogonal projection:

$$v = (v \cdot \hat{n})\hat{n} + \big(v - (v \cdot \hat{n})\hat{n}\big). \tag{179}$$

The second term is orthogonal to $\hat{n}$. Complexification preserves the direct-sum decomposition.

Thus the statement

$$\mathbf{3} \to \mathbf{1} \oplus \mathbf{2} \tag{180}$$

is always valid as a *vector-space splitting*. However, it is not yet a statement about irreducible representations.

D. What is not automatic: protected doublet structure

To obtain a symmetry-protected two-band crossing, one needs more than the kinematic splitting (177). One needs the transverse plane to carry a two-dimensional symmetry sector that remains degenerate at $p = 0$.

This requires that the linearized shell-point Hamiltonian H_0 act on the transverse plane as

$$H_0\big|_{\hat{n}^{\perp}_{\mathbb{C}}} = E_* \mathbf{1}_2. \tag{181}$$

The question is whether (181) is forced by the exact little group of the BCC shell point.

In general, the answer is:

$$\boxed{\text{No. Bare discrete cubic little-group symmetry does not by itself force (181).}} \tag{182}$$

E. Why the bare cubic little group is insufficient

For a generic shell direction in a cubic environment, the exact little group is discrete. For the direction [110], the stabilizer inside the cubic group is a finite subgroup, not the continuous group $SO(2)$.

A discrete little group may preserve the transverse plane as a subspace, yet still allow the restriction of H_0 to split into two inequivalent one-dimensional pieces:

$$\hat{n}_{\mathbb{C}}^{\perp} \cong \mathbf{1} \oplus \mathbf{1} \qquad \text{(as representations of the discrete little group).} \tag{183}$$

If this happens, then symmetry permits

$$H_0\big|_{\hat{n}_{\mathbb{C}}^{\perp}} = \begin{pmatrix} E_1 & 0 \\ 0 & E_2 \end{pmatrix}, \qquad E_1 \neq E_2, \tag{184}$$

and the doublet degeneracy is lost.

Therefore:

$$\boxed{\text{transverse-plane invariance } \not\Rightarrow \text{ transverse doublet degeneracy.}} \tag{185}$$

This is the exact reason why the previous step must be refined.

F. The correct sufficient condition: emergent transverse $SO(2)$

The degeneracy (181) is guaranteed if the shell-point linearized problem has an effective continuous rotational symmetry on the transverse plane.

Assume there exists a unitary representation

$$R(\theta) : \hat{n}_{\mathbb{C}}^{\perp} \to \hat{n}_{\mathbb{C}}^{\perp}, \qquad \theta \in [0, 2\pi), \tag{186}$$

of $SO(2)$ (or $O(2)$) such that

$$[H_0, R(\theta)] = 0 \qquad \forall \theta. \tag{187}$$

Then H_0 is rotation-invariant on the transverse plane. Equivalently,

$$H_0\big|_{\hat{n}_{\mathbb{C}}^{\perp}} = E_* \mathbf{1}_2. \tag{188}$$

[Transverse isotropy implies protected doublet] If the shell-point Hamiltonian H_0 commutes with an effective transverse $SO(2)$ action, then the transverse plane is exactly degenerate:

$$H_0\big|_{\hat{n}_{\mathbb{C}}^{\perp}} = E_* \mathbf{1}_2. \tag{189}$$

A Hermitian operator commuting with all planar rotations must be proportional to the identity on that plane. This follows because the commutant of the standard $SO(2)$ action on $\mathbb{R}^2$ is generated by the identity, and Hermiticity removes the antisymmetric generator. After complexification the same conclusion holds.

Thus the correct logic is:

$$\boxed{\text{effective transverse } SO(2) \Rightarrow \text{protected transverse doublet.}} \tag{190}$$

G. TECT interpretation of the extra condition

In TECT, the natural origin of this effective transverse $SO(2)$ is not the exact microscopic cubic symmetry itself, but the already-emergent infrared isotropy of the condensed medium.

More precisely:

1. the BCC background selects a distinguished shell direction $\hat{n}$;

2. the long-wavelength linearized sector splits into one longitudinal and two transverse polarizations;
3. in the infrared, cubic anisotropy is suppressed by higher-order corrections;
4. therefore the leading transverse sector acquires an effective $SO(2)$ symmetry.

Hence the protected doublet is not a purely crystallographic statement. It is a statement about the *infrared isotropic shell-point theory.*

H. Corrected decomposition theorem

We can now formulate the exact theorem that is justified at the present stage.
[Corrected shell-point decomposition theorem] Let $G_* \neq 0$ be a shell point of the BCC reciprocal star.

1. The internal vector space $\mathbb{C}^3$ always admits the canonical orthogonal splitting
$$\mathbb{C}^3 = \mathbb{C}\hat{n} \oplus \hat{n}^{\perp}_{\mathbb{C}}, \qquad \hat{n} = \frac{G_*}{|G_*|}. \tag{191}$$
2. This yields a kinematic longitudinal/transverse decomposition
$$\mathbf{3} \to \mathbf{1} \oplus (\text{2D transverse plane}). \tag{192}$$
3. However, the exact discrete cubic little group of G_* does not by itself force the transverse plane to be a symmetry-protected degenerate doublet.
4. A protected two-band transverse sector is guaranteed if the shell-point Hamiltonian possesses an effective transverse $SO(2)$ symmetry, equivalently
$$H_0\big|_{\hat{n}^{\perp}_{\mathbb{C}}} = E_* \mathbf{1}_2. \tag{193}$$

Item (1) is linear algebra. Item (2) is simply a rephrasing of the direct sum. Item (3) follows because a discrete little group may preserve the plane without forcing scalar action on it. Item (4) follows from the commutant argument for the $SO(2)$ action.

I. Consequences for Weyl emergence

Once the protected transverse doublet exists, the local shell-point Hamiltonian projected to that doublet has the form
$$h_{\mathbf{2}}(p) = E_* \mathbf{1}_2 + \sum_{b,\mu} M_{b\mu} p_\mu \sigma^b + \mathcal{O}(|p|^2), \qquad \mu \in \{\|, 1, 2\}. \tag{194}$$
The Weyl condition is then
$$\det M \neq 0. \tag{195}$$

Therefore the exact remaining nontrivial step is not the kinematic splitting itself, but proving the emergence of the transverse $SO(2)$ shell-point symmetry strongly enough to protect the doublet.

J. True frontier after this refinement

The microscopic frontier is now sharply localized:
$$\boxed{\text{TECT PDE} \Rightarrow \text{infrared shell-point transverse } SO(2) \Rightarrow \text{protected doublet} \Rightarrow \det M \neq 0.} \tag{196}$$
Thus the real missing theorem is:

an infrared shell-point isotropy theorem strong enough to force $H_0|_{\hat{n}^{\perp}_{\mathbb{C}}} = E_* \mathbf{1}_2$.

LXXVII. FINAL SUMMARY

At a BCC shell point G_*, the internal space $\mathbb{C}^3$ always splits canonically as

$$\mathbb{C}^3 = \mathbb{C}\hat{n} \oplus \hat{n}^{\perp}_{\mathbb{C}},$$

that is, into one longitudinal line and one transverse two-plane.

This statement is purely kinematic and always true.

However, the stronger statement that the transverse plane forms a symmetry-protected degenerate two-band doublet is not automatic from the bare discrete cubic little group. A discrete little group may preserve the transverse plane while still allowing it to split into two nondegenerate one-dimensional sectors.

Therefore the correct sufficient condition for a protected transverse doublet is an effective shell-point transverse $SO(2)$ symmetry:

$$[H_0, R(\theta)] = 0 \quad \Rightarrow \quad H_0|_{\hat{n}^{\perp}_{\mathbb{C}}} = E_* \mathbf{1}_2.$$

Hence the true microscopic frontier is not the existence of the longitudinal/transverse splitting, but the derivation of an infrared shell-point isotropy theorem strong enough to enforce transverse degeneracy. Once that is obtained, the projected doublet Hamiltonian takes the Weyl form

$$h_{\mathbf{2}}(p) = E_* \mathbf{1}_2 + \sum_{b,\mu} M_{b\mu} p_\mu \sigma^b + \mathcal{O}(|p|^2),$$

and the Weyl criterion is simply

$$\det M \neq 0.$$

LXXVIII. BCC SHELL VALLEY EXISTENCE THEOREM

Let a TECT condensate arise from a shell instability

$$r(k) = r_0 + c(|k|^2 - q_*^2)^2.$$

Let the condensate select the BCC star

$$S_{BCC} = \frac{q_*}{\sqrt{2}}\{(\pm1, \pm1, 0), (\pm1, 0, \pm1), (0, \pm1, \pm1)\}.$$

Then each shell vector

$$G_* \in S_{BCC}$$

is a stationary point of the dispersion relation.

Near the shell point

$$k = G_* + p$$

the quadratic kernel expands as

$$r(k) = r_0 + 4c(G_* \cdot p)^2 + \mathcal{O}(p^3).$$

Thus the shell forms a degenerate manifold in the transverse directions.

If the internal sector provides a protected two-band doublet, the projected effective Hamiltonian takes the form

$$h(p) = \sum_{i,j} M_{ij} p_j \sigma^i.$$

If

$$\det M \neq 0$$

the shell point is a Weyl valley.

LXXIX. WEYL-COEFFICIENT EXTRACTION FROM A MINIMAL TECT CONDENSATE PDE ANSATZ

A. Statement of scope

We do *not* claim here that the full microscopic TECT condensate PDE has already been derived uniquely from first principles. What *is* already available is the theorem-level reduction of the local Weyl problem to the three coefficients

$$\lambda_\parallel, \qquad \alpha, \qquad \beta,$$

together with the identity

$$\det M = \lambda_\parallel(\alpha^2 + \beta^2).$$

The purpose of this section is to perform the explicit coefficient calculation for the minimal symmetry-compatible shell-enriched ansatz.

B. Selected BCC shell pair and adapted frame

Fix a selected reciprocal shell pair

$$\pm G_* \in S_{\mathrm{BCC}}, \qquad \hat{n} := \frac{G_*}{|G_*|}.$$

Choose an orthonormal adapted frame

$$\{e_1, e_2, \hat{n}\}, \qquad e_a \cdot \hat{n} = 0, \qquad e_a \cdot e_b = \delta_{ab}, \quad a, b = 1, 2.$$

Decompose the small shell-relative momentum as

$$p = p_\parallel \hat{n} + p_1 e_1 + p_2 e_2.$$

C. Minimal shell-enriched first-order symbol

On the enriched shell space

$$H_{\mathrm{enr}} = \mathbb{C}^2_{\mathrm{pair}} \otimes \mathbb{C}^2_{\mathrm{int}},$$

let τ^i act on the shell-pair factor and s_α on the internal doublet. The minimal axial-equivariant first-order ansatz is

$$L_{\min}(G_* + p) = E_*\,\mathbf{1}_4 + v_\parallel p_\parallel\, \tau^3 \otimes \mathbf{1}_2 + \alpha\big(p_1 \tau^1 \otimes s_1 + p_2 \tau^1 \otimes s_2\big) + \beta\big(-p_2 \tau^1 \otimes s_1 + p_1 \tau^1 \otimes s_2\big) + O(|p|^2). \tag{197}$$

This is exactly the minimal shell-enriched axial ansatz.

D. Derivative operators

Define the first derivative operators at the selected shell point by

$$K_i := \partial_{k_i} L_{\min}(k)\big|_{k=G_*}.$$

Using

$$p_\parallel = \hat{n}_i(k_i - G_{*i}), \qquad p_a = e_{ai}(k_i - G_{*i}),$$

we obtain

$$K_i = v_\parallel \hat{n}_i\, \tau^3 \otimes \mathbf{1}_2 + \alpha\big(e_{1i}\tau^1 \otimes s_1 + e_{2i}\tau^1 \otimes s_2\big) + \beta\big(-e_{2i}\tau^1 \otimes s_1 + e_{1i}\tau^1 \otimes s_2\big). \tag{198}$$

Contracting with the adapted frame gives

$$K_\parallel := \hat{n}_i K_i = v_\parallel \tau^3 \otimes \mathbf{1}_2, \tag{199}$$

because $\hat{n} \cdot e_a = 0$, and

$$K_1 := e_{1i} K_i = \alpha\, \tau^1 \otimes s_1 + \beta\, \tau^1 \otimes s_2, \tag{200}$$

$$K_2 := e_{2i} K_i = -\beta\, \tau^1 \otimes s_1 + \alpha\, \tau^1 \otimes s_2. \tag{201}$$

E. Valley projection and reduced Pauli identification

Let P_+ denote the projection onto the selected two-dimensional valley subspace. Under the reduced identification of the valley Pauli basis,

$$P_+(\tau^3 \otimes \mathbf{1}_2)P_+ \mapsto \sigma^3, \qquad P_+(\tau^1 \otimes s_1)P_+ \mapsto \sigma^1, \qquad P_+(\tau^1 \otimes s_2)P_+ \mapsto \sigma^2,$$

the projected first-order Hamiltonian becomes

$$h_+(p) = v_\parallel p_\parallel \sigma^3 + \alpha\,(p_1\sigma^1 + p_2\sigma^2) + \beta\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2). \tag{202}$$

Therefore the transverse 2×2 coefficient matrix is

$$T = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}, \qquad \det T = \alpha^2 + \beta^2. \tag{203}$$

F. Explicit Weyl coefficient matrix

Write the traceless linearized Hamiltonian as

$$h_+^{\mathrm{lin}}(p) = \sum_{\mu,j} M_{\mu j} p_j \sigma^\mu, \qquad j \in \{1, 2, \parallel\}, \quad \mu \in \{1, 2, 3\}.$$

Comparing with (202), we read off

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & v_\parallel \end{pmatrix}. \tag{204}$$

Hence

$$\det M = v_\parallel(\alpha^2 + \beta^2). \tag{205}$$

This is the desired Weyl coefficient matrix extracted from the minimal TECT shell ansatz.

G. Microscopic enriched tensor matching

Now let the enriched microscopic shell coupling be

$$\delta L_{\mathrm{enr}} = \sum_{a=1}^{3} N_{ij}^a\,(\partial_i\phi_0)\,(-i\partial_j) \otimes s_a. \tag{206}$$

Near the selected shell component, the condensate gradient contributes a factor proportional to G_{*i}. Define the shell-reduced vectors

$$C_j^a := G_{*i} N_{ij}^a.$$

Projecting these vectors onto the adapted frame gives the effective coefficients.

The longitudinal coefficient is

$$\lambda_\parallel = \kappa_* |G_*|\, \hat{n}_i \hat{n}_j\, N_{ij}^3. \tag{207}$$

The transverse matrix entries are

$$T_{ab} = \kappa_* |G_*|\, e_{aj} \hat{n}_i\, N_{ij}^b, \qquad a, b = 1, 2. \tag{208}$$

Therefore

$$\alpha = \frac{\kappa_*|G_*|}{2}\Big(e_{1j}\hat{n}_i N_{ij}^1 + e_{2j}\hat{n}_i N_{ij}^2\Big), \tag{209}$$

$$\beta = \frac{\kappa_*|G_*|}{2}\Big(e_{2j}\hat{n}_i N^1_{ij} - e_{1j}\hat{n}_i N^2_{ij}\Big). \tag{210}$$

Thus the complete microscopic-to-Weyl dictionary is

$$\boxed{\lambda_\parallel = \kappa_*|G_*|\,\hat{n}_i\hat{n}_j N^3_{ij},} \tag{211}$$

$$\boxed{\alpha = \frac{\kappa_*|G_*|}{2}\Big(e_{1j}\hat{n}_i N^1_{ij} + e_{2j}\hat{n}_i N^2_{ij}\Big),} \tag{212}$$

$$\boxed{\beta = \frac{\kappa_*|G_*|}{2}\Big(e_{2j}\hat{n}_i N^1_{ij} - e_{1j}\hat{n}_i N^2_{ij}\Big).} \tag{213}$$

Substituting into (205), the local Weyl criterion becomes

$$\boxed{\det M = \lambda_\parallel(\alpha^2+\beta^2).} \tag{214}$$

H. Immediate microscopic criterion

Hence the selected BCC shell crossing is locally Weyl provided

$$\hat{n}_i\hat{n}_j N^3_{ij} \neq 0, \tag{215}$$

and

$$\Big(e_{1j}\hat{n}_i N^1_{ij} + e_{2j}\hat{n}_i N^2_{ij}\Big)^2 + \Big(e_{2j}\hat{n}_i N^1_{ij} - e_{1j}\hat{n}_i N^2_{ij}\Big)^2 \neq 0. \tag{216}$$

If, in addition, an opposite valley exists with the standard pairing symmetry, then opposite-valley doubling yields the Dirac operator.

I. Honest status

What has been computed here is the full coefficient extraction for the minimal symmetry-compatible shell ansatz. What is still open is the final first-principles derivation of the actual TECT condensate PDE and the actual microscopic computation of the tensors N^a_{ij} generated by that PDE.

LXXX. FINAL SUMMARY

For the minimal symmetry-compatible TECT shell-enriched ansatz near a selected BCC shell vector G_*,

$$L_{\min}(G_*+p) = E_*\,\mathbf{1}_4 + v_\parallel p_\parallel\,\tau^3\otimes\mathbf{1}_2 + \alpha\big(p_1\tau^1\otimes s_1 + p_2\tau^1\otimes s_2\big) + \beta\big(-p_2\tau^1\otimes s_1 + p_1\tau^1\otimes s_2\big) + O(|p|^2),$$

the projected valley Hamiltonian is

$$h_+(p) = v_\parallel p_\parallel\sigma^3 + \alpha(p_1\sigma^1 + p_2\sigma^2) + \beta(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2).$$

Therefore the Weyl coefficient matrix is

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & v_\parallel \end{pmatrix}, \qquad \det M = v_\parallel(\alpha^2+\beta^2).$$

For the microscopic enriched tensor coupling

$$\delta L_{\mathrm{enr}} = \sum_{a=1}^{3} N^a_{ij}\,(\partial_i\phi_0)\,(-i\partial_j)\otimes s_a,$$

the effective coefficients are

$$\lambda_\parallel = \kappa_*|G_*|\,\hat{n}_i\hat{n}_j N^3_{ij},$$

$$\alpha = \frac{\kappa_*|G_*|}{2}\Big(e_{1j}\hat{n}_i N^1_{ij} + e_{2j}\hat{n}_i N^2_{ij}\Big),$$

$$\beta = \frac{\kappa_*|G_*|}{2}\Big(e_{2j}\hat{n}_i N^1_{ij} - e_{1j}\hat{n}_i N^2_{ij}\Big).$$

Hence the local BCC shell crossing is Weyl if and only if

$$\hat{n}_i\hat{n}_j N^3_{ij} \neq 0$$

and

$$\Big(e_{1j}\hat{n}_i N^1_{ij} + e_{2j}\hat{n}_i N^2_{ij}\Big)^2 + \Big(e_{2j}\hat{n}_i N^1_{ij} - e_{1j}\hat{n}_i N^2_{ij}\Big)^2 \neq 0.$$

Thus the local Weyl/Dirac problem has been reduced to an explicit finite tensor-contraction test.

LXXXI. EXPLICIT EXTRACTION OF N^a_{ij} FROM A MINIMAL MICROSCOPIC TECT ANSATZ

A. Scope

At this stage, the theorem-level architecture is already known: the local Weyl problem near a selected BCC shell pair is controlled by

$$\lambda_\parallel, \qquad \alpha, \qquad \beta,$$

with

$$\det M = \lambda_\parallel(\alpha^2+\beta^2).$$

What remains is the explicit extraction of the microscopic enriched tensors

$$N^a_{ij}$$

from a concrete symmetry-compatible condensate PDE ansatz.

We do not claim uniqueness of the full microscopic TECT PDE here. Instead, we choose the minimal first-order enriched ansatz that is compatible with the BCC shell axis and the reduced internal doublet.

B. Selected shell pair and adapted frame

Fix a selected reciprocal shell pair

$$\pm G_* \in S_{\mathrm{BCC}}, \qquad \hat{n} := \frac{G_*}{|G_*|}.$$

Choose an orthonormal adapted frame

$$\{e_1, e_2, \hat{n}\}, \qquad e_a\cdot\hat{n} = 0, \qquad e_a\cdot e_b = \delta_{ab}, \quad a,b = 1,2.$$

Write

$$p = p_\parallel\hat{n} + p_1 e_1 + p_2 e_2.$$

C. Minimal enriched microscopic coupling

Consider the enriched first-order microscopic coupling

$$\delta L_{\rm enr} = \sum_{a=1}^{3} N^a_{ij}\,(\partial_i\phi_0)\,(-i\partial_j) \otimes s_a, \tag{217}$$

where:

- ϕ_0 is the condensed BCC background,
- s_a act on the reduced internal doublet,
- N^a_{ij} are the microscopic enriched tensors.

We now choose the smallest symmetry-compatible ansatz that can generate the three effective Weyl coefficients:

$$N^3_{ij} = A\,\hat{n}_i\hat{n}_j, \tag{218}$$

$$N^1_{ij} = B\,\hat{n}_i e_{1j} + C\,\hat{n}_i e_{2j}, \tag{219}$$

$$N^2_{ij} = -C\,\hat{n}_i e_{1j} + B\,\hat{n}_i e_{2j}. \tag{220}$$

This is the minimal axial-equivariant tensor family:

- A is the longitudinal coefficient,
- B is the transverse even coefficient,
- C is the transverse odd coefficient.

D. Why this ansatz is the correct minimal one

The structure (218)–(220) is the tensor version of the operator-level ansatz

$$\lambda_\parallel(\hat{n}\cdot p)\sigma^3 + \alpha(p_1\sigma^1 + p_2\sigma^2) + \beta(-p_2\sigma^1 + p_1\sigma^2).$$

Indeed:

- $\hat{n}_i\hat{n}_j$ is the unique axial longitudinal projector,
- $\hat{n}_i e_{aj}$ gives the mixed longitudinal-to-transverse shell contraction,
- the pair (B, C) is arranged so that the transverse block becomes exactly

$$\begin{pmatrix} B & -C \\ C & B \end{pmatrix}.$$

Thus (218)–(220) is the minimal microscopic realization of the already-established Weyl template.

E. Microscopic-to-effective matching formulas

The shell-level matching formulas are

$$\lambda_\parallel = \kappa_*|G_*|\,\hat{n}_i\hat{n}_j N^3_{ij}, \tag{221}$$

$$\alpha = \frac{\kappa_*|G_*|}{2}\left(e_{1j}\hat{n}_i N^1_{ij} + e_{2j}\hat{n}_i N^2_{ij}\right), \tag{222}$$

$$\beta = \frac{\kappa_*|G_*|}{2}\left(e_{2j}\hat{n}_i N^1_{ij} - e_{1j}\hat{n}_i N^2_{ij}\right). \tag{223}$$

We now evaluate these explicitly.

F. Longitudinal coefficient

Using (218),

$$\hat{n}_i\hat{n}_j N^3_{ij} = \hat{n}_i\hat{n}_j(A\hat{n}_i\hat{n}_j) = A(\hat{n}\cdot\hat{n})^2 = A.$$

Therefore

$$\boxed{\lambda_\parallel = \kappa_*|G_*|\,A.} \tag{224}$$

G. Transverse even coefficient

First compute

$$e_{1j}\hat{n}_i N^1_{ij} = e_{1j}\hat{n}_i(B\hat{n}_i e_{1j} + C\hat{n}_i e_{2j}) = B(\hat{n}\cdot\hat{n})(e_1\cdot e_1) + C(\hat{n}\cdot\hat{n})(e_1\cdot e_2) = B.$$

Similarly,

$$e_{2j}\hat{n}_i N^2_{ij} = e_{2j}\hat{n}_i(-C\hat{n}_i e_{1j} + B\hat{n}_i e_{2j}) = -C(e_2\cdot e_1) + B(e_2\cdot e_2) = B.$$

Thus

$$e_{1j}\hat{n}_i N^1_{ij} + e_{2j}\hat{n}_i N^2_{ij} = 2B,$$

and therefore

$$\boxed{\alpha = \kappa_*|G_*|\,B.} \tag{225}$$

H. Transverse odd coefficient

Now compute

$$e_{2j}\hat{n}_i N^1_{ij} = e_{2j}\hat{n}_i(B\hat{n}_i e_{1j} + C\hat{n}_i e_{2j}) = B(e_2\cdot e_1) + C(e_2\cdot e_2) = C,$$

and

$$e_{1j}\hat{n}_i N^2_{ij} = e_{1j}\hat{n}_i(-C\hat{n}_i e_{1j} + B\hat{n}_i e_{2j}) = -C(e_1\cdot e_1) + B(e_1\cdot e_2) = -C.$$

Hence

$$e_{2j}\hat{n}_i N^1_{ij} - e_{1j}\hat{n}_i N^2_{ij} = C - (-C) = 2C,$$

and therefore

$$\boxed{\beta = \kappa_*|G_*|\,C.} \tag{226}$$

I. Explicit Weyl coefficient matrix

Substituting (224)–(226) into the reduced Weyl block gives

$$h_+(p) = \lambda_\parallel p_\parallel \sigma^3 + \alpha(p_1\sigma^1 + p_2\sigma^2) + \beta(-p_2\sigma^1 + p_1\sigma^2).$$

Thus the coefficient matrix is

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_\parallel \end{pmatrix} = \kappa_*|G_*| \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix}. \tag{227}$$

Therefore

$$\boxed{\det M = (\kappa_*|G_*|)^3\, A(B^2 + C^2).} \tag{228}$$

J. Local Weyl criterion in microscopic tensor form

Hence the selected shell crossing is locally Weyl if and only if

$$A \neq 0, \qquad (B, C) \neq (0, 0). \tag{229}$$

Equivalently,

$$\hat{n}_i \hat{n}_j N^3_{ij} \neq 0, \tag{230}$$

and

$$\left(e_{1j} \hat{n}_i N^1_{ij} + e_{2j} \hat{n}_i N^2_{ij}\right)^2 + \left(e_{2j} \hat{n}_i N^1_{ij} - e_{1j} \hat{n}_i N^2_{ij}\right)^2 \neq 0. \tag{231}$$

K. Interpretation

This proves that the minimal microscopic enriched tensor family already contains the full local Weyl data:

- the longitudinal projector channel $\hat{n}_i \hat{n}_j \otimes s_3$ controls the coefficient $\lambda_\parallel$,
- the transverse even channel controls α,
- the transverse odd channel controls β.

Therefore the Weyl problem is not an uncontrolled PDE existence problem anymore. It is a finite microscopic coefficient test.

L. Dirac consequence

If an opposite shell valley at $-G_*$ exists with the standard pairing relation

$$h_-(p) = -h_+(-p),$$

then the doubled Hamiltonian is Dirac. Thus the only remaining genuinely microscopic steps are:

1. derive the actual full TECT condensate PDE,
2. compute the actual N^a_{ij} generated by that PDE,
3. verify that their coefficients reduce to

$$A \neq 0, \qquad (B, C) \neq (0, 0),$$

4. verify opposite-valley existence and pairing.

LXXXII. FINAL SUMMARY

For the minimal symmetry-compatible microscopic enriched tensors

$$N^3_{ij} = A\, \hat{n}_i \hat{n}_j,$$

$$N^1_{ij} = B\, \hat{n}_i e_{1j} + C\, \hat{n}_i e_{2j},$$

$$N^2_{ij} = -C\, \hat{n}_i e_{1j} + B\, \hat{n}_i e_{2j},$$

the shell-matching formulas give

$$\lambda_\parallel = \kappa_* |G_*|\, A, \qquad \alpha = \kappa_* |G_*|\, B, \qquad \beta = \kappa_* |G_*|\, C.$$

Hence the local Weyl coefficient matrix is

$$M = \kappa_* |G_*| \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix},$$

and therefore

$$\det M = (\kappa_* |G_*|)^3 \, A(B^2 + C^2).$$

Thus the selected BCC shell crossing is locally Weyl if and only if

$$A \neq 0, \qquad (B, C) \neq (0, 0).$$

Equivalently, the local Weyl/Dirac problem has been reduced to the explicit microscopic question: does the true TECT condensate linearization contain a nonzero longitudinal projector channel and at least one nontrivial transverse channel?

LXXXIII. EXTRACTION OF THE WEYL COEFFICIENTS FROM A MINIMAL ACTUAL TECT OPERATOR ANSATZ

A. Honest scope

At the present stage, the full microscopic TECT condensate PDE is still not uniquely derived. What is already available is:

1. the reduction of the local Weyl problem to the three coefficients

$$\lambda_\parallel, \qquad \alpha, \qquad \beta,$$

2. the determinant formula

$$\det M = \lambda_\parallel (\alpha^2 + \beta^2),$$

3. and the explicit shell-level extraction formulas from the enriched microscopic tensors

$$N^a_{ij}.$$

Therefore, the strongest honest next step is to choose a minimal actual symmetry-compatible operator ansatz and read off the coefficients explicitly.

B. Selected shell pair and adapted frame

Fix a selected reciprocal shell pair

$$\pm G_* \in S_{\mathrm{BCC}}, \qquad \hat{n} := \frac{G_*}{|G_*|}.$$

Choose an adapted orthonormal frame

$$\{e_1, e_2, \hat{n}\}, \qquad e_a \cdot \hat{n} = 0, \qquad e_a \cdot e_b = \delta_{ab}, \quad a, b = 1, 2.$$

Write

$$p = p_\parallel \hat{n} + p_1 e_1 + p_2 e_2.$$

C. Minimal actual operator ansatz

The smallest symmetry-compatible first-order enriched operator is

$$L_{\rm act}(G_* + p) = E_* \mathbf{1}_4 + A\,(\hat{n}\cdot p)\,\tau^3 \otimes \mathbf{1}_2 + B\,(p_1\tau^1 \otimes s_1 + p_2\tau^1 \otimes s_2) + C\,(-p_2\tau^1 \otimes s_1 + p_1\tau^1 \otimes s_2) + O(|p|^2). \quad (232)$$

Here:

- A is the longitudinal axial coefficient,
- B is the transverse even coefficient,
- C is the transverse odd coefficient.

This is exactly the minimal shell-enriched axial ansatz used in the explicit extraction calculation.

D. First derivative operators

Define

$$K_i := \partial_{k_i} L_{\rm act}(k)\big|_{k=G_*}.$$

Using

$$p_\parallel = \hat{n}_i(k_i - G_{*i}), \qquad p_a = e_{ai}(k_i - G_{*i}),$$

we obtain

$$K_i = A\,\hat{n}_i\,\tau^3 \otimes \mathbf{1}_2 + B\,(e_{1i}\tau^1 \otimes s_1 + e_{2i}\tau^1 \otimes s_2) + C\,(-e_{2i}\tau^1 \otimes s_1 + e_{1i}\tau^1 \otimes s_2). \quad (233)$$

Hence

$$K_\parallel := \hat{n}_i K_i = A\,\tau^3 \otimes \mathbf{1}_2, \quad (234)$$

$$K_1 := e_{1i}K_i = B\,\tau^1 \otimes s_1 + C\,\tau^1 \otimes s_2, \quad (235)$$

$$K_2 := e_{2i}K_i = -C\,\tau^1 \otimes s_1 + B\,\tau^1 \otimes s_2. \quad (236)$$

E. Valley projection

Project to the selected two-band valley subspace with projector P_+. Under the reduced identification

$$P_+(\tau^3 \otimes \mathbf{1}_2)P_+ \mapsto \sigma^3, \qquad P_+(\tau^1 \otimes s_1)P_+ \mapsto \sigma^1, \qquad P_+(\tau^1 \otimes s_2)P_+ \mapsto \sigma^2,$$

the projected Hamiltonian becomes

$$h_+(p) = A\,p_\parallel \sigma^3 + B\,(p_1\sigma^1 + p_2\sigma^2) + C\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2). \quad (237)$$

Thus, immediately,

$$\lambda_\parallel = A, \qquad \alpha = B, \qquad \beta = C. \quad (238)$$

F. Explicit Weyl matrix

Comparing (237) with

$$h_+^{\mathrm{lin}}(p) = \sum_{\mu,j} M_{\mu j} p_j \sigma^\mu, \qquad j \in \{1, 2, \|\},$$

we read off

$$M = \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix}. \tag{239}$$

Hence

$$\det M = A(B^2 + C^2). \tag{240}$$

Therefore the local Weyl criterion is simply

$$A \neq 0, \qquad (B, C) \neq (0, 0). \tag{241}$$

G. Matching to the enriched microscopic tensors

Now write the enriched microscopic shell coupling as

$$\delta L_{\mathrm{enr}} = \sum_{a=1}^{3} N_{ij}^a \, (\partial_i \phi_0) \, (-i\partial_j) \otimes s_a. \tag{242}$$

The shell-matching formulas are

$$\lambda_\| = \kappa_* |G_*| \, \hat{n}_i \hat{n}_j N_{ij}^3, \tag{243}$$

$$\alpha = \frac{\kappa_* |G_*|}{2} \Big(e_{1j} \hat{n}_i N_{ij}^1 + e_{2j} \hat{n}_i N_{ij}^2 \Big), \tag{244}$$

$$\beta = \frac{\kappa_* |G_*|}{2} \Big(e_{2j} \hat{n}_i N_{ij}^1 - e_{1j} \hat{n}_i N_{ij}^2 \Big). \tag{245}$$

To realize (232), choose the minimal tensor ansatz

$$N_{ij}^3 = \frac{A}{\kappa_* |G_*|} \, \hat{n}_i \hat{n}_j, \tag{246}$$

$$N_{ij}^1 = \frac{1}{\kappa_* |G_*|} \left(B \, \hat{n}_i e_{1j} + C \, \hat{n}_i e_{2j} \right), \tag{247}$$

$$N_{ij}^2 = \frac{1}{\kappa_* |G_*|} \left(-C \, \hat{n}_i e_{1j} + B \, \hat{n}_i e_{2j} \right). \tag{248}$$

Substituting (246)–(248) into (243)–(245) gives

$$\lambda_\| = A, \qquad \alpha = B, \qquad \beta = C,$$

exactly as required.

H. Interpretation

Thus the minimal actual operator ansatz (232) is microscopically realized by the explicit enriched tensors

$$N^3_{ij} \propto \hat n_i \hat n_j, \qquad N^1_{ij} \propto \hat n_i e_{1j}, \hat n_i e_{2j}, \qquad N^2_{ij} \propto \hat n_i e_{1j}, \hat n_i e_{2j}.$$

In other words:

- the longitudinal projector channel produces $A = \lambda_\parallel$,
- the transverse even channel produces $B = \alpha$,
- the transverse odd channel produces $C = \beta$.

Hence the entire local Weyl problem is reduced to the microscopic yes/no question: does the true TECT linearization contain these three channels with

$$A \neq 0, \qquad (B, C) \neq (0, 0)?$$

I. Dirac consequence

If an opposite shell valley at $-G_*$ exists and satisfies the standard pairing relation

$$h_-(p) = -h_+(-p),$$

then the doubled theory is Dirac. Therefore, once the actual TECT operator matches the minimal ansatz with

$$A \neq 0, \qquad (B, C) \neq (0, 0),$$

the only remaining local issue is opposite-valley existence and pairing.

LXXXIV. FINAL SUMMARY

For the minimal actual symmetry-compatible TECT operator

$$L_{\text{act}}(G_* + p) = E_* \mathbf{1}_4 + A(\hat n \cdot p)\tau^3 \otimes \mathbf{1}_2 + B(p_1 \tau^1 \otimes s_1 + p_2 \tau^1 \otimes s_2) + C(-p_2 \tau^1 \otimes s_1 + p_1 \tau^1 \otimes s_2) + O(|p|^2),$$

the projected two-band valley Hamiltonian is

$$h_+(p) = A\, p_\parallel \sigma^3 + B(p_1 \sigma^1 + p_2 \sigma^2) + C(-p_2 \sigma^1 + p_1 \sigma^2) + O(|p|^2).$$

Therefore

$$\lambda_\parallel = A, \qquad \alpha = B, \qquad \beta = C,$$

and the Weyl coefficient matrix is

$$M = \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix}, \qquad \det M = A(B^2 + C^2).$$

A microscopic tensor realization of this ansatz is

$$N^3_{ij} = \frac{A}{\kappa_* |G_*|} \hat n_i \hat n_j,$$

$$N^1_{ij} = \frac{1}{\kappa_* |G_*|} \left(B \hat n_i e_{1j} + C \hat n_i e_{2j} \right),$$

$$N^2_{ij} = \frac{1}{\kappa_* |G_*|} \left(-C \hat n_i e_{1j} + B \hat n_i e_{2j} \right).$$

Hence the local BCC shell crossing is Weyl if and only if

$$A \neq 0, \qquad (B, C) \neq (0, 0).$$

Thus the remaining microscopic TECT task is not structural: it is to verify that the true condensate linearization indeed generates these three channels with nonzero coefficients.

LXXXV. PROJECTOR-FIELD TECT PDE AND EXPLICIT EXTRACTION OF THE WEYL COEFFICIENTS

A. Honest scope

We do not yet claim a unique final microscopic TECT condensate PDE. What is already theorem-level closed is the following:

1. the local BCC-shell Weyl problem reduces to the three coefficients
$$\lambda_{\|}, \qquad \alpha, \qquad \beta,$$
2. the determinant identity is
$$\det M = \lambda_{\|}(\alpha^2+\beta^2),$$
3. and the coefficients are extracted from the microscopic enriched tensors
$$N^a_{ij}$$
by explicit shell contractions.

The purpose of the present section is therefore to choose a minimal actual projector-field ansatz and compute the induced coefficients A, B, C explicitly.

B. Projective internal field

Let
$$z(x,t)\in\mathbb{C}^3, \qquad z^\dagger z = 1, \qquad z\sim e^{i\chi(x,t)}z,$$
and define the rank-one projector
$$P(x,t) := z(x,t)z^\dagger(x,t), \qquad P^2=P, \qquad P^\dagger = P, \qquad P = 1.$$
Thus
$$P(x,t)\in\mathbb{CP}^2.$$

The natural gauge-covariant derivative is
$$D_iP := \partial_i P - i[A_i, P],$$
where $A_i = A^a_i T^a \in su(3)$ is the already-emergent internal connection.

C. Minimal projector-field PDE ansatz

We now write the smallest shell-compatible first-order enriched PDE ansatz near a selected BCC shell pair. Let $\phi_0(x)$ denote the BCC condensate background and let
$$\hat n := \frac{G_*}{|G_*|}, \qquad \{e_1, e_2, \hat n\}$$
be the adapted shell frame.

We assume that the reduced internal doublet sector is generated by three Hermitian matrices
$$\Sigma_3, \qquad \Sigma_1, \qquad \Sigma_2,$$

which, after projection to the valley subspace, reduce to the Pauli basis

$$\Sigma_3 \mapsto \sigma^3, \qquad \Sigma_1 \mapsto \sigma^1, \qquad \Sigma_2 \mapsto \sigma^2.$$

The minimal first-order projector-field operator is

$$i\partial_t \Psi = \Big[E_* + \mathcal{L}_{\text{shell}} + \mathcal{L}_{\text{enr}}\Big]\Psi, \tag{249}$$

with

$$\mathcal{L}_{\text{shell}} = A\,(\hat{n}\cdot(-i\nabla))\,\Sigma_3, \tag{250}$$

and

$$\mathcal{L}_{\text{enr}} = B\Big((e_1\cdot(-i\nabla))\Sigma_1 + (e_2\cdot(-i\nabla))\Sigma_2\Big) + C\Big(-(e_2\cdot(-i\nabla))\Sigma_1 + (e_1\cdot(-i\nabla))\Sigma_2\Big). \tag{251}$$

This is exactly the minimal axial-equivariant first-order ansatz: A is the longitudinal channel, B the transverse-even channel, and C the transverse-odd channel.

D. Microscopic tensor form

To match with the theorem-level shell-reduction formulas, rewrite (251) in tensor form:

$$\delta L_{\text{enr}} = \sum_{a=1}^{3} N^a_{ij}\,(\partial_i\phi_0)\,(-i\partial_j)\otimes s_a. \tag{252}$$

Near the selected shell component, the condensate gradient contributes a factor proportional to G_{*i}. Hence the shell-reduced vectors are

$$C^a_j := G_{*i}N^a_{ij}.$$

The minimal tensor realization that reproduces (250)–(251) is

$$N^3_{ij} = \frac{A}{\kappa_*|G_*|}\,\hat{n}_i\hat{n}_j, \tag{253}$$

$$N^1_{ij} = \frac{1}{\kappa_*|G_*|}\Big(B\,\hat{n}_i e_{1j} + C\,\hat{n}_i e_{2j}\Big), \tag{254}$$

$$N^2_{ij} = \frac{1}{\kappa_*|G_*|}\Big(-C\,\hat{n}_i e_{1j} + B\,\hat{n}_i e_{2j}\Big), \tag{255}$$

where κ_* is the shell-projection normalization factor.

E. Direct extraction of $\lambda_\parallel$, α, β

The shell-level extraction formulas are

$$\lambda_\parallel = \kappa_*|G_*|\,\hat{n}_i\hat{n}_j N^3_{ij}, \tag{256}$$

$$\alpha = \frac{\kappa_*|G_*|}{2}\Big(e_{1j}\hat{n}_i N^1_{ij} + e_{2j}\hat{n}_i N^2_{ij}\Big), \tag{257}$$

$$\beta = \frac{\kappa_*|G_*|}{2}\Big(e_{2j}\hat{n}_i N^1_{ij} - e_{1j}\hat{n}_i N^2_{ij}\Big). \tag{258}$$

Substituting (253) gives

$$\hat{n}_i\hat{n}_j N^3_{ij} = \frac{A}{\kappa_*|G_*|}(\hat{n}\cdot\hat{n})^2 = \frac{A}{\kappa_*|G_*|},$$

hence

$$\boxed{\lambda_\parallel = A.} \tag{259}$$

Next, using (254) and (255),

$$e_{1j}\hat{n}_i N^1_{ij} = \frac{1}{\kappa_*|G_*|}\Big(B\,(\hat{n}\cdot\hat{n})(e_1\cdot e_1) + C\,(\hat{n}\cdot\hat{n})(e_1\cdot e_2)\Big) = \frac{B}{\kappa_*|G_*|},$$

and

$$e_{2j}\hat{n}_i N^2_{ij} = \frac{1}{\kappa_*|G_*|}\Big(-C\,(\hat{n}\cdot\hat{n})(e_2\cdot e_1) + B\,(\hat{n}\cdot\hat{n})(e_2\cdot e_2)\Big) = \frac{B}{\kappa_*|G_*|}.$$

Therefore

$$\boxed{\alpha = B.} \tag{260}$$

Similarly,

$$e_{2j}\hat{n}_i N^1_{ij} = \frac{C}{\kappa_*|G_*|}, \qquad e_{1j}\hat{n}_i N^2_{ij} = -\frac{C}{\kappa_*|G_*|},$$

so

$$\boxed{\beta = C.} \tag{261}$$

F. Projected two-band Hamiltonian

After projection to the selected two-band valley subspace, the reduced linear Hamiltonian is

$$h_+(p) = A\,p_\parallel\sigma^3 + B\,(p_1\sigma^1 + p_2\sigma^2) + C\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2). \tag{262}$$

Hence the Weyl coefficient matrix is

$$M = \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix}. \tag{263}$$

Therefore

$$\boxed{\det M = A(B^2 + C^2).} \tag{264}$$

G. Microscopic Weyl criterion

Thus the local BCC shell crossing is Weyl if and only if

$$\boxed{A \neq 0, \qquad (B, C) \neq (0, 0).} \tag{265}$$

Equivalently, in microscopic tensor language,

$$\hat{n}_i\hat{n}_j N^3_{ij} \neq 0, \tag{266}$$

and

$$\Big(e_{1j}\hat{n}_i N^1_{ij} + e_{2j}\hat{n}_i N^2_{ij}\Big)^2 + \Big(e_{2j}\hat{n}_i N^1_{ij} - e_{1j}\hat{n}_i N^2_{ij}\Big)^2 \neq 0. \tag{267}$$

H. What this proves and what it does not

What is proved here is:

1. a minimal actual projector-field operator ansatz exists whose first-order coefficients are exactly A, B, C,
2. its enriched microscopic tensors N^a_{ij} are explicitly known,
3. the induced Weyl matrix is explicitly

$$M = \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix},$$

4. and local Weyl nondegeneracy is equivalent to

$$A \neq 0, \qquad (B, C) \neq (0, 0).$$

What remains open is:

1. the final unique derivation of the full TECT condensate PDE,
2. the proof that the actual microscopic linearization reduces to this minimal ansatz,
3. and the existence plus pairing of the opposite valley at $-G_*$.

LXXXVI. FINAL SUMMARY

For the minimal projector-field TECT operator

$$i\partial_t \Psi = \Big[E_* + A(\hat{n} \cdot (-i\nabla))\Sigma_3 + B\big((e_1 \cdot (-i\nabla))\Sigma_1 + (e_2 \cdot (-i\nabla))\Sigma_2\big) + C\big(-(e_2 \cdot (-i\nabla))\Sigma_1 + (e_1 \cdot (-i\nabla))\Sigma_2\big)\Big]\Psi,$$

the microscopic enriched tensors are realized by

$$N^3_{ij} = \frac{A}{\kappa_* |G_*|} \hat{n}_i \hat{n}_j,$$

$$N^1_{ij} = \frac{1}{\kappa_* |G_*|} \big(B\hat{n}_i e_{1j} + C\hat{n}_i e_{2j}\big),$$

$$N^2_{ij} = \frac{1}{\kappa_* |G_*|} \big(-C\hat{n}_i e_{1j} + B\hat{n}_i e_{2j}\big).$$

The shell-level contraction formulas give

$$\lambda_\parallel = A, \qquad \alpha = B, \qquad \beta = C.$$

Therefore the projected Weyl coefficient matrix is

$$M = \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix}, \qquad \det M = A(B^2 + C^2).$$

Hence the selected BCC shell crossing is locally Weyl if and only if

$$A \neq 0, \qquad (B, C) \neq (0, 0).$$

Thus the remaining microscopic task is now precise: derive the actual TECT condensate linearization and verify that it indeed generates the three channels corresponding to A, B, and C.

LXXXVII. OPPOSITE-VALLEY PAIRING AND DIRAC DOUBLING

A. Goal

We now isolate the exact condition under which two opposite BCC shell valleys

$$\pm G_*$$

combine into a Dirac fermion.

The key point is that the existence of a Weyl valley at $+G_*$ does *not* by itself imply the existence of a Dirac fermion. To obtain a Dirac block, one additionally needs an opposite valley at $-G_*$ with the correct pairing relation.

B. Local Weyl block at $+G_*$

From the previous step, the projected two-band Hamiltonian near the selected shell point $+G_*$ is

$$h_+(p) = A\, p_\parallel \sigma^3 + B\,(p_1\sigma^1 + p_2\sigma^2) + C\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2), \tag{268}$$

where

$$p = p_\parallel \hat{n} + p_1 e_1 + p_2 e_2, \qquad \hat{n} = \frac{G_*}{|G_*|}.$$

Its coefficient matrix is

$$M_+ = \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix}, \qquad \det M_+ = A(B^2 + C^2). \tag{269}$$

If

$$A \neq 0, \qquad (B, C) \neq (0, 0),$$

then $+G_*$ is a Weyl valley.

C. Local Weyl block at $-G_*$

Let the opposite shell vector be

$$-G_*, \qquad \hat{n}_- := -\hat{n}.$$

Choose the opposite adapted frame

$$\{e_1^-, e_2^-, \hat{n}_-\}.$$

There is some freedom in how one chooses the transverse basis at the opposite valley. A natural and convenient convention is

$$e_1^- = e_1, \qquad e_2^- = -e_2, \qquad \hat{n}_- = -\hat{n},$$

so that the ordered frame remains right-handed. Then the local momentum near the opposite valley is

$$k = -G_* + p.$$

The most general projected Hamiltonian near $-G_*$ is

$$h_-(p) = A_-\, p_\parallel \sigma^3 + B_-\,(p_1\sigma^1 + p_2\sigma^2) + C_-\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2). \tag{270}$$

The question is: under what conditions are the coefficients related so that

$$h_-(p) = -h_+(-p)?$$

D. What the pairing relation means explicitly

From (268),

$$h_+(-p) = -A\,p_\parallel \sigma^3 - B\,(p_1\sigma^1 + p_2\sigma^2) - C\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2).$$

Therefore

$$-h_+(-p) = A\,p_\parallel \sigma^3 + B\,(p_1\sigma^1 + p_2\sigma^2) + C\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2). \tag{271}$$

Thus the relation

$$h_-(p) = -h_+(-p)$$

is equivalent to

$$A_- = A, \qquad B_- = B, \qquad C_- = C \tag{272}$$

in a suitably matched valley basis.

So the problem reduces to whether the microscopic symmetry maps the $+G_*$ valley to the $-G_*$ valley while preserving the three reduced coefficients.

E. Sufficient symmetry condition: unitary inversion-type pairing

Assume there exists a unitary symmetry $\mathcal{I}$ such that:

1. $\mathcal{I}$ maps the shell point $+G_*$ to $-G_*$,
$$\mathcal{I} : G_* \mapsto -G_*;$$
2. after projection to the two-band valley subspaces, $\mathcal{I}$ identifies the Pauli basis at the two valleys;
3. the microscopic linearized symbol satisfies
$$\mathcal{I}\,L(G_* + p)\,\mathcal{I}^{-1} = L(-G_* - p).$$

Projecting to the two valley subspaces then gives

$$h_-(p) = U_{\mathcal{I}}\,h_+(-p)\,U_{\mathcal{I}}^{-1},$$

for some unitary $U_{\mathcal{I}}$ acting in the two-band space.

If, moreover, the basis is chosen so that

$$U_{\mathcal{I}}\sigma^\mu U_{\mathcal{I}}^{-1} = -\sigma^\mu$$

for the traceless part, then

$$h_-(p) = -h_+(-p)$$

after subtraction of the common scalar part.

This is a sufficient condition for opposite-valley pairing.

F. Sufficient antiunitary condition: time-reversal-type pairing

Alternatively, suppose there exists an antiunitary symmetry $\mathcal{T}$ such that

$$\mathcal{T}\,L(G_* + p)\,\mathcal{T}^{-1} = L(-G_* - p).$$

After projection, one gets

$$h_-(p) = U_{\mathcal{T}}\,h_+^*(-p)\,U_{\mathcal{T}}^{-1}.$$

If the reduced Hamiltonian is brought to the real canonical Weyl form or the basis is chosen so that complex conjugation acts trivially on the coefficient functions, then this again yields the standard opposite-valley relation up to unitary equivalence.

Thus either inversion-type or time-reversal-type shell pairing symmetry is sufficient.

G. Microscopic tensor criterion for opposite-valley pairing

Let the enriched microscopic coupling near $+G_*$ be described by the tensors

$$N^a_{ij}(+),$$

and near $-G_*$ by

$$N^a_{ij}(-).$$

The effective coefficients are

$$A_\pm = \kappa_*|G_*|\,\hat{n}^i_\pm \hat{n}^j_\pm N^3_{ij}(\pm),$$

$$B_\pm = \frac{\kappa_*|G_*|}{2}\Big(e^\pm_{1j}\hat{n}^i_\pm N^1_{ij}(\pm) + e^\pm_{2j}\hat{n}^i_\pm N^2_{ij}(\pm)\Big),$$

$$C_\pm = \frac{\kappa_*|G_*|}{2}\Big(e^\pm_{2j}\hat{n}^i_\pm N^1_{ij}(\pm) - e^\pm_{1j}\hat{n}^i_\pm N^2_{ij}(\pm)\Big).$$

Hence a direct finite microscopic sufficient condition for standard Dirac doubling is

$$A_+ = A_-, \qquad B_+ = B_-, \qquad C_+ = C_-, \tag{273}$$

in matched opposite-valley frames. Then automatically

$$h_-(p) = -h_+(-p)$$

in the canonical reduced basis.

H. Dirac doubling theorem

[Opposite-valley Dirac doubling] Assume:

1. the selected shell pair $+G_*$ and $-G_*$ both carry nondegenerate Weyl valleys;
2. their reduced Hamiltonians satisfy the standard pairing relation

$$h_-(p) = -h_+(-p)$$

after subtraction of the common scalar part;

3. equivalently, in matched valley bases, the reduced coefficients satisfy

$$A_- = A_+, \qquad B_- = B_+, \qquad C_- = C_+.$$

Then the doubled Hamiltonian

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix} \tag{274}$$

is unitarily equivalent to a massless Dirac Hamiltonian.

Write

$$h_+(p) = d_i(p)\sigma^i, \qquad h_-(p) = -d_i(-p)\sigma^i.$$

Then (274) becomes

$$H_D(p) = \begin{pmatrix} d_i(p)\sigma^i & 0 \\ 0 & -d_i(-p)\sigma^i \end{pmatrix}.$$

To linear order near the crossing, $d_i(p) = M_{ij}p_j$, so

$$H_D^{\text{lin}}(p) = \begin{pmatrix} M_{ij}p_j\sigma^i & 0 \\ 0 & -M_{ij}p_j\sigma^i \end{pmatrix}.$$

Define the valley Pauli matrices ρ^μ acting on the $(+,-)$ valley index. Then

$$H_D^{\text{lin}}(p) = \rho^3 \otimes (M_{ij}p_j\sigma^i).$$

After a standard Clifford-basis rearrangement, this is exactly the massless Dirac Hamiltonian

$$H_D^{\text{lin}}(p) = \Gamma^i q_i, \qquad \{\Gamma^i, \Gamma^j\} = 2\delta^{ij},$$

with $q_i = M_{ij}p_j$.

I. Chirality cancellation and Dirac neutrality

A single Weyl node has momentum-space topological charge

$$\chi = \text{sgn}(\det M).$$

Under the standard opposite-valley relation, the partner node has opposite chirality:

$$\chi_- = -\chi_+.$$

Therefore the doubled Dirac block is topologically neutral as a whole, even though each valley separately is topologically nontrivial.

This is exactly the usual Weyl-to-Dirac doubling mechanism.

J. What is automatic and what is not

We now separate the exact logical status.

a. Automatic once pairing holds:

- the doubled block is Dirac;
- the two Weyl nodes have opposite chirality;
- the local four-component theory is massless Dirac to linear order.

b. Not automatic:

- existence of the opposite valley itself;
- equality of the reduced coefficients (A_-, B_-, C_-) and (A_+, B_+, C_+);
- the precise symmetry that enforces the pairing relation.

So the remaining opposite-valley problem is not conceptual but finite and explicit.

K. Final reduction of the local Dirac problem

The local Dirac problem is now reduced to the following finite microscopic checks:

1. compute the reduced coefficients (A_+, B_+, C_+) at $+G_*$;
2. compute the reduced coefficients (A_-, B_-, C_-) at $-G_*$;
3. verify the matching relation
$$A_+ = A_-, \qquad B_+ = B_-, \qquad C_+ = C_-;$$
4. verify that both valleys are nondegenerate, i.e.
$$A_\pm \neq 0, \qquad (B_\pm, C_\pm) \neq (0,0).$$

Once these hold, opposite-valley doubling is Dirac.

LXXXVIII. FINAL SUMMARY

The local Weyl block at $+G_*$ has the form

$$h_+(p) = A\, p_\parallel \sigma^3 + B\,(p_1\sigma^1 + p_2\sigma^2) + C\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2).$$

The opposite valley at $-G_*$ has the analogous form

$$h_-(p) = A_-\, p_\parallel \sigma^3 + B_-\,(p_1\sigma^1 + p_2\sigma^2) + C_-\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2).$$

The standard opposite-valley pairing relation

$$h_-(p) = -h_+(-p)$$

is equivalent, in matched valley bases, to

$$A_- = A_+, \qquad B_- = B_+, \qquad C_- = C_+.$$

If both valleys are nondegenerate Weyl nodes and this pairing relation holds, then the doubled block

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix}$$

is unitarily equivalent to a massless Dirac Hamiltonian.

Therefore the remaining local Dirac problem is reduced to a finite coefficient-matching check between the two opposite shell valleys.

LXXXIX. EXPLICIT OPPOSITE-VALLEY COEFFICIENT MATCHING

A. Goal

We now compute the reduced coefficients at the opposite shell point

$$-G_*$$

and compare them with those at $+G_*$. The purpose is to verify explicitly, for the minimal symmetry-compatible microscopic tensor ansatz, that

$$A_- = A_+, \qquad B_- = B_+, \qquad C_- = C_+,$$

which is equivalent to the standard pairing relation

$$h_-(p) = -h_+(-p).$$

B. Data at the positive valley

At $+G_*$, we fixed the adapted orthonormal frame

$$\{e_1, e_2, \hat{n}\}, \qquad \hat{n} = \frac{G_*}{|G_*|}.$$

For the minimal microscopic tensor ansatz,

$$N^3_{ij} = \frac{A}{\kappa_*|G_*|}\,\hat{n}_i\hat{n}_j, \tag{275}$$

$$N^1_{ij} = \frac{1}{\kappa_*|G_*|}\left(B\,\hat{n}_i e_{1j} + C\,\hat{n}_i e_{2j}\right), \tag{276}$$

$$N^2_{ij} = \frac{1}{\kappa_*|G_*|}\left(-C\,\hat{n}_i e_{1j} + B\,\hat{n}_i e_{2j}\right), \tag{277}$$

the shell contractions gave

$$A_+ = A, \qquad B_+ = B, \qquad C_+ = C.$$

C. Choice of adapted frame at the negative valley

At the opposite shell point $-G_*$, define

$$\hat{n}_- := -\hat{n}.$$

To preserve a right-handed orthonormal frame, choose

$$e_1^- := e_1, \qquad e_2^- := -e_2, \qquad \hat{n}_- := -\hat{n}. \tag{278}$$

Indeed,

$$e_1^- \times e_2^- = e_1 \times (-e_2) = -(e_1 \times e_2) = -\hat{n} = \hat{n}_-.$$

This is the natural matched opposite-valley frame.

D. Minimal tensor ansatz at the negative valley

Now write the same minimal microscopic ansatz, but in the $-G_*$ adapted frame:

$$N_{ij}^{3(-)} = \frac{A_-}{\kappa_*|G_*|}\, \hat{n}_i^- \hat{n}_j^-, \tag{279}$$

$$N_{ij}^{1(-)} = \frac{1}{\kappa_*|G_*|} \left(B_-\, \hat{n}_i^- e_{1j}^- + C_-\, \hat{n}_i^- e_{2j}^-\right), \tag{280}$$

$$N_{ij}^{2(-)} = \frac{1}{\kappa_*|G_*|} \left(-C_-\, \hat{n}_i^- e_{1j}^- + B_-\, \hat{n}_i^- e_{2j}^-\right). \tag{281}$$

Substituting (278),

$$\hat{n}^- = -\hat{n}, \qquad e_1^- = e_1, \qquad e_2^- = -e_2,$$

we obtain

$$N_{ij}^{3(-)} = \frac{A_-}{\kappa_*|G_*|}\, (-\hat{n}_i)(-\hat{n}_j) = \frac{A_-}{\kappa_*|G_*|}\, \hat{n}_i \hat{n}_j, \tag{282}$$

$$N_{ij}^{1(-)} = \frac{1}{\kappa_*|G_*|} \left(-B_-\, \hat{n}_i e_{1j} + C_-\, \hat{n}_i e_{2j}\right), \tag{283}$$

$$N_{ij}^{2(-)} = \frac{1}{\kappa_*|G_*|} \left(C_-\, \hat{n}_i e_{1j} + B_-\, \hat{n}_i e_{2j}\right). \tag{284}$$

E. Shell contractions at the negative valley

The shell-level extraction formulas at the negative valley are

$$A_-^{\mathrm{eff}} = \kappa_*|G_*|\, \hat{n}_i^- \hat{n}_j^- N_{ij}^{3(-)}, \tag{285}$$

$$B_-^{\mathrm{eff}} = \frac{\kappa_*|G_*|}{2} \Big(e_{1j}^- \hat{n}_i^- N_{ij}^{1(-)} + e_{2j}^- \hat{n}_i^- N_{ij}^{2(-)} \Big), \tag{286}$$

$$C_-^{\mathrm{eff}} = \frac{\kappa_*|G_*|}{2} \Big(e_{2j}^- \hat{n}_i^- N_{ij}^{1(-)} - e_{1j}^- \hat{n}_i^- N_{ij}^{2(-)} \Big). \tag{287}$$

We evaluate them explicitly.

a. Longitudinal coefficient. Using (282),

$$\hat{n}_i^- \hat{n}_j^- N_{ij}^{3(-)} = (-\hat{n}_i)(-\hat{n}_j)\, \frac{A_-}{\kappa_*|G_*|} \hat{n}_i \hat{n}_j = \frac{A_-}{\kappa_*|G_*|}.$$

Hence

$$A_-^{\mathrm{eff}} = A_-. \tag{288}$$

b. Transverse even coefficient. First,

$$e_{1j}^- \hat{n}_i^- N_{ij}^{1(-)} = e_{1j}(-\hat{n}_i)\, \frac{1}{\kappa_*|G_*|}\, (-B_-\, \hat{n}_i e_{1j} + C_-\, \hat{n}_i e_{2j}) = \frac{B_-}{\kappa_*|G_*|}.$$

Indeed,

$$(-\hat{n}_i)(-\hat{n}_i) = 1, \qquad e_1 \cdot e_1 = 1, \qquad e_1 \cdot e_2 = 0.$$

Second,

$$e_{2j}^- \hat{n}_i^- N_{ij}^{2(-)} = (-e_{2j})(-\hat{n}_i)\, \frac{1}{\kappa_*|G_*|}\, (C_-\, \hat{n}_i e_{1j} + B_-\, \hat{n}_i e_{2j}) = \frac{B_-}{\kappa_*|G_*|}.$$

Therefore

$$B_-^{\mathrm{eff}} = B_-. \tag{289}$$

c. Transverse odd coefficient. Now,

$$e_{2j}^- \hat{n}_i^- N_{ij}^{1(-)} = (-e_{2j})(-\hat{n}_i)\, \frac{1}{\kappa_*|G_*|}\, (-B_-\, \hat{n}_i e_{1j} + C_-\, \hat{n}_i e_{2j}) = \frac{C_-}{\kappa_*|G_*|},$$

and

$$e_{1j}^- \hat{n}_i^- N_{ij}^{2(-)} = e_{1j}(-\hat{n}_i)\, \frac{1}{\kappa_*|G_*|}\, (C_-\, \hat{n}_i e_{1j} + B_-\, \hat{n}_i e_{2j}) = -\frac{C_-}{\kappa_*|G_*|}.$$

Hence

$$C_-^{\mathrm{eff}} = C_-. \tag{290}$$

Thus the negative-valley shell contractions reproduce exactly the coefficients (A_-, B_-, C_-) appearing in the ansatz.

F. Matched opposite-valley condition

Now suppose the microscopic opposite-valley tensor family is obtained from the same coefficient triple:

$$A_- = A, \qquad B_- = B, \qquad C_- = C.$$

Then the reduced negative-valley Hamiltonian is

$$h_-(p) = A\, p_\parallel \sigma^3 + B\,(p_1\sigma^1 + p_2\sigma^2) + C\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2). \tag{291}$$

On the other hand, from the positive valley,

$$h_+(-p) = -A\, p_\parallel \sigma^3 - B\,(p_1\sigma^1 + p_2\sigma^2) - C\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2),$$

so

$$-h_+(-p) = A\, p_\parallel \sigma^3 + B\,(p_1\sigma^1 + p_2\sigma^2) + C\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2). \tag{292}$$

Comparing (291) and (292), we obtain

$$\boxed{h_-(p) = -h_+(-p).} \tag{293}$$

So in the matched opposite-valley frame (278), equality of the three coefficients is sufficient and exact.

G. Doubled Dirac block

Once (293) holds, the doubled Hamiltonian

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix} \tag{294}$$

takes the standard Weyl-pair form

$$H_D(p) = \begin{pmatrix} d_i(p)\sigma^i & 0 \\ 0 & -d_i(-p)\sigma^i \end{pmatrix}.$$

To linear order,

$$d_i(p) = M_{ij}p_j,$$

with

$$M = \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix}.$$

Therefore

$$H_D^{\text{lin}}(p) = \rho^3 \otimes (M_{ij}p_j\sigma^i),$$

which is unitarily equivalent to the massless Dirac Hamiltonian.

H. Final microscopic criterion

Thus, in the minimal ansatz, local Dirac doubling reduces to the following finite conditions:

1. $A \neq 0$,
2. $(B, C) \neq (0, 0)$,
3. the opposite valley exists,
4. in the matched opposite-valley frame,

$$A_- = A_+, \qquad B_- = B_+, \qquad C_- = C_+.$$

Once these hold, the local doubled shell theory is Dirac.

I. What is proved and what remains open

What is proved here:

1. the negative-valley adapted-frame computation is explicit;
2. the standard pairing relation

   $$h_-(p) = -h_+(-p)$$

   follows exactly if the opposite valley carries the same coefficient triple (A, B, C) in the matched frame;
3. the doubled block is Dirac.

What remains open:

1. the actual microscopic symmetry guaranteeing the existence of the opposite valley;
2. the derivation that the true TECT linearization produces the same coefficient triple at $\pm G_*$;
3. the spectral isolation of this doubled block from remote modes in the fully microscopic theory.

XC. FINAL SUMMARY

At the positive shell valley $+G_*$, the reduced Weyl Hamiltonian is

$$h_+(p) = A\, p_\parallel \sigma^3 + B\,(p_1\sigma^1 + p_2\sigma^2) + C\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2).$$

At the opposite valley $-G_*$, choose the matched right-handed frame

$$\hat{n}_- = -\hat{n}, \qquad e_1^- = e_1, \qquad e_2^- = -e_2.$$

If the microscopic tensor ansatz at the opposite valley carries the same coefficient triple,

$$A_- = A, \qquad B_- = B, \qquad C_- = C,$$

then the reduced negative-valley Hamiltonian is

$$h_-(p) = A\, p_\parallel \sigma^3 + B\,(p_1\sigma^1 + p_2\sigma^2) + C\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2) = -h_+(-p).$$

Therefore the doubled Hamiltonian

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix}$$

is the standard Weyl-pair block and is unitarily equivalent to a massless Dirac Hamiltonian.

Thus, in the minimal TECT shell ansatz, local Dirac doubling reduces to the finite coefficient-matching condition

$$A_- = A_+, \qquad B_- = B_+, \qquad C_- = C_+,$$

together with local Weyl nondegeneracy

$$A_\pm \neq 0, \qquad (B_\pm, C_\pm) \neq (0,0).$$

XCI. SPECTRAL ISOLATION OF THE LOCAL DIRAC BLOCK BUILT FROM THE COEFFICIENTS A, B, C

A. Goal

We now attach the spectral-isolation argument directly to the previously constructed local Weyl/Dirac block. The purpose is to show that once the opposite-valley doubled block exists, it remains a genuine low-energy Dirac sector of the *full* microscopic theory, provided the remote sector is gapped and the inter-sector mixing is sufficiently weak.

B. Local doubled Dirac block

From the previous steps, the positive-valley reduced Weyl block is

$$h_+(p) = A\, p_\parallel \sigma^3 + B\,(p_1\sigma^1 + p_2\sigma^2) + C\,(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2). \tag{295}$$

In the matched opposite-valley frame, if

$$A_- = A, \qquad B_- = B, \qquad C_- = C,$$

then

$$h_-(p) = -h_+(-p) + O(|p|^2). \tag{296}$$

Hence the doubled four-dimensional block is

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix}. \tag{297}$$

To linear order this is a massless Dirac block. Indeed, define

$$d(p) = Mp, \qquad M = \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix},$$

with momentum coordinates ordered as $(p_1, p_2, p_\parallel)$. Then the doubled linearized Hamiltonian is unitarily equivalent to

$$H_D^{\mathrm{lin}}(p) = \Gamma^i q_i, \qquad q_i = (Mp)_i, \qquad \{\Gamma^i, \Gamma^j\} = 2\delta^{ij}\mathbf{1}_4. \tag{298}$$

Its eigenvalues are

$$E_D^{\pm}(p) = \pm\sqrt{A^2 p_\parallel^2 + (B^2 + C^2)(p_1^2 + p_2^2)}. \tag{299}$$

Thus the local Dirac velocities are

$$v_\parallel = |A|, \qquad v_\perp = \sqrt{B^2 + C^2}.$$

C. Full Hilbert-space decomposition

Let the full microscopic one-particle Hilbert space decompose as

$$\mathcal{H}_{\mathrm{full}} = \mathcal{H}_D \oplus \mathcal{H}_R, \tag{300}$$

where:

- $\mathcal{H}_D$ is the four-dimensional local Dirac sector generated by the opposite shell pair $\pm G_*$,
- $\mathcal{H}_R$ is the orthogonal complement containing all remote states.

With respect to this decomposition, the full Hamiltonian takes block form

$$H_{\mathrm{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V^\dagger(p) & H_R(p) \end{pmatrix}. \tag{301}$$

D. Remote-gap assumption

Assume there exists $\Delta > 0$ and a neighborhood $\mathcal{N} \ni 0$ in momentum space such that

$$\mathrm{spec}\big(H_R(p)\big) \cap (-\Delta, \Delta) = \varnothing \qquad (p \in \mathcal{N}). \tag{302}$$

Equivalently,

$$\|H_R(p)^{-1}\| \le \frac{1}{\Delta} \qquad (p \in \mathcal{N}). \tag{303}$$

This is the remote-gap condition. It means that all non-Dirac modes remain uniformly massive in the low-energy window.

E. Zero-mixing assumption at the crossing

Assume next that the inter-sector mixing vanishes at the crossing:

$$V(0) = 0. \tag{304}$$

A natural sufficient reason is little-group mismatch: the Dirac block and the remote sector transform in distinct irreducible little-group sectors at $p = 0$, so constant inter-sector mixing is symmetry-forbidden. :contentReference[oaicite:1]index=1

Assume moreover that

$$\|V(p)\| \le c\,|p| \qquad (p \in \mathcal{N}) \tag{305}$$

for some constant $c > 0$.

F. Schur-complement reduction

For $|E| < \Delta/2$, the remote resolvent exists:

$$H_R(p) - E$$

is invertible. Therefore the low-energy spectrum of $H_{\mathrm{full}}(p)$ is determined by the Schur complement

$$F(E, p) = H_D(p) - E - V(p)\big(H_R(p) - E\big)^{-1}V^\dagger(p). \tag{306}$$

Equivalently,

$$E \in \mathrm{spec}(H_{\mathrm{full}}(p)) \iff 0 \in \mathrm{spec}(F(E, p)). \tag{307}$$

Define the self-energy

$$\Sigma(E, p) := V(p)\big(H_R(p) - E\big)^{-1}V^\dagger(p). \tag{308}$$

Then

$$F(E, p) = H_D(p) - E - \Sigma(E, p). \tag{309}$$

G. Quadratic bound on the self-energy

Using (302) and (305), for $|E| < \Delta/2$ one has

$$\|(H_R(p) - E)^{-1}\| \le \frac{2}{\Delta}.$$

Hence

$$\|\Sigma(E, p)\| \le \|V(p)\|^2\,\|(H_R(p) - E)^{-1}\| \le \frac{2c^2}{\Delta}|p|^2. \tag{310}$$

Thus the self-energy correction is only quadratic in momentum.
This is the central stability estimate.

H. Effective low-energy Dirac operator

Therefore the exact low-energy effective Hamiltonian on $\mathcal{H}_D$ is

$$H_{\mathrm{eff}}(E, p) = H_D(p) - \Sigma(E, p) = H_D^{\mathrm{lin}}(p) + O(|p|^2) + O\left(\frac{|p|^2}{\Delta}\right). \tag{311}$$

Using (298), we obtain

$$H_{\mathrm{eff}}(E, p) = \Gamma^i q_i + O\left(\frac{|p|^2}{\Delta}\right), \qquad q_i = (Mp)_i. \tag{312}$$

Hence the leading linear Dirac cone is unchanged by the remote sector. Remote states only generate subleading curvature corrections.

I. Mass protection

Spectral isolation alone does not automatically forbid an induced constant mass term. So we also assume a residual low-energy symmetry that excludes a Dirac mass.

A sufficient condition is the existence of a grading operator Γ_5 on $\mathcal{H}_D$ such that

$$\{\Gamma_5, \Gamma^i\} = 0, \qquad \Gamma_5^2 = \mathbf{1}_4, \tag{313}$$

and that the low-energy theory preserves this grading:

$$\Gamma_5 H_D(p)\Gamma_5 = -H_D(p), \qquad \Gamma_5 V(p)\Gamma_5 = -V(p). \tag{314}$$

Then the induced self-energy is also odd under Γ_5, so no constant Dirac mass is generated. In particular,

$$H_{\mathrm{eff}}(0, 0) = 0. \tag{315}$$

J. Spectral-isolation theorem for the A, B, C Dirac block

[Spectral isolation of the local A, B, C Dirac block] Assume:

1. the opposite shell pair $\pm G_*$ exists and yields the doubled local Dirac block (297);
2. the local Weyl coefficients satisfy
$$A \neq 0, \qquad (B, C) \neq (0, 0);$$
3. the remote sector obeys the uniform gap condition (302);
4. the inter-sector mixing satisfies
$$V(0) = 0, \qquad \|V(p)\| \leq c|p|;$$
5. a residual symmetry forbids a constant Dirac mass term.

Then, for sufficiently small p, the full microscopic Hamiltonian $H_{\mathrm{full}}(p)$ contains a spectrally isolated four-dimensional low-energy sector whose effective Hamiltonian is

$$H_{\mathrm{eff}}(p) = \Gamma^i (Mp)_i + O\left(\frac{|p|^2}{\Delta}\right), \tag{316}$$

with

$$M = \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix}, \qquad \det M = A(B^2 + C^2) \neq 0. \tag{317}$$

Consequently, the full theory contains a stable massless Dirac cone.

Assumptions (1) and (2) give a nondegenerate local Dirac block on $\mathcal{H}_D$. Assumption (3) makes the remote resolvent well-defined in a fixed low-energy window. Assumption (4) implies the quadratic self-energy bound (310), so the Schur-complement effective Hamiltonian differs from the local Dirac block only by $O(|p|^2/\Delta)$. Assumption (5) excludes a generated constant mass term. Hence the leading linear Dirac cone persists and remains spectrally isolated from all remote states.

K. Microscopic checklist in coefficient language

In the present A, B, C language, the full local Dirac-stability problem reduces to the finite checklist:

$$A \neq 0, \qquad (B, C) \neq (0, 0), \qquad A_- = A_+,\ B_- = B_+,\ C_- = C_+, \tag{318}$$

together with

$$\mathrm{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing, \qquad V(0) = 0, \qquad \|V(p)\| \leq c|p|, \tag{319}$$

and a mass-forbidding symmetry.

Thus the full local Dirac problem is no longer conceptual. It is an explicit finite coefficient-and-gap verification problem.

L. What is proved and what remains open

What is now proved:

1. the local doubled A, B, C shell block is Dirac;
2. remote states only modify it at order $O(|p|^2/\Delta)$;
3. the Dirac cone is stable if the remote gap and zero-mixing conditions hold.

What remains open:

1. the actual microscopic proof of the remote gap $\Delta > 0$,
2. the actual microscopic little-group mismatch implying $V(0) = 0$,
3. the final symmetry argument forbidding a generated mass term,
4. and the embedding of this isolated Dirac block into the full chiral gauge-matter bundle.

XCII. FINAL SUMMARY

The local Weyl coefficients extracted from the microscopic shell ansatz are

$$M = \begin{pmatrix} B & -C & 0 \\ C & B & 0 \\ 0 & 0 & A \end{pmatrix}, \qquad \det M = A(B^2 + C^2).$$

If

$$A \neq 0, \qquad (B, C) \neq (0, 0),$$

then each valley is Weyl. If, in addition, the opposite valley satisfies

$$A_- = A_+, \qquad B_- = B_+, \qquad C_- = C_+,$$

then the doubled shell block is Dirac.

Now decompose the full Hilbert space as

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R,$$

with $\mathcal{H}_D$ the four-dimensional Dirac block and $\mathcal{H}_R$ the remote sector. If the remote sector is gapped,

$$\text{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing,$$

and the inter-sector mixing obeys

$$V(0) = 0, \qquad \|V(p)\| \leq c|p|,$$

then the Schur complement gives the effective low-energy Hamiltonian

$$H_{\text{eff}}(p) = \Gamma^i (Mp)_i + O(|p|^2/\Delta).$$

Hence the full microscopic theory contains a stable massless Dirac cone, with only quadratic curvature corrections from remote states.

XCIII. LITTLE-GROUP MISMATCH IMPLIES VANISHING CONSTANT MIXING

A. Goal

We now prove the key representation-theoretic statement used in the spectral-isolation argument:

$$\boxed{\text{If the local Dirac sector and the remote sector belong to inequivalent little-group representations at } p = 0, \text{ then } V(0) = 0.}$$

This is the precise meaning of the "little-group mismatch" condition appearing in the spectral-isolation checklist.

B. Setup

Let H_* denote the little group of the selected shell point G_*, or more generally the residual symmetry group acting on the linearized microscopic theory at the crossing point $p = 0$.

Let the full Hilbert space decompose as

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R,$$

where

- $\mathcal{H}_D$ is the four-dimensional local Dirac sector built from the opposite shell pair $\pm G_*$,
- $\mathcal{H}_R$ is the orthogonal complement containing all remote modes.

Assume that H_* acts unitarily on $\mathcal{H}_{\text{full}}$, and that this action preserves the decomposition:

$$U(h)\mathcal{H}_D \subset \mathcal{H}_D, \qquad U(h)\mathcal{H}_R \subset \mathcal{H}_R, \qquad h \in H_*.$$

Thus $\mathcal{H}_D$ and $\mathcal{H}_R$ are H_*-modules.

At $p = 0$, write the full Hamiltonian in block form

$$H(0) = \begin{pmatrix} H_D(0) & V(0) \\ V(0)^\dagger & H_R(0) \end{pmatrix}. \tag{320}$$

The claim is that $V(0)$ must vanish if the two sectors are representation-theoretically mismatched.

C. Equivariance condition on the mixing block

Assume the microscopic Hamiltonian is H_*-equivariant at $p = 0$:

$$U(h)H(0)U(h)^{-1} = H(0) \qquad \forall h \in H_*.$$

Writing this in block form relative to $\mathcal{H}_D \oplus \mathcal{H}_R$, we obtain

$$\begin{pmatrix} U_D(h) & 0 \\ 0 & U_R(h) \end{pmatrix} \begin{pmatrix} H_D(0) & V(0) \\ V(0)^\dagger & H_R(0) \end{pmatrix} \begin{pmatrix} U_D(h)^{-1} & 0 \\ 0 & U_R(h)^{-1} \end{pmatrix} = \begin{pmatrix} H_D(0) & V(0) \\ V(0)^\dagger & H_R(0) \end{pmatrix}.$$

Comparing off-diagonal blocks gives

$$U_D(h)\, V(0)\, U_R(h)^{-1} = V(0) \qquad \forall h \in H_*. \tag{321}$$

Equivalently,

$$V(0) \in \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D). \tag{322}$$

Thus the constant mixing block is not an arbitrary operator. It must be an H_*-intertwiner.

D. Schur-type vanishing criterion

We now use the standard representation-theoretic fact:

> If two H_*-modules share no common irreducible component, then the intertwiner space between them is zero.

More precisely, decompose

$$\mathcal{H}_D \cong \bigoplus_\lambda m_\lambda^D\, W_\lambda, \qquad \mathcal{H}_R \cong \bigoplus_\lambda m_\lambda^R\, W_\lambda,$$

where W_λ runs over irreducible H_*-modules and $m_\lambda^{D,R}$ are multiplicities. Then

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) \cong \bigoplus_\lambda \mathrm{Mat}_{m_\lambda^D \times m_\lambda^R}(\mathbb{C}). \tag{323}$$

Therefore,

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \quad \iff \quad m_\lambda^D m_\lambda^R = 0 \text{ for all } \lambda. \tag{324}$$

In words: if $\mathcal{H}_D$ and $\mathcal{H}_R$ have no common irreducible sector, then there is no nonzero equivariant map from $\mathcal{H}_R$ to $\mathcal{H}_D$.

Combining (322) and (324), we immediately get $V(0) = 0$.

E. Main theorem

[Little-group mismatch implies zero constant mixing] Let H_* be the little group at the selected shell crossing $p = 0$. Assume:

1. the full microscopic Hamiltonian $H(0)$ is H_*-equivariant;
2. the full Hilbert space decomposes into H_*-invariant subspaces
$$\mathcal{H}_{\mathrm{full}} = \mathcal{H}_D \oplus \mathcal{H}_R;$$
3. the H_*-modules $\mathcal{H}_D$ and $\mathcal{H}_R$ share no common irreducible component, i.e.
$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0.$$

Then the constant inter-sector mixing vanishes:

$$\boxed{V(0) = 0.} \tag{325}$$

By H_*-equivariance of $H(0)$, the off-diagonal block $V(0)$ must satisfy

$$U_D(h) V(0) U_R(h)^{-1} = V(0) \qquad \forall h \in H_*.$$

Hence $V(0) \in \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D)$. By assumption, this intertwiner space is zero. Therefore $V(0) = 0$.

F. Irreducible special case

A particularly transparent corollary occurs when both sectors are irreducible.

[Irreducible mismatch] If $\mathcal{H}_D$ and $\mathcal{H}_R$ are irreducible H_*-modules and

$$\mathcal{H}_D \ncong \mathcal{H}_R,$$

then

$$V(0) = 0.$$

By Schur's lemma,

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0$$

for inequivalent irreducible modules. Then the theorem applies.

G. Why this is the right condition for the Dirac block

In the TECT setting, $\mathcal{H}_D$ is the enriched shell block built from the opposite shell pair $\pm G_*$. The remote sector $\mathcal{H}_R$ consists of all other microscopic modes.

If the enriched Dirac block transforms in a distinguished little-group sector that is absent from the remote states, then constant hybridization at the crossing is symmetry-forbidden. This is exactly the content of $V(0) = 0$.

Hence the little-group mismatch is not merely heuristic. It is the precise representation-theoretic reason why the Dirac block decouples at leading order.

H. Linear mixing and the next nontrivial order

Once $V(0) = 0$, the leading possible mixing begins at first order in momentum:

$$V(p) = V_i p_i + O(|p|^2).$$

Whether the linear term is allowed depends on the representation content of the momentum vector itself.

Indeed, since p transforms as the spatial little-group representation R_p, the linear coefficient must belong to

$$\mathrm{Hom}_{H_*}(R_p \otimes \mathcal{H}_R, \mathcal{H}_D).$$

Therefore:

- if this Hom-space also vanishes, then

$$V(p) = O(|p|^2);$$

- if it is nonzero, then the generic situation is

$$V(p) = O(|p|).$$

For the spectral-isolation theorem, $V(0) = 0$ together with $V(p) = O(|p|)$ is already sufficient, because the induced self-energy is then $O(|p|^2/\Delta)$.

I. Compatibility with the Schur-complement bound

The spectral-isolation argument used previously assumes:

1. a remote gap

$$\mathrm{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing;$$

2. zero constant mixing

$$V(0) = 0;$$

3. a bound

$$\|V(p)\| \leq c|p|.$$

The theorem proved here gives the exact representation-theoretic origin of item (2). Together with the remote gap, it implies that the self-energy correction satisfies

$$\Sigma(E, p) = V(p)\big(H_R(p) - E\big)^{-1} V(p)^\dagger = O(|p|^2/\Delta),$$

so the leading linear Dirac cone is preserved.

J. Microscopic checklist in representation language

Therefore the zero-mixing check is now completely explicit:

1. determine the little group H_* of the selected shell crossing;
2. decompose the enriched shell Dirac block $\mathcal{H}_D$ into irreducible H_*-representations;
3. decompose the remote sector $\mathcal{H}_R$ into irreducible H_*-representations;
4. verify that no irreducible representation appears in both.

If that holds, then $V(0) = 0$ follows automatically.

K. What is proved and what remains open

What is now proved rigorously:

1. constant mixing $V(0)$ is an H_*-intertwiner;
2. if $\mathcal{H}_D$ and $\mathcal{H}_R$ have no common irreducible component, then $V(0) = 0$;
3. this is the exact content of the "little-group mismatch" condition used in spectral isolation.

What remains open microscopically:

1. identify the actual little group H_* of the TECT shell crossing in the full microscopic model;
2. decompose the actual enriched Dirac block into H_*-irreps;
3. decompose the remote sector into H_*-irreps;
4. verify the absence of common irreducible sectors.

XCIV. FINAL SUMMARY

At the selected shell crossing $p = 0$, the constant inter-sector mixing block $V(0)$ is not arbitrary. If the microscopic Hamiltonian is little-group equivariant, then $V(0)$ must satisfy

$$U_D(h)V(0)U_R(h)^{-1} = V(0), \qquad h \in H_*,$$

so

$$V(0) \in \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

Therefore, if the local Dirac sector $\mathcal{H}_D$ and the remote sector $\mathcal{H}_R$ share no common irreducible representation of the little group H_*, then

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0,$$

and hence

$$V(0) = 0.$$

This is the rigorous form of the little-group mismatch criterion. It explains why the local Dirac block decouples at the crossing point and why, together with the remote gap condition, the Schur-complement self-energy is only

$$O(|p|^2/\Delta),$$

so the linear Dirac cone remains stable.

XCV. ACTUAL LITTLE GROUP FOR THE SELECTED BCC AXIS AND THE REPRESENTATION CONTENT OF THE LOCAL DIRAC BLOCK

A. Goal

We now fix the actual little group of the selected BCC shell direction and clarify the exact status of the local Dirac block.

The key point is that two distinct symmetry levels must be separated:

1. the exact discrete cubic little group of the shell direction;
2. the infrared axial remnant acting on the transverse plane in the $k \cdot p$ / continuum limit.

Confusing these two levels leads to an overstatement. The bare cubic little group does *not* by itself force an ordinary two-dimensional irreducible representation.

B. Selected shell axis

Fix the BCC shell direction

$$\hat{n} = \frac{1}{\sqrt{2}}(1, 1, 0), \qquad G_* = |G_*|\hat{n}.$$

There are two related stabilizer groups:

1. the stabilizer of the *oriented* vector $+\hat{n}$;
2. the stabilizer of the *unoriented pair* $\{\pm\hat{n}\}$.

C. Exact discrete cubic little groups

At the level of the cubic point group, the exact statements are:

$$\mathrm{Stab}(+\hat{n}) \cong C_{2v}, \qquad \mathrm{Stab}(\{\pm\hat{n}\}) \cong D_{2h}. \tag{326}$$

Moreover, all ordinary complex irreducible representations of C_{2v} and D_{2h} are one-dimensional. Therefore:

$$\boxed{\text{the bare cubic little group does not force an ordinary complex 2D irrep.}} \tag{327}$$

This is the exact group-theoretic obstruction.

D. What remains true at the kinematic level

Even though the discrete cubic little group has only one-dimensional ordinary irreps, the spatial vector space still splits canonically as

$$\mathbb{R}^3 = \mathbb{R}\hat{n} \oplus \hat{n}^\perp.$$

Hence there is always a longitudinal/transverse decomposition:

$$\mathbf{3} \to \mathbf{1} \oplus (\text{2D transverse plane}).$$

This is a vector-space splitting, not yet an irreducible-representation statement.

E. Infrared axial remnant

For the continuum $k \cdot p$ analysis, the relevant symmetry is the infrared axial remnant acting on the transverse plane:

$$\hat{n}^{\perp}.$$

At that level, the two transverse polarizations furnish a real two-dimensional representation, equivalently a complex two-component low-energy envelope.

Thus the correct statement is:

$$\boxed{\text{the 2D transverse structure is forced at the IR axial level, not by bare cubic ordinary irreps alone.}} \tag{328}$$

F. Shell-level transverse doublet

Let

$$\Pi_{\perp} = \hat{n}^{\perp} = \mathrm{span}\{e_1, e_2\}.$$

For the selected shell direction, define the spatial witness tensors

$$W_{ij}^{(\alpha)} = e_i^{\alpha} \hat{n}_j, \qquad \alpha = 1, 2.$$

Under the residual shell symmetry,

$$h \cdot \hat{n} = \hat{n}, \qquad h \cdot e_{\alpha} = R(h)_{\alpha}{}^{\beta} e_{\beta},$$

so

$$h \cdot W^{(\alpha)} = R(h)_{\alpha}{}^{\beta} W^{(\beta)}.$$

Hence the space

$$W_{\mathrm{mix}} = \mathrm{span}\{W^{(1)}, W^{(2)}\} = \Pi_{\perp} \otimes \hat{n}$$

transforms as the transverse shell doublet.

This is the correct shell-level 2D representation entering the Dirac coupling test.

G. Internal matching doublet

Now let the internal-channel space contain a two-dimensional subrepresentation

$$U_D = \mathrm{span}\{u^{(1)}, u^{(2)}\}$$

transforming in the same transverse way. Then the tensor product

$$W_{\mathrm{mix}} \otimes U_D$$

contains an invariant line. This is exactly the representation-theoretic reason why the shell couplings survive once a matching internal doublet exists.

So the true local Dirac carrier is not

$$\text{bare cubic 2D irrep}$$

but rather

$$\text{transverse shell doublet} \otimes \text{matching internal doublet}.$$

H. Actual structure of the local Dirac block

Therefore the local enriched shell Dirac block should be understood as follows:

$$\mathcal{H}_D \cong \mathbb{C}^2_{\mathrm{pair}} \otimes \mathbb{C}^2_{\mathrm{int}}, \tag{329}$$

where

- $\mathbb{C}^2_{\mathrm{pair}}$ is the opposite-shell pair index $\pm G_*$,
- $\mathbb{C}^2_{\mathrm{int}}$ is the internal transverse doublet sector.

The first factor is not an ordinary irrep of the single-wavevector little group; it comes from pairing the opposite valleys. The second factor is the actual doublet carrier relevant for the shell-level Dirac theorem.

Thus the four-dimensional Dirac block is

$$\mathcal{H}_D = \mathbb{C}^2_{\mathrm{valley}} \otimes \mathbb{C}^2_{\mathrm{int}}.$$

I. What is actually proved

We may now state the strongest honest theorem.

[Actual representation content of the local Dirac block] For the selected BCC shell axis $\hat{n} \parallel (1,1,0)$:

1. the exact oriented cubic stabilizer is C_{2v}, and the pair stabilizer is D_{2h};
2. the ordinary complex irreducible representations of these exact discrete groups are one-dimensional, so bare cubic symmetry alone does not force a 2D ordinary irrep;
3. nevertheless, the shell analysis canonically supplies a 2D transverse plane $\Pi_\perp$, and the infrared axial remnant acts on it as a real two-dimensional transverse representation;
4. if the microscopic enriched sector contains a matching internal transverse doublet, then the shell-coupling test closes and the local reduced block is carried by

$$\mathbb{C}^2_{\mathrm{pair}} \otimes \mathbb{C}^2_{\mathrm{int}}.$$

J. Consequence for the zero-mixing lemma

This also clarifies the zero-mixing statement. The "little-group mismatch" used in

$$V(0) = 0$$

should not be interpreted as mismatch between ordinary bare-cubic 2D irreps. Rather, it means mismatch between the actual shell-relevant representation sectors:

- the enriched Dirac block sector

$$\mathbb{C}^2_{\mathrm{pair}} \otimes \mathbb{C}^2_{\mathrm{int}},$$

- and the remote sectors appearing in the microscopic linearization.

If no common shell-relevant irrep sector is shared, then

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0$$

and therefore

$$V(0) = 0.$$

K. What remains open

The remaining open microscopic problem is now sharp:

1. identify the actual internal doublet inside the TECT enriched linearization;
2. identify the actual remote-sector irrep content;
3. verify that the enriched Dirac block and remote sector share no common shell-relevant representation;
4. verify the remote spectral gap.

XCVI. FINAL SUMMARY

For the selected BCC shell axis

$$\hat{n} = \frac{1}{\sqrt{2}}(1, 1, 0),$$

the exact cubic little groups are

$$\mathrm{Stab}(+\hat{n}) \cong C_{2v}, \qquad \mathrm{Stab}(\{\pm\hat{n}\}) \cong D_{2h}.$$

Their ordinary complex irreducible representations are one-dimensional. Therefore the bare cubic little group alone does not force an ordinary two-dimensional irrep.

However, the shell analysis still gives the canonical longitudinal/transverse splitting

$$\mathbb{R}^3 = \mathbb{R}\hat{n} \oplus \hat{n}^{\perp},$$

and in the infrared $k \cdot p$ / continuum analysis the transverse plane $\hat{n}^{\perp}$ carries the relevant axial two-dimensional representation.

Thus the local Dirac block is not built from a bare cubic 2D irrep. It is built from

$$\mathbb{C}^2_{\mathrm{pair}} \otimes \mathbb{C}^2_{\mathrm{int}},$$

namely the opposite-shell pair factor times the matching internal transverse doublet.

Accordingly, the little-group mismatch condition used in the zero-mixing lemma means mismatch between this shell-relevant enriched Dirac sector and the remote-sector representation content, not mismatch between ordinary bare-cubic 2D irreps.

XCVII. REMOTE-SECTOR REPRESENTATION CHECKLIST FOR THE ZERO-MIXING CONDITION

A. Goal

The purpose of this section is to turn the abstract zero-mixing statement

$$V(0) = 0$$

into a practical finite verification procedure.

The key theorem already proved is:

$$V(0) \in \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D), \tag{330}$$

so if

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0,$$

then

$$V(0) = 0.$$

What remains is to identify the shell-relevant representation content of the local Dirac block $\mathcal{H}_D$ and compare it with the remote sector $\mathcal{H}_R$.

B. Shell-relevant form of the Dirac block

The local Dirac block is not to be understood as a bare cubic 2D irrep. Its actual shell-relevant structure is

$$\mathcal{H}_D \cong \mathbb{C}^2_{\text{pair}} \otimes \mathbb{C}^2_{\text{int}}, \tag{331}$$

where:

- $\mathbb{C}^2_{\text{pair}}$ is the opposite-shell pair sector built from $\pm G_*$,
- $\mathbb{C}^2_{\text{int}}$ is the matching internal transverse doublet.

Thus the representation content relevant for zero-mixing is the shell-enriched sector carried by (331).

C. Remote-sector decomposition

Let the remote sector decompose into shell-relevant irreducible sectors:

$$\mathcal{H}_R \cong \bigoplus_\lambda m^R_\lambda W_\lambda. \tag{332}$$

Likewise decompose the Dirac block as

$$\mathcal{H}_D \cong \bigoplus_\lambda m^D_\lambda W_\lambda. \tag{333}$$

Then

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) \cong \bigoplus_\lambda \mathrm{Mat}_{m^D_\lambda \times m^R_\lambda}(\mathbb{C}). \tag{334}$$

Therefore the constant mixing vanishes if and only if

$$m^D_\lambda m^R_\lambda = 0 \qquad \forall\lambda. \tag{335}$$

D. Practical checklist

To verify $V(0) = 0$, it is sufficient to carry out the following finite steps.

a. Step 1: fix the shell data. Choose the shell pair

$$\pm G_*, \qquad \hat{n} = \frac{G_*}{|G_*|},$$

and fix the matched adapted frames at $\pm G_*$.

b. Step 2: identify the shell-relevant symmetry group. Determine the actual residual symmetry H_* acting on the selected shell sector. At the exact cubic level this is the appropriate stabilizer subgroup; at the shell-reduction level one uses the shell-relevant axial/transverse representation data.

c. Step 3: identify the Dirac carrier. Construct the enriched Dirac block

$$\mathcal{H}_D \cong \mathbb{C}^2_{\text{pair}} \otimes \mathbb{C}^2_{\text{int}}.$$

Then decompose $\mathcal{H}_D$ into irreducible H_*-sectors.

d. Step 4: identify the remote carriers. Decompose the remote modes into:

- volumetric sector,
- non-enriched shell sectors,
- higher-shell sectors,
- internal sectors not belonging to the matched transverse doublet.

Then decompose each into irreducible H_*-sectors.

e. Step 5: compare the sectors. Check whether any irrep appearing in $\mathcal{H}_D$ also appears in $\mathcal{H}_R$. If none does, then

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0.$$

f. Step 6: conclude zero mixing. By equivariance,

$$V(0) \in \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D),$$

hence

$$V(0) = 0.$$

E. Sufficient exclusion rules

In practice, the no-common-sector condition is usually guaranteed if the remote states differ from the Dirac block in at least one of the following labels:

1. opposite-shell pair label is absent;
2. internal transverse-doublet label is absent;
3. parity/inversion label differs;
4. chirality or grading label differs;
5. shell-order label differs;
6. volumetric/transverse character differs.

Any such mismatch can be enough to force the intertwiner space to vanish.

F. Minimal operational theorem

[Operational zero-mixing criterion] Let H_* be the shell-relevant residual symmetry group at $p = 0$. Assume:

1. the full Hamiltonian is H_*-equivariant at $p = 0$;
2. the local Dirac block $\mathcal{H}_D$ and the remote sector $\mathcal{H}_R$ are H_*-invariant;
3. after decomposition into irreducible shell-relevant sectors, $\mathcal{H}_D$ and $\mathcal{H}_R$ share no common irreducible component.

Then

$$V(0) = 0.$$

By equivariance, $V(0)$ is an H_*-intertwiner from $\mathcal{H}_R$ to $\mathcal{H}_D$. If there is no common irreducible sector, the intertwiner space vanishes. Hence $V(0) = 0$.

G. Failure mode

The only way the zero-mixing test can fail is if a remote mode carries the same shell-relevant representation sector as the Dirac block. Then

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) \neq 0,$$

and constant mixing is symmetry-allowed.

In that case, one must either:

1. enlarge the low-energy block to include that mode, or
2. use an additional symmetry/grading to rule out the mixing, or
3. show dynamically that the corresponding matrix element vanishes.

H. Connection to the master checklist

This completes the rigorous form of the checklist item:

$$\text{Little-group mismatch check} \Longrightarrow V(0) = 0.$$

Together with the remote gap condition and the mass-protection symmetry, this feeds directly into the spectral-isolation theorem for the local Dirac block.

XCVIII. FINAL SUMMARY

To verify the zero-mixing condition $V(0) = 0$, one must compare the shell-relevant representation content of the local Dirac block

$$\mathcal{H}_D \cong \mathbb{C}^2_{\text{pair}} \otimes \mathbb{C}^2_{\text{int}}$$

with that of the remote sector

$$\mathcal{H}_R.$$

If the Hamiltonian is H_*-equivariant at $p = 0$, then the constant mixing block satisfies

$$V(0) \in \text{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D).$$

Therefore

$$V(0) = 0$$

whenever $\mathcal{H}_D$ and $\mathcal{H}_R$ share no common irreducible shell-relevant sector.

Operationally, the check is finite: fix the shell pair, identify the shell-relevant symmetry group, decompose both the Dirac block and the remote sector into irreducible sectors, and verify that no sector appears in both. If that holds, then zero constant mixing follows automatically.

XCIX. REMOTE-SECTOR DECOMPOSITION AND THE PRACTICAL MISMATCH CHECKLIST

A. Goal

We now decompose the remote sector into physically meaningful components and determine, sector by sector, whether constant mixing with the local Dirac block is symmetry-allowed.

The local Dirac carrier is

$$\mathcal{H}_D \cong \mathbb{C}^2_{\text{pair}} \otimes \mathbb{C}^2_{\text{int}}, \tag{336}$$

where

- $\mathbb{C}^2_{\text{pair}}$ is the $\pm G_*$ shell-pair space,
- $\mathbb{C}^2_{\text{int}}$ is the matched internal transverse doublet.

The question is whether any remote sector contains the same shell-relevant representation content.

B. Canonical decomposition of the remote sector

The most useful practical decomposition is

$$\mathcal{H}_R = \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non–enr}} \oplus \mathcal{H}_{\text{wrong–int}} \oplus \mathcal{H}_{\text{extra}}, \tag{337}$$

where:

1. $\mathcal{H}_{\text{vol}}$: volumetric / longitudinal sector;
2. $\mathcal{H}_{\text{high}}$: higher-shell sector not built from the selected $\pm G_*$ pair;
3. $\mathcal{H}_{\text{non–enr}}$: shell states without the enriched internal doublet;
4. $\mathcal{H}_{\text{wrong–int}}$: states with internal content not matching the selected doublet;
5. $\mathcal{H}_{\text{extra}}$: any additional sector specific to the microscopic realization.

C. Sector 1: volumetric sector

The local Dirac block is built from the transverse shell sector. Hence the volumetric sector differs already at the level of polarization character.

Therefore, unless the microscopic theory contains an accidental symmetry identifying longitudinal and transverse modes, one expects

$$\mathrm{Hom}_{H_*}(\mathcal{H}_{\text{vol}}, \mathcal{H}_D) = 0. \tag{338}$$

So the default status of the volumetric sector is:

$$\boxed{\text{volumetric/longitudinal character mismatch} \implies V_{\text{vol}\to D}(0) = 0.} \tag{339}$$

D. Sector 2: higher-shell sector

The Dirac block is built from the selected shell pair $\pm G_*$. Any state belonging to a different shell orbit carries a different shell label. Thus, unless the microscopic Hamiltonian collapses shell labels at $p = 0$, one has a shell-label mismatch.

Therefore the default expectation is

$$\mathrm{Hom}_{H_*}(\mathcal{H}_{\text{high}}, \mathcal{H}_D) = 0. \tag{340}$$

So:

$$\boxed{\text{different shell orbit} \implies V_{\text{high}\to D}(0) = 0.} \tag{341}$$

E. Sector 3: non-enriched shell sector

These are states that may live on the same shell but do not carry the enriched internal doublet needed for the Dirac block. Thus they fail the internal matching condition.

Hence

$$\mathrm{Hom}_{H_*}(\mathcal{H}_{\text{non–enr}}, \mathcal{H}_D) = 0 \tag{342}$$

provided the internal doublet is genuinely absent.

So:

$$\boxed{\text{same shell but no enriched internal doublet} \implies V_{\text{non–enr}\to D}(0) = 0.} \tag{343}$$

F. Sector 4: wrong internal sector

This sector may have a nontrivial internal structure, but not the one matching the selected transverse doublet. For example, it may carry:

- a singlet instead of a doublet,
- a different doublet not identified with the shell-relevant one,
- or a larger representation whose relevant multiplicity does not match.

If there is no common matching internal sector, then

$$\mathrm{Hom}_{H_*}(\mathcal{H}_{\text{wrong–int}}, \mathcal{H}_D) = 0. \tag{344}$$

Thus:

$$\boxed{\text{wrong internal representation} \implies V_{\text{wrong–int}\to D}(0) = 0.} \tag{345}$$

G. Sector 5: extra microscopic sectors

Any additional sector must be tested case by case. However, the same criterion applies:

$$V(0) = 0 \quad \Longleftrightarrow \quad \mathrm{Hom}_{H_*}(\mathcal{H}_{\text{extra}}, \mathcal{H}_D) = 0.$$

So no new conceptual ingredient is needed.

H. Compact checklist

The practical test can be compressed into the following decision rule.

A remote sector $\mathcal{K} \subset \mathcal{H}_R$ can mix at $p = 0$ with the Dirac block only if it matches $\mathcal{H}_D$ in *all* of the following shell-relevant labels:

1. selected shell-pair label $\pm G_*$,
2. transverse/non-volumetric character,
3. enriched internal doublet label,
4. residual parity / grading / chirality labels, if present,
5. any extra microscopic conserved label carried by $\mathcal{H}_D$.

If even one label differs, then that sector is excluded from constant mixing.

I. Practical theorem

[Sectorwise zero-mixing criterion] Let

$$\mathcal{H}_R = \bigoplus_a \mathcal{K}_a$$

be any decomposition of the remote sector into H_*-invariant sectors. Assume the full Hamiltonian is H_*-equivariant at $p = 0$. If each $\mathcal{K}_a$ fails to match the Dirac block $\mathcal{H}_D$ in at least one shell-relevant label, then

$$\mathrm{Hom}_{H_*}(\mathcal{K}_a, \mathcal{H}_D) = 0 \qquad \forall a,$$

hence

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0$$

and therefore

$$V(0) = 0.$$

Since the decomposition is H_*-invariant,

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) \cong \bigoplus_a \mathrm{Hom}_{H_*}(\mathcal{K}_a, \mathcal{H}_D).$$

If every summand vanishes, then the full Hom-space vanishes. By the previously proved equivariance theorem, $V(0)$ must belong to this Hom-space. Hence $V(0) = 0$.

J. Failure mode

The only dangerous case is when a remote sector $\mathcal{K}$ shares the full shell-relevant label set of $\mathcal{H}_D$. Then

$$\mathrm{Hom}_{H_*}(\mathcal{K}, \mathcal{H}_D) \neq 0$$

may occur, and constant mixing becomes symmetry-allowed.

If that happens, one must do one of the following:

1. enlarge the Dirac block to include $\mathcal{K}$,
2. invoke an additional symmetry to forbid the matrix element,
3. or show dynamically that the matrix element vanishes.

K. Operational research order

The most efficient order of execution is:

1. identify the enriched Dirac carrier $\mathcal{H}_D$,
2. split $\mathcal{H}_R$ into the four canonical sectors in (337),
3. test shell-pair label mismatch,
4. test volumetric/transverse mismatch,
5. test enriched-internal-doublet mismatch,
6. then only for surviving cases test the finer grading/parity labels.

In generic situations, the first three tests already eliminate all constant mixing channels.

C. FINAL SUMMARY

The local Dirac block is carried by

$$\mathcal{H}_D \cong \mathbb{C}^2_{\mathrm{pair}} \otimes \mathbb{C}^2_{\mathrm{int}}.$$

To verify the zero-mixing condition $V(0) = 0$, decompose the remote sector as

$$\mathcal{H}_R = \mathcal{H}_{\mathrm{vol}} \oplus \mathcal{H}_{\mathrm{high}} \oplus \mathcal{H}_{\mathrm{non\text{–}enr}} \oplus \mathcal{H}_{\mathrm{wrong\text{–}int}} \oplus \mathcal{H}_{\mathrm{extra}}.$$

A remote sector can mix constantly with $\mathcal{H}_D$ only if it matches the Dirac block in all shell-relevant labels: the selected shell pair, transverse character, enriched internal doublet, and any additional residual labels. If any one of these labels differs, then the corresponding intertwiner space vanishes and that sector gives no constant mixing.

Thus the practical zero-mixing test is finite and sectorwise: check each remote sector against the Dirac carrier label set. If all sectors mismatch in at least one label, then

$$V(0) = 0.$$

CI. MINIMAL DERIVATION OF THE TECT CONDENSATE PDE

A. Honest scope

At the current stage of TECT, the full unique microscopic condensate equation

$$\partial_t \Phi = F(\Phi, \nabla\Phi)$$

has not yet been derived from first principles. What can be derived rigorously now is the *minimal effective condensate PDE* compatible with the already-closed structural requirements:

1. translational invariance,
2. spatial isotropy before shell selection,
3. finite-shell instability at $|k| = q_*$,
4. quartic stabilization,
5. BCC shell selection at the nonlinear level.

Thus the present result is an effective microscopic PDE derivation in the Landau–Brazovskii class, not yet the final unique UV completion.

B. Minimal symmetry requirements

Let $\Phi(x,t)$ denote the condensate order parameter field. The weakest assumptions compatible with the already established TECT shell architecture are:

1. translation invariance,
2. rotational invariance before condensation,
3. instability at a nonzero shell radius q_*,
4. quartic nonlinear stabilization.

Under these assumptions, the quadratic kernel must attain its minimum on the momentum shell

$$|k| = q_*,$$

rather than at $k = 0$.

C. Minimal symmetry requirements

Let $\Phi(x,t)$ denote the condensate order parameter field. The weakest assumptions compatible with the already established TECT shell architecture are:

1. translation invariance,
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3. instability at a nonzero shell radius q_*,
4. quartic nonlinear stabilization.

Under these assumptions, the quadratic kernel must attain its minimum on the momentum shell

$$|k| = q_*,$$

rather than at $k = 0$.

D. Minimal Landau–Brazovskii functional

The minimal local free-energy functional with the required shell instability is

$$\mathcal{F}[\Phi] = \int d^3x \left[\frac{r}{2}|\Phi|^2 + \frac{\kappa}{2}|\nabla\Phi|^2 + \frac{\lambda}{2}|\nabla^2\Phi|^2 + \frac{u}{4}|\Phi|^4 \right], \tag{346}$$

with

$$\lambda > 0, \qquad u > 0.$$

To obtain a finite-shell instability one needs

$$\kappa < 0.$$

Then the quadratic kernel is minimized at a nonzero shell radius.

E. Shell radius

In momentum space, the quadratic dispersion is

$$\omega(k) = r + \kappa|k|^2 + \lambda|k|^4. \tag{347}$$

Its stationary points satisfy

$$\frac{d\omega}{d|k|} = 2|k|(\kappa + 2\lambda|k|^2) = 0.$$

Besides the trivial solution $|k| = 0$, the nonzero minimum occurs at

$$q_*^2 = -\frac{\kappa}{2\lambda}, \qquad (\kappa < 0,\ \lambda > 0). \tag{348}$$

Thus the instability shell is precisely

$$|k| = q_*.$$

F. Condensate PDE

Using relaxational dynamics,

$$\partial_t\Phi = -\frac{\delta\mathcal{F}}{\delta\Phi^\dagger}, \tag{349}$$

we compute

$$\frac{\delta\mathcal{F}}{\delta\Phi^\dagger} = r\Phi - \kappa\nabla^2\Phi + \lambda\nabla^4\Phi + u|\Phi|^2\Phi. \tag{350}$$

Therefore the minimal TECT condensate PDE is

$$\boxed{\partial_t\Phi = -r\Phi + \kappa\nabla^2\Phi - \lambda\nabla^4\Phi - u|\Phi|^2\Phi.} \tag{351}$$

Equivalently, one may write the conservative form

$$i\partial_t\Phi = r\Phi - \kappa\nabla^2\Phi + \lambda\nabla^4\Phi + u|\Phi|^2\Phi, \tag{352}$$

but for condensation and shell selection the gradient-flow form (351) is the natural minimal choice.

G. Linearization around the BCC background

Write

$$\Phi = \Phi_0 + \eta.$$

Linearizing (351) gives

$$\partial_t \eta = -\mathbb{L}_{\Phi_0} \eta, \tag{353}$$

with

$$\mathbb{L}_{\Phi_0} = r - \kappa \nabla^2 + \lambda \nabla^4 + 2u|\Phi_0|^2 \tag{354}$$

in the simplest real scalar reduction.

In the complex or enriched case, the fluctuation operator is of Bogoliubov type,

$$i\partial_t \eta = \mathbb{L}_{\Phi_0} \eta + \mathbb{M}_{\Phi_0} \bar{\eta}, \tag{355}$$

and Bloch reduction of (358) around the selected shell pair $\pm G_*$ produces the reduced shell operator whose first-order coefficients are the already-identified Weyl data

$$\lambda_\parallel, \qquad \alpha, \qquad \beta.$$

H. Linearization around the BCC background

Write

$$\Phi = \Phi_0 + \eta.$$

Linearizing (351) gives

$$\partial_t \eta = -\mathbb{L}_{\Phi_0} \eta, \tag{356}$$

with

$$\mathbb{L}_{\Phi_0} = r - \kappa \nabla^2 + \lambda \nabla^4 + 2u|\Phi_0|^2 \tag{357}$$

in the simplest real scalar reduction.

In the complex or enriched case, the fluctuation operator is of Bogoliubov type,

$$i\partial_t \eta = \mathbb{L}_{\Phi_0} \eta + \mathbb{M}_{\Phi_0} \bar{\eta}, \tag{358}$$

and Bloch reduction of (358) around the selected shell pair $\pm G_*$ produces the reduced shell operator whose first-order coefficients are the already-identified Weyl data

$$\lambda_\parallel, \qquad \alpha, \qquad \beta.$$

I. What is closed and what is still open

The following is now effectively closed:

$$\boxed{\text{The minimal effective TECT condensate PDE is of Landau–Brazovskii type.}} \tag{359}$$

Namely,

$$\partial_t \Phi = -r\Phi + \kappa \nabla^2 \Phi - \lambda \nabla^4 \Phi - u|\Phi|^2 \Phi$$

is the minimal PDE forced by translation invariance, pre-condensation isotropy, finite-shell instability, and quartic stabilization.

What remains open is the fully enriched microscopic completion:

1. the unique internal multi-component completion,
2. the actual microscopic origin of the enriched tensors N^a_{ij},
3. the proof of opposite-valley existence and pairing,
4. the full UV derivation of the shell-enriched Dirac block.

CII. FINAL SUMMARY

Starting from the minimal structural requirements of TECT — translation invariance, pre-condensation isotropy, finite-shell instability, and quartic stabilization — the minimal effective condensate free energy is

$$\mathcal{F}[\Phi] = \int d^3x \left[\frac{r}{2}|\Phi|^2 + \frac{\kappa}{2}|\nabla\Phi|^2 + \frac{\lambda}{2}|\nabla^2\Phi|^2 + \frac{u}{4}|\Phi|^4\right],$$

with

$$\kappa < 0, \qquad \lambda > 0, \qquad u > 0.$$

Its quadratic kernel is minimized on the shell

$$|k| = q_*, \qquad q_*^2 = -\frac{\kappa}{2\lambda}.$$

The corresponding minimal condensate PDE is

$$\boxed{\partial_t\Phi = -r\Phi + \kappa\nabla^2\Phi - \lambda\nabla^4\Phi - u|\Phi|^2\Phi.}$$

Projecting onto the BCC shell gives the BCC condensate amplitude equation

$$\left(r - \frac{\kappa^2}{4\lambda}\right)A + \beta_{\mathrm{BCC}}uA^3 = 0.$$

Thus the Landau–Brazovskii class is the minimal effective condensate PDE class compatible with the already-closed TECT shell architecture.

The remaining open step is not the existence of this minimal PDE class, but the fully enriched microscopic completion that generates the tensors N^a_{ij}, the local Weyl coefficients, and opposite-valley Dirac doubling.

A. Minimal condensate PDE

The minimal effective condensate PDE is

$$\partial_t\Phi = -r\Phi + \kappa\nabla^2\Phi - \lambda\nabla^4\Phi - u|\Phi|^2\Phi, \tag{360}$$

with

$$\kappa < 0, \qquad \lambda > 0, \qquad u > 0.$$

Its quadratic kernel

$$\omega(k) = r + \kappa|k|^2 + \lambda|k|^4$$

is minimized on the shell

$$|k| = q_*, \qquad q_*^2 = -\frac{\kappa}{2\lambda}. \tag{361}$$

B. Minimal condensate PDE

The minimal effective condensate PDE is

$$\partial_t\Phi = -r\Phi + \kappa\nabla^2\Phi - \lambda\nabla^4\Phi - u|\Phi|^2\Phi, \tag{362}$$

with

$$\kappa < 0, \qquad \lambda > 0, \qquad u > 0.$$

Its quadratic kernel

$$\omega(k) = r + \kappa|k|^2 + \lambda|k|^4$$

is minimized on the shell

$$|k| = q_*, \qquad q_*^2 = -\frac{\kappa}{2\lambda}. \tag{363}$$

C. Internal projector field

Introduce the projective internal field

$$z(x,t) \in \mathbb{C}^3, \qquad z^\dagger z = 1, \qquad z \sim e^{i\chi(x,t)} z,$$

and the rank-one projector

$$P(x,t) = z(x,t) z^\dagger(x,t), \qquad P^2 = P, \quad P^\dagger = P, \quad P = 1.$$

Hence P takes values in $\mathbb{CP}^2$.

Let the relevant reduced internal channels be denoted by Hermitian generators

$$s_1, \qquad s_2, \qquad s_3,$$

where s_1, s_2 span the shell-relevant internal transverse doublet and s_3 is the longitudinal/internal axial channel.

D. Minimal enriched shell coupling

The minimal shell-enriched first-order coupling is

$$\delta L_{\text{enr}} = \sum_{a=1}^{3} N_{ij}^a \, (\partial_i \phi_0) \, (-i\partial_j) \otimes s_a. \tag{364}$$

This is precisely the shell-reduced enriched form controlling the local Weyl coefficients. :contentReference[oaicite:3]index=3

Fix the selected shell direction

$$\hat{n} = \frac{G_*}{|G_*|},$$

and an adapted orthonormal frame

$$\{e_1, e_2, \hat{n}\}.$$

The minimal axial tensor structures are generated by

$$\hat{n}_i \hat{n}_j, \qquad \hat{n}_i e_{1j}, \qquad \hat{n}_i e_{2j}.$$

Equivalently, on the transverse plane one may use the identity and complex-structure basis

$$I_2, \qquad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

E. Minimal projector/enriched operator ansatz

The smallest actual enriched completion compatible with the shell-reduced Weyl template is

$$\begin{aligned} \delta L_{\text{enr}} = g_\parallel \, (\partial_i \phi_0) \hat{n}_i \, (\hat{n} \cdot (-i\nabla)) \otimes s_3 \\ + g_I \, (\partial_i \phi_0) \hat{n}_i \Big[(e_1 \cdot (-i\nabla)) \otimes s_1 + (e_2 \cdot (-i\nabla)) \otimes s_2 \Big] \\ + g_J \, (\partial_i \phi_0) \hat{n}_i \Big[-(e_2 \cdot (-i\nabla)) \otimes s_1 + (e_1 \cdot (-i\nabla)) \otimes s_2 \Big]. \end{aligned} \tag{365}$$

Here:

- $g_\parallel$ is the longitudinal enriched coupling,
- g_I is the transverse-even coupling,
- g_J is the transverse-odd coupling.

F. Direct extraction of the microscopic tensors

Comparing (365) with the master tensor form

$$\delta L_{\mathrm{enr}} = \sum_{a=1}^{3} N^a_{ij}\,(\partial_i \phi_0)\,(-i\partial_j) \otimes s_a,$$

we read off immediately:

$$\boxed{N^3_{ij} = g_\parallel\, \hat{n}_i \hat{n}_j,} \tag{366}$$

$$\boxed{N^1_{ij} = g_I\, \hat{n}_i e_{1j} - g_J\, \hat{n}_i e_{2j},} \tag{367}$$

$$\boxed{N^2_{ij} = g_I\, \hat{n}_i e_{2j} + g_J\, \hat{n}_i e_{1j}.} \tag{368}$$

These are the explicit microscopic enriched tensors generated by the minimal projector/enriched completion.

G. Extraction of the Weyl coefficients

Using the shell-level contraction formulas already established in the shell-reduction theorem, :contentReference[oaicite:5]index=5

$$\lambda_\parallel = \kappa_* |G_*|\, \hat{n}_i \hat{n}_j N^3_{ij}, \tag{369}$$

$$\alpha = \frac{\kappa_* |G_*|}{2} \left(e_{1j} \hat{n}_i N^1_{ij} + e_{2j} \hat{n}_i N^2_{ij} \right), \tag{370}$$

$$\beta = \frac{\kappa_* |G_*|}{2} \left(e_{2j} \hat{n}_i N^1_{ij} - e_{1j} \hat{n}_i N^2_{ij} \right), \tag{371}$$

we substitute (366)–(368).

For the longitudinal coefficient:

$$\hat{n}_i \hat{n}_j N^3_{ij} = g_\parallel (\hat{n} \cdot \hat{n})^2 = g_\parallel,$$

hence

$$\boxed{\lambda_\parallel = \kappa_* |G_*|\, g_\parallel.} \tag{372}$$

For α:

$$e_{1j} \hat{n}_i N^1_{ij} = g_I (e_1 \cdot e_1) - g_J (e_1 \cdot e_2) = g_I,$$

$$e_{2j} \hat{n}_i N^2_{ij} = g_I (e_2 \cdot e_2) + g_J (e_2 \cdot e_1) = g_I,$$

so

$$\boxed{\alpha = \kappa_* |G_*|\, g_I.} \tag{373}$$

For β:

$$e_{2j} \hat{n}_i N^1_{ij} = g_I (e_2 \cdot e_1) - g_J (e_2 \cdot e_2) = -g_J,$$

$$e_{1j} \hat{n}_i N^2_{ij} = g_I (e_1 \cdot e_2) + g_J (e_1 \cdot e_1) = g_J,$$

therefore

$$\boxed{\beta = -\kappa_* |G_*|\, g_J.} \tag{374}$$

The overall sign of β depends on the transverse orientation convention; only β^2 is invariant.

H. Explicit Weyl matrix and determinant

The reduced local Weyl block is

$$h_+(p) = \lambda_\parallel p_\parallel \sigma^3 + \alpha(p_1\sigma^1 + p_2\sigma^2) + \beta(-p_2\sigma^1 + p_1\sigma^2). \tag{375}$$

Hence the coefficient matrix is

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_\parallel \end{pmatrix}. \tag{376}$$

Therefore

$$\det M = \lambda_\parallel(\alpha^2 + \beta^2). \tag{377}$$

Substituting (372)–(374),

$$\boxed{\det M = (\kappa_*|G_*|)^3\, g_\parallel\, (g_I^2 + g_J^2).} \tag{378}$$

Thus the local Weyl nondegeneracy condition is

$$\boxed{g_\parallel \neq 0, \qquad (g_I, g_J) \neq (0,0).} \tag{379}$$

I. What has now been achieved

We have now explicitly combined:

1. the minimal Landau–Brazovskii condensate PDE,
2. the minimal projector-field internal sector,
3. the shell-enriched first-order coupling,
4. and the shell-reduction matching theorem.

The result is that the microscopic enriched tensors are explicitly

$$N^3_{ij} = g_\parallel\, \hat{n}_i\hat{n}_j, \qquad N^1_{ij} = g_I\, \hat{n}_i e_{1j} - g_J\, \hat{n}_i e_{2j}, \qquad N^2_{ij} = g_I\, \hat{n}_i e_{2j} + g_J\, \hat{n}_i e_{1j},$$

and therefore

$$\lambda_\parallel = \kappa_*|G_*|\, g_\parallel, \qquad \alpha = \kappa_*|G_*|\, g_I, \qquad \beta = -\kappa_*|G_*|\, g_J.$$

So the local Weyl problem has been reduced to the explicit finite coefficient test

$$g_\parallel \neq 0, \qquad (g_I, g_J) \neq (0,0).$$

J. What is closed and what remains open

What is now effectively closed:

$$\boxed{\text{Minimal Landau–Brazovskii PDE} + \text{minimal enriched projector coupling} \Longrightarrow N^a_{ij},\ \lambda_\parallel,\ \alpha,\ \beta,\ M.} \tag{380}$$

What remains open:

1. the proof that the actual microscopic TECT dynamics generates precisely this minimal enriched coupling;
2. the existence and pairing of the opposite shell valley;
3. the remote-sector gap;
4. the final mass-protection symmetry.

CIII. FINAL SUMMARY

Starting from the minimal shell-enriched projector ansatz

$$\delta L_{\rm enr} = g_\parallel(\partial_i\phi_0)\hat{n}_i(\hat{n}\cdot(-i\nabla))\otimes s_3+g_I(\partial_i\phi_0)\hat{n}_i\Big[(e_1\cdot(-i\nabla))\otimes s_1+(e_2\cdot(-i\nabla))\otimes s_2\Big]+g_J(\partial_i\phi_0)\hat{n}_i\Big[-(e_2\cdot(-i\nabla))\otimes s_1+(e_1\cdot(-i\nabla))\otimes s_2\Big],$$

the enriched microscopic tensors are explicitly

$$N^3_{ij} = g_\parallel\,\hat{n}_i\hat{n}_j,$$

$$N^1_{ij} = g_I\,\hat{n}_ie_{1j} - g_J\,\hat{n}_ie_{2j},$$

$$N^2_{ij} = g_I\,\hat{n}_ie_{2j} + g_J\,\hat{n}_ie_{1j}.$$

The shell-contraction formulas then give

$$\lambda_\parallel = \kappa_*|G_*|\,g_\parallel, \qquad \alpha = \kappa_*|G_*|\,g_I, \qquad \beta = -\kappa_*|G_*|\,g_J.$$

Hence the Weyl coefficient matrix is

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_\parallel \end{pmatrix},$$

and therefore

$$\det M = (\kappa_*|G_*|)^3\,g_\parallel\,(g_I^2+g_J^2).$$

Thus the local BCC shell crossing is Weyl whenever

$$g_\parallel \neq 0, \qquad (g_I, g_J) \neq (0,0).$$

So the local Weyl problem is now reduced to an explicit finite coefficient check on the three enriched couplings $g_\parallel, g_I, g_J$.

CIV. LANDAU–BRAZOVSKII CONDENSATE PDE WITH MINIMAL ENRICHED PROJECTOR COUPLING

A. Scope

At the present stage of TECT, the full unique UV microscopic condensate equation has not yet been derived from first principles. However, the following two facts are already structurally fixed:

1. the minimal condensate PDE must be of Landau–Brazovskii type, as a consequence of translation invariance, pre-condensation isotropy, finite-shell instability, and quartic stabilization;
2. the local Weyl/Dirac problem near a selected BCC shell pair has already been reduced to the enriched microscopic tensors

   $$N^a_{ij},$$

   through the shell-contraction formulas for

   $$\lambda_\parallel, \qquad \alpha, \qquad \beta.$$

The purpose of this section is to combine these two ingredients into a single minimal enriched effective theory and to extract the tensors N^a_{ij} explicitly.

B. Minimal Landau–Brazovskii condensate PDE

Let $\Phi(x,t)$ denote the condensate order parameter. The minimal effective free-energy functional compatible with finite-shell instability is

$$\mathcal{F}[\Phi] = \int d^3x \left[\frac{r}{2}|\Phi|^2 + \frac{\kappa}{2}|\nabla\Phi|^2 + \frac{\lambda}{2}|\nabla^2\Phi|^2 + \frac{u}{4}|\Phi|^4 \right], \tag{381}$$

with

$$\kappa < 0, \qquad \lambda > 0, \qquad u > 0.$$

The quadratic dispersion is

$$\omega(k) = r + \kappa|k|^2 + \lambda|k|^4. \tag{382}$$

Its nonzero minimum occurs at

$$q_*^2 = -\frac{\kappa}{2\lambda}. \tag{383}$$

Thus the instability shell is

$$|k| = q_*.$$

Using relaxational dynamics,

$$\partial_t\Phi = -\frac{\delta\mathcal{F}}{\delta\Phi^\dagger}, \tag{384}$$

we obtain

$$\frac{\delta\mathcal{F}}{\delta\Phi^\dagger} = r\Phi - \kappa\nabla^2\Phi + \lambda\nabla^4\Phi + u|\Phi|^2\Phi. \tag{385}$$

Hence the minimal effective condensate PDE is

$$\boxed{\partial_t\Phi = -r\Phi + \kappa\nabla^2\Phi - \lambda\nabla^4\Phi - u|\Phi|^2\Phi.} \tag{386}$$

C. BCC shell background

Let the first BCC reciprocal shell be

$$S_{\text{BCC}} = \frac{q_*}{\sqrt{2}} \{(\pm1,\pm1,0),\ (\pm1,0,\pm1),\ (0,\pm1,\pm1)\}. \tag{387}$$

Choose one representative from each antipodal pair, denoted Q_a, $a = 1,\ldots,6$, and consider the standing-wave ansatz

$$\Phi_0(x) = 2A\sum_{a=1}^{6}\cos(Q_a\cdot x). \tag{388}$$

Substituting (388) into the static Euler–Lagrange equation

$$r\Phi - \kappa\nabla^2\Phi + \lambda\nabla^4\Phi + u|\Phi|^2\Phi = 0$$

and projecting back to the BCC shell gives

$$\left(r - \frac{\kappa^2}{4\lambda}\right)A + \beta_{\text{BCC}}uA^3 = 0, \tag{389}$$

where $\beta_{\text{BCC}} > 0$ is the BCC shell quartic projection coefficient. Therefore

$$A^2 = -\frac{1}{\beta_{\text{BCC}}u}\left(r - \frac{\kappa^2}{4\lambda}\right). \tag{390}$$

D. Internal projector field

Introduce the internal projective field

$$z(x,t) \in \mathbb{C}^3, \qquad z^\dagger z = 1, \qquad z \sim e^{i\chi(x,t)} z, \tag{391}$$

and define the rank-one projector

$$P(x,t) = z(x,t) z^\dagger(x,t), \qquad P^2 = P, \qquad P^\dagger = P, \qquad P = 1. \tag{392}$$

Thus P takes values in $\mathbb{CP}^2$.

Let the shell-relevant reduced internal generators be

$$s_1, \qquad s_2, \qquad s_3,$$

where s_1, s_2 span the internal transverse doublet and s_3 is the axial/longitudinal internal channel.

E. Selected shell direction and adapted frame

Fix a selected shell vector

$$G_* \in S_{\mathrm{BCC}}, \qquad \hat{n} := \frac{G_*}{|G_*|}. \tag{393}$$

Choose an adapted orthonormal frame

$$\{e_1, e_2, \hat{n}\}, \qquad e_a \cdot \hat{n} = 0, \qquad e_a \cdot e_b = \delta_{ab}, \quad a, b = 1, 2. \tag{394}$$

Write the shell-relative momentum as

$$p = p_\parallel \hat{n} + p_1 e_1 + p_2 e_2. \tag{395}$$

F. Minimal enriched shell coupling

The minimal shell-enriched first-order operator consistent with the already established Weyl template is

$$\begin{aligned} \delta L_{\mathrm{enr}} = {} & g_\parallel \, (\partial_i \phi_0) \hat{n}_i \, (\hat{n} \cdot (-i\nabla)) \otimes s_3 \\ & + g_I \, (\partial_i \phi_0) \hat{n}_i \Big[(e_1 \cdot (-i\nabla)) \otimes s_1 + (e_2 \cdot (-i\nabla)) \otimes s_2 \Big] \\ & + g_J \, (\partial_i \phi_0) \hat{n}_i \Big[-(e_2 \cdot (-i\nabla)) \otimes s_1 + (e_1 \cdot (-i\nabla)) \otimes s_2 \Big]. \end{aligned} \tag{396}$$

Here:

- $g_\parallel$ is the longitudinal enriched coupling,
- g_I is the transverse-even coupling,
- g_J is the transverse-odd coupling.

G. Microscopic tensor extraction

The master shell-enriched tensor form is

$$\delta L_{\mathrm{enr}} = \sum_{a=1}^{3} N^a_{ij} \, (\partial_i \phi_0) \, (-i\partial_j) \otimes s_a. \tag{397}$$

Comparing (396) with (397), we read off

$$\boxed{N^3_{ij} = g_\parallel\, \hat{n}_i \hat{n}_j,} \tag{398}$$

$$\boxed{N^1_{ij} = g_I\, \hat{n}_i e_{1j} - g_J\, \hat{n}_i e_{2j},} \tag{399}$$

$$\boxed{N^2_{ij} = g_I\, \hat{n}_i e_{2j} + g_J\, \hat{n}_i e_{1j}.} \tag{400}$$

These are the explicit enriched microscopic tensors generated by the minimal projector-enriched completion of the Landau–Brazovskii condensate sector.

H. Shell-contraction formulas

The shell-reduction theorem gives the effective Weyl coefficients through the contractions

$$\lambda_\parallel = \kappa_* |G_*|\, \hat{n}_i \hat{n}_j N^3_{ij}, \tag{401}$$

$$\alpha = \frac{\kappa_* |G_*|}{2} \left(e_{1j} \hat{n}_i N^1_{ij} + e_{2j} \hat{n}_i N^2_{ij} \right), \tag{402}$$

$$\beta = \frac{\kappa_* |G_*|}{2} \left(e_{2j} \hat{n}_i N^1_{ij} - e_{1j} \hat{n}_i N^2_{ij} \right). \tag{403}$$

We now compute them explicitly.

a. Longitudinal coefficient. Using (398),

$$\hat{n}_i \hat{n}_j N^3_{ij} = g_\parallel (\hat{n} \cdot \hat{n})^2 = g_\parallel,$$

hence

$$\boxed{\lambda_\parallel = \kappa_* |G_*|\, g_\parallel.} \tag{404}$$

b. Transverse-even coefficient. Using (399) and (400),

$$e_{1j} \hat{n}_i N^1_{ij} = g_I (e_1 \cdot e_1) - g_J (e_1 \cdot e_2) = g_I,$$

$$e_{2j} \hat{n}_i N^2_{ij} = g_I (e_2 \cdot e_2) + g_J (e_2 \cdot e_1) = g_I.$$

Therefore

$$\boxed{\alpha = \kappa_* |G_*|\, g_I.} \tag{405}$$

c. Transverse-odd coefficient. Again,

$$e_{2j} \hat{n}_i N^1_{ij} = g_I (e_2 \cdot e_1) - g_J (e_2 \cdot e_2) = -g_J,$$

$$e_{1j} \hat{n}_i N^2_{ij} = g_I (e_1 \cdot e_2) + g_J (e_1 \cdot e_1) = g_J.$$

Hence

$$\boxed{\beta = -\kappa_* |G_*|\, g_J.} \tag{406}$$

The sign of β depends on the transverse orientation convention; only β^2 is invariant.

I. Reduced Weyl Hamiltonian

The local reduced Weyl Hamiltonian therefore takes the canonical form

$$h_+(p) = \lambda_{\|} p_{\|} \sigma^3 + \alpha(p_1\sigma^1 + p_2\sigma^2) + \beta(-p_2\sigma^1 + p_1\sigma^2). \tag{407}$$

Its coefficient matrix is

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_{\|} \end{pmatrix}. \tag{408}$$

Hence

$$\det M = \lambda_{\|}(\alpha^2 + \beta^2). \tag{409}$$

Substituting (404)–(406), we obtain

$$\boxed{\det M = (\kappa_* |G_*|)^3 \, g_{\|} \, (g_I^2 + g_J^2).} \tag{410}$$

J. Local Weyl criterion

Therefore the local BCC shell crossing is Weyl if and only if

$$\boxed{g_{\|} \neq 0, \qquad (g_I, g_J) \neq (0,0).} \tag{411}$$

Thus the local Weyl problem is no longer an abstract existence problem. It has been reduced to a finite coefficient test on the three enriched couplings

$$g_{\|}, \qquad g_I, \qquad g_J.$$

K. What is closed and what remains open

What is now effectively closed is the following chain:

$$\text{Landau–Brazovskii condensate PDE} + \text{minimal enriched projector coupling} \Longrightarrow N^a_{ij} \Longrightarrow \lambda_{\|}, \alpha, \beta \Longrightarrow M, \det M.$$

What remains open is not this extraction itself, but the final microscopic completion:

1. the proof that the actual TECT microscopic dynamics generates precisely this minimal enriched coupling;
2. the existence and pairing of the opposite shell valley at $-G_*$;
3. the remote-sector gap;
4. the final symmetry forbidding a constant Dirac mass.

CV. FINAL SUMMARY

Starting from the minimal shell-enriched projector ansatz

$$\delta L_{\text{enr}} = g_{\|}(\partial_i\phi_0)\hat{n}_i(\hat{n}\cdot(-i\nabla))\otimes s_3 + g_I(\partial_i\phi_0)\hat{n}_i\Big[(e_1\cdot(-i\nabla))\otimes s_1 + (e_2\cdot(-i\nabla))\otimes s_2\Big] + g_J(\partial_i\phi_0)\hat{n}_i\Big[-(e_2\cdot(-i\nabla))\otimes s_1 + (e_1\cdot(-i\nabla))\otimes s_2\Big],$$

the enriched microscopic tensors are explicitly

$$N^3_{ij} = g_{\|} \, \hat{n}_i \hat{n}_j,$$

$$N^1_{ij} = g_I\,\hat{n}_i e_{1j} - g_J\,\hat{n}_i e_{2j},$$

$$N^2_{ij} = g_I\,\hat{n}_i e_{2j} + g_J\,\hat{n}_i e_{1j}.$$

The shell-contraction formulas give

$$\lambda_\parallel = \kappa_* |G_*|\, g_\parallel, \qquad \alpha = \kappa_* |G_*|\, g_I, \qquad \beta = -\kappa_* |G_*|\, g_J.$$

Hence the Weyl coefficient matrix is

$$M = \begin{pmatrix} \alpha & -\beta & 0 \\ \beta & \alpha & 0 \\ 0 & 0 & \lambda_\parallel \end{pmatrix},$$

and therefore

$$\det M = (\kappa_* |G_*|)^3\, g_\parallel\, (g_I^2 + g_J^2).$$

Thus the local BCC shell crossing is Weyl whenever

$$g_\parallel \neq 0, \qquad (g_I, g_J) \neq (0,0).$$

So the local Weyl problem has been reduced to an explicit finite coefficient test on the three enriched couplings $g_\parallel, g_I, g_J$.

CVI. WITNESS THEOREM FOR THE MINIMAL ENRICHED COUPLINGS $g_\parallel, g_I, g_J$

A. Goal

The purpose of this section is to determine, at the level of symmetry alone, whether the three minimal enriched couplings

$$g_\parallel, \qquad g_I, \qquad g_J$$

can be forbidden identically.

The strongest honest statement is the following:

1. symmetry determines the *allowed invariant tensor space*;

2. this invariant space is exactly three-dimensional in the minimal shell-enriched setting;

3. the three basis directions correspond precisely to the longitudinal, transverse-even, and transverse-odd couplings;

4. therefore symmetry does *not* force

$$g_\parallel = g_I = g_J = 0;$$

5. however, symmetry alone does *not* guarantee that each coefficient is nonzero in the actual microscopic realization.

Thus the correct result is an *allowance theorem* plus a *witness criterion*, not a purely symmetry-based nonvanishing theorem.

B. Shell-relevant representation data

Fix a selected shell direction

$$\hat{n} = \frac{G_*}{|G_*|},$$

with adapted orthonormal frame

$$\{e_1, e_2, \hat{n}\}.$$

Let

$$\Pi_\perp = \mathrm{span}\{e_1, e_2\}$$

denote the transverse plane.

We work at the shell-relevant infrared level, where the residual symmetry acting on $\Pi_\perp$ is the axial group

$$O(2)_\perp \quad \text{or, at least, its connected part } SO(2)_\perp.$$

The shell-relative momentum decomposes as

$$p = p_\parallel \hat{n} + p_1 e_1 + p_2 e_2.$$

Hence the momentum representation splits as

$$V_p \cong \mathbf{1}_\parallel \oplus \mathbf{2}_\perp,$$

where:

- $\mathbf{1}_\parallel$ is the longitudinal scalar channel carried by $p_\parallel$,
- $\mathbf{2}_\perp$ is the transverse vector channel carried by (p_1, p_2).

Now let the shell-relevant internal sector contain:

- one axial/internal scalar channel s_3,
- one transverse internal doublet (s_1, s_2).

Thus the internal representation also splits as

$$V_{\mathrm{int}} \cong \mathbf{1}_3 \oplus \mathbf{2}_s,$$

where $\mathbf{2}_s$ transforms in the same transverse way as $\mathbf{2}_\perp$.

C. Problem reformulation

A first-order enriched shell coupling is an invariant linear map from the shell-gradient/momentum/internal data to a scalar operator coefficient. Equivalently, it belongs to

$$\mathrm{Hom}_{O(2)_\perp}(V_p \otimes V_{\mathrm{int}}, \mathbf{1}). \tag{412}$$

Using the above decomposition,

$$V_p \otimes V_{\mathrm{int}} \cong (\mathbf{1}_\parallel \oplus \mathbf{2}_\perp) \otimes (\mathbf{1}_3 \oplus \mathbf{2}_s).$$

Expanding,

$$V_p \otimes V_{\mathrm{int}} \cong (\mathbf{1}_\parallel \otimes \mathbf{1}_3) \oplus (\mathbf{1}_\parallel \otimes \mathbf{2}_s) \oplus (\mathbf{2}_\perp \otimes \mathbf{1}_3) \oplus (\mathbf{2}_\perp \otimes \mathbf{2}_s).$$

Now:

- $\mathbf{1}_\parallel \otimes \mathbf{1}_3 \cong \mathbf{1}$ contributes one invariant;
- $\mathbf{1}_\parallel \otimes \mathbf{2}_s \cong \mathbf{2}_s$ contributes no scalar;
- $\mathbf{2}_\perp \otimes \mathbf{1}_3 \cong \mathbf{2}_\perp$ contributes no scalar;
- $\mathbf{2}_\perp \otimes \mathbf{2}_s$ contributes two scalar structures, namely the symmetric Euclidean contraction and the antisymmetric J-contraction.

Therefore the invariant space is three-dimensional.

D. Dimension count

We now state this precisely.

[Dimension of the minimal invariant coupling space] Let $V_p \cong \mathbf{1}_{\|} \oplus \mathbf{2}_{\perp}$ and $V_{\mathrm{int}} \cong \mathbf{1}_3 \oplus \mathbf{2}_s$, with $\mathbf{2}_{\perp}$ and $\mathbf{2}_s$ isomorphic transverse $O(2)_{\perp}$-representations. Then

$$\dim \mathrm{Hom}_{O(2)_{\perp}}(V_p \otimes V_{\mathrm{int}}, \mathbf{1}) = 3. \tag{413}$$

From the above decomposition,

$$V_p \otimes V_{\mathrm{int}} \cong (\mathbf{1} \otimes \mathbf{1}) \oplus (\mathbf{1} \otimes \mathbf{2}) \oplus (\mathbf{2} \otimes \mathbf{1}) \oplus (\mathbf{2} \otimes \mathbf{2}).$$

The first summand contributes one scalar invariant. The middle two contribute none. For the last summand, $\mathbf{2} \otimes \mathbf{2}$, the invariant bilinear forms on a two-dimensional oriented Euclidean plane are exactly two-dimensional: one generated by the metric δ_{ab}, and one generated by the complex structure / antisymmetric tensor J_{ab}. Hence the total invariant dimension is

$$1 + 2 = 3.$$

E. Explicit invariant basis

We now construct these three invariant couplings explicitly.

a. Longitudinal channel. The scalar $\mathbf{1}_{\|} \otimes \mathbf{1}_3$ is generated by

$$\mathcal{O}_{\|} = (\hat{n} \cdot p)\, s_3. \tag{414}$$

b. Transverse-even channel. Using the Euclidean metric on $\Pi_{\perp}$, the first scalar from $\mathbf{2}_{\perp} \otimes \mathbf{2}_s$ is

$$\mathcal{O}_I = p_1 s_1 + p_2 s_2. \tag{415}$$

c. Transverse-odd channel. Using the transverse complex structure

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

the second scalar is

$$\mathcal{O}_J = -p_2 s_1 + p_1 s_2. \tag{416}$$

Therefore every minimal first-order shell-enriched invariant operator is of the form

$$\mathcal{O}^{(1)}_{\mathrm{enr}} = g_{\|} \mathcal{O}_{\|} + g_I \mathcal{O}_I + g_J \mathcal{O}_J. \tag{417}$$

F. Identification with the previously extracted operator

The explicit minimal enriched operator constructed previously was

$$\begin{aligned} \delta L_{\mathrm{enr}} = {} & g_{\|}\, (\partial_i \phi_0) \hat{n}_i\, (\hat{n} \cdot (-i\nabla)) \otimes s_3 \\ & + g_I\, (\partial_i \phi_0) \hat{n}_i \Big[(e_1 \cdot (-i\nabla)) \otimes s_1 + (e_2 \cdot (-i\nabla)) \otimes s_2 \Big] \\ & + g_J\, (\partial_i \phi_0) \hat{n}_i \Big[-(e_2 \cdot (-i\nabla)) \otimes s_1 + (e_1 \cdot (-i\nabla)) \otimes s_2 \Big]. \end{aligned} \tag{418}$$

This is exactly the operator realization of (417) after identifying

$$p_{\|} \leftrightarrow \hat{n} \cdot (-i\nabla), \qquad p_1 \leftrightarrow e_1 \cdot (-i\nabla), \qquad p_2 \leftrightarrow e_2 \cdot (-i\nabla).$$

Hence the previously introduced coefficients

$$g_{\|}, \qquad g_I, \qquad g_J$$

are not arbitrary ad hoc parameters: they are precisely the coefficients of the three independent first-order invariants.

G. Allowance theorem

We can now formulate the main result.

[Symmetry allowance theorem for the minimal enriched couplings] In the minimal shell-enriched TECT setting with shell-relevant symmetry $O(2)_\perp$, one has exactly three linearly independent first-order invariant couplings. They may be chosen as

$$\mathcal{O}_\parallel = (\hat{n} \cdot p)s_3, \qquad \mathcal{O}_I = p_1 s_1 + p_2 s_2, \qquad \mathcal{O}_J = -p_2 s_1 + p_1 s_2.$$

Therefore the most general first-order enriched coupling is

$$g_\parallel \mathcal{O}_\parallel + g_I \mathcal{O}_I + g_J \mathcal{O}_J.$$

In particular, symmetry does not force

$$g_\parallel = 0, \qquad g_I = 0, \qquad g_J = 0.$$

The result follows from the dimension count

$$\dim \mathrm{Hom}_{O(2)_\perp}(V_p \otimes V_{\mathrm{int}}, \mathbf{1}) = 3$$

and from the explicit construction of the three invariant basis elements $\mathcal{O}_\parallel, \mathcal{O}_I, \mathcal{O}_J$. Since each basis element is nonzero and invariant, none of the corresponding coefficients is symmetry-forbidden.

H. What symmetry can and cannot prove

The previous theorem is an *allowance theorem*, not a *nonvanishing theorem*. This distinction is essential.

a. What symmetry proves. Symmetry proves that:

1. the invariant coupling space is three-dimensional;
2. the couplings $g_\parallel, g_I, g_J$ are allowed;
3. there is no representation-theoretic reason forcing them to vanish identically.

b. What symmetry does not prove. Symmetry alone does not prove:

1. $g_\parallel \neq 0$,
2. $g_I \neq 0$,
3. $g_J \neq 0$,

for the actual microscopic model. Any of these could still vanish accidentally by dynamical cancellation or parameter tuning.

Thus the correct nonvanishing statement requires witnesses.

I. Witness criterion

We now formulate the correct finite criterion.

[Witness] A witness for a coupling coefficient is any microscopic matrix element, tensor contraction, or projected shell amplitude which evaluates nontrivially on the corresponding invariant basis direction.

For example:

- a longitudinal witness is any shell-projected matrix element detecting $\mathcal{O}_\parallel$;
- a transverse-even witness detects $\mathcal{O}_I$;
- a transverse-odd witness detects $\mathcal{O}_J$.

[Witness theorem for the enriched couplings] Let the first-order shell-enriched microscopic operator be expanded as

$$\mathcal{O}^{(1)}_{\rm enr} = g_\parallel \mathcal{O}_\parallel + g_I \mathcal{O}_I + g_J \mathcal{O}_J .$$

Then:

1. if there exists a longitudinal witness with nonzero value, then
$$g_\parallel \neq 0;$$
2. if there exists a transverse-even witness with nonzero value, then
$$g_I \neq 0;$$
3. if there exists a transverse-odd witness with nonzero value, then
$$g_J \neq 0.$$

Each witness is, by definition, a nonzero linear functional on the invariant coupling space which does not annihilate the corresponding basis element. Since $\mathcal{O}_\parallel, \mathcal{O}_I, \mathcal{O}_J$ are linearly independent, a nonzero functional detecting one of them implies that the corresponding coefficient cannot vanish in the operator expansion.

J. Connection to the microscopic tensors

Previously, the enriched microscopic tensors were extracted as

$$N^3_{ij} = g_\parallel\, \hat{n}_i \hat{n}_j, \tag{419}$$

$$N^1_{ij} = g_I\, \hat{n}_i e_{1j} - g_J\, \hat{n}_i e_{2j}, \tag{420}$$

$$N^2_{ij} = g_I\, \hat{n}_i e_{2j} + g_J\, \hat{n}_i e_{1j}. \tag{421}$$

Hence the couplings $g_\parallel, g_I, g_J$ are exactly the coordinates of the microscopic tensor triple (N^1, N^2, N^3) in the invariant basis

$$\hat{n}_i \hat{n}_j, \qquad \hat{n}_i e_{1j}, \qquad \hat{n}_i e_{2j}.$$

Therefore the witness theorem may be read equivalently as a statement about nonvanishing contractions of the tensor N^a_{ij}.

K. Immediate consequence for Weyl nondegeneracy

The previously derived shell-contraction formulas gave

$$\lambda_\parallel = \kappa_* |G_*|\, g_\parallel, \qquad \alpha = \kappa_* |G_*|\, g_I, \qquad \beta = -\kappa_* |G_*|\, g_J.$$

Hence

$$\det M = (\kappa_* |G_*|)^3\, g_\parallel\, (g_I^2 + g_J^2). \tag{422}$$

Therefore:

1. symmetry allowance implies the determinant formula is structurally meaningful;
2. a longitudinal witness plus at least one transverse witness implies
$$\det M \neq 0;$$
3. thus the selected shell crossing is Weyl.

This yields the exact finite criterion:

$$\boxed{\text{Weyl} \quad \Longleftarrow \quad \text{one nonzero longitudinal witness and at least one nonzero transverse witness.}} \tag{423}$$

L. Final honest status

We summarize the exact logical status.

a. Now rigorously proved:

1. the first-order invariant coupling space is exactly three-dimensional;
2. the couplings $g_\parallel, g_I, g_J$ form the complete invariant basis;
3. symmetry does not forbid any of them;
4. nonzero witnesses imply nonzero couplings;
5. a longitudinal witness together with a transverse witness implies Weyl nondegeneracy.

b. Still not proved by symmetry alone:

1. that the actual microscopic TECT dynamics produces nonzero witnesses automatically;
2. that all three coefficients are generically nonzero without further microscopic input;
3. that the opposite valley exists and matches the same coefficient triple.

CVII. FINAL SUMMARY

In the minimal shell-enriched TECT setting, the shell-relevant symmetry is the axial transverse group $O(2)_\perp$, under which

$$V_p \cong \mathbf{1}_\parallel \oplus \mathbf{2}_\perp, \qquad V_{\text{int}} \cong \mathbf{1}_3 \oplus \mathbf{2}_s.$$

The space of first-order invariant enriched couplings is

$$\text{Hom}_{O(2)_\perp}(V_p \otimes V_{\text{int}}, \mathbf{1}),$$

and its dimension is exactly

$$3.$$

A basis of invariant couplings is

$$\mathcal{O}_\parallel = (\hat{n} \cdot p)s_3, \qquad \mathcal{O}_I = p_1 s_1 + p_2 s_2, \qquad \mathcal{O}_J = -p_2 s_1 + p_1 s_2.$$

Therefore the most general minimal first-order enriched coupling is

$$g_\parallel \mathcal{O}_\parallel + g_I \mathcal{O}_I + g_J \mathcal{O}_J.$$

Hence symmetry does not force

$$g_\parallel = 0, \qquad g_I = 0, \qquad g_J = 0.$$

However, symmetry alone does not prove that these coefficients are nonzero in the actual microscopic theory. To prove nonvanishing one needs witnesses.

If there exists one nonzero longitudinal witness and at least one nonzero transverse witness, then

$$g_\parallel \neq 0, \qquad (g_I, g_J) \neq (0,0),$$

and therefore

$$\det M = (\kappa_* |G_*|)^3 \, g_\parallel \, (g_I^2 + g_J^2) \neq 0.$$

Thus the selected BCC shell crossing is Weyl.

CVIII. EXPLICIT WITNESS FUNCTIONALS FOR THE ENRICHED COUPLINGS

A. Goal

The purpose of this section is to replace the abstract witness statement by explicit calculable formulas.

We already know that the minimal first-order shell-enriched coupling space is three-dimensional, with invariant basis

$$\mathcal{O}_{\parallel}, \qquad \mathcal{O}_I, \qquad \mathcal{O}_J.$$

Hence the coefficients

$$g_{\parallel}, \qquad g_I, \qquad g_J$$

can be extracted by dual linear functionals on this invariant space.

The goal is therefore:

1. to define an explicit shell-projected first-order operator space;
2. to construct three dual witness functionals
$$W_{\parallel}, \qquad W_I, \qquad W_J;$$
3. to prove that these functionals extract exactly the coefficients
$$g_{\parallel}, \qquad g_I, \qquad g_J;$$
4. to formulate a rigorous finite nonvanishing criterion for Weyl emergence.

B. Shell-projected first-order enriched operator space

Fix the selected shell direction

$$\hat{n} = \frac{G_*}{|G_*|},$$

with adapted orthonormal frame

$$\{e_1, e_2, \hat{n}\}.$$

Let the shell-relative momentum be

$$p = p_{\parallel}\hat{n} + p_1 e_1 + p_2 e_2.$$

Let the reduced internal generators be

$$s_1, \qquad s_2, \qquad s_3.$$

Define the three invariant basis operators

$$\mathcal{O}_{\parallel} := p_{\parallel} s_3, \tag{424}$$

$$\mathcal{O}_I := p_1 s_1 + p_2 s_2, \tag{425}$$

$$\mathcal{O}_J := -p_2 s_1 + p_1 s_2. \tag{426}$$

The shell-projected first-order enriched operator space is

$$\mathcal{V}^{(1)}_{\text{enr}} := \text{span}\{\mathcal{O}_{\parallel}, \mathcal{O}_I, \mathcal{O}_J\}. \tag{427}$$

Every minimal first-order enriched operator is uniquely written as

$$\delta L^{(1)}_{\text{enr}} = g_{\parallel}\mathcal{O}_{\parallel} + g_I \mathcal{O}_I + g_J \mathcal{O}_J. \tag{428}$$

C. Hilbert–Schmidt pairing on the reduced operator space

To construct explicit dual witnesses, we equip the reduced operator space with a natural bilinear pairing. Let $\langle \cdot, \cdot \rangle_p$ denote the Euclidean pairing on the linear momentum forms generated by

$$p_{\|}, \qquad p_1, \qquad p_2,$$

and let $\langle \cdot, \cdot \rangle_s$ denote the Hilbert–Schmidt pairing on the reduced internal generators:

$$\langle A, B \rangle_s := \frac{1}{2}(AB).$$

Assume the reduced generators are normalized by

$$\frac{1}{2}(s_a s_b) = \delta_{ab}, \qquad a, b \in \{1, 2, 3\}. \tag{429}$$

Then define the tensor-product pairing on $\mathcal{V}_{\text{enr}}^{(1)}$ by

$$\langle p_\mu s_a,\ p_\nu s_b \rangle := \langle p_\mu, p_\nu \rangle_p \, \langle s_a, s_b \rangle_s = \delta_{\mu\nu} \delta_{ab}, \tag{430}$$

where $\mu, \nu \in \{\|, 1, 2\}$.

With this normalization, the three basis elements are orthogonal:

$$\langle \mathcal{O}_{\|}, \mathcal{O}_I \rangle = \langle \mathcal{O}_{\|}, \mathcal{O}_J \rangle = \langle \mathcal{O}_I, \mathcal{O}_J \rangle = 0. \tag{431}$$

Moreover,

$$\langle \mathcal{O}_{\|}, \mathcal{O}_{\|} \rangle = 1, \qquad \langle \mathcal{O}_I, \mathcal{O}_I \rangle = 2, \qquad \langle \mathcal{O}_J, \mathcal{O}_J \rangle = 2. \tag{432}$$

D. Dual witnesses

We now define the three explicit witness functionals by normalized projection onto the basis directions:

$$W_{\|}(X) := \langle \mathcal{O}_{\|}, X \rangle, \tag{433}$$

$$W_I(X) := \frac{1}{2} \langle \mathcal{O}_I, X \rangle, \tag{434}$$

$$W_J(X) := \frac{1}{2} \langle \mathcal{O}_J, X \rangle. \tag{435}$$

These are linear maps

$$W_{\|},\ W_I,\ W_J : \mathcal{V}_{\text{enr}}^{(1)} \to \mathbb{C}.$$

E. Exact coefficient extraction

We now prove that these witnesses recover the coefficients exactly.

[Dual witness extraction] For

$$\delta L_{\text{enr}}^{(1)} = g_{\|} \mathcal{O}_{\|} + g_I \mathcal{O}_I + g_J \mathcal{O}_J,$$

the witness functionals satisfy

$$W_{\|}(\delta L_{\text{enr}}^{(1)}) = g_{\|}, \tag{436}$$

$$W_I(\delta L_{\text{enr}}^{(1)}) = g_I, \tag{437}$$

$$W_J(\delta L^{(1)}_{\mathrm{enr}}) = g_J. \tag{438}$$

By linearity and orthogonality,

$$\langle \mathcal{O}_\parallel, \delta L^{(1)}_{\mathrm{enr}} \rangle = g_\parallel \langle \mathcal{O}_\parallel, \mathcal{O}_\parallel \rangle = g_\parallel.$$

Hence

$$W_\parallel(\delta L^{(1)}_{\mathrm{enr}}) = g_\parallel.$$

Similarly,

$$\langle \mathcal{O}_I, \delta L^{(1)}_{\mathrm{enr}} \rangle = g_I \langle \mathcal{O}_I, \mathcal{O}_I \rangle = 2g_I,$$

so

$$W_I(\delta L^{(1)}_{\mathrm{enr}}) = \frac{1}{2}(2g_I) = g_I.$$

Likewise,

$$\langle \mathcal{O}_J, \delta L^{(1)}_{\mathrm{enr}} \rangle = g_J \langle \mathcal{O}_J, \mathcal{O}_J \rangle = 2g_J,$$

hence

$$W_J(\delta L^{(1)}_{\mathrm{enr}}) = g_J.$$

F. Tensor-form witness extraction

We now rewrite the witnesses directly in terms of the microscopic tensors

$$N^a_{ij}.$$

The previously extracted tensor form was

$$N^3_{ij} = g_\parallel \, \hat{n}_i \hat{n}_j, \tag{439}$$

$$N^1_{ij} = g_I \, \hat{n}_i e_{1j} - g_J \, \hat{n}_i e_{2j}, \tag{440}$$

$$N^2_{ij} = g_I \, \hat{n}_i e_{2j} + g_J \, \hat{n}_i e_{1j}. \tag{441}$$

Therefore the dual witnesses may equivalently be written as tensor contractions:

$$\boxed{W_\parallel[N] := \hat{n}_i \hat{n}_j N^3_{ij} = g_\parallel,} \tag{442}$$

$$\boxed{W_I[N] := \frac{1}{2}\left(e_{1j}\hat{n}_i N^1_{ij} + e_{2j}\hat{n}_i N^2_{ij}\right) = g_I,} \tag{443}$$

$$\boxed{W_J[N] := \frac{1}{2}\left(e_{1j}\hat{n}_i N^2_{ij} - e_{2j}\hat{n}_i N^1_{ij}\right) = g_J.} \tag{444}$$

Substitute (439)–(441) directly.
For the longitudinal witness:

$$\hat{n}_i \hat{n}_j N^3_{ij} = g_\parallel (\hat{n} \cdot \hat{n})^2 = g_\parallel.$$

For the transverse-even witness:

$$e_{1j}\hat{n}_i N^1_{ij} = g_I(e_1 \cdot e_1) - g_J(e_1 \cdot e_2) = g_I,$$

$$e_{2j}\hat{n}_i N^2_{ij} = g_I(e_2 \cdot e_2) + g_J(e_2 \cdot e_1) = g_I.$$

Hence

$$\frac{1}{2}\left(e_{1j}\hat{n}_i N^1_{ij} + e_{2j}\hat{n}_i N^2_{ij}\right) = g_I.$$

For the transverse-odd witness:

$$e_{1j}\hat{n}_i N^2_{ij} = g_I(e_1 \cdot e_2) + g_J(e_1 \cdot e_1) = g_J,$$

$$e_{2j}\hat{n}_i N^1_{ij} = g_I(e_2 \cdot e_1) - g_J(e_2 \cdot e_2) = -g_J.$$

Thus

$$\frac{1}{2}\left(e_{1j}\hat{n}_i N^2_{ij} - e_{2j}\hat{n}_i N^1_{ij}\right) = g_J.$$

G. Shell-contraction and Weyl criterion

The reduced Weyl coefficients were

$$\lambda_\parallel = \kappa_* |G_*|\, g_\parallel, \tag{445}$$

$$\alpha = \kappa_* |G_*|\, g_I, \tag{446}$$

$$\beta = -\kappa_* |G_*|\, g_J, \tag{447}$$

and therefore

$$\det M = (\kappa_* |G_*|)^3\, g_\parallel\, (g_I^2 + g_J^2). \tag{448}$$

Using the witness formulas, this becomes

$$\boxed{\det M = (\kappa_* |G_*|)^3\, W_\parallel[N]\, \Big(W_I[N]^2 + W_J[N]^2\Big).} \tag{449}$$

This yields an explicit finite witness criterion for Weyl nondegeneracy.
[Explicit witness criterion for Weyl] Let the shell-enriched tensor triple N^a_{ij} be given. If

$$W_\parallel[N] \neq 0, \tag{450}$$

and

$$\big(W_I[N], W_J[N]\big) \neq (0, 0), \tag{451}$$

then

$$\det M \neq 0, \tag{452}$$

and the selected shell crossing is Weyl.
By (449),

$$\det M = (\kappa_* |G_*|)^3 W_\parallel[N]\left(W_I[N]^2 + W_J[N]^2\right).$$

If $W_\parallel[N] \neq 0$ and $W_I[N]^2 + W_J[N]^2 \neq 0$, then $\det M \neq 0$. Hence the local reduced Hamiltonian is Weyl.

H. Matrix-element realization of the witnesses

The abstract pairing above can also be realized as shell-projected matrix elements. Let P_{shell} be the projection to the selected reduced shell subspace, and let

$$\delta L^{(1)}_{\mathrm{enr}} := P_{\mathrm{shell}}\, \delta L_{\mathrm{enr}}\, P_{\mathrm{shell}}.$$

Then the witnesses are equivalently

$$W_{\parallel} = \frac{\langle \mathcal{O}_{\parallel}, \delta L^{(1)}_{\mathrm{enr}} \rangle}{\langle \mathcal{O}_{\parallel}, \mathcal{O}_{\parallel} \rangle}, \tag{453}$$

$$W_{I} = \frac{\langle \mathcal{O}_{I}, \delta L^{(1)}_{\mathrm{enr}} \rangle}{\langle \mathcal{O}_{I}, \mathcal{O}_{I} \rangle}, \tag{454}$$

$$W_{J} = \frac{\langle \mathcal{O}_{J}, \delta L^{(1)}_{\mathrm{enr}} \rangle}{\langle \mathcal{O}_{J}, \mathcal{O}_{J} \rangle}. \tag{455}$$

Thus the witnesses are not merely formal; they are actual projected observables on the reduced shell operator.

I. What is now proved

We summarize the exact logical status.

a. Now rigorously proved:

1. the first-order shell-enriched invariant space is three-dimensional;
2. the dual witness functionals $W_{\parallel}, W_I, W_J$ extract exactly the coefficients $g_{\parallel}, g_I, g_J$;
3. these witnesses can be written either as dual projections on the invariant operator basis or as explicit tensor contractions of N^a_{ij};
4. Weyl nondegeneracy is equivalent to one nonzero longitudinal witness and at least one nonzero transverse witness.

b. Still not proved automatically:

1. that the actual microscopic TECT model necessarily yields nonzero witness values;
2. that the opposite valley exists and carries the matching coefficient triple;
3. that the remote gap and mass-protection conditions hold in the final UV completion.

CIX. FINAL SUMMARY

Let

$$\delta L^{(1)}_{\mathrm{enr}} = g_{\parallel}\mathcal{O}_{\parallel} + g_I \mathcal{O}_I + g_J \mathcal{O}_J$$

be the shell-projected first-order enriched operator, where

$$\mathcal{O}_{\parallel} = p_{\parallel} s_3, \qquad \mathcal{O}_I = p_1 s_1 + p_2 s_2, \qquad \mathcal{O}_J = -p_2 s_1 + p_1 s_2.$$

Define the dual witness functionals by

$$W_{\parallel}(X) = \langle \mathcal{O}_{\parallel}, X \rangle, \qquad W_I(X) = \frac{1}{2}\langle \mathcal{O}_I, X \rangle, \qquad W_J(X) = \frac{1}{2}\langle \mathcal{O}_J, X \rangle.$$

Then

$$W_{\parallel}(\delta L^{(1)}_{\rm enr}) = g_{\parallel}, \qquad W_I(\delta L^{(1)}_{\rm enr}) = g_I, \qquad W_J(\delta L^{(1)}_{\rm enr}) = g_J.$$

Equivalently, in tensor language,

$$W_{\parallel}[N] = \hat n_i \hat n_j N^3_{ij},$$

$$W_I[N] = \frac{1}{2}\left(e_{1j}\hat n_i N^1_{ij} + e_{2j}\hat n_i N^2_{ij}\right),$$

$$W_J[N] = \frac{1}{2}\left(e_{1j}\hat n_i N^2_{ij} - e_{2j}\hat n_i N^1_{ij}\right).$$

These satisfy

$$W_{\parallel}[N] = g_{\parallel}, \qquad W_I[N] = g_I, \qquad W_J[N] = g_J.$$

Therefore the Weyl determinant becomes

$$\det M = (\kappa_*|G_*|)^3\, W_{\parallel}[N]\,\big(W_I[N]^2 + W_J[N]^2\big).$$

Hence the selected BCC shell crossing is Weyl whenever

$$W_{\parallel}[N] \neq 0, \qquad \big(W_I[N], W_J[N]\big) \neq (0,0).$$

So the local Weyl problem has now been reduced to an explicit finite witness test on the shell-projected enriched tensor.

CX. WITNESS FORMULATION OF OPPOSITE-VALLEY PAIRING AND DIRAC DOUBLING

A. Goal

The purpose of this section is to reformulate the opposite-valley pairing condition entirely in the witness language. Previously, the local Weyl coefficients at the positive valley were extracted by the witness functionals

$$W^+_{\parallel}, \qquad W^+_I, \qquad W^+_J.$$

We now define the corresponding witness functionals at the opposite valley

$$W^-_{\parallel}, \qquad W^-_I, \qquad W^-_J,$$

and prove that, in a matched opposite-valley frame, equality of the two witness triples is equivalent to the standard pairing relation

$$h_-(p) = -h_+(-p).$$

This yields a fully finite witness criterion for local Dirac doubling.

B. Positive-valley witness data

Fix the selected positive shell vector

$$G_*, \qquad \hat n_+ := \frac{G_*}{|G_*|} = \hat n,$$

with adapted right-handed frame

$$\{e^+_1, e^+_2, \hat n_+\} = \{e_1, e_2, \hat n\}.$$

The first-order shell-projected enriched operator at $+G_*$ is

$$\delta L^{(1)}_{\mathrm{enr},+} = g^+_\parallel \mathcal{O}^+_\parallel + g^+_I \mathcal{O}^+_I + g^+_J \mathcal{O}^+_J, \tag{456}$$

where

$$\mathcal{O}^+_\parallel := p_\parallel s_3, \tag{457}$$

$$\mathcal{O}^+_I := p_1 s_1 + p_2 s_2, \tag{458}$$

$$\mathcal{O}^+_J := -p_2 s_1 + p_1 s_2. \tag{459}$$

The corresponding witness functionals satisfy

$$W^+_\parallel(\delta L^{(1)}_{\mathrm{enr},+}) = g^+_\parallel, \qquad W^+_I(\delta L^{(1)}_{\mathrm{enr},+}) = g^+_I, \qquad W^+_J(\delta L^{(1)}_{\mathrm{enr},+}) = g^+_J. \tag{460}$$

C. Matched negative-valley frame

Now consider the opposite shell point

$$-G_*.$$

Define

$$\hat{n}_- := -\hat{n}_+ = -\hat{n}.$$

To preserve a right-handed adapted frame, choose

$$e^-_1 := e^+_1 = e_1, \qquad e^-_2 := -e^+_2 = -e_2, \qquad \hat{n}_- := -\hat{n}. \tag{461}$$

Indeed,

$$e^-_1 \times e^-_2 = e_1 \times (-e_2) = -(e_1 \times e_2) = -\hat{n} = \hat{n}_-.$$

This is the canonical matched opposite-valley frame.

D. Negative-valley invariant basis

Let the shell-relative momentum near $-G_*$ again be written as

$$p = p_\parallel \hat{n}_- + p_1 e^-_1 + p_2 e^-_2.$$

Define the negative-valley invariant basis operators

$$\mathcal{O}^-_\parallel := p_\parallel s_3, \tag{462}$$

$$\mathcal{O}^-_I := p_1 s_1 + p_2 s_2, \tag{463}$$

$$\mathcal{O}^-_J := -p_2 s_1 + p_1 s_2. \tag{464}$$

The first-order shell-projected enriched operator at $-G_*$ is then

$$\delta L^{(1)}_{\mathrm{enr},-} = g^-_\parallel \mathcal{O}^-_\parallel + g^-_I \mathcal{O}^-_I + g^-_J \mathcal{O}^-_J. \tag{465}$$

Define the corresponding negative-valley witnesses

$$W^-_\parallel(X) := \langle \mathcal{O}^-_\parallel, X\rangle, \tag{466}$$

$$W^-_I(X) := \frac{1}{2}\langle \mathcal{O}^-_I, X\rangle, \tag{467}$$

$$W^-_J(X) := \frac{1}{2}\langle \mathcal{O}^-_J, X\rangle. \tag{468}$$

Then

$$W^-_\parallel(\delta L^{(1)}_{\mathrm{enr},-}) = g^-_\parallel, \qquad W^-_I(\delta L^{(1)}_{\mathrm{enr},-}) = g^-_I, \qquad W^-_J(\delta L^{(1)}_{\mathrm{enr},-}) = g^-_J. \tag{469}$$

E. Negative-valley tensor witnesses

Let the enriched tensor triple at $-G_*$ be denoted by

$$N_{ij}^{a(-)}.$$

Then, in the matched frame (461), the negative-valley tensor witnesses are

$$\boxed{W_\parallel^-[N^-] := \hat{n}_-^i \hat{n}_-^j N_{ij}^{3(-)} = g_\parallel^-,} \tag{470}$$

$$\boxed{W_I^-[N^-] := \frac{1}{2}\left(e_{1j}^- \hat{n}_-^i N_{ij}^{1(-)} + e_{2j}^- \hat{n}_-^i N_{ij}^{2(-)}\right) = g_I^-,} \tag{471}$$

$$\boxed{W_J^-[N^-] := \frac{1}{2}\left(e_{1j}^- \hat{n}_-^i N_{ij}^{2(-)} - e_{2j}^- \hat{n}_-^i N_{ij}^{1(-)}\right) = g_J^-.} \tag{472}$$

F. Positive and negative reduced Weyl blocks

The positive-valley reduced Hamiltonian is

$$h_+(p) = \lambda_\parallel^+ p_\parallel \sigma^3 + \alpha^+(p_1\sigma^1 + p_2\sigma^2) + \beta^+(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2), \tag{473}$$

with

$$\lambda_\parallel^+ = \kappa_*|G_*|\, W_\parallel^+, \qquad \alpha^+ = \kappa_*|G_*|\, W_I^+, \qquad \beta^+ = -\kappa_*|G_*|\, W_J^+. \tag{474}$$

Similarly, the negative-valley reduced Hamiltonian is

$$h_-(p) = \lambda_\parallel^- p_\parallel \sigma^3 + \alpha^-(p_1\sigma^1 + p_2\sigma^2) + \beta^-(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2), \tag{475}$$

with

$$\lambda_\parallel^- = \kappa_*|G_*|\, W_\parallel^-, \qquad \alpha^- = \kappa_*|G_*|\, W_I^-, \qquad \beta^- = -\kappa_*|G_*|\, W_J^-. \tag{476}$$

G. Pairing relation in witness language

We now compute

$$-h_+(-p).$$

From (473),

$$h_+(-p) = -\lambda_\parallel^+ p_\parallel \sigma^3 - \alpha^+(p_1\sigma^1 + p_2\sigma^2) - \beta^+(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2).$$

Hence

$$-h_+(-p) = \lambda_\parallel^+ p_\parallel \sigma^3 + \alpha^+(p_1\sigma^1 + p_2\sigma^2) + \beta^+(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2). \tag{477}$$

Comparing (475) and (477), we obtain the exact equivalence:

$$h_-(p) = -h_+(-p) \quad \Longleftrightarrow \quad \lambda_\parallel^- = \lambda_\parallel^+,\ \alpha^- = \alpha^+,\ \beta^- = \beta^+. \tag{478}$$

Using (474) and (476), this is equivalent to

$$\boxed{W_\parallel^- = W_\parallel^+, \qquad W_I^- = W_I^+, \qquad W_J^- = W_J^+.} \tag{479}$$

H. Witness theorem for opposite-valley pairing

We may now state the main theorem.

[Witness theorem for opposite-valley pairing] Let the positive and negative shell-projected first-order enriched operators be

$$\delta L^{(1)}_{\mathrm{enr},+}, \qquad \delta L^{(1)}_{\mathrm{enr},-},$$

with witness triples

$$(W^{+}_{\parallel}, W^{+}_{I}, W^{+}_{J}), \qquad (W^{-}_{\parallel}, W^{-}_{I}, W^{-}_{J}).$$

Then, in the matched opposite-valley frame (461), the following are equivalent:

1. the reduced Hamiltonians satisfy the standard pairing relation
$$h_-(p) = -h_+(-p);$$
2. the reduced coefficients satisfy
$$\lambda^{-}_{\parallel} = \lambda^{+}_{\parallel}, \qquad \alpha^- = \alpha^+, \qquad \beta^- = \beta^+;$$
3. the witness triples are equal,
$$W^{-}_{\parallel} = W^{+}_{\parallel}, \qquad W^{-}_{I} = W^{+}_{I}, \qquad W^{-}_{J} = W^{+}_{J}.$$

The equivalence of (1) and (2) follows by direct comparison of (475) with (477). The equivalence of (2) and (3) follows immediately from the coefficient-witness identifications (474) and (476).

I. Witness formulation of Dirac doubling

Assume both valleys are locally Weyl, i.e.

$$W^{\pm}_{\parallel} \neq 0, \qquad \left(W^{\pm}_{I}, W^{\pm}_{J}\right) \neq (0,0). \tag{480}$$

If, in addition, the witness triples match as in (479), then the doubled Hamiltonian

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix} \tag{481}$$

is unitarily equivalent to a massless Dirac Hamiltonian.

[Explicit witness criterion for Dirac doubling] Suppose:

1. $$W^{+}_{\parallel} \neq 0, \qquad \left(W^{+}_{I}, W^{+}_{J}\right) \neq (0,0),$$
   so the positive valley is Weyl;
2. $$W^{-}_{\parallel} \neq 0, \qquad \left(W^{-}_{I}, W^{-}_{J}\right) \neq (0,0),$$
   so the negative valley is Weyl;
3. the witness triples match,
$$W^{-}_{\parallel} = W^{+}_{\parallel}, \qquad W^{-}_{I} = W^{+}_{I}, \qquad W^{-}_{J} = W^{+}_{J}.$$

Then the opposite-valley doubled block is Dirac.

By assumptions (1) and (2), each valley has nondegenerate Weyl form. By assumption (3) and the previous theorem, the pairing relation

$$h_-(p) = -h_+(-p)$$

holds. Therefore the doubled block (481) is the standard Weyl-pair block and is unitarily equivalent to a massless Dirac Hamiltonian.

J. Determinant form of the witness criteria

At each valley,

$$\det M_\pm = (\kappa_* |G_*|)^3 \, W_\parallel^\pm \Big((W_I^\pm)^2 + (W_J^\pm)^2 \Big). \tag{482}$$

Thus the full local Dirac criterion can be written as:

$$\boxed{W_\parallel^\pm \neq 0, \qquad \big(W_I^\pm, W_J^\pm\big) \neq (0,0), \qquad (W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+).} \tag{483}$$

This is a completely finite witness test for local Dirac doubling.

K. What is now proved

We summarize the exact logical status.
a. Now rigorously proved:

1. the negative-valley witness triple is defined exactly analogously to the positive one;
2. in the matched opposite-valley frame, equality of witness triples is equivalent to the standard pairing relation
$$h_-(p) = -h_+(-p);$$
3. local Dirac doubling is equivalent to:
   - Weyl nondegeneracy at both valleys,
   - equality of the two witness triples.

b. Still open microscopically:

1. whether the actual microscopic TECT dynamics guarantees existence of the opposite valley;
2. whether the actual microscopic enriched tensors at $\pm G_*$ produce equal witness triples automatically;
3. whether remote-gap and mass-protection conditions hold in the full theory.

CXI. FINAL SUMMARY

At the positive valley $+G_*$, the shell-projected first-order enriched operator has witness triple

$$(W_\parallel^+, W_I^+, W_J^+),$$

which determines the Weyl coefficients by

$$\lambda_\parallel^+ = \kappa_* |G_*| \, W_\parallel^+, \qquad \alpha^+ = \kappa_* |G_*| \, W_I^+, \qquad \beta^+ = -\kappa_* |G_*| \, W_J^+.$$

At the negative valley $-G_*$, in the matched right-handed frame

$$\hat{n}_- = -\hat{n}, \qquad e_1^- = e_1, \qquad e_2^- = -e_2,$$

the corresponding witness triple

$$(W_\parallel^-, W_I^-, W_J^-)$$

determines

$$\lambda_\parallel^- = \kappa_* |G_*| \, W_\parallel^-, \qquad \alpha^- = \kappa_* |G_*| \, W_I^-, \qquad \beta^- = -\kappa_* |G_*| \, W_J^-.$$

Then the standard opposite-valley pairing relation

$$h_-(p) = -h_+(-p)$$

is equivalent to the equality of witness triples

$$W_\parallel^- = W_\parallel^+, \qquad W_I^- = W_I^+, \qquad W_J^- = W_J^+.$$

Moreover, each valley is Weyl precisely when

$$W_\parallel^\pm \neq 0, \qquad \left(W_I^\pm, W_J^\pm\right) \neq (0,0),$$

equivalently

$$\det M_\pm = (\kappa_*|G_*|)^3\, W_\parallel^\pm \Big((W_I^\pm)^2 + (W_J^\pm)^2\Big) \neq 0.$$

Therefore the full local Dirac criterion is the finite witness condition

$$W_\parallel^\pm \neq 0, \qquad \left(W_I^\pm, W_J^\pm\right) \neq (0,0), \qquad (W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+).$$

Thus local Dirac doubling has been reduced to a completely explicit finite witness-matching test at the two opposite shell valleys.

CXII. WITNESS FORM OF THE FULL LOCAL DIRAC STABILITY CRITERION

A. Goal

The purpose of this section is to combine the previously established witness formulation of local Weyl/Dirac pairing with the spectral-isolation theorem.

The result will be a single finite criterion for the existence of a stable local massless Dirac cone in the full microscopic theory.

The logic has two layers:

1. the *local shell layer*, where the two opposite valleys must each be Weyl and must satisfy the pairing relation;
2. the *full spectral layer*, where the doubled Dirac block must remain isolated from the remote sector.

We now formulate both layers in a single witness-compatible theorem.

B. Positive and negative valley witness data

Fix a selected opposite shell pair

$$\pm G_*, \qquad \hat{n}_+ = \hat{n}, \qquad \hat{n}_- = -\hat{n}.$$

Let the matched adapted frames be

$$\{e_1^+, e_2^+, \hat{n}_+\} = \{e_1, e_2, \hat{n}\}, \qquad \{e_1^-, e_2^-, \hat{n}_-\} = \{e_1, -e_2, -\hat{n}\}.$$

Let the positive and negative shell-projected first-order enriched operators be

$$\delta L^{(1)}_{\mathrm{enr},+}, \qquad \delta L^{(1)}_{\mathrm{enr},-}.$$

Their corresponding witness triples are

$$(W_\parallel^+, W_I^+, W_J^+), \qquad (W_\parallel^-, W_I^-, W_J^-).$$

The reduced Weyl coefficients are defined by

$$\lambda_\parallel^\pm = \kappa_*|G_*|\, W_\parallel^\pm, \tag{484}$$

$$\alpha^\pm = \kappa_*|G_*|\, W_I^\pm, \tag{485}$$

$$\beta^\pm = -\kappa_*|G_*|\, W_J^\pm. \tag{486}$$

Hence the local reduced Hamiltonians are

$$h_\pm(p) = \lambda_\parallel^\pm p_\parallel \sigma^3 + \alpha^\pm(p_1\sigma^1 + p_2\sigma^2) + \beta^\pm(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2). \tag{487}$$

C. Local Weyl criterion in witness form

At each valley, the Weyl determinant is

$$\det M_{\pm} = (\kappa_* |G_*|)^3 \, W_{\parallel}^{\pm} \Big((W_I^{\pm})^2 + (W_J^{\pm})^2 \Big). \tag{488}$$

Therefore each valley is Weyl if and only if

$$W_{\parallel}^{\pm} \neq 0, \qquad \big(W_I^{\pm}, W_J^{\pm} \big) \neq (0,0). \tag{489}$$

D. Pairing condition in witness form

In the matched opposite-valley frame, the standard pairing relation

$$h_-(p) = -h_+(-p) \tag{490}$$

is equivalent to equality of the two witness triples:

$$W_{\parallel}^{-} = W_{\parallel}^{+}, \qquad W_I^{-} = W_I^{+}, \qquad W_J^{-} = W_J^{+}. \tag{491}$$

Hence the doubled shell block

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix} \tag{492}$$

is a local massless Dirac block whenever (489) and (491) hold.

E. Full Hilbert-space decomposition

Now decompose the full microscopic Hilbert space as

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R, \tag{493}$$

where:

- $\mathcal{H}_D$ is the local four-dimensional doubled shell block (492),
- $\mathcal{H}_R$ is the remote sector.

The full Hamiltonian takes block form

$$H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V^\dagger(p) & H_R(p) \end{pmatrix}. \tag{494}$$

F. Remote-gap hypothesis

Assume there exists $\Delta > 0$ and a neighborhood $\mathcal{N} \ni 0$ such that

$$\operatorname{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing \qquad (p \in \mathcal{N}). \tag{495}$$

Equivalently,

$$\|(H_R(p) - E)^{-1}\| \leq \frac{2}{\Delta} \qquad (|E| < \Delta/2, \; p \in \mathcal{N}). \tag{496}$$

G. Zero-mixing and linear-mixing hypotheses

Assume the constant inter-sector mixing vanishes:

$$V(0) = 0. \tag{497}$$

A sufficient representation-theoretic reason is little-group mismatch:

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0.$$

Assume further that the leading nontrivial mixing is at most linear:

$$\|V(p)\| \leq c|p| \qquad (p \in \mathcal{N}) \tag{498}$$

for some $c > 0$.

H. Schur-complement effective Hamiltonian

For $|E| < \Delta/2$, the remote resolvent exists and the low-energy problem is governed by the Schur complement

$$F(E, p) = H_D(p) - E - V(p)(H_R(p) - E)^{-1}V^\dagger(p). \tag{499}$$

Define the self-energy

$$\Sigma(E, p) := V(p)(H_R(p) - E)^{-1}V^\dagger(p). \tag{500}$$

Then

$$F(E, p) = H_D(p) - E - \Sigma(E, p). \tag{501}$$

Using (496) and (498), we obtain

$$\|\Sigma(E, p)\| \leq \|V(p)\|^2 \, \|(H_R(p) - E)^{-1}\| \leq \frac{2c^2}{\Delta}|p|^2. \tag{502}$$

Hence the remote sector modifies the local block only at order $O(|p|^2/\Delta)$.

I. Effective Dirac Hamiltonian in witness form

Under the local witness conditions, define the common witness triple

$$W_\parallel := W_\parallel^+ = W_\parallel^-, \qquad W_I := W_I^+ = W_I^-, \qquad W_J := W_J^+ = W_J^-. \tag{503}$$

Then the common reduced coefficient matrix is

$$M = \kappa_*|G_*| \begin{pmatrix} W_I & W_J & 0 \\ -W_J & W_I & 0 \\ 0 & 0 & W_\parallel \end{pmatrix}, \tag{504}$$

up to the sign convention chosen for the transverse odd basis. Equivalently, in the canonical Weyl form,

$$\det M = (\kappa_*|G_*|)^3 W_\parallel \big(W_I^2 + W_J^2\big). \tag{505}$$

Therefore the effective full low-energy Hamiltonian on $\mathcal{H}_D$ is

$$H_{\mathrm{eff}}(E, p) = \Gamma^i (Mp)_i + O(|p|^2) + O\left(\frac{|p|^2}{\Delta}\right), \tag{506}$$

where Γ^i are Dirac gamma matrices in a suitable basis.

J. Mass protection

To ensure a massless Dirac cone rather than a massive one, one needs a residual symmetry forbidding a constant Dirac mass term. A sufficient condition is the existence of a grading operator Γ_5 on $\mathcal{H}_D$ such that

$$\{\Gamma_5, \Gamma^i\} = 0, \qquad \Gamma_5^2 = \mathbf{1}_4, \tag{507}$$

and the effective low-energy theory preserves this grading. Then no constant term proportional to a Dirac mass matrix can be generated, so

$$H_{\text{eff}}(0,0) = 0. \tag{508}$$

K. Main theorem

[Full local Dirac stability criterion in witness form] Assume the following:

1. **Positive-valley Weyl condition:**

$$W_{\parallel}^{+} \neq 0, \qquad \big(W_I^{+}, W_J^{+}\big) \neq (0,0).$$

2. **Negative-valley Weyl condition:**

$$W_{\parallel}^{-} \neq 0, \qquad \big(W_I^{-}, W_J^{-}\big) \neq (0,0).$$

3. **Opposite-valley witness matching:**

$$W_{\parallel}^{-} = W_{\parallel}^{+}, \qquad W_I^{-} = W_I^{+}, \qquad W_J^{-} = W_J^{+}.$$

4. **Remote spectral gap:**

$$\operatorname{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing \qquad (p \in \mathcal{N})$$

for some $\Delta > 0$.

5. **Zero constant mixing and linear bound:**

$$V(0) = 0, \qquad \|V(p)\| \leq c|p|.$$

6. **Mass protection:** a residual symmetry forbids a constant Dirac mass term in the effective theory.

Then, for sufficiently small p, the full microscopic theory contains a spectrally isolated four-dimensional low-energy sector whose effective Hamiltonian is

$$H_{\text{eff}}(p) = \Gamma^i (Mp)_i + O\!\left(\frac{|p|^2}{\Delta}\right), \tag{509}$$

with

$$\det M = (\kappa_* |G_*|)^3 W_{\parallel} \big(W_I^2 + W_J^2\big) \neq 0. \tag{510}$$

Consequently, the full microscopic theory contains a stable massless local Dirac cone.

By assumptions (1) and (2), both valleys are Weyl by the witness determinant criterion (488). By assumption (3), the two witness triples match, hence $h_-(p) = -h_+(-p)$, so the doubled shell block is local Dirac. By assumptions (4) and (5), the Schur-complement self-energy obeys the quadratic bound (502), so the remote sector changes the local Dirac block only at order $O(|p|^2/\Delta)$. Assumption (6) excludes an induced constant mass term. Therefore the effective low-energy Hamiltonian retains a massless Dirac cone with the same linear coefficient matrix M as the local shell block.

L. Compressed checklist

The theorem gives the following finite operational checklist:

$$\boxed{W_{\parallel}^{\pm} \neq 0, \qquad \left(W_I^{\pm}, W_J^{\pm}\right) \neq (0,0), \qquad (W_{\parallel}^{-}, W_I^{-}, W_J^{-}) = (W_{\parallel}^{+}, W_I^{+}, W_J^{+}),} \tag{511}$$

together with

$$\boxed{\mathrm{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing, \qquad V(0) = 0, \qquad \|V(p)\| \leq c|p|,} \tag{512}$$

and a mass-forbidding symmetry.

This is the complete finite witness formulation of local Dirac stability.

M. What is now proved and what remains open

a. Now rigorously proved:

1. local Weyl existence at each valley is a finite witness condition;
2. opposite-valley pairing is equality of witness triples;
3. full local Dirac stability is obtained by combining witness matching with remote gap and zero-mixing conditions;
4. the remote sector modifies the local Dirac cone only at order $O(|p|^2/\Delta)$.

b. Still open microscopically:

1. the actual computation of the witness triples from the final microscopic TECT dynamics;
2. the proof of the remote gap $\Delta > 0$ in the full model;
3. the proof of little-group mismatch for the actual remote-sector content;
4. the final microscopic symmetry implementing mass protection.

CXIII. FINAL SUMMARY

Let

$$(W_{\parallel}^{+}, W_I^{+}, W_J^{+}), \qquad (W_{\parallel}^{-}, W_I^{-}, W_J^{-})$$

be the positive- and negative-valley witness triples extracted from the shell-projected first-order enriched operators.

Each valley is Weyl exactly when

$$W_{\parallel}^{\pm} \neq 0, \qquad \left(W_I^{\pm}, W_J^{\pm}\right) \neq (0,0),$$

equivalently

$$\det M_{\pm} = (\kappa_* |G_*|)^3 W_{\parallel}^{\pm} \Big((W_I^{\pm})^2 + (W_J^{\pm})^2 \Big) \neq 0.$$

In the matched opposite-valley frame, the standard pairing relation

$$h_-(p) = -h_+(-p)$$

is equivalent to

$$W_{\parallel}^{-} = W_{\parallel}^{+}, \qquad W_I^{-} = W_I^{+}, \qquad W_J^{-} = W_J^{+}.$$

If, in addition, the full Hamiltonian admits a decomposition

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R,$$

with remote gap

$$\operatorname{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing,$$

zero constant mixing

$$V(0) = 0,$$

and linear mixing bound

$$\|V(p)\| \le c|p|,$$

then the Schur complement yields

$$H_{\text{eff}}(p) = \Gamma^i (Mp)_i + O\left(\frac{|p|^2}{\Delta}\right).$$

With an additional symmetry forbidding a constant Dirac mass, the full microscopic theory contains a stable massless local Dirac cone.

Thus the complete local Dirac stability criterion is the finite witness-and-gap condition

$$W_\parallel^\pm \neq 0, \qquad \big(W_I^\pm, W_J^\pm\big) \neq (0,0), \qquad (W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+),$$

together with

$$\operatorname{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing, \qquad V(0) = 0, \qquad \|V(p)\| \le c|p|.$$

CXIV. FINAL SUMMARY

TECT currently has a mathematically closed shell-reduced local fermion framework.

The minimal effective condensate sector is of Landau–Brazovskii type:

$$\partial_t \Phi = -r\Phi + \kappa \nabla^2 \Phi - \lambda \nabla^4 \Phi - u|\Phi|^2 \Phi, \qquad q_*^2 = -\frac{\kappa}{2\lambda}.$$

The minimal shell-enriched projector coupling yields explicit microscopic tensors

$$N_{ij}^3 = g_\parallel\, \hat{n}_i \hat{n}_j, \qquad N_{ij}^1 = g_I\, \hat{n}_i e_{1j} - g_J\, \hat{n}_i e_{2j}, \qquad N_{ij}^2 = g_I\, \hat{n}_i e_{2j} + g_J\, \hat{n}_i e_{1j}.$$

These determine the witness triple

$$W_\parallel, \qquad W_I, \qquad W_J,$$

and hence the local Weyl determinant

$$\det M = (\kappa_* |G_*|)^3 W_\parallel \big(W_I^2 + W_J^2\big).$$

Thus each valley is Weyl exactly when

$$W_\parallel \neq 0, \qquad (W_I, W_J) \neq (0,0).$$

Two opposite valleys form a local Dirac block exactly when, in the matched frame,

$$(W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+).$$

If, moreover, the remote sector satisfies

$$\operatorname{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing, \qquad V(0) = 0, \qquad \|V(p)\| \le c|p|,$$

and a mass-protection symmetry forbids a constant Dirac mass, then the full low-energy effective Hamiltonian is

$$H_{\text{eff}}(p) = \Gamma^i (Mp)_i + O(|p|^2/\Delta),$$

so the massless local Dirac cone is stable.

Therefore the true remaining microscopic frontier is finite and explicit: compute the actual witness triples, prove opposite-valley existence, prove the remote gap, prove the little-group mismatch for the remote sector, and prove the final mass-protection symmetry.

CXV. FINAL SUMMARY

The proof program is not fully complete, but the local shell-reduced fermion architecture is now essentially closed. What has been established is the following conditional theorem chain:

$$\text{BCC shell background} \Longrightarrow N_{ij}^{a} \Longrightarrow (W_{\parallel}, W_I, W_J) \Longrightarrow \text{Weyl criterion} \Longrightarrow \text{opposite-valley pairing} \Longrightarrow \text{local Dirac block} \Longrightarrow \text{spect}$$

In particular, the local Weyl determinant has been reduced to the explicit witness formula

$$\det M = (\kappa_*|G_*|)^3 W_{\parallel}\big(W_I^2 + W_J^2\big).$$

Thus a shell crossing is Weyl exactly when

$$W_{\parallel} \neq 0, \qquad (W_I, W_J) \neq (0,0).$$

Likewise, local Dirac doubling is equivalent to equality of the positive- and negative-valley witness triples:

$$(W_{\parallel}^{-}, W_I^{-}, W_J^{-}) = (W_{\parallel}^{+}, W_I^{+}, W_J^{+}).$$

Finally, if the remote sector satisfies

$$\operatorname{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing, \qquad V(0) = 0, \qquad \|V(p)\| \leq c|p|,$$

and if a mass-protection symmetry forbids a constant Dirac mass term, then the full low-energy effective Hamiltonian is

$$H_{\text{eff}}(p) = \Gamma^i (Mp)_i + O(|p|^2/\Delta),$$

so the local Dirac cone is stable.

Therefore, the remaining frontier is no longer conceptual. It is the explicit microscopic verification of:

$$\text{nonzero witnesses}, \qquad \text{opposite-valley existence}, \qquad \text{remote gap}, \qquad \text{mass protection}.$$

CXVI. HOW TO DETERMINE THE REMAINING MICROSCOPIC INPUTS

A. Scope

The remaining questions are no longer vague conceptual issues. They are finite microscopic verification problems.

At the current stage, the shell-level Dirac problem has already been reduced to the following finite conditions:

$$\widehat{U}(G_*) = 0, \qquad v_{\parallel} \neq 0, \qquad A_{G_*} \neq 0, \qquad C^1 \cdot G_* \neq 0, \qquad C^2 \cdot G_* \neq 0,$$

together with spectral isolation of the enriched shell block. [**?**]

Equivalently, in the more recent witness language, the local Weyl/Dirac problem is controlled by finite linear or polynomial functionals on the symmetry-allowed microscopic tensor space V. [**?**]

Thus the remaining tasks are:

1. determine whether the required witnesses are nonzero;
2. determine whether the opposite valley survives in the actual microscopic realization;
3. determine whether the enriched shell block is spectrally isolated from remote modes;
4. determine whether a residual symmetry forbids a constant Dirac mass.

B. I. Nonvanishing of the witnesses

1. 1. Tensor-space formulation

Let V be the finite-dimensional real vector space of all symmetry-allowed microscopic enriched tensors

$$N = \{N^a_{ij}\}$$

surviving the microscopic linearization around the condensed TECT background. For each shell-Dirac channel $\alpha = 1, 2$, define the shell contraction

$$L_\alpha(N) = G^j_* e^\alpha_i N^{a_\alpha}_{ij}. \tag{513}$$

This is a linear functional on V. [**?**]

Hence the question

$$L_\alpha(N) \neq 0?$$

is not an infinite-dimensional PDE problem, but a finite-dimensional linear-algebra problem.

2. 2. Exact dichotomy

Let $W = \mathbb{R}^3_{\text{out}} \otimes \mathbb{R}^3_{\text{grad}} \otimes U_{\text{int}}$ be the ambient tensor space and let

$$V \subset W$$

be the symmetry-allowed subspace. Define the target covector

$$T^{(\alpha)}_{ija} = e^\alpha_i G^j_* u^{(\alpha)}_a. \tag{514}$$

Then

$$L_\alpha(N) = \langle T^{(\alpha)}, N \rangle. \tag{515}$$

If $P_V : W \to V$ denotes the orthogonal projection onto the allowed tensor space, then

$$L_\alpha|_V \equiv 0 \quad \Longleftrightarrow \quad P_V T^{(\alpha)} = 0. \tag{516}$$

Therefore exactly one of the following holds: [**?**]

1. $P_V T^{(\alpha)} = 0$: the channel is symmetry-forbidden;
2. $P_V T^{(\alpha)} \neq 0$: the channel is generically nonvanishing on an open dense full-measure subset of V.

So the practical way to know whether the witness is nonzero is:

$$\boxed{\text{Compute } P_V T^{(\alpha)}.} \tag{517}$$

If it is nonzero, the witness survives generically.

3. 3. Seed/witness test

A stronger practical sufficient condition is: if one can construct even a single symmetry-allowed tensor

$$N^{(\alpha)}_{\text{seed}} \in V$$

such that

$$\langle T^{(\alpha)}, N^{(\alpha)}_{\text{seed}} \rangle \neq 0, \tag{518}$$

then automatically

$$P_V T^{(\alpha)} \neq 0,$$

so the channel is generically nonzero. [**?**]

Therefore, to determine whether nature makes the witness nonzero, one should:

1. determine the residual symmetry group H;
2. determine the allowed tensor space $V \subset W$;
3. either compute $P_V T^{(\alpha)}$ directly, or construct one explicit admissible seed tensor with nonzero overlap.

C. II. Existence of the opposite valley

1. 1. What is already known

At the shell level, the enriched Dirac block is built on the opposite shell pair

$$\pm G_*,$$

and the local shell space is

$$\mathbb{C}^2_{\text{pair}} \otimes \mathbb{C}^2_{\text{int}}.$$

This is already the correct carrier space for shell-level Dirac emergence. [**? ?**]

The shell-projected Dirac theorem requires that both opposite shell sectors contribute and that the projected block satisfy:

$$B_0 = 0, \qquad v_\parallel \neq 0, \qquad B(p) = p_1 W^1 + p_2 W^2 + O(|p|^2),$$

with W^1, W^2 nonzero, traceless, and independent after internal reduction. [**?**]

2. 2. What remains to be checked

The remaining microscopic question is not whether opposite-valley doubling would produce Dirac if present; that is already closed. The actual question is whether the microscopic linearization really leaves a nontrivial projected block at $-G_*$ with matching reduced data.

In practice, this is checked by:

1. constructing the reduced shell block at $+G_*$;
2. constructing the reduced shell block at $-G_*$;
3. verifying that both survive projection;
4. verifying the matching relation between the two reduced blocks.

In the witness formulation, this means computing the negative-valley witness triple and checking whether it matches the positive one.

D. III. Remote-sector gap

1. 1. What it means

Let the full microscopic Hilbert space be decomposed as

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R,$$

where $\mathcal{H}_D$ is the enriched local Dirac block and $\mathcal{H}_R$ is the remote sector. Spectral isolation means that the remote sector has a nonzero gap:

$$\text{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing \qquad \text{for some } \Delta > 0. \tag{519}$$

This is precisely the remaining spectral-isolation problem identified in the current TECT program. [**? ?**]

2. 2. How to determine it

This is a concrete band/spectral calculation:

1. construct the full Bloch or shell-Bloch operator in a finite reciprocal-shell truncation;
2. isolate the enriched $\pm G_*$ block;
3. diagonalize the complement;
4. determine whether all remote eigenvalues stay outside a fixed low-energy window.

Equivalently, one can prove an operator inequality showing that all states orthogonal to the enriched shell block have energy bounded away from zero by Δ.

Thus:

$$\boxed{\text{remote gap} = \text{finite-dimensional Bloch/spectral computation.}} \tag{520}$$

E. IV. Mass-protection symmetry

1. 1. What it means

Even if the local doubled shell block is Dirac and spectrally isolated, a constant Dirac mass term could in principle appear unless forbidden by symmetry.

So one must determine whether the reduced low-energy theory admits a residual symmetry (for example a grading or chiral-odd symmetry) excluding any constant mass operator.

2. 2. How to determine it

This is a symmetry-classification problem:

1. identify the residual symmetry group acting on the reduced Dirac block;
2. classify all allowed constant Hermitian 4×4 operators;
3. determine whether a Dirac mass matrix belongs to the allowed symmetry class.

If no symmetry-allowed constant mass matrix exists, then the Dirac cone is symmetry-protected.

Thus:

$$\boxed{\text{mass protection} = \text{classification of symmetry-allowed constant operators.}} \tag{521}$$

F. V. Practical research order

The optimal order of attack is now clear.

a. Step 1. Determine the actual symmetry-allowed tensor space V and compute

$$P_V T^{(1)}, \qquad P_V T^{(2)}.$$

This answers whether the shell witnesses are generically nonzero.

b. Step 2. Construct one explicit seed tensor for each channel. This immediately proves generic shell survival if the overlap is nonzero.

c. Step 3. Build the reduced shell blocks at $+G_*$ and $-G_*$, and compare their reduced coefficients or witness triples.

d. Step 4. Compute the remote spectrum and test whether a uniform gap $\Delta > 0$ exists.

e. Step 5. Classify the symmetry-allowed constant operators on the doubled block and determine whether a Dirac mass is forbidden.

G. VI. Honest conclusion

At the theorem level, the local shell-Dirac mechanism is already reduced to a finite problem. What remains is not a conceptual gap, but the explicit microscopic verification of four finite statements:

$$\boxed{P_V T^{(\alpha)} \neq 0 \text{ for the required channels;}} \tag{522}$$

$$\boxed{\text{the opposite shell valley survives projection;}} \tag{523}$$

$$\boxed{\operatorname{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing;} \tag{524}$$

$$\boxed{\text{no symmetry-allowed constant Dirac mass exists.}} \tag{525}$$

Once these are checked, the local Dirac part of TECT is effectively closed.

CXVII. FINAL SUMMARY

The remaining TECT microscopic questions are all finite verification problems.

First, the nonvanishing of the shell witnesses is controlled by the linear functionals

$$L_\alpha(N) = G_*^j e_i^\alpha N_{ij}^{a_\alpha}.$$

Let V be the symmetry-allowed tensor space and P_V the orthogonal projector onto V. Then

$$L_\alpha|_V \equiv 0 \quad \Longleftrightarrow \quad P_V T^{(\alpha)} = 0.$$

Hence a witness survives if and only if the corresponding projected target covector is nonzero. Equivalently, one may construct a single admissible seed tensor with nonzero overlap.

Second, opposite-valley existence is checked by constructing the reduced shell blocks at $\pm G_*$ and verifying that both survive projection and match appropriately.

Third, the remote-sector gap is a Bloch/spectral calculation:

$$\operatorname{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing$$

must be verified for the complement of the enriched shell block.

Fourth, mass protection is a symmetry-classification problem: one must determine whether any constant symmetry-allowed Hermitian operator on the doubled block can serve as a Dirac mass. If none exists, the massless Dirac cone is protected.

Therefore the remaining frontier is fully explicit: compute the projected witness covectors, verify opposite-valley survival, verify the remote gap, and classify the allowed constant operators on the local Dirac block.

CXVIII. RESIDUAL SYMMETRY CALCULATION FOR THE WITNESS TENSORS

Fix the BCC shell vector

$$G_* = \frac{1}{\sqrt{2}}(1, 1, 0).$$

Define

$$\hat{n} = \frac{G_*}{|G_*|}, \qquad e_1 = \frac{1}{\sqrt{2}}(1, -1, 0), \qquad e_2 = (0, 0, 1).$$

The stabilizer group of this axis inside the cubic group is

$$H = C_{2v}.$$

The microscopic tensor space is

$$W = \mathbb{R}^3 \otimes \mathbb{R}^3 \otimes U_{\text{int}}, \qquad U_{\text{int}} = \text{span}(s_1, s_2, s_3).$$

Imposing H-invariance gives the allowed tensor space

$$N^3_{ij} = g_{\parallel} \hat{n}_i \hat{n}_j,$$

$$N^1_{ij} = g_I \hat{n}_i e_{1j} - g_J \hat{n}_i e_{2j},$$

$$N^2_{ij} = g_I \hat{n}_i e_{2j} + g_J \hat{n}_i e_{1j}.$$

Hence

$$\dim V = 3.$$

For the target tensors

$$T^{(\alpha)}_{ija} = e^\alpha_i G^j_* u_a,$$

the contractions give

$$L_1(N) = |G_*| g_I,$$

$$L_2(N) = |G_*| g_J.$$

Therefore

$$P_V T^{(1)} \neq 0, \qquad P_V T^{(2)} \neq 0.$$

Thus symmetry does not forbid the transverse witnesses.
Consequently

$$\det M = (\kappa_* |G_*|)^3 g_{\parallel} (g_I^2 + g_J^2),$$

and Weyl nodes are symmetry-allowed generically.

CXIX. REMOTE-SECTOR GAP AS A FINITE BLOCH-SPECTRAL PROBLEM

A. Goal

The purpose of this section is to reduce the remaining remote-gap question

$$\mathrm{spec}(H_R) \cap (-\Delta, \Delta) = \varnothing$$

to a finite Bloch-spectral calculation and to give rigorous sufficient conditions guaranteeing $\Delta > 0$.

At the present stage, the local Dirac block is already reduced to a finite shell-enriched problem. What remains is to show that all states outside this local Dirac block stay uniformly away from zero energy in a neighborhood of the selected shell crossing.

We do not yet compute a numerical value of Δ, because that depends on the final microscopic operator and on the chosen reciprocal-shell truncation. What we do prove here is the exact finite-dimensional criterion and a rigorous lower-bound theorem.

B. Bloch decomposition and block splitting

Let $H_{\mathrm{full}}(k)$ denote the Bloch-reduced microscopic Hamiltonian near a selected BCC shell pair $\pm G_*$. Fix a small shell-relative momentum p and write

$$k = G_* + p.$$

After reciprocal-shell truncation and internal reduction, the Bloch operator becomes a finite Hermitian matrix. Choose the basis so that the local Dirac carrier appears first, and decompose the full Hilbert space as

$$\mathcal{H}_{\mathrm{full}} = \mathcal{H}_D \oplus \mathcal{H}_R,$$

where

- $\mathcal{H}_D$ is the selected four-dimensional local Dirac block built from the opposite shell pair $\pm G_*$,
- $\mathcal{H}_R$ is the orthogonal complement, called the remote sector.

Then

$$H_{\mathrm{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V^\dagger(p) & H_R(p) \end{pmatrix}. \tag{526}$$

The remote-gap problem concerns only the spectrum of $H_R(p)$.

C. Remote block as diagonal part plus residual mixing

Write the remote block as

$$H_R(p) = D_R(p) + R_R(p), \tag{527}$$

where:

- $D_R(p)$ is the diagonal part in the chosen truncated reciprocal-shell basis,
- $R_R(p)$ is the off-diagonal remote-remote mixing.

Let the diagonal entries of $D_R(p)$ be denoted by

$$E_r^{(0)}(p), \qquad r \in \mathcal{I}_R,$$

where $\mathcal{I}_R$ indexes the remote basis states.

Define the unperturbed remote spectral distance from zero by

$$\delta_0(p) := \min_{r\in\mathcal{I}_R} |E_r^{(0)}(p)|. \tag{528}$$

If

$$\delta_0(p) > 0,$$

then the diagonal remote energies are already separated from zero before remote-remote mixing is included.

The question is whether the off-diagonal perturbation $R_R(p)$ can close this gap.

D. Basic operator-norm lower bound

We now state the most elementary sufficient condition.

[Norm lower bound for the remote block] Let $H_R(p) = D_R(p) + R_R(p)$ with $D_R(p)$ diagonal and Hermitian, and $R_R(p)$ Hermitian. If

$$\delta_0(p) := \min_r |E_r^{(0)}(p)| > \|R_R(p)\|, \tag{529}$$

then

$$\operatorname{spec}(H_R(p)) \cap \big(-(\delta_0(p) - \|R_R(p)\|),\, \delta_0(p) - \|R_R(p)\|\big) = \varnothing. \tag{530}$$

In particular, the remote gap satisfies

$$\Delta(p) \geq \delta_0(p) - \|R_R(p)\|. \tag{531}$$

For any normalized vector $\psi \in \mathcal{H}_R$,

$$|\langle\psi, H_R(p)\psi\rangle| = |\langle\psi, D_R(p)\psi\rangle + \langle\psi, R_R(p)\psi\rangle|.$$

Since $D_R(p)$ is diagonal and every diagonal entry has absolute value at least $\delta_0(p)$, one has

$$|\langle\psi, D_R(p)\psi\rangle| \geq \delta_0(p).$$

Also,

$$|\langle\psi, R_R(p)\psi\rangle| \leq \|R_R(p)\|.$$

Therefore

$$|\langle\psi, H_R(p)\psi\rangle| \geq \delta_0(p) - \|R_R(p)\|.$$

By the variational characterization of the spectrum of a Hermitian operator, every eigenvalue of $H_R(p)$ has absolute value at least $\delta_0(p) - \|R_R(p)\|$. This proves (530) and (531).

E. Uniform remote gap in a momentum neighborhood

To use this in the spectral-isolation theorem, one needs a gap uniform in p near the crossing. Define a small neighborhood

$$\mathcal{N}_\rho := \{p : |p| \leq \rho\}.$$

Set

$$\delta_{0,\rho} := \inf_{|p|\leq\rho} \delta_0(p), \qquad \eta_\rho := \sup_{|p|\leq\rho} \|R_R(p)\|. \tag{532}$$

Then:

[Uniform remote-gap criterion] If

$$\delta_{0,\rho} > \eta_\rho, \tag{533}$$

then for all $|p| \leq \rho$,

$$\operatorname{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \varnothing, \qquad \Delta_\rho := \delta_{0,\rho} - \eta_\rho > 0. \tag{534}$$

Apply Proposition 1 pointwise in p, then take the infimum over $|p| \leq \rho$.

Thus the remote gap is reduced to the finite verification of the inequality

$$\delta_{0,\rho} > \eta_\rho.$$

F. Gershgorin refinement

In finite dimensions, one can also use Gershgorin discs. Let the remote matrix entries be

$$(H_R)_{rs}(p), \qquad r, s \in \mathcal{I}_R.$$

Define the row radii

$$\mathcal{R}_r(p) := \sum_{s \neq r} |(H_R)_{rs}(p)|. \tag{535}$$

Then every eigenvalue of $H_R(p)$ lies in the union of discs

$$\bigcup_{r \in \mathcal{I}_R} \{z \in \mathbb{C} : \; |z - (H_R)_{rr}(p)| \leq \mathcal{R}_r(p)\}.$$

Hence a sufficient condition for a positive remote gap is

$$\min_{r \in \mathcal{I}_R} \left(|(H_R)_{rr}(p)| - \mathcal{R}_r(p)\right) > 0. \tag{536}$$

Uniformly on $|p| \leq \rho$, define

$$\Delta_{\mathrm{G},\rho} := \inf_{|p| \leq \rho} \min_{r \in \mathcal{I}_R} \left(|(H_R)_{rr}(p)| - \mathcal{R}_r(p)\right). \tag{537}$$

If $\Delta_{\mathrm{G},\rho} > 0$, then

$$\operatorname{spec}(H_R(p)) \cap (-\Delta_{\mathrm{G},\rho}, \Delta_{\mathrm{G},\rho}) = \varnothing \qquad (|p| \leq \rho). \tag{538}$$

This is often more practical than the norm estimate because it uses the actual sparse matrix structure.

G. Application to the shell-truncated Bloch operator

In the TECT shell-reduced setting, remote states arise from:

1. higher-shell harmonics,
2. non-enriched shell states,
3. volumetric/longitudinal channels,
4. wrong internal sectors,
5. any remaining modes outside the selected $\pm G_*$ enriched Dirac block.

Thus the actual procedure is:

1. build the truncated Bloch matrix $H_{\mathrm{full}}(p)$;
2. identify the four-dimensional local Dirac block $\mathcal{H}_D$;
3. delete the rows and columns belonging to $\mathcal{H}_D$, thereby obtaining $H_R(p)$;
4. compute either

   $$\delta_{0,\rho}, \qquad \eta_\rho$$

   or the Gershgorin lower bound $\Delta_{\mathrm{G},\rho}$.

Any positive lower bound obtained this way is a rigorous remote gap.

H. Compatibility with the witness formulation

The witness data determine only the local Dirac block H_D, namely whether the selected two opposite valleys form a massless Dirac cone. The remote gap is logically independent of the witness nonvanishing.

Thus the full local Dirac stability criterion separates into two independent finite verifications:

a. Local witness block:

$$W_{\parallel}^{\pm} \neq 0, \qquad (W_I^{\pm}, W_J^{\pm}) \neq (0,0), \qquad (W_{\parallel}^{-}, W_I^{-}, W_J^{-}) = (W_{\parallel}^{+}, W_I^{+}, W_J^{+}).$$

b. Remote spectral block:

$$\Delta_\rho > 0 \quad \text{or equivalently} \quad \delta_{0,\rho} > \eta_\rho,$$

or any equivalent rigorous lower bound such as $\Delta_{\mathrm{G},\rho} > 0$.

I. Schur-complement consequence

Once the remote gap is established, and once

$$V(0) = 0, \qquad \|V(p)\| \leq c|p|$$

also hold, the self-energy

$$\Sigma(E,p) = V(p)(H_R(p) - E)^{-1} V^\dagger(p)$$

satisfies

$$\|\Sigma(E,p)\| \leq \frac{2c^2}{\Delta_\rho}|p|^2 \qquad (|E| < \Delta_\rho/2,\ |p| \leq \rho).$$

Therefore the effective low-energy Dirac Hamiltonian is

$$H_{\mathrm{eff}}(p) = \Gamma^i (Mp)_i + O(|p|^2/\Delta_\rho),$$

as already proved in the spectral-isolation theorem.

Thus the remote-gap calculation is exactly the last nontrivial spectral ingredient needed for full local Dirac stability.

J. Main theorem

[Remote-gap reduction theorem] Let $H_{\mathrm{full}}(p)$ be a finite Hermitian Bloch truncation near a selected shell pair $\pm G_*$, and let

$$\mathcal{H}_{\mathrm{full}} = \mathcal{H}_D \oplus \mathcal{H}_R$$

be the decomposition into the local Dirac block and the remote sector. Assume the remote block is written as

$$H_R(p) = D_R(p) + R_R(p),$$

with $D_R(p)$ diagonal and $R_R(p)$ Hermitian.

Define

$$\delta_{0,\rho} := \inf_{|p|\leq\rho} \min_r |E_r^{(0)}(p)|, \qquad \eta_\rho := \sup_{|p|\leq\rho} \|R_R(p)\|.$$

If

$$\delta_{0,\rho} > \eta_\rho,$$

then the remote sector has a uniform gap

$$\Delta_\rho := \delta_{0,\rho} - \eta_\rho > 0,$$

namely

$$\mathrm{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \varnothing \qquad (|p| \leq \rho).$$

Hence the remote-gap problem is reduced to a finite matrix calculation.

Immediate from the uniform remote-gap criterion.

K. What is now proved and what remains open

What is now rigorously established is:

1. the remote-gap problem is a finite Hermitian-matrix problem on the remote block $H_R(p)$;
2. a sufficient lower bound is
$$\Delta_\rho \geq \delta_{0,\rho} - \eta_\rho;$$
3. an alternative sufficient lower bound is given by Gershgorin discs;
4. once $\Delta_\rho > 0$ is obtained, the Schur-complement correction is $O(|p|^2/\Delta_\rho)$.

What remains open is not the form of the criterion, but its evaluation in the actual microscopic truncation chosen for TECT. That requires building the concrete Bloch matrix and computing the corresponding lower bound.

CXX. FINAL SUMMARY

The remote-sector gap is not a vague infinite-dimensional issue. After shell truncation and block decomposition

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R,$$

it becomes a finite Hermitian-matrix problem for the remote block

$$H_R(p).$$

Writing

$$H_R(p) = D_R(p) + R_R(p),$$

with $D_R(p)$ diagonal and $R_R(p)$ the remote-remote mixing, define

$$\delta_{0,\rho} := \inf_{|p|\leq\rho} \min_r |E_r^{(0)}(p)|, \qquad \eta_\rho := \sup_{|p|\leq\rho} \|R_R(p)\|.$$

If

$$\delta_{0,\rho} > \eta_\rho,$$

then the remote sector has a uniform gap

$$\Delta_\rho = \delta_{0,\rho} - \eta_\rho > 0,$$

so

$$\operatorname{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \varnothing \qquad (|p| \leq \rho).$$

Equivalently, one may use the Gershgorin lower bound

$$\Delta_{\mathrm{G},\rho} = \inf_{|p|\leq\rho} \min_r \left(|(H_R)_{rr}(p)| - \sum_{s\neq r} |(H_R)_{rs}(p)| \right).$$

Any positive lower bound gives a rigorous remote gap.

Therefore the remote-gap problem has been fully reduced to a finite Bloch-spectral calculation on the remote block. Once such a gap is established, the Schur-complement correction to the local Dirac block is only

$$O(|p|^2/\Delta_\rho),$$

which is exactly what is needed for full local Dirac stability.

CXXI. MASS-PROTECTION SYMMETRY AS A FINITE CLASSIFICATION PROBLEM

A. Goal

The purpose of this section is to reduce the mass-protection question to a finite algebraic classification problem on the doubled local Dirac block.

The key question is:

Does there exist a symmetry-allowed constant Hermitian operator on the local doubled block that acts as a Dirac mass?

This is no longer an infinite-dimensional PDE problem. Once the local shell-reduced doubled block has been identified, the problem reduces to classifying the allowed constant 4×4 Hermitian operators under the residual low-energy symmetry.

B. Local doubled Dirac block

Assume that the positive and negative valleys have already been matched in the witness sense, so that the local doubled block is Dirac. Then, to linear order, the effective Hamiltonian has the form

$$H_D(p) = \Gamma^i q_i, \qquad q_i = (Mp)_i, \tag{539}$$

where M is the 3×3 real coefficient matrix determined by the common witness triple, and the 4×4 Dirac matrices satisfy

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij}\mathbf{1}_4, \qquad i, j = 1, 2, 3. \tag{540}$$

The question is whether a constant Hermitian perturbation

$$\delta H = K \tag{541}$$

is symmetry-allowed and, if so, whether it can gap the Dirac node.

C. Classification of constant Hermitian operators

Let Herm(4) denote the real vector space of 4×4 Hermitian matrices. Its dimension is 16. Choose a basis adapted to the Dirac structure:

$$\mathrm{Herm}(4) = \mathrm{span}\left\{\mathbf{1}_4,\, \Gamma^1, \Gamma^2, \Gamma^3,\, \Gamma^{12}, \Gamma^{23}, \Gamma^{31},\, \Gamma^5,\, \Gamma^1\Gamma^5, \Gamma^2\Gamma^5, \Gamma^3\Gamma^5,\, \Gamma^{123},\, \Gamma^{12}\Gamma^5, \Gamma^{23}\Gamma^5, \Gamma^{31}\Gamma^5,\, \Gamma^{123}\Gamma^5\right\}, \tag{542}$$

where

$$\Gamma^{ij} := \frac{i}{2}[\Gamma^i, \Gamma^j], \qquad \Gamma^{123} := -i\Gamma^1\Gamma^2\Gamma^3,$$

and Γ^5 is an additional Hermitian matrix satisfying

$$\{\Gamma^5, \Gamma^i\} = 0, \qquad (\Gamma^5)^2 = \mathbf{1}_4. \tag{543}$$

For the mass-protection problem, only a small part of this basis matters.

D. Definition of a Dirac mass

A constant Hermitian operator K is called a Dirac mass if

$$\{K, \Gamma^i\} = 0 \qquad \forall i = 1, 2, 3, \tag{544}$$

and

$$K^2 = \mathbf{1}_4 \quad \text{(up to positive rescaling).} \tag{545}$$

Then the perturbed Hamiltonian

$$H_D^{(m)}(p) = \Gamma^i q_i + mK \tag{546}$$

has spectrum

$$E_\pm(p) = \pm\sqrt{|q|^2 + m^2}, \tag{547}$$

and is therefore gapped at $p = 0$.

So the existence of a symmetry-allowed mass matrix is exactly the obstruction to mass protection.

E. Existence of abstract mass matrices

Without imposing any residual symmetry, a Dirac mass generically exists. Indeed, by construction Γ^5 satisfies

$$\{\Gamma^5, \Gamma^i\} = 0,$$

so $K = \Gamma^5$ is a valid Dirac mass matrix.

Therefore:

$$\boxed{\text{mass protection never follows from the Dirac algebra alone;}} \tag{548}$$

it requires an additional symmetry that forbids the allowed mass matrices.

F. Residual symmetry and allowed operator algebra

Let G_{res} be the residual low-energy symmetry group acting on the doubled local Dirac block. A constant Hermitian operator K is symmetry-allowed if

$$U(g)KU(g)^{-1} = K \qquad \forall g \in G_{\text{res}}. \tag{549}$$

Define the symmetry-allowed operator algebra

$$\mathcal{A}_{\text{res}} := \{K \in \text{Herm}(4) \ : \ U(g)KU(g)^{-1} = K \ \forall g \in G_{\text{res}}\}. \tag{550}$$

Then the mass-protection question is equivalent to:

$$\boxed{\text{Does } \mathcal{A}_{\text{res}} \text{ contain any } K \text{ satisfying } \{K, \Gamma^i\} = 0 \text{ ?}} \tag{551}$$

Thus the problem is finite and purely algebraic.

G. Chiral/grading symmetry as a sufficient protection mechanism

A particularly important sufficient condition is a grading or chiral symmetry. Assume there exists a unitary Hermitian operator S on the doubled block such that

$$S^2 = \mathbf{1}_4, \tag{552}$$

and

$$S\Gamma^i S^{-1} = -\Gamma^i \qquad \forall i = 1, 2, 3. \tag{553}$$

Then the kinetic Hamiltonian is odd under S:

$$SH_D(p)S^{-1} = -H_D(p).$$

Suppose the low-energy theory is required to preserve this oddness condition, i.e. any allowed perturbation K must also satisfy

$$SKS^{-1} = -K. \tag{554}$$

Now let K_m be a candidate Dirac mass. If it is even under S, i.e.

$$SK_mS^{-1} = +K_m,$$

then it is forbidden.

Conversely, if all mass matrices are even under the grading while the symmetry only allows odd operators, then no Dirac mass is allowed.

This yields a sufficient criterion for mass protection.

H. Mass-protection theorem

[Finite mass-protection criterion] Let

$$H_D(p) = \Gamma^i q_i$$

be the local doubled Dirac block. Let G_{res} be the residual low-energy symmetry group and

$$\mathcal{A}_{\text{res}} = \{K \in \text{Herm}(4) : U(g)KU(g)^{-1} = K \ \forall g \in G_{\text{res}}\}$$

the corresponding allowed constant operator algebra.

If

$$\mathcal{A}_{\text{res}} \cap \{K \in \text{Herm}(4) : \{K, \Gamma^i\} = 0 \ \forall i\} = \{0\}, \tag{555}$$

then no symmetry-allowed constant Dirac mass exists, and the local Dirac cone is mass-protected.

Any constant gap-opening perturbation of Dirac type must be a Hermitian operator K anticommuting with all three kinetic gamma matrices. If no such operator belongs to the allowed constant symmetry algebra $\mathcal{A}_{\text{res}}$, then no symmetry-allowed Dirac mass can be added. Therefore the crossing remains massless under all symmetry-preserving constant perturbations.

I. Equivalent basis-level formulation

Choose any finite basis

$$\{B_A\}_{A=1}^{16}$$

of Herm(4). Then every constant Hermitian perturbation can be written

$$K = \sum_A c_A B_A.$$

The mass-protection test can then be performed in two finite steps:

a. Step 1: symmetry filter. Solve the linear equations

$$U(g)KU(g)^{-1} = K \qquad \forall g \in G_{\text{res}}, \tag{556}$$

to determine the allowed subspace $\mathcal{A}_{\text{res}} \subset \text{Herm}(4)$.

b. Step 2: mass filter. Inside that allowed subspace, solve the linear equations

$$\{K, \Gamma^i\} = 0 \qquad i = 1, 2, 3. \tag{557}$$

If the only solution is $K = 0$, then the Dirac mass is symmetry-forbidden.

Thus mass protection is a finite system of linear equations on 4×4 Hermitian matrices.

J. Relation to the spectral-isolation theorem

The spectral-isolation theorem already reduced full local Dirac stability to:

1. local witness-based Dirac formation;
2. remote-sector gap $\Delta > 0$;
3. zero constant mixing $V(0) = 0$;
4. linear mixing bound $\|V(p)\| \le c|p|$;
5. absence of a symmetry-allowed constant mass.

The present section closes the last item in the following sense: the mass-protection question is exactly the finite algebraic problem (555).

Hence the full local Dirac stability problem is now completely reduced to finite witness, spectral, intertwiner, and constant-operator classification checks.

K. Practical research procedure

To determine mass protection in the actual TECT realization, one should:

1. identify the actual residual low-energy symmetry group G_{res} acting on the local doubled Dirac block;
2. construct its 4×4 representation $U(g)$;
3. solve the invariance equations
$$U(g)KU(g)^{-1} = K$$
to obtain the allowed constant operator algebra $\mathcal{A}_{\text{res}}$;
4. solve the anticommutation equations
$$\{K, \Gamma^i\} = 0$$
inside $\mathcal{A}_{\text{res}}$;
5. conclude mass protection if and only if the only solution is $K = 0$.

L. Main consequence

Therefore the final unresolved mass question is not conceptual. It is exactly the finite classification problem:

$$\boxed{\text{Classify the symmetry-allowed constant } 4 \times 4 \text{ Hermitian operators on } \mathcal{H}_D.} \tag{558}$$

If none of them is a Dirac mass, the local Dirac cone is protected.

M. What is now proved and what remains open

What is now rigorously established:

1. the existence of a mass matrix is a finite 4×4 algebraic question;
2. mass protection is equivalent to the emptiness of the intersection
$$\mathcal{A}_{\text{res}} \cap \{K : \{K, \Gamma^i\} = 0\};$$

3. this is a finite linear classification problem once the residual symmetry representation is known.

What remains open microscopically:

1. the actual residual symmetry $G_{\rm res}$ of the TECT doubled block;
2. the explicit 4×4 representation $U(g)$;
3. the corresponding allowed constant operator algebra $\mathcal{A}_{\rm res}$.

CXXII. FINAL SUMMARY

The mass-protection question for the local doubled Dirac block is a finite 4×4 Hermitian-matrix classification problem.

Let the kinetic local Dirac Hamiltonian be

$$H_D(p) = \Gamma^i q_i, \qquad \{\Gamma^i, \Gamma^j\} = 2\delta^{ij}\mathbf{1}_4.$$

A constant Hermitian operator K is a Dirac mass precisely when

$$\{K, \Gamma^i\} = 0 \qquad (i = 1, 2, 3).$$

Without additional symmetry, such a mass generally exists.

Let $G_{\rm res}$ be the residual low-energy symmetry group acting on the doubled block, and let

$$\mathcal{A}_{\rm res} = \{K \in \mathrm{Herm}(4) : U(g)KU(g)^{-1} = K \ \forall g \in G_{\rm res}\}$$

be the algebra of symmetry-allowed constant Hermitian operators.

Then the local Dirac cone is mass-protected if and only if

$$\mathcal{A}_{\rm res} \cap \{K \in \mathrm{Herm}(4) : \{K, \Gamma^i\} = 0 \ \forall i\} = \{0\}.$$

Thus mass protection is not a vague dynamical question. It is exactly the finite problem of classifying the symmetry-allowed constant 4×4 Hermitian operators and checking whether any of them can serve as a Dirac mass.

CXXIII. MICROSCOPIC VERIFICATION ROADMAP

A. Purpose

The purpose of this roadmap is to compress the remaining microscopic verification tasks of TECT into a finite and operational research program.

At the present stage, the local shell-reduced fermion architecture is already reduced to the chain

$$\text{BCC shell background} \Longrightarrow N^a_{ij} \Longrightarrow (W_\parallel, W_I, W_J) \Longrightarrow \text{Weyl} \Longrightarrow \text{Dirac pairing} \Longrightarrow \text{spectral isolation.}$$

Therefore the remaining frontier is no longer conceptual. It is the explicit verification of four finite microscopic statements:

$$\boxed{\text{(I) witness nonvanishing}} \tag{559}$$

$$\boxed{\text{(II) opposite-valley survival and matching}} \tag{560}$$

$$\boxed{\text{(III) remote-sector spectral gap}} \tag{561}$$

$$\boxed{\text{(IV) mass-protection symmetry}} \tag{562}$$

B. Global logic

The complete local Dirac stability theorem now has the form:

1. if the positive and negative shell valleys each satisfy the Weyl witness condition,
2. and if their witness triples match,
3. and if the remote sector is gapped,
4. and if constant mixing vanishes,
5. and if no symmetry-allowed constant Dirac mass exists,

then the full low-energy theory contains a stable massless local Dirac cone.

Thus the remaining microscopic work is to verify the corresponding hypotheses one by one.

C. Task I: witness nonvanishing

1. Statement

The selected shell crossing is Weyl if

$$W_\parallel^\pm \neq 0, \qquad \big(W_I^\pm, W_J^\pm\big) \neq (0,0).$$

Equivalently,

$$\det M_\pm = (\kappa_* |G_*|)^3 W_\parallel^\pm \Big((W_I^\pm)^2 + (W_J^\pm)^2\Big) \neq 0.$$

2. Input

Let $V \subset W$ be the symmetry-allowed microscopic tensor space, where

$$W = \mathbb{R}^3_{\mathrm{out}} \otimes \mathbb{R}^3_{\mathrm{grad}} \otimes U_{\mathrm{int}}.$$

For each required channel α, define the shell target covector

$$T^{(\alpha)}_{ija} = e^\alpha_i G^j_* u^{(\alpha)}_a.$$

3. Operational test

Compute the orthogonal projection

$$P_V T^{(\alpha)}.$$

Then:

$$P_V T^{(\alpha)} = 0 \quad \Longleftrightarrow \quad \text{the channel is symmetry-forbidden,}$$

whereas

$$P_V T^{(\alpha)} \neq 0 \quad \Longleftrightarrow \quad \text{the channel is generically nonzero on an open dense full-measure subset of } V.$$

4. Minimal deliverable

For each required channel, produce one of the following:

1. an explicit computation of $P_V T^{(\alpha)} \neq 0$, or
2. one explicit admissible seed tensor $N^{(\alpha)}_{\mathrm{seed}} \in V$ such that

$$\langle T^{(\alpha)}, N^{(\alpha)}_{\mathrm{seed}} \rangle \neq 0.$$

5. Completion criterion

Task I is complete once the necessary channels satisfy

$$P_V T^{(\alpha)} \neq 0.$$

D. Task II: opposite-valley survival and matching

1. Statement

The two opposite shell valleys form a local Dirac block if the two witness triples satisfy

$$(W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+).$$

Equivalently,

$$h_-(p) = -h_+(-p)$$

in the matched opposite-valley frame.

2. Input

Construct the reduced shell blocks at

$$+G_* \qquad \text{and} \qquad -G_*.$$

Use the matched frame

$$\hat{n}_- = -\hat{n}_+, \qquad e_1^- = e_1^+, \qquad e_2^- = -e_2^+.$$

3. Operational test

Compute the witness triples

$$(W_\parallel^+, W_I^+, W_J^+), \qquad (W_\parallel^-, W_I^-, W_J^-)$$

from the two shell-projected first-order enriched operators.

4. Minimal deliverable

Show that:

1. both valleys survive the shell projection;
2. both valleys satisfy the Weyl nondegeneracy test;
3. the two witness triples are equal.

5. Completion criterion

Task II is complete once

$$W_\parallel^- = W_\parallel^+, \qquad W_I^- = W_I^+, \qquad W_J^- = W_J^+.$$

E. Task III: remote-sector spectral gap

1. Statement

Let

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R, \qquad H_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V(p) \\ V^\dagger(p) & H_R(p) \end{pmatrix}.$$

The remote sector must satisfy

$$\operatorname{spec}(H_R(p)) \cap (-\Delta, \Delta) = \varnothing$$

for some $\Delta > 0$ in a neighborhood of the crossing.

2. Input

Build a finite reciprocal-shell truncation of the Bloch-reduced microscopic Hamiltonian and isolate the remote block $H_R(p)$.

3. Operational test

Write

$$H_R(p) = D_R(p) + R_R(p),$$

with $D_R(p)$ diagonal and $R_R(p)$ off-diagonal. Define

$$\delta_{0,\rho} := \inf_{|p|\le\rho} \min_r |E_r^{(0)}(p)|, \qquad \eta_\rho := \sup_{|p|\le\rho} \|R_R(p)\|.$$

If

$$\delta_{0,\rho} > \eta_\rho,$$

then

$$\Delta_\rho := \delta_{0,\rho} - \eta_\rho > 0$$

is a rigorous remote gap.

Alternatively, use the Gershgorin lower bound

$$\Delta_{\mathrm{G},\rho} = \inf_{|p|\le\rho} \min_r \left(|(H_R)_{rr}(p)| - \sum_{s\neq r} |(H_R)_{rs}(p)| \right).$$

4. Minimal deliverable

Produce one explicit positive lower bound:

$$\Delta_\rho > 0 \qquad \text{or} \qquad \Delta_{\mathrm{G},\rho} > 0.$$

5. Completion criterion

Task III is complete once a strictly positive remote-gap lower bound is proved.

F. Task IV: mass-protection symmetry

1. Statement

Let the local doubled block be

$$H_D(p) = \Gamma^i q_i.$$

A constant Hermitian matrix K is a Dirac mass if

$$\{K, \Gamma^i\} = 0 \qquad (i = 1, 2, 3).$$

Mass protection means that no such mass matrix is symmetry-allowed.

2. Input

Let G_{res} be the residual low-energy symmetry group acting on the doubled block, with representation $U(g)$ on $\mathbb{C}^4$.

3. Operational test

Define the allowed constant Hermitian operator algebra

$$\mathcal{A}_{\text{res}} = \{K \in \text{Herm}(4) : U(g)KU(g)^{-1} = K \ \forall g \in G_{\text{res}}\}.$$

Then solve the finite system

$$\{K, \Gamma^i\} = 0 \qquad (i = 1, 2, 3)$$

inside $\mathcal{A}_{\text{res}}$.

4. Minimal deliverable

Show that

$$\mathcal{A}_{\text{res}} \cap \{K \in \text{Herm}(4) : \{K, \Gamma^i\} = 0 \ \forall i\} = \{0\}.$$

5. Completion criterion

Task IV is complete once the only symmetry-allowed mass candidate is $K = 0$.

G. Task V: zero constant mixing

1. Statement

At the shell crossing $p = 0$, the inter-sector mixing must satisfy

$$V(0) = 0.$$

2. Input

Let H_* be the shell-relevant residual symmetry at the crossing and let

$$\mathcal{H}_{\text{full}} = \mathcal{H}_D \oplus \mathcal{H}_R$$

be the corresponding decomposition.

3. Operational test

Compute the shell-relevant irrep decomposition of $\mathcal{H}_D$ and $\mathcal{H}_R$. If

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0,$$

then

$$V(0) = 0.$$

4. Minimal deliverable

Show that the local Dirac block and the remote sector share no common shell-relevant irreducible representation.

5. Completion criterion

Task V is complete once

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0$$

is proved.

H. Optimal order of execution

The most efficient order is:

1. Task I: compute $P_V T^{(\alpha)}$ or construct explicit seed tensors;
2. Task II: compute and match the two witness triples at $\pm G_*$;
3. Task V: verify the little-group mismatch and conclude $V(0) = 0$;
4. Task III: prove the remote gap $\Delta > 0$;
5. Task IV: classify the allowed constant 4×4 Hermitian operators and exclude Dirac masses.

This order is optimal because the first two steps confirm that the local shell-reduced Dirac block actually exists, while the last three certify its stability inside the full theory.

I. Global completion theorem

Once Tasks I–V are completed, the local fermion sector of TECT is effectively closed.
More precisely, one obtains:

$$\boxed{\text{actual nonzero shell witnesses}} \tag{563}$$

$$\Longrightarrow$$

$$\boxed{\text{actual Weyl valleys at } \pm G_*} \tag{564}$$

$$\Longrightarrow$$

$$\boxed{\text{actual local Dirac doubling}} \tag{565}$$

$$\Longrightarrow$$

$$\boxed{\text{stable spectrally isolated massless local Dirac cone}} \tag{566}$$

Thus the remaining frontier is no longer conceptual. It is a finite microscopic verification program.

J. Honest final status

At the present stage, the following are already closed:

1. the shell-reduced Weyl/Dirac architecture;
2. the witness formulation of local Weyl nondegeneracy;
3. the witness formulation of opposite-valley pairing;
4. the zero-mixing lemma;
5. the remote-gap reduction theorem;
6. the mass-protection reduction theorem.

What remains open is the explicit completion of Tasks I–V in the final microscopic realization adopted for TECT.

CXXIV. FINAL SUMMARY

Fix the selected BCC shell axis

$$\hat{n} = \frac{1}{\sqrt{2}}(1,1,0), \qquad e_1 = \frac{1}{\sqrt{2}}(1,-1,0), \qquad e_2 = (0,0,1).$$

Under the shell-relevant residual symmetry

$$H_{\rm sh} = SO(2)_\perp,$$

the shell-relevant tensor space reduces to

$$W_{\rm sh} = (\mathbb{R}\hat{n}) \otimes \mathbb{R}^3_{\rm mom} \otimes U_{\rm int}, \qquad U_{\rm int} \cong \mathbf{2}_s \oplus \mathbf{1}_3.$$

The symmetry-allowed invariant tensor space is exactly

$$V = \mathrm{span}\{B_\parallel, B_I, B_J\},$$

where

$$(B_\parallel)^3_{ij} = \hat{n}_i \hat{n}_j,$$

$$(B_I)^1_{ij} = \hat{n}_i e_{1j}, \qquad (B_I)^2_{ij} = \hat{n}_i e_{2j},$$

$$(B_J)^1_{ij} = -\hat{n}_i e_{2j}, \qquad (B_J)^2_{ij} = \hat{n}_i e_{1j}.$$

Hence

$$\dim V = 3.$$

For the transverse target tensors

$$T^{(1)} = B_I, \qquad T^{(2)} = B_J,$$

their projections satisfy

$$P_V T^{(1)} = T^{(1)} \neq 0, \qquad P_V T^{(2)} = T^{(2)} \neq 0.$$

Likewise the longitudinal target also survives projection.

Therefore none of the three shell-enriched channels is symmetry-forbidden. In particular, the two transverse witness channels are generically allowed by symmetry.

CXXV. TASK II: EXPLICIT COMPARISON OF THE POSITIVE AND NEGATIVE VALLEY WITNESS TRIPLES

A. Goal

The purpose of this section is to complete Task II inside the minimal projector/enriched ansatz.

More precisely, we shall prove that in the minimal shell-enriched projector completion, the witness triples at the two opposite shell points $+G_*$ and $-G_*$ are identical once the natural matched opposite-valley frame is chosen.

Equivalently, we shall prove that the standard pairing relation

$$h_-(p) = -h_+(-p)$$

holds automatically in the minimal ansatz.

B. Positive valley data

Fix the positive shell point

$$G_*, \qquad \hat{n}_+ := \frac{G_*}{|G_*|} = \hat{n},$$

with adapted right-handed frame

$$\{e_1^+, e_2^+, \hat{n}_+\} = \{e_1, e_2, \hat{n}\}. \tag{567}$$

The minimal shell-enriched tensor ansatz at the positive valley is

$$N_{ij}^{3(+)} = g_\parallel\, \hat{n}_i \hat{n}_j, \tag{568}$$

$$N_{ij}^{1(+)} = g_I\, \hat{n}_i e_{1j} - g_J\, \hat{n}_i e_{2j}, \tag{569}$$

$$N_{ij}^{2(+)} = g_I\, \hat{n}_i e_{2j} + g_J\, \hat{n}_i e_{1j}. \tag{570}$$

Define the positive-valley witness functionals by

$$W_\parallel^+[N^+] := \hat{n}_+^i \hat{n}_+^j N_{ij}^{3(+)}, \tag{571}$$

$$W_I^+[N^+] := \frac{1}{2}\left(e_{1j}^+ \hat{n}_+^i N_{ij}^{1(+)} + e_{2j}^+ \hat{n}_+^i N_{ij}^{2(+)}\right), \tag{572}$$

$$W_J^+[N^+] := \frac{1}{2}\left(e_{1j}^+ \hat{n}_+^i N_{ij}^{2(+)} - e_{2j}^+ \hat{n}_+^i N_{ij}^{1(+)}\right). \tag{573}$$

Substituting (568)–(570), one finds immediately

$$W_\parallel^+ = g_\parallel, \qquad W_I^+ = g_I, \qquad W_J^+ = g_J. \tag{574}$$

C. Matched opposite-valley frame

Now consider the opposite shell point

$$-G_*.$$

Define

$$\hat{n}_- := -\hat{n}.$$

To keep the adapted frame right-handed, choose

$$e_1^- := e_1, \qquad e_2^- := -e_2, \qquad \hat{n}_- := -\hat{n}. \tag{575}$$

Indeed,

$$e_1^- \times e_2^- = e_1 \times (-e_2) = -(e_1 \times e_2) = -\hat{n} = \hat{n}_-.$$

This is the canonical matched frame used throughout the opposite-valley analysis.

D. Negative valley minimal ansatz

We now write the minimal projector/enriched tensor ansatz at the negative valley in its own adapted frame:

$$N^{3(-)}_{ij} = g^-_\parallel\, \hat{n}^-_i \hat{n}^-_j\,, \tag{576}$$

$$N^{1(-)}_{ij} = g^-_I\, \hat{n}^-_i e^-_{1j} - g^-_J\, \hat{n}^-_i e^-_{2j}, \tag{577}$$

$$N^{2(-)}_{ij} = g^-_I\, \hat{n}^-_i e^-_{2j} + g^-_J\, \hat{n}^-_i e^-_{1j}. \tag{578}$$

Now substitute the matched frame (575):

$$\hat{n}^- = -\hat{n}, \qquad e^-_1 = e_1, \qquad e^-_2 = -e_2.$$

Then

$$N^{3(-)}_{ij} = g^-_\parallel\, (-\hat{n}_i)(-\hat{n}_j) = g^-_\parallel\, \hat{n}_i\hat{n}_j, \tag{579}$$

$$N^{1(-)}_{ij} = g^-_I\, (-\hat{n}_i)e_{1j} - g^-_J\, (-\hat{n}_i)(-e_{2j}) = -\, g^-_I\, \hat{n}_i e_{1j} - g^-_J\, \hat{n}_i e_{2j}, \tag{580}$$

$$N^{2(-)}_{ij} = g^-_I\, (-\hat{n}_i)(-e_{2j}) + g^-_J\, (-\hat{n}_i)e_{1j} = g^-_I\, \hat{n}_i e_{2j} - g^-_J\, \hat{n}_i e_{1j}. \tag{581}$$

This is simply the same invariant tensor pattern written in the common ambient basis $\{e_1, e_2, \hat{n}\}$.

E. Negative-valley witnesses

Define the negative-valley witnesses by

$$W^-_\parallel[N^-] := \hat{n}^i_- \hat{n}^j_- N^{3(-)}_{ij}, \tag{582}$$

$$W^-_I[N^-] := \frac{1}{2}\left(e^-_{1j}\hat{n}^i_- N^{1(-)}_{ij} + e^-_{2j}\hat{n}^i_- N^{2(-)}_{ij}\right), \tag{583}$$

$$W^-_J[N^-] := \frac{1}{2}\left(e^-_{1j}\hat{n}^i_- N^{2(-)}_{ij} - e^-_{2j}\hat{n}^i_- N^{1(-)}_{ij}\right). \tag{584}$$

We now compute them explicitly.

a. Longitudinal witness. Using (579),

$$\hat{n}^i_- \hat{n}^j_- N^{3(-)}_{ij} = (-\hat{n}_i)(-\hat{n}_j)\, g^-_\parallel \hat{n}_i\hat{n}_j = g^-_\parallel\,.$$

Hence

$$W^-_\parallel = g^-_\parallel\,. \tag{585}$$

b. Transverse-even witness. Using (580) and (581),

$$e^-_{1j}\hat{n}^i_- N^{1(-)}_{ij} = e_{1j}(-\hat{n}_i)\left(-\, g^-_I\, \hat{n}_i e_{1j} - g^-_J\, \hat{n}_i e_{2j}\right) = g^-_I\,,$$

because

$$(-\hat{n}_i)(-\hat{n}_i) = 1, \qquad e_1 \cdot e_1 = 1, \qquad e_1 \cdot e_2 = 0.$$

Likewise,

$$e^-_{2j}\hat{n}^i_- N^{2(-)}_{ij} = (-e_{2j})(-\hat{n}_i)\left(g^-_I\, \hat{n}_i e_{2j} - g^-_J\, \hat{n}_i e_{1j}\right) = g^-_I\,.$$

Therefore

$$W^-_I = g^-_I\,. \tag{586}$$

c. Transverse-odd witness. Again,

$$e^-_{1j}\hat{n}^i_- N^{2(-)}_{ij} = e_{1j}(-\hat{n}_i)\left(g^-_I\,\hat{n}_i e_{2j} - g^-_J\,\hat{n}_i e_{1j}\right) = g^-_J\,,$$

while

$$e^-_{2j}\hat{n}^i_- N^{1(-)}_{ij} = (-e_{2j})(-\hat{n}_i)\left(-\,g^-_I\,\hat{n}_i e_{1j} - g^-_J\,\hat{n}_i e_{2j}\right) = -\,g^-_J\,.$$

Hence

$$W^-_J = g^-_J\,. \tag{587}$$

Thus the negative-valley witness triple is

$$W^-_\parallel = g^-_\parallel\,, \qquad W^-_I = g^-_I\,, \qquad W^-_J = g^-_J\,. \tag{588}$$

F. Minimal ansatz equality of the two valleys

In the minimal projector/enriched completion, the same microscopic coupling constants multiply the same three invariant channels at both opposite valleys. This means precisely

$$g^-_\parallel = g^+_\parallel =: g_\parallel, \qquad g^-_I = g^+_I =: g_I, \qquad g^-_J = g^+_J =: g_J. \tag{589}$$

Combining (574), (588), and (589), we obtain

$$\boxed{W^-_\parallel = W^+_\parallel, \qquad W^-_I = W^+_I, \qquad W^-_J = W^+_J.} \tag{590}$$

Thus the witness triples match automatically in the minimal ansatz.

G. Consequence for the reduced Weyl blocks

At the positive valley, the reduced block is

$$h_+(p) = \lambda^+_\parallel p_\parallel \sigma^3 + \alpha^+(p_1\sigma^1 + p_2\sigma^2) + \beta^+(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2), \tag{591}$$

with

$$\lambda^+_\parallel = \kappa_*|G_*|\,W^+_\parallel, \qquad \alpha^+ = \kappa_*|G_*|\,W^+_I, \qquad \beta^+ = -\kappa_*|G_*|\,W^+_J. \tag{592}$$

At the negative valley,

$$h_-(p) = \lambda^-_\parallel p_\parallel \sigma^3 + \alpha^-(p_1\sigma^1 + p_2\sigma^2) + \beta^-(-p_2\sigma^1 + p_1\sigma^2) + O(|p|^2), \tag{593}$$

with

$$\lambda^-_\parallel = \kappa_*|G_*|\,W^-_\parallel, \qquad \alpha^- = \kappa_*|G_*|\,W^-_I, \qquad \beta^- = -\kappa_*|G_*|\,W^-_J. \tag{594}$$

Using (590), we obtain

$$\lambda^-_\parallel = \lambda^+_\parallel, \qquad \alpha^- = \alpha^+, \qquad \beta^- = \beta^+.$$

Hence

$$h_-(p) = -h_+(-p). \tag{595}$$

So the standard opposite-valley pairing relation is automatic in the minimal projector/enriched ansatz.

H. Task II completion theorem

[Completion of Task II in the minimal ansatz] Consider the minimal shell-enriched projector completion with invariant coupling constants

$$g_\parallel, \qquad g_I, \qquad g_J.$$

Let the positive and negative shell points $\pm G_*$ be equipped with the matched opposite-valley frame

$$\hat{n}_- = -\hat{n}_+, \qquad e_1^- = e_1^+, \qquad e_2^- = -e_2^+.$$

Then the corresponding witness triples satisfy

$$W_\parallel^- = W_\parallel^+, \qquad W_I^- = W_I^+, \qquad W_J^- = W_J^+.$$

Equivalently, the reduced shell Hamiltonians obey

$$h_-(p) = -h_+(-p).$$

Thus opposite-valley witness matching is automatic in the minimal projector/enriched ansatz.

The positive-valley witness triple equals $(g_\parallel^+, g_I^+, g_J^+)$, and the negative-valley witness triple equals $(g_\parallel^-, g_I^-, g_J^-)$, by direct calculation. In the minimal ansatz the same invariant coupling constants appear at the two opposite valleys, so

$$(g_\parallel^-, g_I^-, g_J^-) = (g_\parallel^+, g_I^+, g_J^+).$$

Therefore the witness triples match. The equivalence with the pairing relation follows from the coefficient-witness dictionary for the reduced Weyl blocks.

I. What is proved and what remains open

What is now rigorously established:

1. in the minimal projector/enriched completion, the negative-valley witness triple is computed explicitly;
2. in the matched opposite-valley frame, it equals the positive-valley witness triple;
3. therefore the local opposite-valley pairing condition is automatic in the minimal ansatz.

What remains open:

1. whether the full microscopic TECT dynamics realizes precisely this minimal ansatz without extra symmetry-breaking corrections;
2. whether remote-sector gap and mass-protection conditions hold in the final microscopic model.

CXXVI. FINAL SUMMARY

For the minimal shell-enriched projector completion, the positive-valley enriched tensors are

$$N_{ij}^{3(+)} = g_\parallel\, \hat{n}_i \hat{n}_j, \qquad N_{ij}^{1(+)} = g_I\, \hat{n}_i e_{1j} - g_J\, \hat{n}_i e_{2j}, \qquad N_{ij}^{2(+)} = g_I\, \hat{n}_i e_{2j} + g_J\, \hat{n}_i e_{1j},$$

and the corresponding witness triple is

$$(W_\parallel^+, W_I^+, W_J^+) = (g_\parallel, g_I, g_J).$$

At the negative valley, using the matched frame

$$\hat{n}_- = -\hat{n}, \qquad e_1^- = e_1, \qquad e_2^- = -e_2,$$

the minimal ansatz gives

$$(W_{\parallel}^{-}, W_{I}^{-}, W_{J}^{-}) = (g_{\parallel}, g_I, g_J)$$

as well.
Hence

$$W_{\parallel}^{-} = W_{\parallel}^{+}, \qquad W_{I}^{-} = W_{I}^{+}, \qquad W_{J}^{-} = W_{J}^{+}.$$

Therefore the reduced opposite-valley blocks satisfy

$$h_{-}(p) = -h_{+}(-p).$$

So Task II is completed at the level of the minimal projector/enriched ansatz: opposite-valley witness matching, and hence local Dirac pairing, is automatic.

CXXVII. MINIMAL TRUNCATION ANALYSIS FOR THE REMOTE GAP AND ZERO CONSTANT MIXING

A. Goal

The purpose of this section is to treat Task III (remote-sector gap) and Task V (zero constant mixing) simultaneously in the first nontrivial finite truncation.

We construct a minimal shell-relevant truncated Hilbert space containing:

1. the local doubled Dirac block $\mathcal{H}_D$,
2. a finite set of remote sectors carrying the nearest possible competing modes.

We then prove:

1. sectorwise shell-relevant label mismatch implies
$$V(0) = 0;$$
2. if the remote diagonal energies stay uniformly away from zero and remote-remote mixing is sufficiently small, then the remote sector has a uniform spectral gap
$$\Delta > 0.$$

Thus, within the minimal truncation, Tasks III and V reduce to explicit finite inequalities.

B. Minimal truncated Hilbert space

Let

$$\mathcal{H}_{\rm full}^{\rm tr} = \mathcal{H}_D \oplus \mathcal{H}_{\rm vol} \oplus \mathcal{H}_{\rm high} \oplus \mathcal{H}_{\rm non\text{–}enr} \oplus \mathcal{H}_{\rm wrong\text{–}int}.$$

Here:

1. $\mathcal{H}_D \cong \mathbb{C}^2_{\rm pair} \otimes \mathbb{C}^2_{\rm int}$ is the local doubled shell Dirac carrier;
2. $\mathcal{H}_{\rm vol}$ is the volumetric/longitudinal sector;
3. $\mathcal{H}_{\rm high}$ is the higher-shell sector not belonging to the selected shell pair $\pm G_*$;
4. $\mathcal{H}_{\rm non\text{–}enr}$ is the same-shell but non-enriched sector;
5. $\mathcal{H}_{\rm wrong\text{–}int}$ contains modes with internal content not matching the selected shell doublet.

This is the smallest truncation in which all previously identified dangerous remote channels are explicitly present.

C. Shell-relevant labels

Associate to each sector the following shell-relevant labels:

1. shell-pair label,
2. transverse vs. volumetric character,
3. enriched internal-doublet label,
4. any additional grading/parity label carried by the local Dirac block.

The local Dirac block $\mathcal{H}_D$ carries the label set

$$\mathfrak{L}_D = (\pm G_*,\ \text{transverse},\ \text{enriched doublet},\ \text{grading}_D). \tag{596}$$

The remote sectors carry, respectively,

$$\mathfrak{L}_{\text{vol}} = (\text{not shell-pair matched},\ \text{volumetric},\ \cdots), \tag{597}$$
$$\mathfrak{L}_{\text{high}} = (\text{different shell orbit},\ \cdots), \tag{598}$$
$$\mathfrak{L}_{\text{non–enr}} = (\pm G_*,\ \text{transverse},\ \text{no enriched doublet},\ \cdots), \tag{599}$$
$$\mathfrak{L}_{\text{wrong–int}} = (\pm G_*,\ \text{transverse},\ \text{wrong internal irrep},\ \cdots). \tag{600}$$

Each remote sector differs from $\mathfrak{L}_D$ in at least one shell-relevant label.

D. Zero constant mixing in the minimal truncation

Let H_* denote the shell-relevant residual symmetry group at the crossing $p = 0$. Assume the truncated Hamiltonian is H_*-equivariant at $p = 0$. Then the constant inter-sector mixing satisfies

$$V(0) \in \mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D), \qquad \mathcal{H}_R := \mathcal{H}_{\text{vol}} \oplus \mathcal{H}_{\text{high}} \oplus \mathcal{H}_{\text{non–enr}} \oplus \mathcal{H}_{\text{wrong–int}}.$$

We now verify sectorwise vanishing.

[Sectorwise zero constant mixing] In the minimal truncation (CXXVII B), suppose each remote sector differs from $\mathcal{H}_D$ in at least one shell-relevant label. Then

$$\mathrm{Hom}_{H_*}(\mathcal{H}_{\text{vol}}, \mathcal{H}_D) = 0, \qquad \mathrm{Hom}_{H_*}(\mathcal{H}_{\text{high}}, \mathcal{H}_D) = 0, \qquad \mathrm{Hom}_{H_*}(\mathcal{H}_{\text{non–enr}}, \mathcal{H}_D) = 0, \qquad \mathrm{Hom}_{H_*}(\mathcal{H}_{\text{wrong–int}}, \mathcal{H}_D) = 0. \tag{601}$$

Consequently,

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \qquad \Longrightarrow \qquad V(0) = 0. \tag{602}$$

The full remote space is a direct sum of the four sectoral subspaces, hence

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) \cong \bigoplus_{a \in \{\text{vol},\text{high},\text{non–enr},\text{wrong–int}\}} \mathrm{Hom}_{H_*}(\mathcal{H}_a, \mathcal{H}_D).$$

If a sector $\mathcal{H}_a$ differs from $\mathcal{H}_D$ in at least one shell-relevant label, no nonzero H_*-intertwiner can preserve all labels simultaneously. Hence each summand vanishes, and therefore the full Hom-space vanishes. The implication $V(0) = 0$ then follows from H_*-equivariance of the truncated Hamiltonian.

E. Minimal truncated Hamiltonian

Write the truncated Hamiltonian in block form

$$H^{\text{tr}}_{\text{full}}(p) = \begin{pmatrix} H_D(p) & V_{DR}(p) \\ V^{\dagger}_{DR}(p) & H_R(p) \end{pmatrix}, \tag{603}$$

with

$$H_R(p) = \begin{pmatrix} H_{\mathrm{vol}}(p) & R_{\mathrm{vol,high}}(p) & \cdots \\ R_{\mathrm{high,vol}}(p) & H_{\mathrm{high}}(p) & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}. \tag{604}$$

At $p = 0$, the Dirac-remote mixing vanishes by Proposition 1:

$$V_{DR}(0) = 0.$$

Thus the only spectral issue is whether the remote block $H_R(p)$ can approach zero energy.

F. Diagonal remote energies

Let

$$\mu_{\mathrm{vol}}(p), \qquad \mu_{\mathrm{high}}(p), \qquad \mu_{\mathrm{non\text{–}enr}}(p), \qquad \mu_{\mathrm{wrong\text{–}int}}(p)$$

denote the diagonal or sector-center energies of the four remote blocks. Define the pointwise unperturbed remote distance from zero:

$$\delta_0(p) := \min\left\{|\mu_{\mathrm{vol}}(p)|,\, |\mu_{\mathrm{high}}(p)|,\, |\mu_{\mathrm{non\text{–}enr}}(p)|,\, |\mu_{\mathrm{wrong\text{–}int}}(p)|\right\}. \tag{605}$$

Let

$$R_R(p) := H_R(p) - D_R(p), \tag{606}$$

where $D_R(p)$ is the block-diagonal or diagonal part containing the sector-center energies. Then define

$$\eta_R(p) := \|R_R(p)\|. \tag{607}$$

G. Pointwise remote-gap bound

[Pointwise remote-gap bound in the minimal truncation] If

$$\delta_0(p) > \eta_R(p), \tag{608}$$

then the remote sector has a positive pointwise gap

$$\Delta(p) \geq \delta_0(p) - \eta_R(p) > 0. \tag{609}$$

Apply the standard norm lower bound to the Hermitian remote matrix $H_R(p) = D_R(p) + R_R(p)$. Every diagonal remote eigenvalue is at least $\delta_0(p)$ away from zero, and the total remote-remote mixing has norm at most $\eta_R(p)$. Thus no remote eigenvalue can enter the interval

$$(-(\delta_0(p) - \eta_R(p)),\, \delta_0(p) - \eta_R(p)).$$

H. Uniform local remote gap

Fix a neighborhood

$$\mathcal{N}_\rho := \{p : |p| \leq \rho\}.$$

Define

$$\delta_{0,\rho} := \inf_{|p|\leq\rho} \delta_0(p), \qquad \eta_{R,\rho} := \sup_{|p|\leq\rho} \eta_R(p). \tag{610}$$

[Uniform remote gap in the minimal truncation] If

$$\delta_{0,\rho} > \eta_{R,\rho}, \tag{611}$$

then

$$\operatorname{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \varnothing \qquad (|p| \le \rho), \tag{612}$$

with

$$\Delta_\rho := \delta_{0,\rho} - \eta_{R,\rho} > 0. \tag{613}$$

Apply Proposition 2 pointwise and take the infimum over $|p| \le \rho$.
Thus the remote-gap problem in the minimal truncation is reduced to the finite inequality (611).

I. Gershgorin alternative

Let the remote matrix entries be $(H_R)_{ab}(p)$ in any fixed basis of $\mathcal{H}_R$. Define the row sums

$$\mathcal{R}_a(p) := \sum_{b \ne a} |(H_R)_{ab}(p)|. \tag{614}$$

Then a sufficient condition for a remote gap is

$$\min_a \left(|(H_R)_{aa}(p)| - \mathcal{R}_a(p)\right) > 0. \tag{615}$$

Uniformly on $|p| \le \rho$, define

$$\Delta_{\mathrm{G},\rho} := \inf_{|p| \le \rho} \min_a \left(|(H_R)_{aa}(p)| - \mathcal{R}_a(p)\right). \tag{616}$$

If $\Delta_{\mathrm{G},\rho} > 0$, then

$$\operatorname{spec}(H_R(p)) \cap (-\Delta_{\mathrm{G},\rho}, \Delta_{\mathrm{G},\rho}) = \varnothing \qquad (|p| \le \rho).$$

This is often easier to check in a sparse minimal truncation.

J. Consequence for Schur complement

Suppose, in addition to Theorem 1, that the linear Dirac-remote mixing obeys

$$\|V_{DR}(p)\| \le c|p|. \tag{617}$$

Then for $|E| < \Delta_\rho/2$,

$$(H_R(p) - E)^{-1}$$

exists and

$$\|(H_R(p) - E)^{-1}\| \le \frac{2}{\Delta_\rho}.$$

Hence the self-energy correction satisfies

$$\Sigma(E,p) = V_{DR}(p)(H_R(p) - E)^{-1} V_{DR}^\dagger(p) = O(|p|^2/\Delta_\rho). \tag{618}$$

Therefore the local Dirac block is stable up to quadratic corrections.

K. Combined Task III + Task V theorem

[Minimal-truncation completion of Tasks III and V] Consider the minimal truncated Hilbert space

$$\mathcal{H}_{\rm full}^{\rm tr} = \mathcal{H}_D \oplus \mathcal{H}_{\rm vol} \oplus \mathcal{H}_{\rm high} \oplus \mathcal{H}_{\rm non\text{–}enr} \oplus \mathcal{H}_{\rm wrong\text{–}int}.$$

Assume:

1. each remote sector differs from $\mathcal{H}_D$ in at least one shell-relevant label;
2. the truncated Hamiltonian is shell-relevant symmetry-equivariant at $p = 0$;
3. the uniform remote-gap inequality
$$\delta_{0,\rho} > \eta_{R,\rho}$$
holds in a neighborhood $|p| \leq \rho$;
4. the Dirac-remote mixing satisfies
$$\|V_{DR}(p)\| \leq c|p|.$$

Then:

1. the constant Dirac-remote mixing vanishes,
$$V_{DR}(0) = 0;$$
2. the remote sector has a uniform gap
$$\Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0;$$
3. the Schur-complement correction to the local Dirac block is
$$\Sigma(E, p) = O(|p|^2/\Delta_\rho).$$

Thus, in the minimal truncation, Tasks III and V are completed once the finite inequalities above are checked.
Item (1) follows from Proposition 1. Item (2) follows from Theorem 1. Item (3) follows from the resolvent bound and the quadratic self-energy estimate.

L. What is proved and what remains open

What is now rigorously established:

1. in the minimal truncation, the zero constant mixing statement is reduced to sectorwise shell-label mismatch;
2. the remote gap is reduced to the finite inequality
$$\delta_{0,\rho} > \eta_{R,\rho};$$
3. once this holds, the Schur-complement correction is only $O(|p|^2/\Delta_\rho)$.

What remains open:

1. the actual numerical or analytic evaluation of the remote energies $\mu_a(p)$ in the final microscopic truncation;
2. the actual value of the lower bound Δ_ρ;
3. the final mass-protection symmetry classification.

CXXVIII. FINAL SUMMARY

In the first nontrivial truncation,

$$\mathcal{H}_{\mathrm{full}}^{\mathrm{tr}} = \mathcal{H}_D \oplus \mathcal{H}_{\mathrm{vol}} \oplus \mathcal{H}_{\mathrm{high}} \oplus \mathcal{H}_{\mathrm{non\text{–}enr}} \oplus \mathcal{H}_{\mathrm{wrong\text{–}int}},$$

each remote sector differs from the local Dirac block $\mathcal{H}_D$ in at least one shell-relevant label.
Therefore, if the truncated Hamiltonian is shell-relevant symmetry-equivariant at $p = 0$, then

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0 \qquad \Longrightarrow \qquad V_{DR}(0) = 0.$$

Writing the remote block as

$$H_R(p) = D_R(p) + R_R(p),$$

and defining

$$\delta_{0,\rho} := \inf_{|p|\le\rho} \min_a |\mu_a(p)|, \qquad \eta_{R,\rho} := \sup_{|p|\le\rho} \|R_R(p)\|,$$

one obtains a uniform remote gap whenever

$$\delta_{0,\rho} > \eta_{R,\rho}.$$

In that case

$$\Delta_\rho = \delta_{0,\rho} - \eta_{R,\rho} > 0,$$

and hence

$$\mathrm{spec}(H_R(p)) \cap (-\Delta_\rho, \Delta_\rho) = \varnothing \qquad (|p| \le \rho).$$

If moreover

$$\|V_{DR}(p)\| \le c|p|,$$

then the Schur-complement self-energy is

$$\Sigma(E, p) = O(|p|^2/\Delta_\rho).$$

Thus, in the minimal truncation, zero constant mixing and remote spectral isolation are both reduced to explicit finite checks.

CXXIX. TASK IV: EXPLICIT CLASSIFICATION OF CONSTANT HERMITIAN OPERATORS ON THE MINIMAL DOUBLED DIRAC BLOCK

A. Goal

The purpose of this section is to complete Task IV for the minimal doubled Dirac ansatz.
We will:

1. choose an explicit representation of the local doubled Dirac block;
2. classify all constant Hermitian 4×4 operators;
3. determine exactly which of them are Dirac masses;
4. show that the minimal doubled Dirac algebra alone does *not* forbid mass terms;
5. identify a sufficient residual symmetry that forbids all mass matrices.

Thus the mass-protection problem is reduced to an explicit finite classification problem.

B. Minimal doubled Dirac representation

After opposite-valley matching, the local doubled block is

$$H_D(p) = \begin{pmatrix} h_+(p) & 0 \\ 0 & h_-(p) \end{pmatrix}, \qquad h_-(p) = -h_+(-p).$$

To linear order, after the linear change of variables $q_i = (Mp)_i$, this is unitarily equivalent to

$$H_D(q) = \rho^3 \otimes (q_1\sigma^1 + q_2\sigma^2 + q_3\sigma^3). \tag{619}$$

Here:

- ρ^a are Pauli matrices acting on the valley-pair index $(+,-)$,
- σ^a are Pauli matrices acting on the internal two-component Weyl spinor.

Thus the three kinetic gamma matrices may be taken as

$$\Gamma^i = \rho^3 \otimes \sigma^i, \qquad i = 1,2,3. \tag{620}$$

They satisfy

$$\{\Gamma^i, \Gamma^j\} = 2\delta^{ij}\mathbf{1}_4. \tag{621}$$

C. Full basis of constant Hermitian operators

Every constant Hermitian 4×4 matrix can be expanded in the basis

$$K = \sum_{a=0}^{3}\sum_{b=0}^{3} c_{ab}\, \rho^a \otimes \sigma^b, \qquad c_{ab} \in \mathbb{R}, \tag{622}$$

where

$$\rho^0 = \mathbf{1}_2, \qquad \sigma^0 = \mathbf{1}_2.$$

Hence

$$\dim_{\mathbb{R}} \mathrm{Herm}(4) = 16.$$

Our task is to determine which such K satisfy the Dirac-mass condition

$$\{K, \Gamma^i\} = 0 \qquad (i = 1,2,3). \tag{623}$$

D. Mass-space classification

Let

$$K = \rho^a \otimes \sigma^b$$

be a basis element. We compute

$$\{\rho^a \otimes \sigma^b,\ \rho^3 \otimes \sigma^i\} = \rho^a\rho^3 \otimes \sigma^b\sigma^i + \rho^3\rho^a \otimes \sigma^i\sigma^b.$$

There are two qualitatively distinct possibilities:

a. Case 1: ρ^a commutes with ρ^3. Then $a = 0$ or $a = 3$, and the anticommutator becomes

$$\rho^3\rho^a \otimes \{\sigma^b, \sigma^i\}.$$

For this to vanish for all $i = 1,2,3$, one would need a 2×2 matrix σ^b anticommuting with all three Pauli matrices simultaneously. No such nonzero matrix exists. Therefore no mass arises from $a = 0, 3$.

b. Case 2: ρ^a anticommutes with ρ^3. Then $a = 1$ or $a = 2$, and the anticommutator becomes

$$\rho^a \rho^3 \otimes [\sigma^b, \sigma^i].$$

For this to vanish for all $i = 1, 2, 3$, one needs σ^b to commute with all three Pauli matrices. Hence σ^b must be proportional to the identity, i.e. $b = 0$.

Therefore the only nonzero Hermitian matrices anticommuting with all three kinetic gamma matrices are

$$\rho^1 \otimes \mathbf{1}_2, \qquad \rho^2 \otimes \mathbf{1}_2. \tag{624}$$

We summarize this in the following theorem.

[Mass-space classification for the minimal doubled Dirac block] For the minimal doubled Dirac Hamiltonian

$$H_D(q) = \rho^3 \otimes (q_i \sigma^i),$$

the real vector space of constant Hermitian Dirac masses is exactly

$$\mathcal{M} = \mathrm{span}_{\mathbb{R}}\{\rho^1 \otimes \mathbf{1}_2,\ \rho^2 \otimes \mathbf{1}_2\}. \tag{625}$$

In particular,

$$\dim_{\mathbb{R}} \mathcal{M} = 2.$$

The above case-by-case analysis shows that only $a = 1, 2$ and $b = 0$ solve

$$\{K, \rho^3 \otimes \sigma^i\} = 0 \qquad (i = 1, 2, 3).$$

Hence the mass space is exactly (625).

E. Consequence: no automatic mass protection

Because $\mathcal{M} \neq \{0\}$, the local Dirac algebra by itself does not protect the massless node. Indeed, for any nonzero real pair (m_1, m_2), the operator

$$K_m = m_1\, \rho^1 \otimes \mathbf{1}_2 + m_2\, \rho^2 \otimes \mathbf{1}_2 \tag{626}$$

is a valid constant mass perturbation. The perturbed Hamiltonian

$$H_D^{(m)}(q) = \rho^3 \otimes (q_i \sigma^i) + m_1\, \rho^1 \otimes \mathbf{1}_2 + m_2\, \rho^2 \otimes \mathbf{1}_2 \tag{627}$$

has spectrum

$$E_\pm(q) = \pm\sqrt{|q|^2 + m_1^2 + m_2^2}, \tag{628}$$

so it is gapped at $q = 0$.

Hence:

$$\boxed{\text{the minimal doubled Dirac block alone does not forbid a mass.}} \tag{629}$$

F. Valley-charge symmetry as a sufficient protection mechanism

We now identify a natural residual symmetry that forbids all such masses.

Define the valley-charge generator

$$Q_v := \rho^3 \otimes \mathbf{1}_2. \tag{630}$$

Consider the unitary valley $U(1)$ symmetry

$$U_v(\theta) = e^{i\theta Q_v} = e^{i\theta \rho^3} \otimes \mathbf{1}_2. \tag{631}$$

The kinetic Hamiltonian (619) is invariant under this symmetry because

$$[Q_v, \rho^3 \otimes \sigma^i] = 0 \qquad (i = 1, 2, 3). \tag{632}$$

Now examine the two mass generators. Using Pauli-matrix algebra,

$$U_v(\theta)\,(\rho^1 \otimes \mathbf{1})\,U_v(\theta)^{-1} = (\cos 2\theta)\,\rho^1 \otimes \mathbf{1} + (\sin 2\theta)\,\rho^2 \otimes \mathbf{1}, \tag{633}$$

$$U_v(\theta)\,(\rho^2 \otimes \mathbf{1})\,U_v(\theta)^{-1} = -(\sin 2\theta)\,\rho^1 \otimes \mathbf{1} + (\cos 2\theta)\,\rho^2 \otimes \mathbf{1}. \tag{634}$$

Thus the mass space $\mathcal{M}$ transforms as a nontrivial doublet under valley $U(1)$, rather than as an invariant scalar.
Therefore no nonzero element of $\mathcal{M}$ is invariant under all θ, and we obtain:

$$\mathcal{A}_{U(1)_v} \cap \mathcal{M} = \{0\}, \tag{635}$$

where

$$\mathcal{A}_{U(1)_v} := \{K \in \mathrm{Herm}(4) : U_v(\theta) K U_v(\theta)^{-1} = K \ \forall \theta\}. \tag{636}$$

[Valley $U(1)$ mass protection] If the doubled local Dirac block preserves valley-charge $U(1)$ generated by $Q_v = \rho^3 \otimes \mathbf{1}_2$, then every constant Dirac mass is forbidden. Equivalently,

$$\mathcal{A}_{U(1)_v} \cap \mathcal{M} = \{0\}.$$

Hence the local Dirac node is mass-protected against all constant symmetry-preserving perturbations.
The kinetic Hamiltonian commutes with Q_v, so the symmetry is compatible with the local Dirac block. The mass generators $\rho^1 \otimes \mathbf{1}$ and $\rho^2 \otimes \mathbf{1}$ rotate nontrivially under $U_v(\theta)$, by (633) and (634). Therefore no nonzero linear combination of them is invariant under all θ. Hence no constant Dirac mass belongs to the allowed symmetry algebra.

G. Equivalent finite classification of the allowed constant algebra

For completeness, let us classify the constant Hermitian operators invariant under valley $U(1)$.
A basis element $\rho^a \otimes \sigma^b$ is $U(1)_v$-invariant iff it commutes with $Q_v = \rho^3 \otimes \mathbf{1}$. Hence only $a = 0$ and $a = 3$ survive. Therefore

$$\mathcal{A}_{U(1)_v} = \mathrm{span}_{\mathbb{R}}\{\mathbf{1} \otimes \mathbf{1},\, \mathbf{1} \otimes \sigma^1,\, \mathbf{1} \otimes \sigma^2,\, \mathbf{1} \otimes \sigma^3,\, \rho^3 \otimes \mathbf{1},\, \rho^3 \otimes \sigma^1,\, \rho^3 \otimes \sigma^2,\, \rho^3 \otimes \sigma^3\}. \tag{637}$$

This is an 8-dimensional real algebra. Its intersection with the mass space $\mathcal{M}$ is clearly zero.

H. Interpretation

The two mass matrices

$$\rho^1 \otimes \mathbf{1}, \qquad \rho^2 \otimes \mathbf{1}$$

are precisely intervalley constant mixing terms. They convert the $+$ and $-$ valley sectors into one another. Thus valley-charge conservation forbids them exactly because it forbids constant intervalley mixing.
This is the cleanest finite interpretation of mass protection in the minimal doubled Dirac ansatz.

I. General finite mass-protection criterion

We now state the final criterion in the present explicit basis.
[Explicit Task IV completion criterion] For the minimal doubled local Dirac block

$$H_D(q) = \rho^3 \otimes (q_i \sigma^i),$$

the constant Dirac mass space is

$$\mathcal{M} = \mathrm{span}_{\mathbb{R}}\{\rho^1 \otimes \mathbf{1},\, \rho^2 \otimes \mathbf{1}\}.$$

Hence Task IV is completed once one proves that the residual symmetry-allowed constant algebra $\mathcal{A}_{\mathrm{res}}$ satisfies

$$\mathcal{A}_{\mathrm{res}} \cap \mathrm{span}_{\mathbb{R}}\{\rho^1 \otimes \mathbf{1},\, \rho^2 \otimes \mathbf{1}\} = \{0\}. \tag{638}$$

A sufficient condition is preservation of valley-charge $U(1)$.

The mass-space classification theorem gives the full list of possible constant Dirac masses. Therefore mass protection is equivalent to excluding these two matrices from the allowed constant algebra. The valley $U(1)$ theorem provides one explicit sufficient mechanism.

J. What is proved and what remains open

What is now rigorously established:

1. the constant mass space of the minimal doubled Dirac block is exactly two-dimensional;
2. without extra symmetry, mass terms are allowed;
3. valley-charge $U(1)$ forbids all constant masses;
4. Task IV is reduced to checking whether the actual residual symmetry algebra excludes $\rho^1 \otimes \mathbf{1}$ and $\rho^2 \otimes \mathbf{1}$.

What remains open microscopically:

1. whether the final TECT microscopic realization indeed preserves valley-charge $U(1)$ or an equivalent symmetry;
2. whether additional symmetry-breaking terms generate intervalley constant operators;
3. whether the final allowed constant algebra $\mathcal{A}_{\mathrm{res}}$ coincides with or is contained in $\mathcal{A}_{U(1)_v}$.

CXXX. FINAL SUMMARY

For the minimal doubled local Dirac block

$$H_D(q) = \rho^3 \otimes (q_1\sigma^1 + q_2\sigma^2 + q_3\sigma^3),$$

every constant Hermitian 4×4 operator can be expanded in the basis

$$\rho^a \otimes \sigma^b, \qquad a, b = 0, 1, 2, 3.$$

The constant Hermitian operators that anticommute with all three kinetic gamma matrices

$$\Gamma^i = \rho^3 \otimes \sigma^i$$

form the two-dimensional mass space

$$\mathcal{M} = \mathrm{span}_{\mathbb{R}}\{\rho^1 \otimes \mathbf{1},\, \rho^2 \otimes \mathbf{1}\}.$$

Therefore the minimal doubled Dirac algebra alone does not protect the massless node: without extra symmetry, a constant Dirac mass is allowed.

However, if the residual symmetry includes valley-charge $U(1)$ generated by

$$Q_v = \rho^3 \otimes \mathbf{1},$$

then the two mass generators transform nontrivially and are not symmetry-invariant. Hence

$$\mathcal{A}_{U(1)_v} \cap \mathcal{M} = \{0\},$$

so all constant Dirac masses are forbidden.

Thus Task IV is reduced to the explicit finite check: determine the actual residual constant symmetry algebra $\mathcal{A}_{\mathrm{res}}$ and verify whether it excludes

$$\rho^1 \otimes \mathbf{1}, \qquad \rho^2 \otimes \mathbf{1}.$$

A sufficient condition is preservation of valley-charge $U(1)$.

CXXXI. FINAL SEPARATION BETWEEN WHAT IS AUTOMATIC IN THE MINIMAL ANSATZ AND WHAT MUST BE VERIFIED IN THE ACTUAL MICROSCOPIC TECT REALIZATION

A. Goal

The purpose of this section is to make a definitive separation between:

1. statements that are automatic inside the minimal shell-enriched projector ansatz;
2. statements that still require explicit verification in the actual microscopic TECT realization.

This separation is essential. At the present stage, the local shell-reduced fermion framework is already structurally closed, but the final microscopic realization is not yet fully verified.

Thus the correct logical status is:

$$\boxed{\text{The local Weyl/Dirac architecture is closed at the minimal-ansatz level,}}$$

whereas

$$\boxed{\text{the final microscopic TECT realization still requires explicit finite checks.}}$$

B. I. What is automatic in the minimal ansatz

We first collect the statements that follow automatically once one adopts the minimal shell-enriched projector completion.

1. 1. Minimal condensate PDE class

Under translation invariance, pre-condensation isotropy, finite-shell instability, and quartic stabilization, the minimal effective condensate PDE is of Landau–Brazovskii type:

$$\partial_t \Phi = -r\Phi + \kappa\nabla^2\Phi - \lambda\nabla^4\Phi - u|\Phi|^2\Phi, \qquad q_*^2 = -\frac{\kappa}{2\lambda}.$$

Thus the existence of a finite-shell instability and the corresponding BCC shell background are automatic at the minimal effective level.

2. 2. Allowed shell-enriched tensor space

For the shell-relevant residual symmetry

$$H_{\rm sh} = SO(2)_\perp,$$

the allowed shell-relevant tensor space is exactly three-dimensional:

$$V = \mathrm{span}\{B_\parallel, B_I, B_J\}.$$

Therefore none of the three shell-enriched channels is symmetry-forbidden:

$$P_V T^{(\parallel)} \neq 0, \qquad P_V T^{(1)} \neq 0, \qquad P_V T^{(2)} \neq 0.$$

3. 3. Explicit enriched tensor form

The minimal shell-enriched projector coupling produces the explicit tensor triple

$$N^3_{ij} = g_\parallel\, \hat n_i \hat n_j,$$

$$N^1_{ij} = g_I\, \hat n_i e_{1j} - g_J\, \hat n_i e_{2j},$$

$$N^2_{ij} = g_I\, \hat n_i e_{2j} + g_J\, \hat n_i e_{1j}.$$

Thus the microscopic shell-enriched tensor structure is explicit in the minimal ansatz.

4. 4. Exact witness extraction

The witness functionals

$$W_\parallel, \qquad W_I, \qquad W_J$$

are exactly dual to the invariant tensor basis and satisfy

$$W_\parallel = g_\parallel, \qquad W_I = g_I, \qquad W_J = g_J.$$

Equivalently,

$$W_\parallel[N] = \hat n_i \hat n_j N^3_{ij},$$

$$W_I[N] = \frac{1}{2}\left(e_{1j}\hat n_i N^1_{ij} + e_{2j}\hat n_i N^2_{ij}\right),$$

$$W_J[N] = \frac{1}{2}\left(e_{1j}\hat n_i N^2_{ij} - e_{2j}\hat n_i N^1_{ij}\right).$$

5. 5. Local Weyl criterion

In the minimal ansatz, the reduced Weyl coefficients are

$$\lambda_\parallel = \kappa_*|G_*|\, g_\parallel, \qquad \alpha = \kappa_*|G_*|\, g_I, \qquad \beta = -\kappa_*|G_*|\, g_J,$$

and therefore

$$\det M = (\kappa_*|G_*|)^3\, g_\parallel\, (g_I^2 + g_J^2) = (\kappa_*|G_*|)^3\, W_\parallel\, (W_I^2 + W_J^2).$$

Hence local Weyl nondegeneracy is automatic whenever

$$g_\parallel \neq 0, \qquad (g_I, g_J) \neq (0,0),$$

or equivalently

$$W_\parallel \neq 0, \qquad (W_I, W_J) \neq (0,0).$$

6. 6. Opposite-valley pairing in the minimal ansatz

In the matched opposite-valley frame

$$\hat n_- = -\hat n_+, \qquad e_1^- = e_1^+, \qquad e_2^- = -e_2^+,$$

the minimal ansatz uses the same three invariant coupling constants at $\pm G_*$. Therefore the witness triples satisfy

$$(W_\parallel^-, W_I^-, W_J^-) = (W_\parallel^+, W_I^+, W_J^+),$$

and hence

$$h_-(p) = -h_+(-p).$$

Thus local opposite-valley pairing is automatic in the minimal ansatz.

7. 7. Constant mass-space classification

For the minimal doubled Dirac block

$$H_D(q) = \rho^3 \otimes (q_i \sigma^i),$$

the constant Hermitian Dirac mass space is exactly

$$\mathcal{M} = \mathrm{span}_{\mathbb{R}}\{\rho^1 \otimes \mathbf{1},\ \rho^2 \otimes \mathbf{1}\}.$$

Thus the mass problem is completely reduced to a finite 4×4 matrix classification.

C. II. What must still be verified in the actual microscopic TECT realization

We now list the statements that do *not* follow automatically from the minimal ansatz and must still be checked in the final microscopic model.

1. 1. Actual nonvanishing of the witness triple

The minimal ansatz shows that the three channels are symmetry-allowed. However, symmetry allowance does not imply actual nonvanishing. In the full microscopic theory one must still verify that

$$W_{\parallel}^{\pm} \neq 0, \qquad (W_I^{\pm}, W_J^{\pm}) \neq (0,0).$$

Equivalently, one must compute the actual microscopic projected tensor coefficients and confirm that they do not vanish by accidental cancellation or parameter tuning.

Thus:

$$\boxed{\text{symmetry allowance is automatic, actual witness nonvanishing is not.}}$$

2. 2. Actual realization of the minimal ansatz

The minimal shell-enriched projector completion is a symmetry-compatible effective ansatz. What is not yet proved is that the final microscopic TECT dynamics generates precisely this minimal operator structure with no additional symmetry-breaking first-order corrections.

Thus one must still verify that the actual microscopic linearization reduces to the same invariant basis

$$B_{\parallel}, \quad B_I, \quad B_J$$

with the same shell-relevant coupling pattern.

3. 3. Actual opposite-valley survival

Within the minimal ansatz, opposite-valley matching is automatic. However, in the final microscopic realization one must still verify that the reduced shell projection actually leaves a nontrivial surviving block at both $+G_*$ and $-G_*$.

So the remaining question is not whether matched opposite-valley pairing implies Dirac – that is already closed. The remaining question is whether the actual microscopic theory indeed realizes the two required shell blocks.

4. 4. Zero constant mixing in the actual remote-sector decomposition

In the abstract theorem, if

$$\mathrm{Hom}_{H_*}(\mathcal{H}_R, \mathcal{H}_D) = 0,$$

then

$$V(0) = 0.$$

In the minimal truncation, the remote sectors were classified by shell-relevant labels, and this gave a sufficient sectorwise mismatch criterion.

What remains to be checked microscopically is that the actual remote-sector content of the final TECT truncation indeed shares no common shell-relevant irreducible component with the local Dirac block.

So:

$$\boxed{\text{the zero-mixing theorem is closed, but the actual remote-sector irrep input must still be checked.}}$$

5. 5. Actual remote spectral gap

The remote-gap problem has already been reduced to a finite Bloch-spectral inequality:

$$\delta_{0,\rho} > \eta_{R,\rho} \quad \Longrightarrow \quad \Delta_\rho > 0.$$

However, the numerical or analytic value of Δ_ρ depends on the actual microscopic truncation and has not yet been computed.

Thus:

$$\boxed{\text{the gap criterion is closed, but the actual gap value is not yet verified.}}$$

6. 6. Actual residual symmetry for mass protection

The mass-protection problem is fully reduced to the finite check

$$\mathcal{A}_{\text{res}} \cap \operatorname{span}_{\mathbb{R}}\{\rho^1 \otimes \mathbf{1},\, \rho^2 \otimes \mathbf{1}\} = \{0\}.$$

A sufficient condition is preservation of valley-charge $U(1)$. What remains open is whether the actual microscopic TECT realization preserves such a symmetry, or an equivalent one, after all shell-enriched corrections are included.

Thus:

$$\boxed{\text{the mass-classification theorem is closed, but the actual residual symmetry algebra is not yet fixed.}}$$

D. III. Final truth table

We can now summarize the exact current logical status.

Statement	Minimal ansatz	Actual microscopic TECT
Finite-shell condensate PDE class	automatic	must be realized dynamically
Allowed tensor basis $B_\parallel, B_I, B_J$	automatic	must appear in actual linearization
Symmetry allowance of transverse channels	automatic	yes, inherited if same symmetry survives
Witness extraction formulas	automatic	automatic once the same shell basis is realized
Local Weyl criterion formula	automatic	automatic once actual witnesses are known
Opposite-valley witness matching	automatic	must verify actual $-G_*$ block survives and matches
Zero-mixing theorem	automatic theorem	must verify actual irrep mismatch input
Remote-gap theorem	automatic theorem	must verify actual finite gap inequality
Mass-space classification	automatic theorem	must verify actual residual symmetry excludes masses
Stable massless local Dirac cone	conditional	requires all actual checks

E. IV. Precise current status

The most honest global statement is therefore:

$$\boxed{\text{At the minimal shell-enriched projector level, the local Weyl/Dirac architecture is essentially complete.}} \tag{639}$$

But:

$$\boxed{\text{At the final microscopic TECT level, four explicit checks still remain:}} \tag{640}$$

namely

1. actual witness nonvanishing,
2. actual opposite-valley survival,
3. actual remote gap,
4. actual residual symmetry excluding the two mass generators.

F. V. Final theorem-level formulation

We may summarize the situation in theorem form.
[Final separation theorem] The following are fully closed at the level of the minimal shell-enriched projector ansatz:

1. the shell-relevant invariant tensor basis;
2. the explicit witness extraction;
3. the local Weyl determinant formula;
4. the opposite-valley witness matching mechanism;
5. the zero-mixing theorem;
6. the remote-gap reduction theorem;
7. the mass-space classification theorem.

The following are not yet automatic and must still be verified in the actual microscopic TECT realization:

1. actual nonvanishing of the positive and negative witness triples;
2. actual survival of the opposite shell valley;
3. actual irrep mismatch between the remote sector and the local Dirac block;
4. actual remote spectral gap;
5. actual residual symmetry forbidding the two intervalley mass matrices.

All items in the first list were proved in the shell-reduced minimal ansatz by explicit invariant-basis construction, shell projection, opposite-valley matching, block decomposition, spectral reduction, and finite 4×4 mass classification. The second list consists precisely of the remaining input data required to transfer those abstract closed theorems into the final microscopic realization.

G. VI. Honest conclusion

The correct final conclusion is not "the entire proof is finished," but rather:

$$\boxed{\text{the theorem framework is finished, and the remaining work is explicit microscopic verification.}}$$

This is already a major achievement, because the unresolved part is no longer a conceptual uncertainty. It is a finite checklist.

CXXXII. FINAL SUMMARY

At the minimal shell-enriched projector level, the local fermion architecture is essentially closed.

Automatic in the minimal ansatz are:

$$\text{the allowed tensor basis } \{B_{\parallel}, B_I, B_J\},$$

$$\text{the witness extraction formulas } (W_{\parallel}, W_I, W_J),$$

$$\det M = (\kappa_* |G_*|)^3 W_{\parallel} \big(W_I^2 + W_J^2\big),$$

$$(W_{\parallel}^-, W_I^-, W_J^-) = (W_{\parallel}^+, W_I^+, W_J^+),$$

and the finite reduction of both the remote-gap problem and the mass-protection problem.

What is not yet automatic in the final microscopic TECT realization is:

actual witness nonvanishing, actual opposite-valley survival, actual remote spectral gap, actual residual symmetry e

Therefore the present honest status is: the theorem structure is finished, while the remaining work is an explicit finite microscopic verification program.